



GE Medical Systems

Technical Publications

Direction 2294004

Revision 8

Signa® 1.5T *TwinSpeed* System Installation

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Operating Documentation

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- Contact your local service coordinator for more information on this process.

Rev. 08/15/2003



GE Medical Systems

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WARNING

- THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.
- IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.
- DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.
- FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.

AVERTISSEMENT

- CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.
- SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.
- NE PAS TENTER D'INTERVENTION SUR LES ÉQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ÉTÉ CONSULTÉ ET COMPRIS.
- LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.

WARNUNG

- DIESES KUNDENDIENST-HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.
- FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.
- VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH ZU RATE GEZOGEN UND VERSTANDEN WURDE.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

AVISO

- ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.
- SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.
- NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
- LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.

ATENÇÃO

- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.
- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENHA TENTADO REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.
- O NÃO CUMPRIMENTO DESTE AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

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- IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.
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- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

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- 本维修手册仅存有英文本。
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- 忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。

REVISION HISTORY

REV	DATE	PRIMARY REASON FOR CHANGE
0	Aug 22, 2001	Initial Release.
1	Feb 28, 2002	Revised title by adding 1.5T to title page and headers on internal pages. Overview Tab, Sec 1 – Added new Product Locator information submittal via web for U.S. sites. In flowchart, removed SEC 4. LCOIL & GAIN BALANCE, from Gram Tuning & Grafidy ACGD TwinSpeed procedure. Sec 2 – Added customer provided ethernet cable note to cable map. Tab1 – Dir. 2308287, corrected label errors in Procedures A3 and C2. Corrected ground wire installation in Procedure C3. Tab 2 – Revised connection of coolant hoses, installation of split bridge, added gradient cable label connection table, and added alternate connection of vacuum hose to rear end bell. Tab 8 – Updated System cabinet instructions. Tab 13 – Updated Operator Workspace installation to include SCIM/ keyboard and information on customer provided ethernet cable for connection to InSite. Tab 15 – Updated ACGD/PDU Installation to include ferrite filter for incoming power and ribbon cable 2315915 installation. Tab 17 – Updated TAC Install by showing the correct orientation of cable terminals for connecting to module.
2	Jul 8, 2002	Overview Tab, Section 1 – Updated catalog with the addition of Release 10.x (MGD), EXCITE Technology information. Section 2 – Added new MGD cable information to tables and added two new cable maps for MGD. Tab 1 – Added Direction 338872 for MGD fiber optics. Tab 2 – Updated existing enclosure installation instructions for MGD. Tab 4 – Updated penetration Panel installation for MGD. Tab 8 – Changed title to existing System Cabinet instructions by adding 8.x/9.x. Added Direction 2338502, Signa 10.x (MGD) System Cabinet Installation Instructions. Tab 13 – Updated Octane Workspace installation with MGD parts and information. Tab 15 – Updated ACGD/PDU circuit breaker information by adding "I" DIP switch settings. Tab 16 – Added Direction 2346942, SRFD2 Cabinet installation instructions.
3	Sep 27, 2002	Overview Tab, Section 1 – Added SRFD2 cabinet to system configuration. Added TwinSpeed to MDP description in Illustration 1–15. Removed Head/Body systems Gain & SNR from flowchart calibration. Section 2 – Corrected gradient cable wire connection designators at the gradient coil on the cable maps. Tab 1 – Revised Direction 2308287 to reflect recent PCN's that changed cable part numbers due to cable length changes. Also added new preterminated cable that connects to TAC cabinet from ACGD/PDU cabinet. Tab 2 – Added End Bell bridge leveling mark on page 4. Tab 4 – Added more information on attaching right angle heliax connectors to Pen Panel showing correct installation of O-rings (CQA 1029849). Tab 13 – Revised information on DIP switch settings on serial expansion box and replaced existing monitor with LCD monitor. Tab 15 – Added "I" DIP switch setting information for circuit breaker. Removed ferrite filter from main power and ground installation (CQA 1026740). Tab 16 – Added alternative power connection for SRF cabinet.
4	Jun 16, 2003	Tab 4 – Added O-ring installation for heliax at penetration panel. Tab 15 – Added neutral connection for SSM and TAC in PDU power cable connections table (SPR# MRIge82869). Revised wording on DIP switch settings. Tab 16 – Removed runs 745 and 748 from title in procedure C. These are 1.0T cables and are not used on TwinSpeed. Added note for ation of UFI Label kit.
5	Oct 28, 2003	Overview Tab, Section 1 – Revised Product Locator to emphasize preferred method of submitting FE Site Verification information. Updated catalog offerings for EXCITE II product. Added new Common Chiller and Gradient Water Heat Exchanger information. Updated flow chart to include EXCITE II equipment, Times Microwave cables, Linux OW, and System Calibration. Section 2 – Added EXCITE II cables to information tables. Added EXCITE II cable map. Added Common Chiller and Heat Exchanger cable maps. Tab 1 – Added Direction 238919, 1.5T Times Microwave RF cable installation and Direction 2398342, Signa EXCITE II Fiber Optic cable installation. Added EXCITE II Gradient Output cable installation information to Direction 2308327. Tab 2 – Added EXCITE Magnet Enclosure Installation, Direction 2398593. Tab 4 – Revised Penetration Panel installation for alternate Times Microwave RF cable installation. Tab 8 – Added Direction 2398344, EXCITE II System Cabinet installation. Tab 13 – Added Direction 2386129, Linux Operator Workspace installation instructions. Tab 15 – Revised attachment procedure of fan to ACGD/PDU cabinet. Tab 16 – Added connection point for Run 038 ground wire.
6	Jan 14, 2005	Section 1 – Updated flowchart LVSHIM value from 500 Hz to 200Hz and added Generate IIS Key instruction. Section 2 – Corrected Run 1042 cable description (SPR# MRIhc01910). Tab 2 – Direction 2398604: Added grounding wire to coolant system, 2mm–4mm adjustment height for bridge, and Laser Alignment Light Warning Label Installation instructions
7	Jun 16, 2009	Tab 2 – Directions 2309132 and 2398593: Removed instructions for installation of rating plates on end of bridge. (PQR 13246367). Added instructions for applying laser alignment light warning labels. (PQR 13246299).
8	Nov 10, 2010	Overview Tab, Section 1 – Revised Ferrous Material Handling warning. Section 3: Added step in Section 3–5 for confirming blower box anchoring. Tab 1 – Directions 2308287 and 2287106: Revised Ferrous Material Handling warnings. Tab 2 – Directions 2398342, 2309132 and 2398593: Revised instructions for installing blower box. (CAPA 1859293)

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DIRECTION 2346942 SIGNA SRFD2 CABINET INSTALLATION

TAB 17 – TAC CABINET

DIRECTION 2308633 SIGNA *TwinSpeed* TAC (TWIN ACCESSORY CABINET) INSTALLATION

SECTION 1 – GETTING STARTED

1-1 SIGNA 1.5T TwinSpeed SYSTEM CONFIGURATION

This manual provides instructions to install a new 1.5T TwinSpeed System.

Additional options are installed according to the installation manual delivered with the option.

Note

There are some naming differences between the sales literature documents and the technical documentation. Listed below are the corresponding names. The Installation document uses the technical document names.

SALES LITERATURE NAME

Signa Infinity 1.5T TwinSpeed with EXCITE Technology

Signa Infinity 1.5T TwinSpeed

CX 150 Magnet with K4 Technology

= TECHNICAL DOCUMENT NAME

= Signa 1.5T TwinSpeed with EXCITE Technology

= Signa 1.5T TwinSpeed

= 1.5T LCC (Active Shield) Magnet

1-2 SITE READY CHECK

Pre-installation work must be completed before equipment is delivered to avoid delays and confusion. Refer to *Direction 2294003, Signa TwinSpeed Pre-installation*, Preinstallation Checklist tab.

Tools and test equipment to install and calibrate the Signa MR/i system are listed in *Direction 2294003, Signa TwinSpeed Pre-installation*, Section 10, TOOLS AND TEST EQUIPMENT.

WARNING!

FERROUS MATERIAL HAZARD! THE CRIMP TOOL, BLOWER BOX, AND OTHER TOOLS AND PARTS REQUIRED FOR THIS INSTALLATION CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. KEEP ALL FERROUS TOOLS AS FAR FROM THE MAGNET AS POSSIBLE.

Service installation tools used in this manual that may not be included in above references lists include:

- 46-301450G1 Fiber Optic Cable Connector Repair kit
- 46-320362G1 Helix Termination kit
- 2134776 Gradient Cable Termination kit

1-3 PRODUCT DELIVERY INSTRUCTIONS AND PACKING

The “Shipping Document” lists catalog numbers delivered. Review and confirm that order is delivered complete. Check impact on installation schedule if Catalogs and/or packing boxes are missing and/or noted as shipped short.

Labels that are attached to the outside of packing boxes summarize box contents. The labels are color coded for different installation areas as follows:

- Green – Equipment Room
- Purple – Operator’s Room
- Orange – Magnet Room
- Yellow – Post-installation (Phantoms, Coils).

“Product Delivery Instructions” (PDI) specify box contents, part numbers, and shipping procedure. The PDI is numbered according to the catalog number (for example, PDI – M1000PM is for the Fixed Site Installation Kit). Lists of items included with each box are detailed by separate checklists, or a separate sheet that provides a summary of that box contents. Refer to PDI and packing lists for information specific to your shipment.

A set of service and operator manuals is delivered with the Fixed Site Kit. Refer to checklists packed with “Technical Publication” boxes for a list of delivered documents.

1-4 PRODUCT LOCATOR

The Global Install Base Database (also known as Product Locator System) tracks shipment, trans-shipment, and field location of the serialized models. There are two methods for submitting Product Locator information.

1. PREFERRED METHOD – Submit information to the *FE Site Verification Web Site* at:
<http://gein2.med.ge.com/gib>

The FE Site Verification consists of three components that are available on the web from the main menu. They are:

- Global Install Base Entries: Install/deinstall product locator model and serial numbers
- Add/modify ship to address information
- Update CARES FE data for primary/secondary FEs

To obtain a copy of the FE training tutorial for using this website, a downloadable copy is available at the following:
Product Locator Support Central Page – <http://supportcentral.ge.com/15563>

2. One “Shipping Card” is filled out and submitted when shipped (extra cards are supplied for trans-shipments between storage and distribution points), and the “Installation Card” and extra shipment cards are attached.

Verify that serial and model number on each rating plate matches installation card numbers before removing installation card. Note that there may be one or more shipment cards and bar code labels with the installation card. These shipment cards are used to trace the transfer of serialized units between various inventory storage and distribution points until the product reaches its final installation destination. Process just the installation card and discard any extra shipment cards and labels.

1-5 BASIC SYSTEM

The basic Signa 1.5T *TwinSpeed* system for fixed site operation consists of the following major equipment:

- System cooling equipment (See Illustration 1–15 flowchart for system cooling equipment configurations options and relationship to other system equipment/options selections.)

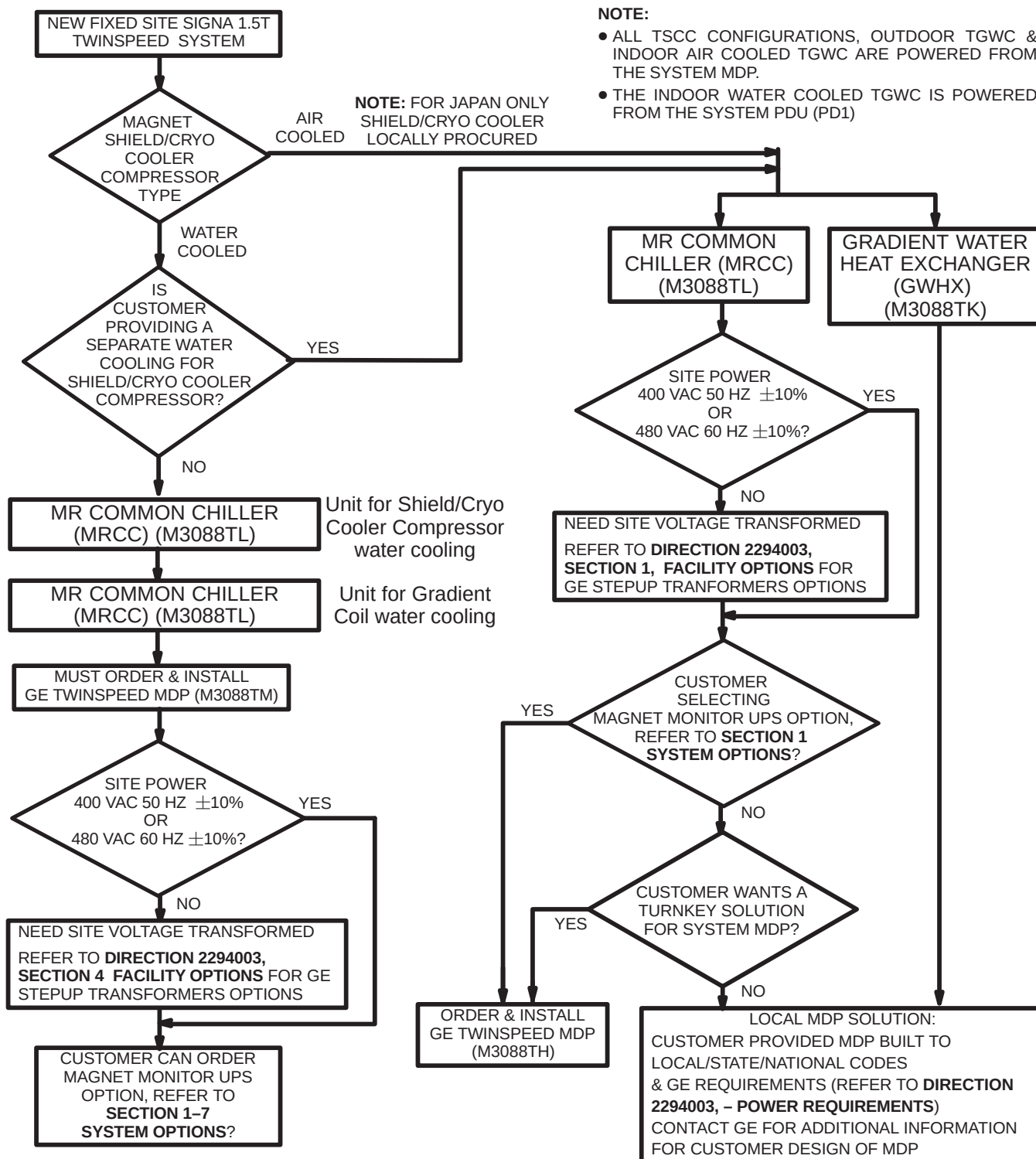
Note

True Mobile system water cooling for the Gradient Coil and the Shield/Cryo Cooler Compressor is provided by the Mobile Van equipment.

- 15 kilogauss (1.5T) LCC (Active Shield) Magnet & Magnet Enclosure
- Magnet Accessories
- Shield/Cryo Cooler Compressor Cabinet
- *TwinSpeed* Gradient (TRM) and RF body coils and 1.5T Quad Head Coil
- Main Disconnect Panel (GE supplied or customer supplied, dependent on Illustration 1–15 flowchart decisions)
- System electronics cabinets: ACGD/PDU Cabinet, SRFD2 Cabinet, System Cabinet, Twin Accessory Cabinet
 - For Signa EXCITE II 1.5T *TwinSpeed*: the System Cabinet contains the Multi Generational Data Acquisition (MGD) chassis, Remote RF chassis, Driver Module, and Terminal Server
 - For Signa EXCITE II 1.5T *TwinSpeed*: the System Cabinet contains the Multi Generational Data Acquisition (MGD) chassis, Remote RF chassis, and Driver Module,
 - For Signa *TwinSpeed* the System Cabinet contains the Integrated System Electronics (ISE) chassis and Transient Noise Suppression Module
- Operator Workspace equipment
- Pneumatic Patient Alert System
- Patient Transport Table
- Patient accessories

The tables in this section list the catalogs for the hardware equipment which comprise the Signa 1.5T *TwinSpeed* system. Illustrations 1–16 and 1–17 show the major equipment of the system. Refer to Section **1-6-1, System Cooling Equipment Selections** for explanation of available configurations.

1-5 BASIC SYSTEM (Continued)



FIXED SITE SYSTEM COOLING EQUIPMENT & MAIN DISCONNECT PANEL SELECTIONS

ILLUSTRATION 1-15

1–5 BASIC SYSTEM (Continued)

TABLE 1–1
SYSTEM ELECTRONICS MAJOR EQUIPMENT

CATALOG	DESCRIPTION
M3000TD Signa EXCITE II 1.5T TwinSpeed	System Cabinet consisting of MGD chassis, Remote RF chassis, and Driver Module. ACGD/PDU Cabinet containing ACGD Gradient drivers and Power Distribution Unit module with unregulated transformer 200/380/400/415/480Volt, 50/60 Hz with power filter. Twin Accessory Cabinet containing the Gradient Switch and the vacuum pump for Quiet Technology Magnet Enclosure. Operator Workspace (OW) equipment: Workspace Cabinet with Linux PC, LCD Display, Mouse and Mouse Pad, SCSI Tower, 15 inch LCD panel, and chair. Note, refer to Tables 1–7, 1–8, and 1–9 for catalog choices required to complete Operator Workspace equipment. Pneumatic Patient Alert System. Pneumatic Patient Alert System. 1.5T Quad Head Coil Patient Transport Table and cradle. Patient accessories such as: a phantom kit, patient log book, head cushion and sponges, chin and forehead straps, body wedges, knee cushions, and security/restraint straps. Gating accessories which include: patient cardiac leads, peripheral gating probe, and respiratory bellows.
M3000TC Signa EXCITE I 1.5T TwinSpeed	System Cabinet consisting of MGD chassis and Remote RF chassis. ACGD/PDU Cabinet containing ACGD Gradient drivers and Power Distribution Unit module with unregulated transformer 200/380/400/415/480Volt, 50/60 Hz with power filter. 1.5T SRFD2 Cabinet containing the RF Amplifiers, RF Power Monitor, power supplies for the Magnet Enclosure system components. Twin Accessory Cabinet containing the Gradient Switch and the vacuum pump for Quiet Technology Magnet Enclosure. Operator Workspace equipment: Octane 2 Computer, Workspace Cabinet, Mouse and Mouse Pad, LCD panel, and chair. Note, refer to Tables 1–7, 1–8, and 1–9 for catalog choices required to complete Operator Workspace equipment. Pneumatic Patient Alert System. 1.5T Quad Head Coil Patient Transport Table and cradle. Patient accessories such as: a phantom kit, patient log book, head cushion and sponges, chin and forehead straps, body wedges, knee cushions, and security/restraint straps. Gating accessories which include: patient cardiac leads, peripheral gating probe, and respiratory bellows.
M3000TA Signa 1.5T TwinSpeed	System Cabinet consisting of Integrated Systems Electronics subsystem, Universal Combined Exciter/Receiver (UCERD), Phase Array hardware integrated, and with Fast Receiver. ACGD/PDU Cabinet containing ACGD Gradient drivers and Power Distribution Unit module with unregulated transformer 200/380/400/415/480Volt, 50/60 Hz with power filter. 1.5T SRFD2 Cabinet containing the RF Amplifiers, RF Power Monitor, power supplies for the Magnet Enclosure system components. Twin Accessory Cabinet containing the Gradient Switch and the vacuum pump for Quiet Technology Magnet Enclosure. Operator Workspace equipment: Octane Computer, Workspace Cabinet, Mouse and Mouse Pad, LCD panel, and chair. Note, refer to Tables 1–7, 1–8, and 1–9 for catalog choices required to complete Operator Workspace equipment. Pneumatic Patient Alert System. 1.5T Quad Head Coil Patient Transport Table and cradle. Patient accessories such as: a phantom kit, patient log book, head cushion and sponges, chin and forehead straps, body wedges, knee cushions, and security/restraint straps. Gating accessories which include: patient cardiac leads, peripheral gating probe, and respiratory bellows.

TABLE 1–2
EXCITE II SRFD CABINET (SELECT ONE)

CATALOG	DESCRIPTION
M3090LK EXCITE II SRFD Cabinet	For Signa EXCITE II 1.5T TwinSpeed: 1.5T SRFD II Cabinet containing the RF Amplifiers, RF Power Monitor, power supplies for the Magnet Enclosure system components.

1–5 BASIC SYSTEM (Continued)

TABLE 1–3
1.5T MAGNET SUBSYSTEM (SELECT ONE)

CATALOG	DESCRIPTION
M3060TA	15 kilogauss (1.5T) LCC (Active Shield) Magnet for Fixed Site or Transportable/Relocatable Mobile (See Note 1). Includes Quiet Technology Enclosure partially installed on magnet, Magnet Rundown Unit, Magnet Monitor (performs cryogen meter function and magnet pressure control), and Global Modem. (See Notes 2 & 3)
M3061TA	15 kilogauss (1.5T) LCC (Active Shield) Magnet for True Mobile (See Note 1). Includes Quiet Technology Enclosure partially installed on magnet, Magnet Rundown Unit, Magnet Monitor (performs cryogen meter function and magnet pressure control), Magnet Monitor Remote Alarm and Global Modem. (See Note 2).
Note 1 For Mobiles (Transportable, Relocatable, and True Mobile) contact your selected Van Manufacturer for site pre-installation. 2 M1060JW Shield/Cryo Cooler Compressor (water cooled) must be ordered for LCC Magnet. The Shield/Cryo Cooler Compressor is not part of the LCC Magnet Catalogs but is required at magnet delivery. 3 If M1060TA Magnet is to be shipped by air then must order M1060SC CX Magnet Air Transport Crate (not required within Continental USA).	

TABLE 1–4
LCC MAGNET 2.5 METER CEILING HEIGHT KIT (OPTIONAL)

CATALOG	DESCRIPTION
M1060SR	LCC Magnet Low Ceiling Height (2.5 Meter) Siting Option Note: This option reduces the minimum required ceiling height for the Magnet Room to 2.5 meters from 2.67 meters. Check with GE Installation Specialist to determine if this is a required option.

TABLE 1–5
INTEGRATED BODY MODULE (SELECT ONE)

CATALOG	DESCRIPTION
M3085TA	1.5T Advanced 60 cm Twin Integrated Body Module (TRM)

TABLE 1–6
MAIN DISCONNECT PANEL (MDP) (SELECT ONE)

CATALOG	DESCRIPTION
M3088TM	Signa TwinSpeed Main Disconnect Panel compatible with MRCC or GWHX system cooling equipment.
M3088TH	Signa TwinSpeed Main Disconnect Panel compatible with TSCC or TGWC system cooling equipment.
Note 1 See Illustration 1–15 flowchart to determine when GE MDP is to be utilized or customer supplied MDP which meets requirements can be considered.	

1–5 BASIC SYSTEM (Continued)

TABLE 1–7
OPERATOR WORKSPACE MONITOR (SELECT ONE)

CATALOG	DESCRIPTION
M1000NT	18 inch LCD Color Monitor

TABLE 1–8
OPERATOR WORKSPACE TABLE (ONE REQUIRED)

CATALOG	DESCRIPTION
M1000MW	Table for Operator Workspace

TABLE 1–9
COUNTRY KITS (SELECT ONE)

CATALOG	DESCRIPTION
M1000MN	English Keyboard
M1000MP	French Keyboard
M1000MR	German Keyboard
M1000MS	Scandinavian Keyboard
M1000NH	Italian Keyboard
M1000NJ	Portuguese Keyboard
M1000NK	Spanish Keyboard

TABLE 1–10
PHYSICIAN CHAIR (SELECT ONE)

CATALOG	DESCRIPTION
E8803BD	Physician Chair for Fixed Site or Transportable/Relocatable Mobile Site
E8803BE	Physician Chair for True Mobile Site

TABLE 1–11
FILMING INTERFACE (SEE NOTE 1 IN TABLE)

CATALOG	DESCRIPTION
FOR SIGNA EXCITE II 1.5T TWIN SPEED	
M1000MK	Digital DASM
FOR SIGNA 1.5T TWIN SPEED & EXCITE I TWIN SPEED	
M1000MJ	Analog DASM
M1000MK	Digital DASM
Note 1 Must choose one Filming DASM Board unless DICOM Print will be used exclusively for software filming to DICOM print peripheral devices.	

1–5 BASIC SYSTEM (Continued)

TABLE 1–12
RECON PROCESSOR WITH BAM MEMORY (SELECT ONE)

CATALOG	DESCRIPTION
FOR SIGNA EXCITE II 1.5T TWIN SPEED	
M3000RH	EXCITE ReFlex200
M3000RJ	EXCITE ReFlex400
FOR SIGNA EXCITE I 1.5T TWIN SPEED	
M3000RD	EXCITE ReFlex100
M3000RE	EXCITE ReFlex200
M3000RF	EXCITE ReFlex400
FOR SIGNA 1.5T TWIN SPEED	
M3000RA	ReFlex100 – Hex Power PC Processor with four 128 MB Bulk Acquisition Memory (BAM) boards (512 MB)

TABLE 1–13
CABLE COLLECTOR FOR SIGNA EXCITE II 1.5T TWIN SPEED (SELECT ONE)

CATALOG	DESCRIPTION
M3143TD	Fixed Site system interconnect cables for configuration A lengths (refer to Section Signa EXCITE II 1.5T TwinSpeed with Cable Groups Length Provided)
M3144TD	Fixed Site system interconnect cables for configuration B lengths (refer to Section Signa EXCITE II 1.5T TwinSpeed with Cable Groups Length Provided)
M3145TD	Fixed Site system interconnect cables for configuration C lengths (refer to Section Signa EXCITE II 1.5T TwinSpeed with Cable Groups Length Provided)
M3153TD	True Mobile system interconnect cables
M3163TD	Transportable/Relocatable Mobile system interconnect cables

TABLE 1–14
BROADBAND CONNECTIVITY (SELECT ONE)

CATALOG	DESCRIPTION
E1000LG	Initial Insite Broadband installation
E1001LG	Additional system for site that already has Insite Broadband

TABLE 1–15
MR INSITE (MUST SELECT ONE IF BROADBAND NOT AVAILABLE)

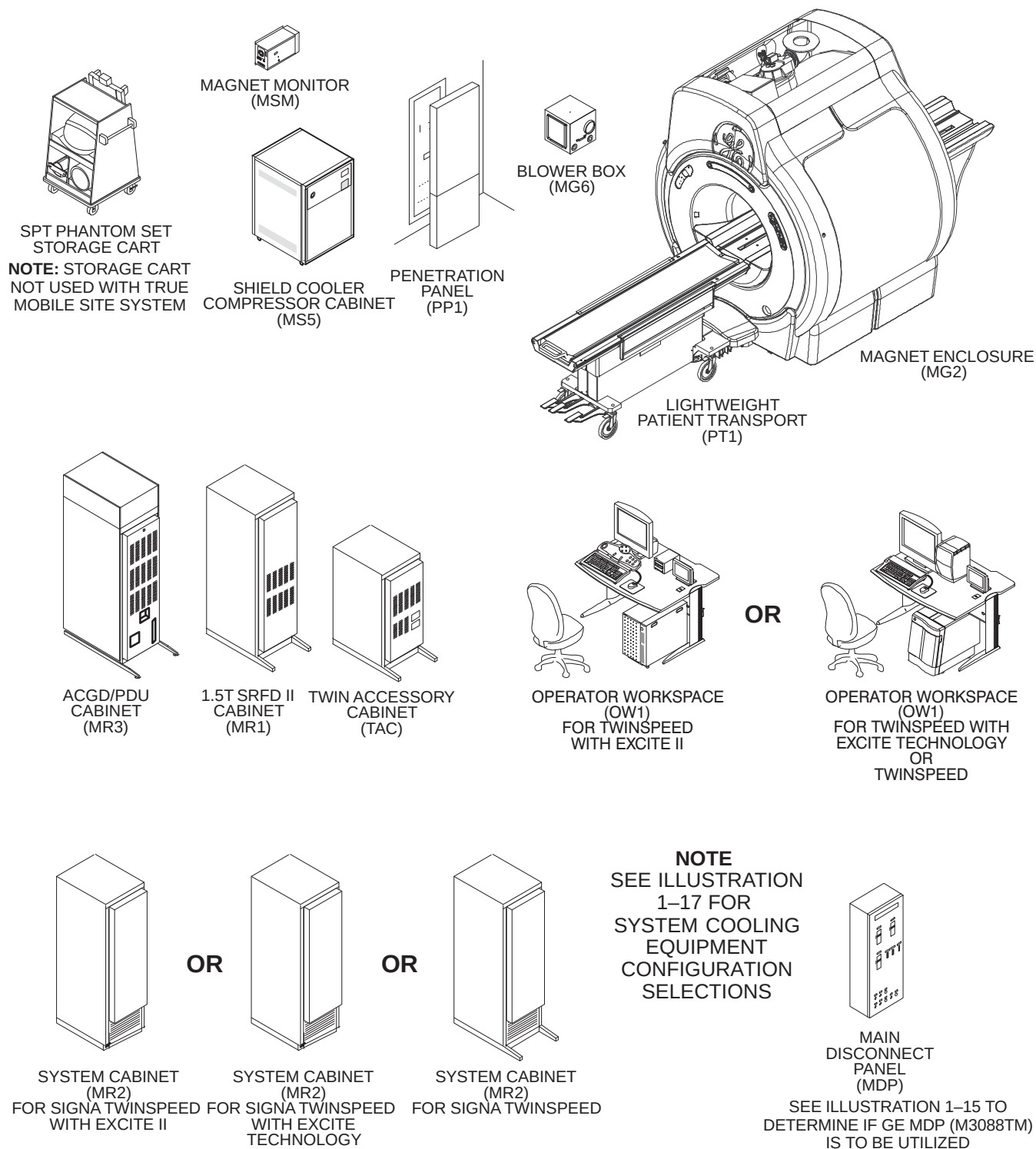
CATALOG	DESCRIPTION
M10000NW	IIP Global Modem

1–5 BASIC SYSTEM (Continued)

TABLE 1–16
SITE COLLECTOR KIT (SELECT ONE)

CATALOG	DESCRIPTION
SITE COLLECTOR WITHOUT CABLES FOR SIGNA EXCITE II 1.5T TWIN SPEED	
M3043TD	LCC (Active Shield) Magnet Fixed Site Quiet Technology Enclosure Penetration Panel with Phase Array hardware integrated and Penetration Panel Covers SPT Phantom Set with Shipping/Storage Cart
M3053TD	LCC (Active Shield) Magnet True Mobile Site Quiet Technology Enclosure Penetration Panel with Phase Array hardware integrated SPT Phantom Set
M3063TD	LCC (Active Shield) Magnet Transportable/Relocatable Mobile Site Quiet Technology Enclosure Penetration Panel with Phase Array hardware integrated SPT Phantom Set with Shipping/Storage Cart
SITE COLLECTOR WITH CABLES FOR SIGNA EXCITE I 1.5T TWIN SPEED	
M3043TC Refer to Note 3	LCC (Active Shield) Magnet Fixed Site Quiet Technology Enclosure Penetration Panel with Phase Array hardware integrated and Penetration Panel Covers Fixed Site system interconnect cables including Phased Array Option & Power Distribution Unit cables SPT Phantom Set with Shipping/Storage Cart
M3053TC	LCC (Active Shield) Magnet True Mobile Site Quiet Technology Enclosure Penetration Panel with Phase Array hardware integrated True Mobile Site system interconnect cables including Phased Array Option & Power Distribution Unit cables SPT Phantom Set
M3063TC	LCC (Active Shield) Magnet Transportable/Relocatable Mobile Site Quiet Technology Enclosure Penetration Panel with Phase Array hardware integrated Transportable/Relocatable Mobile Site system interconnect cables including Phased Array Option & PDU cables SPT Phantom Set with Shipping/Storage Cart
SITE COLLECTOR WITH CABLES FOR SIGNA 1.5T TWIN SPEED	
M3043TA Refer to Notes 1 & 2	LCC (Active Shield) Magnet Fixed Site Quiet Technology Enclosure Penetration Panel with Phase Array hardware integrated and Penetration Panel Covers Fixed Site system interconnect cables including Phased Array Option & Power Distribution Unit cables SPT Phantom Set with Shipping/Storage Cart
M3053TA	LCC (Active Shield) Magnet True Mobile Site Quiet Technology Enclosure Penetration Panel with Phase Array hardware integrated True Mobile Site system interconnect cables including Phased Array Option True Mobile Site Power Distribution Unit cables SPT Phantom Set
M3063TA	LCC (Active Shield) Magnet Transportable/Relocatable Mobile Site Quiet Technology Enclosure Penetration Panel with Phase Array hardware integrated Transportable/Relocatable Mobile Site system interconnect cables including Phased Array Option Transportable/Relocatable Mobile Site Power Distribution Unit cables SPT Phantom Set with Shipping/Storage Cart
Note	1 Operator Workspace Long Cables (M1090ZN) is available to order only if the GE Installation Specialist determines the standard length cables (supplied in listed catalogs) for interconnects from OW to other system components will not work at the site. M1090ZN provides long length interconnects for OW to PDU, OW to System Cabinet, and OW to Penetration Panel ONLY. 2 Signa TwinSpeed Site Collector Kit with Long Cables (M3044TA) is available to order only if the GE Installation Specialist determines the standard length cables in M3043TA will not work at the site. M3044TA includes OW Long Cables with the long cables for the Equipment Room. 3 Signa TwinSpeed with EXCITE Technology Site Collector Kit with Long Cables (M3044TC) is available to order only if the GE Installation Specialist determines the standard length cables in M3043TA will not work at the site.

1-5 BASIC SYSTEM (Continued)



SIGNA 1.5T TWINSPEED SYSTEM
ILLUSTRATION 1-16

1-6 SYSTEM COOLING EQUIPMENT SELECTIONS

The Signa EXCITE II 1.5T TwinSpeed, Signa EXCITE I 1.5T TwinSpeed, and the Signa 1.5T TwinSpeed system requires water cooling for the LCC Magnet Shield/Cryo Cooler Compressor Cabinet and the TRM Gradient Coil located in the Magnet bore. The following system cooling equipment is available with the MR system.

1-6-1 TSCC and TGWC

The MR Engineering Team together with Ellis & Watts International Inc. have designed several different water cooling cabinet options to optimize siting power requirements and monthly maintenance costs for the customer. The configuration options include:

- Twin System Cooling Cabinet (TSCC)
TSCC units provide water cooling for the LCC Magnet Shield/Cryo Cooler Compressor Cabinet and the Gradient Coil located in the Magnet bore. See Table 1-17 for listing of TSCC offerings.
- Twin Gradient Water Cooler (TGWC)
TGWC units provide water cooling for ONLY the Gradient Coil located in the Magnet bore. **When a TGWC is selected then the customer site must still provide water cooling for the LCC Magnet Shield/Cryo Cooler Compressor Cabinet.** See Table 1-18 for listing of TGWC offerings.

Note

See Illustration 1-15 flowchart for system cooling equipment configurations options and relationship to other system equipment/options selections.

TSCC Configurations

Refer to Table 1-17 for list of TSCC catalogs offerings.

TABLE 1-17
TWIN SYSTEM COOLING CABINET (TSCC) SEE TABLE NOTE 1

CATALOG	DESCRIPTION
M3088TA	Outdoor Air Cooled TSCC (Water-to-Air Chiller) 50/60 Hz for Gradient Coil Cooling & Shield/Cryo Cooler Compressor cooling <i>See Note 3</i> <i>TSCC installed external to building with Remote Control Panel installed in Equipment Room.</i> <i>The most cost effective for the customer.</i>
M3088TB	Indoor Water Cooled TSCC (Water-to-Water Chiller) 50/60 Hz for Gradient Coil Cooling & Shield/Cryo Cooler Compressor cooling <i>See Note 2</i> <i>Installed in the Equipment Room, low additional Air Conditioning requirements.</i>
M3088TC	Indoor Air Cooled TSCC (Water-to-Air Chiller) 50/60 Hz for Gradient Coil Cooling & Shield/Cryo Cooler Compressor cooling <i>Installed in the Equipment Room, increases Air Conditioning requirements.</i>
Note	1 The TSCC requires utilization of the GE MDP per Illustration 1-15. Therefore must order and install M3088TH TwinSpeed Main Disconnect Panel. 2 M3088TA is standard cooling equipment configuration of Transportable/Relocatable Mobile systems. 3 The water cooled configuration System Cooling Cabinets require customer supplied water for cooling. Refer to Section NO TAG Water Cooled System Cooling Cabinet Configuration Requirements for water cooling specifications.

Note

True Mobile system water cooling for the Gradient Coil and the Shield/Cryo Cooler Compressor is provided by the Mobile Van equipment.

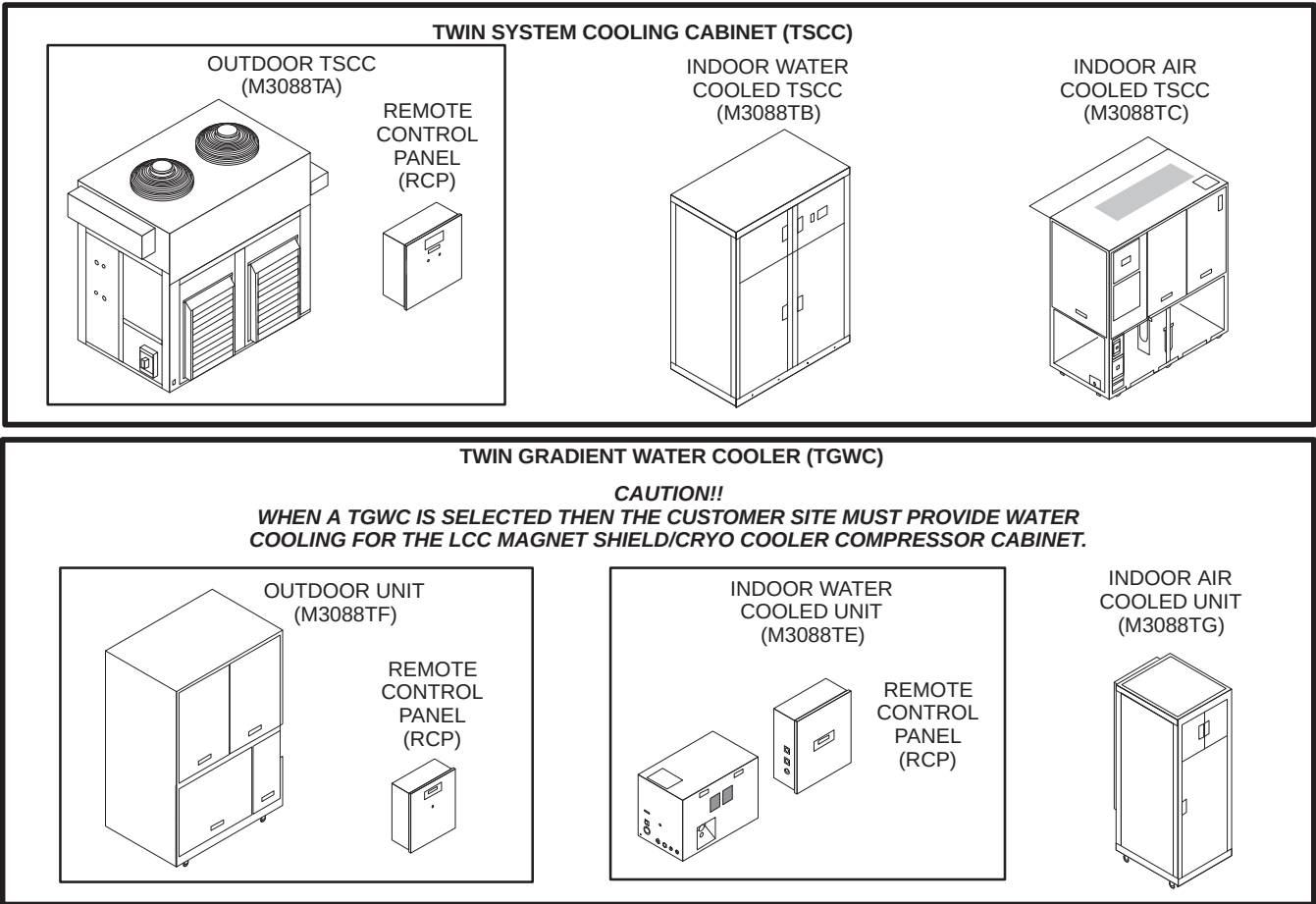
1-6-1 TSCC and TGWC (Continued)

TGWC Configurations

Refer to Table 1-18 for list of TGWC offerings.

TABLE 1-18
TWIN GRADIENT WATER COOLER (TGWC) See Note 1

CATALOG	DESCRIPTION
M3088TF	Outdoor Air Cooled TGWC (Water-to-Air Chiller) 50/60 for Gradient Coil Cooling TGWC installed external to building with Remote Control Panel installed in Equipment Room. The most cost effective for the customer with separate cooling available for the Shield/Cryo Cooler Compressor.
M3088TE	Indoor Water Cooled TGWC (Water-to-Water Heat Exchanger) 50/60 Hz for Gradient Coil Cooling See Note 2 & Unit and Remote Control Panel installed in the Equipment Room, low additional Air Conditioning requirements.
M3088TG	Indoor Air Cooled TGWC (Water-to-Air Chiller) 50/60 Hz for Gradient Coil Cooling Installed in the Equipment Room, increases Air Conditioning requirements.
Note 1 When a TGWC is selected then the customer site must still provide water cooling for the LCC Magnet Shield/Cryo Cooler Compressor Cabinet. 2 The water cooled configuration of TGWC requires customer supplied water for cooling. Refer to Direction 2294003, Water Cooled System Cooling Cabinet Configuration Requirements for water cooling specifications.	



SYSTEM COOLING EQUIPMENT SELECTIONS
ILLUSTRATION 1-17

1–6–2 MRCC and GWHX

The MRCC (Magnetic Resonance Common Chiller) and the GWHX (Gradient Water Heat Exchanger) are new chillers and will replace the Ellis & Watts chillers. They are manufactured by AirSys Inc in China. The MRCC and GWHX configuration options include:

- **MR Common Chiller (MRCC)**
The MRCC is a single-loop 10KW 50/60 Hz water chiller that can be used to cool either Shield/Cryo Cooler Compressor or the TwinSpeed Gradient Coil. The MRCC can be installed either indoors or outdoors. There is a Remote Panel to be installed in the MR system Equipment Room that can stop or restart the chiller. The MRCC power is supplied by the MR system Main Disconnect Panel (MDP). This chiller will be serviced by a third party vendor.
- **Gradient Water Heat Exchanger (GWHX)**
The GWHX is a 10KW 50/60 Hz water-to-water heat exchanger that is used for cooling the TwinSpeed Gradient Coil. The GWHX is for indoor use only. The GWHX requires facility provided chilled water. The GWHX power is supplied by the MR system Power Distribution Unit (PDU). This heat exchanger will be serviced by GE Service. When the GWHX is used then the customer must supply water cooling for the Shield/Cryo Cooler Compressor.

Note

Gradient Coil water cooling must be supplied by cooling equipment (MRCC or GWHX) supplied with the MR system to prevent contamination/damage to the coil and for proper image quality.

There are several possible scenarios which a customer may utilize to meet the system water cooling requirements for the LCC Magnet Shield/Cryo Cooler Compressor Cabinet and the TRM Gradient Coil.

- Two MRCC installed outdoors where one unit provides Shield/Cryo Cooler Compressor cooling and the other provides Gradient Coil cooling. The 2 MRCC Remote Control Panels to be installed in the MR system Equipment Room.
- Two MRCC installed indoors where one unit provides Shield/Cryo Cooler Compressor cooling and the other provides Gradient Coil cooling. The 2 MRCC Remote Control Panels to be installed in the MR system Equipment Room. Customer supplied air cooling is required for the MRCC units.
- One GWHX installed indoors provides water cooling to the Gradient Coil (GWHX requires facility water cooling) and Customer provided facility water cooling for the LCC Magnet Shield/Cryo Cooler Compressor Cabinet.
- One MRCC installed outdoors provides water cooling to the Gradient Coil and Customer provided facility water cooling for the LCC Magnet Shield/Cryo Cooler Compressor Cabinet. The MRCC Remote Control Panel to be installed in the MR system Equipment Room.
- One MRCC installed indoors provides water cooling to the Gradient Coil and Customer provided facility water cooling for the LCC Magnet Shield/Cryo Cooler Compressor Cabinet. The MRCC Remote Control Panel to be installed in the MR system Equipment Room. Customer supplied air cooling is required for the MRCC unit.

1-6-2 MRCC and GWHX (Continued)

Note

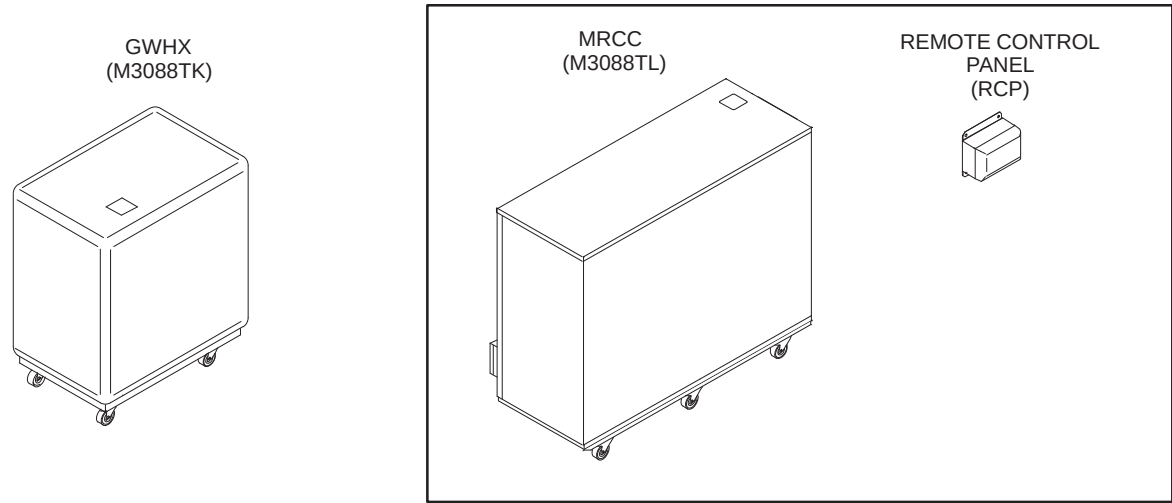
See Illustration 1-15 flowchart for system cooling equipment configurations options and relationship to other system equipment/options selections.

TABLE 1-19
SYSTEM COOLING EQUIPMENT *SEE TABLE NOTE 1*

CATALOG	DESCRIPTION
M3088TK	Gradient Water Heat Exchanger (GWHX) <i>See Note 2</i>
M3088TL	MR Common Chiller (MRCC) <i>See Note 3</i>
Note 1 See Illustration 1-15 to determine system cooling equipment configurations that must order and install the GE MDP. 2 When GWHX is selected, customer supplied water cooling is required for the LCC Magnet Shield/Cryo Cooler Compressor. Refer to Section NO TAG Shield/Cryo Cooler Compressor Requirements For Site Provided Water Cooling for water cooling specifications. 3 Two M3088TL installed outdoors is standard cooling equipment configuration of Transportable/Relocatable Mobile systems.	

Note

True Mobile system water cooling for the Gradient Coil and the Shield/Cryo Cooler Compressor is provided by the Mobile Van equipment.



SYSTEM COOLING EQUIPMENT SELECTIONS
ILLUSTRATION 1-18

1-7 SYSTEM OPTIONS

This section lists options for the Signa 1.5T TwinSpeed system.

Purchased options may also be installed concurrently with the system upgrade. Flow charts provide steering to the Option Installation Direction that is delivered with the system.

Refer to manual shipped with the option for installation of Advantage Window options and Camera Options.

Refer to manual shipped with the option for installation of Surface Coil Options.

Refer to installation manual shipped with the option for the following:

- M1060KM – Oxygen Monitor: *Direction 15336, Oxygen Monitor Subsystem*
- M1090TL – Second Patient Transport: *Direction 15477, Second Patient Transport Option Installation*

PERFORMANCE UPGRADE OPTIONS

- M1090RW – Acquisition Memory – 128MB for Medcam
- M1090RY – Host Memory Kit – 128MB
- M3000RD – EXCITE I ReFlex 100 Array Processor
- M3000RE – EXCITE I ReFlex 200 Array Processor
- M3000RF – EXCITE I ReFlex 400 Array Processor
- M3000RH – EXCITE II ReFlex200
- M3000RJ – EXCITE II ReFlex400

HARDWARE OPTIONS

- M1090TZ – VCR Interface Kit
- M3000JB – 18 GB Disk Drive for Octane
- M3080TA – Resistive Shim for TwinSpeed
- M1033BL – EXCITE I & II only: BrainWaveHW Lite (w/software)
- M3033CB – EXCITE II only: EXCITE 8 Channel

SOFTWARE OPTIONS

- M1033JA – iDrive Pro
- M1033JB – SmartPrep 2000 (Includes SmartStep)
- M1033JD – Probe2000
- M1033JE – EchoPlus Base
- M1033JF – ScanTools 2000
- M1033JH – SmartPrep 2000 Update
- M1033JJ – Probe2000 Update (for customers that have Probe software)
- M1033JK – LX FuncToll 2000
- M1033JL – LX Tools
- M1033JZ – FuncTool 2000 for AW
- M1033MD – Spectro for LX with SRFD
- M1033MG – SAGE 7
- M1033PA – Performed Procedure Step
- M1033PL – ConnectPro
- M1090CR – Cardiac

1–7 System Options (Continued)

- M1090LP – EchoPlus Diffusion Weighted Imaging Software
- M1090LR – Single Shot – FSE
- M1090LT – SmartPrep IA
- M1090MW – 8.x Software Non-Z with Multi-slice
- M1090MZ – Clariview
- M1090NN – Functool
- M1090PM – Bar Code Reader
- M1090PR – Flow Analysis
- M3033JA – Probe 3D Brain
- M3033JB – Functool CSI
- M3033JC – Ultra-Short TR for 3D Fast GRE
- M3033JD – BodyPak
- M3033JE – ACGD Plus
- M3033JG – Phase Sensitive IR
- M3033JF – Flouro Triggered MRA
- M3033JP – FGRET
- M3033JM – Spiral Hi Resolution
- M3033JN – Spiral Real-Time
- M3033JR – FGRET Real-Time
- M3033JJ – Prose Pulse Sequence
- M3033JH – SSFSE-XL
- M3033JL – 3D FSE Variable R1
- M3033LR – HiSpeed Plus

EXCITE II SOFTWARE OPTIONS

- M3033KS – ASSET
- M3033LW – 3D FRFSE
- M3033LZ – Flouro Trigger MRA
- M3033MB – Fiesta C
- M3033MD – 3D Delayed Enhancement
- M3033ME – Navigator 3D
- M3033MF – 3D Fiesta
- M3033MH – Tricks
- M3033MJ – Vibrant
- M3033MM – Spiral Hi-Res
- M3033MP – FGRE-ET
- M3033MR – Real-Time FGRE-ET
- M3033MZ – Real-Time Spiral
- M3033MT – iDrivePro
- M3033MW – iDrivePro Plus

1–7 System Options (Continued)

- M3033MY – Cardiac Tagging
- M3033PJ – Performed Proc Step
- M3090KJ – 2D Fiesta
- M3090KK – 3D fiesta
- M3090KR – 2D Delayed Enhancement
- M3090LF – HiSpeed Option
- M3090LG – EchoSpeed Option
- M3090LP – EchoPlus
- M3090LR – Diffusion Tensor
- M3090MA – Probe 2000
- M3090MB – Probe 2000 Upgrade
- M3090MC – Sage 7
- M3090MD – Probe 3D Brain
- M3090PL – ConnectPro

1–8 FACILITY OPTIONS

- The GE MDP, all configurations of TSCC, the OutdoorTGWC, and the Indoor Air Cooled TGWC require high voltage power. If system input voltage configuration does not meet specifications in **Direction 2294003 – POWER REQUIREMENTS** then customer provided transformer may be needed. Refer to Table 1–20 for low voltage step up transformer options.

TABLE 1–20
LOW VOLTAGE (200 or 208 VOLTS) STEP UP TRANSFORMER OPTIONS

CATALOG See Note 1	DESCRIPTION
R4500AS	150 KVA 208 to 480Y277 Volt, 60 Hz transformer
R4500BD	150 KVA 200 to 400Y230 Volt, 50/60 Hz transformer
Note 1 A step-up transformer is required if the site input voltage configuration does not meet specifications in DIRECTION 2294003 – POWER REQUIREMENTS . The GE Main Disconnect Panel (MDP) & Twin System Cooling Cabinet (TSCC) or Twin Gradient Water Cooler (TGWC) configurations require high voltage power.	

- Signa System Seismic Anchorage Service (R4390JA).

Note

Magnet Seismic anchoring is the customer's responsibility to coordinate magnet mounting methods with the RF shielded room vendor to prevent RF leaks and secondary grounding problems. Refer to DIRECTION 2294003 – ANCHOR HARDWARE REQUIREMENTS FOR MR EQUIPMENT INSIDE RF SHIELDED ROOM for details.

- Oxygen Monitor Kit (M1060KM) which includes Oxygen Monitor and Remote Oxygen Monitor Module.

1-9 INSTALLATION PROCEDURE

1-9-1 Installation Flow

Note

All on-site construction must be completed before equipment is delivered and installation starts. Attempting to install the system while construction is being completed will impact installation efficiency and further delay site completion. Making sure that all preinstallation and construction work is completed before equipment is delivered will usually result in an earlier turnover date.

The flow chart in Illustration 1-2 should be followed for an orderly and efficient installation of the Signa TwinSpeed system. Note that many procedures may be performed in parallel and may be performed in any order according to the specific situation of each site.

Note

The flow chart references the Tab number and the PIP (Pictorial Installation Procedure) Direction number for the procedure. PIP Directions are similar to cable maps in that they are to be removed from this binder and used at the location where hardware installation is being performed. The PIP Direction may be returned to the binder for future reference upon completion of the installation. Other Directions and installation drawings delivered with kits are also referenced by the flowchart.

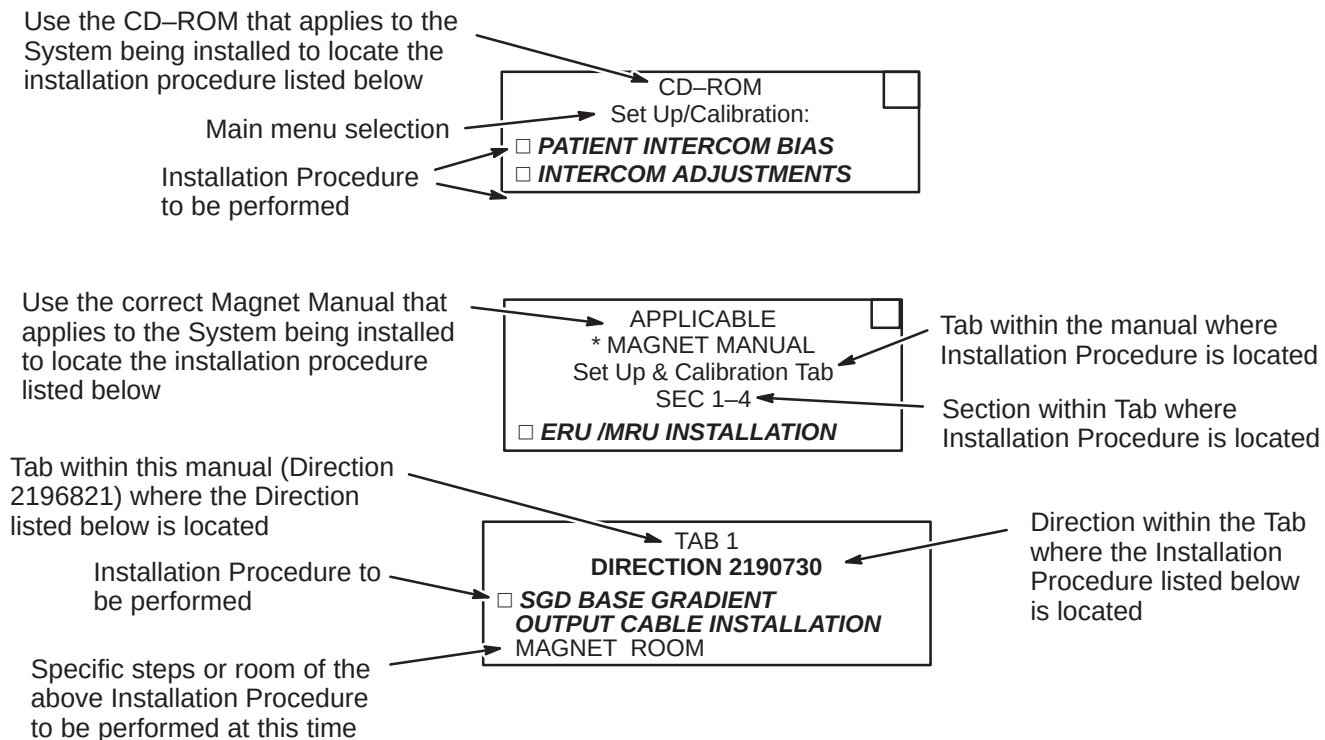
This installation flowchart has been developed assuming that all Signa TwinSpeed equipment has been delivered together. Ensure that every part required is available before starting.

The general installation flow is as follows:

- Hardware installation per applicable Pictorial Installation Procedure (PIP) Direction
- Power-up and other pre-software load checks (this can be done while Magnet is being ramped)
- Software reconfiguration (this can be done while Magnet is being ramped)
- Function checks, Set-up and Calibration, and other activity requiring software running that can be done while magnet field is stabilizing
- Mechanical shim of Magnet
- System calibrations
- System performance checks
- Cover replacement
- Final sign-off and records
- Customer turn over

1-9-2 Installation Flowchart Explanation

Shown below are three examples of the flow chart Installation Procedures and an explanation of the areas within the procedures.



LEGEND FOR FOLLOWING INSTALLATION FLOWCHART



INDICATES MILESTONE COMPLETION GOAL IN DAYS



INDICATES AND OPTIONAL PATH DEPENDING ON SYSTEM CONFIGURATION



INDICATES GE PROPRIETARY PROCEDURE



INDICATES AN ADDITIONAL PROPRIETARY TEST OR ALTERNATIVE PROPRIETARY CALIBRATION PROCEDURE AVAILABLE FOR GE USE OR TO CUSTOMERS WITH AN ADVANCED SERVICE PACKAGE LIMITED LICENSE.



AFTER GRADIENT SHIM (OR LV SHIM) IT MAY BE NECESSARY TO REPAIR THE MAGNET.

NOTE 1: STANDARD CORRELATED NOISE CHECK NOT REQUIRED IF PROPRIETARY SPT FULL TEST MODE PROCEDURE PERFORMED EARLIER DURING INSTALLATION (HOWEVER, MULTICOIL AND FAST RECEIVER OPTION CORRELATED NOISE CHECKS IN PROCEDURE STILL REQUIRED).

NOTE 2: IF TROUBLESHOOTING NECESSARY.

NOTE 3: SRF PROCEDURES ARE TO BE USED IF SITE HAS SRFD2 CABINET INSTALLED.

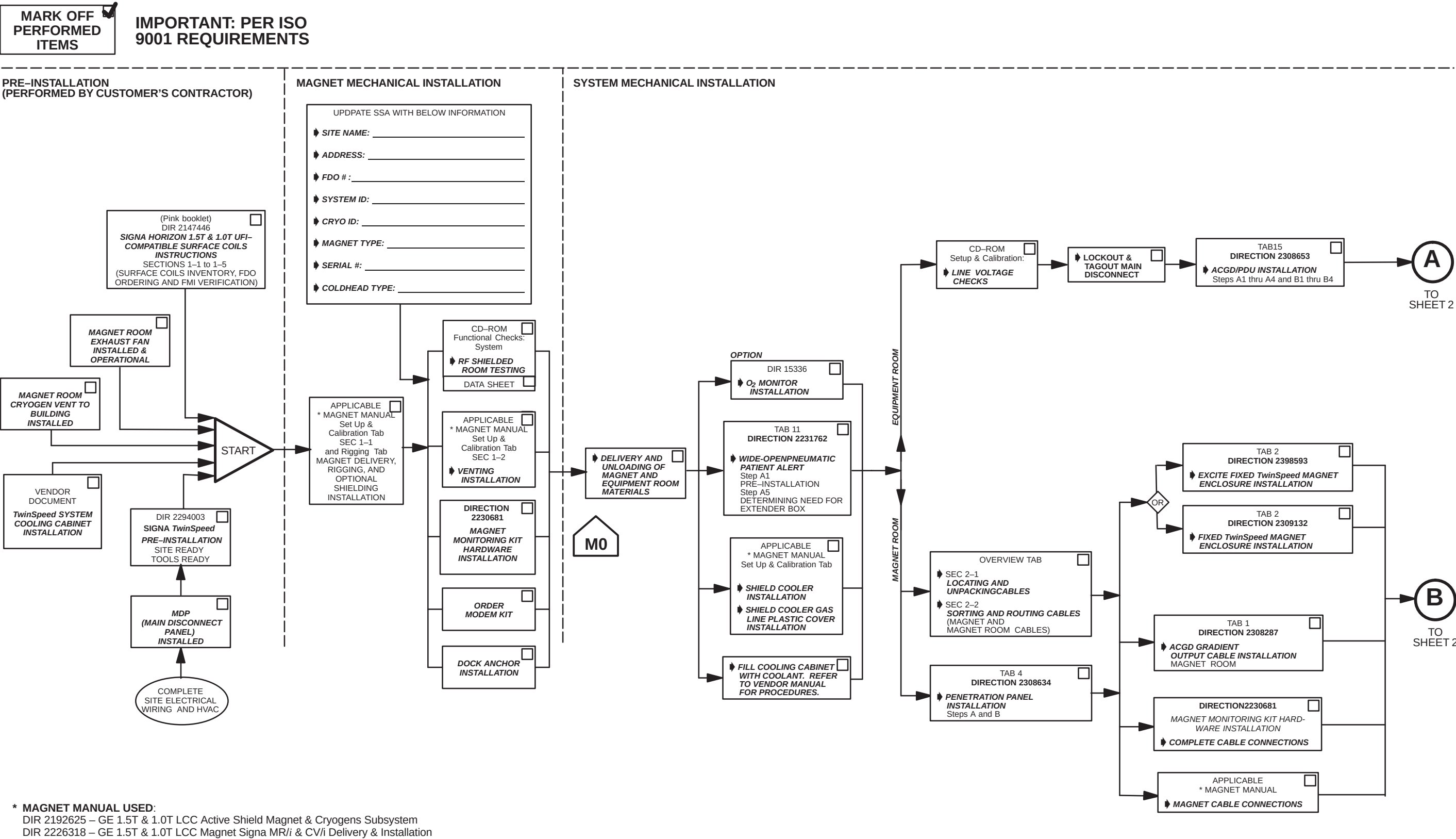
NOTE 4: A "saveINFO" SHOULD BE DONE BEFORE SPT IS RUN. IF YOU HAVE THE EPI OPTION, IT IS NECESSARY TO "saveINFO" AGAIN. IN THIS WAY, ANY DATA COLLECTED FOR CALIBRATION WILL NOT BE LOST IF A PROBLEM OCCURS.

NOTE 5: VALUES OF LVSHIM ARE DEPENDENT UPON EDDY CURRENT COMPENSATION.

NOTE 6: VALUES OF EDDY CURRENT COMPENSATION ARE INDEPENDENT OF MAGNET SHIM.

NOTE 7: OVERNIGHT DIAGS MAY BE RUN AT END OF ANY DAY TO DETECT POSSIBLE ERRORS.

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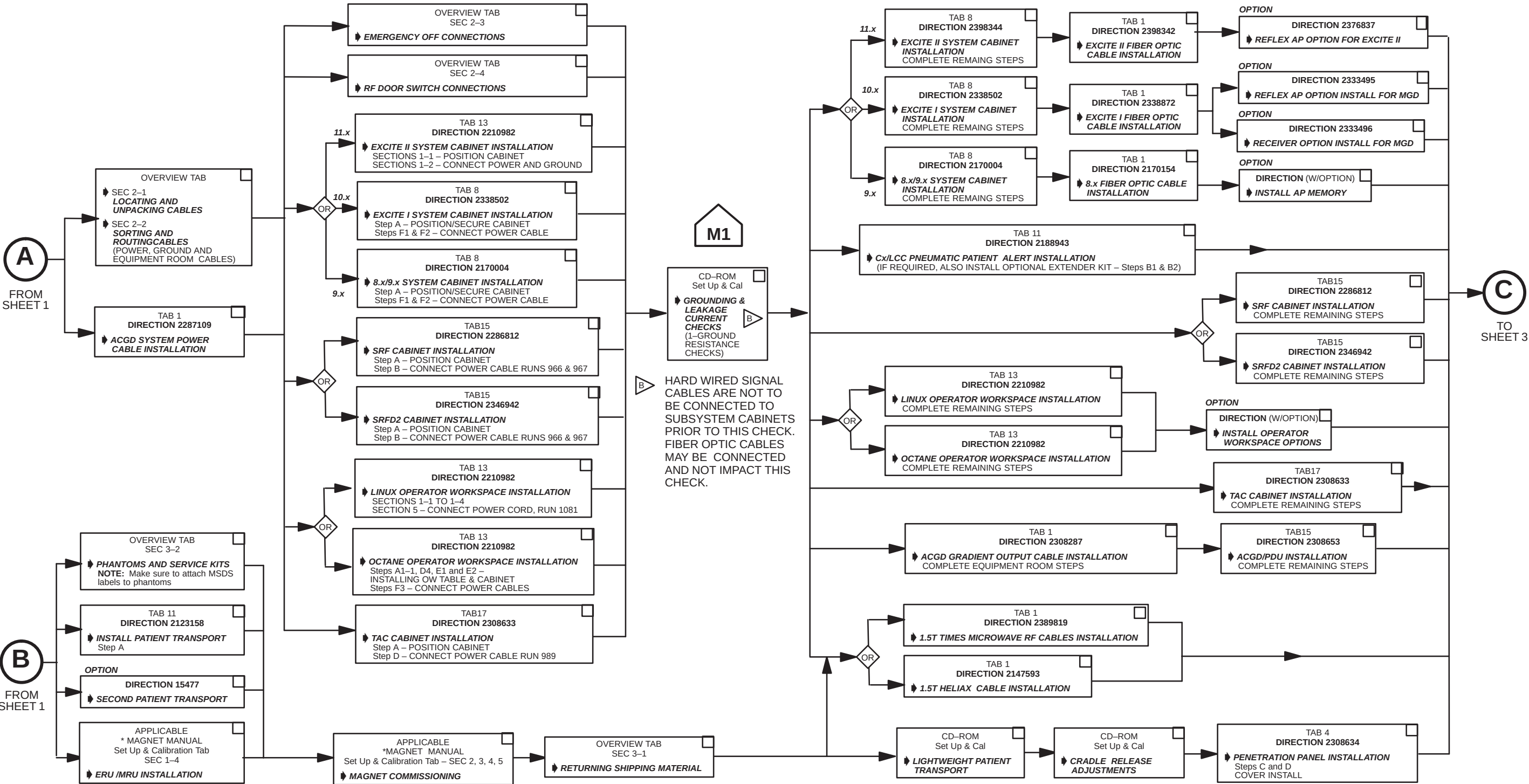


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IMPORTANT: PER ISO
9001 REQUIREMENTS

SYSTEM MECHANICAL INSTALLATION



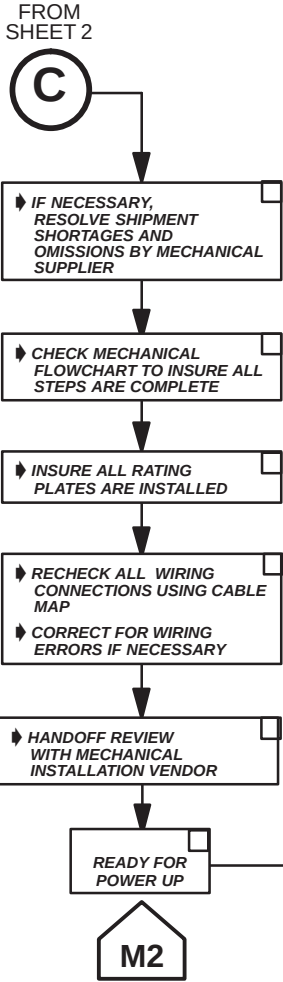
SIGNA TwinSpeed INSTALLATION FLOW CHART
ILLUSTRATION 1-2 (SHEET 2 OF 5)

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IMPORTANT: PER ISO
9001 REQUIREMENTS

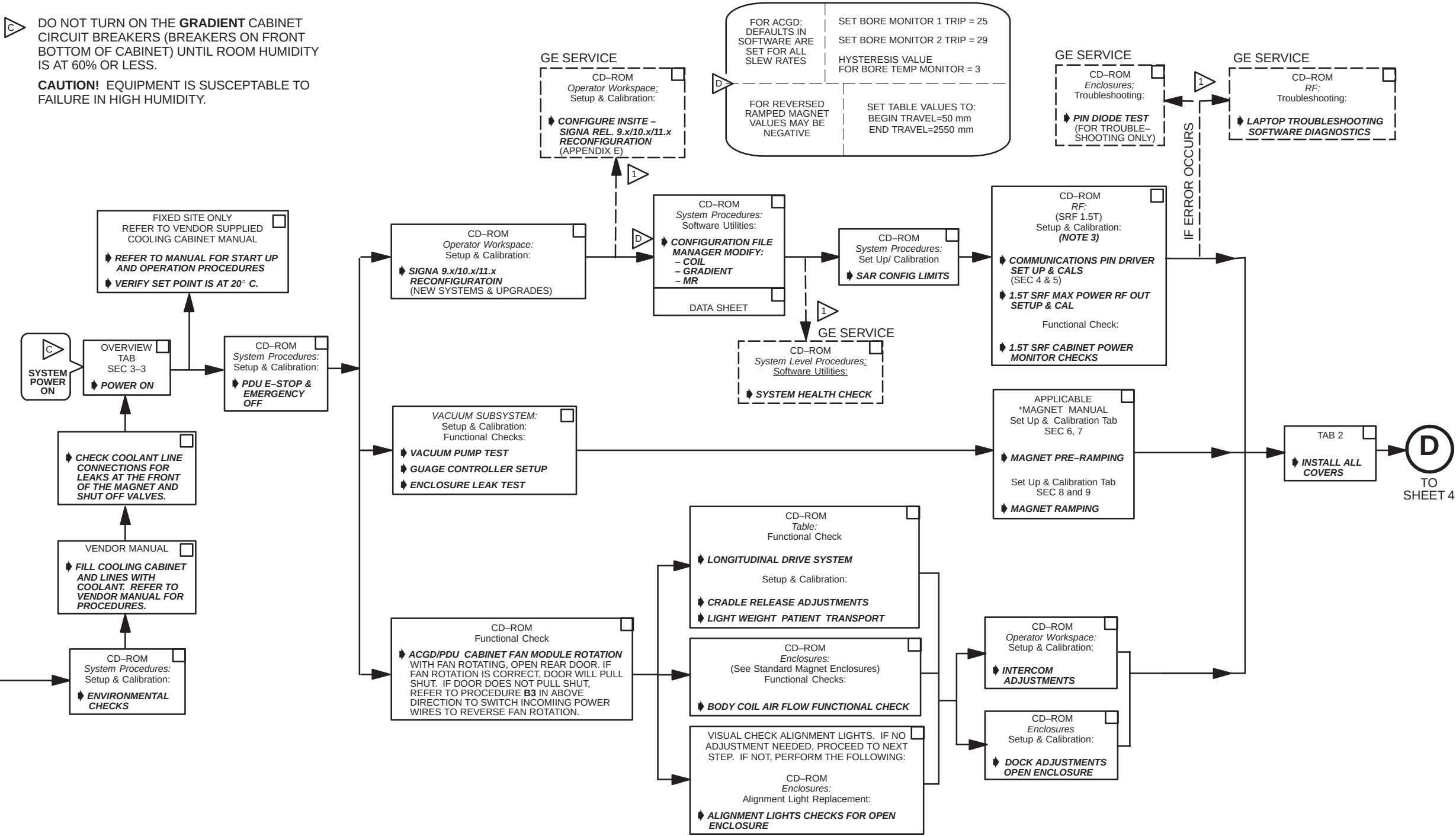
SYSTEM PREPARATION FOR
CALIBRATION



SYSTEM CALIBRATION

DO NOT TURN ON THE **GRADIENT** CABINET
CIRCUIT BREAKERS (BREAKERS ON FRONT
BOTTOM OF CABINET) UNTIL ROOM HUMIDITY
IS AT 60% OR LESS.

CAUTION! EQUIPMENT IS SUSCEPTABLE TO
FAILURE IN HIGH HUMIDITY.



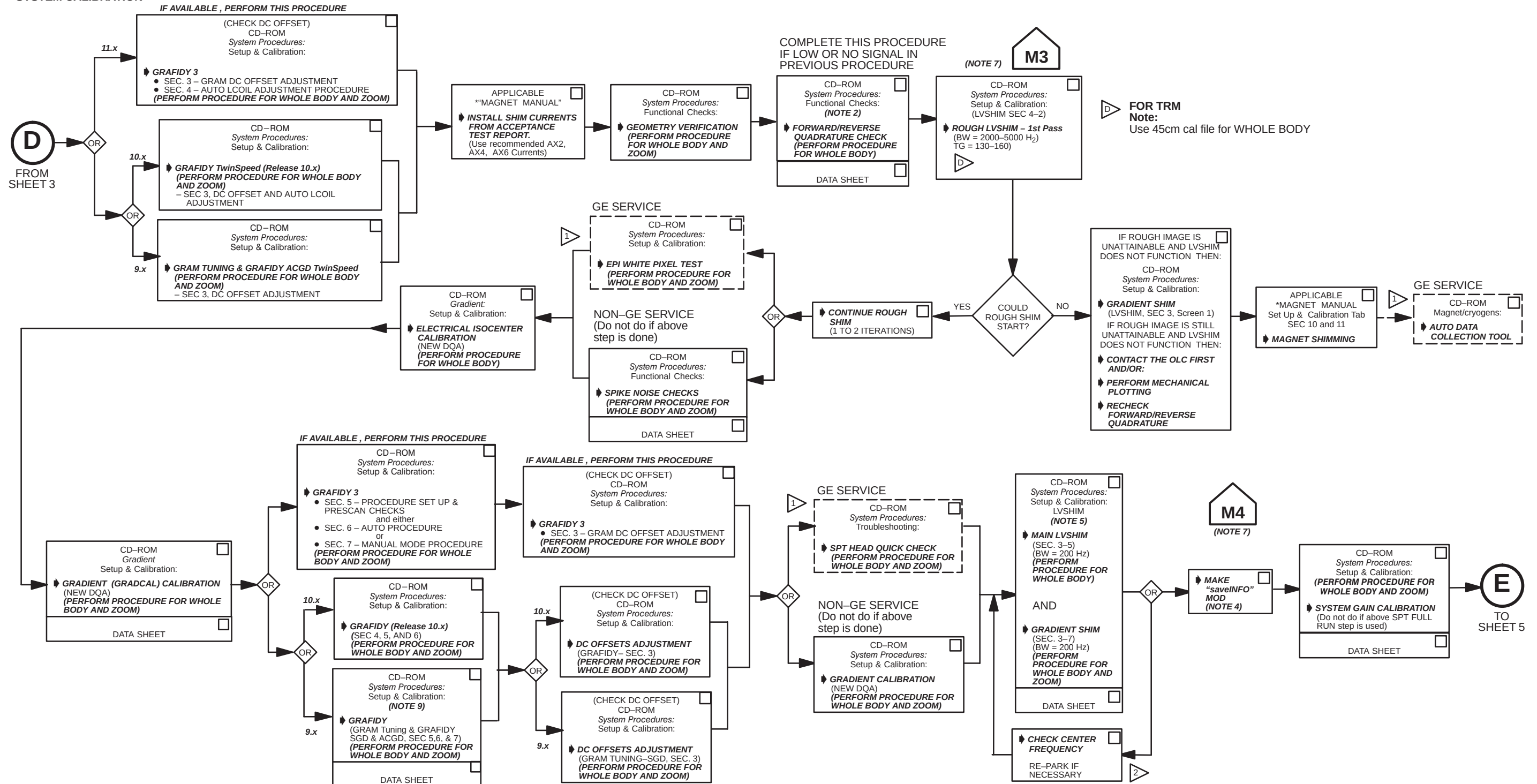
SIGNA TwinSpeed INSTALLATION FLOW CHART
ILLUSTRATION 1-2 (SHEET 3 OF 5)

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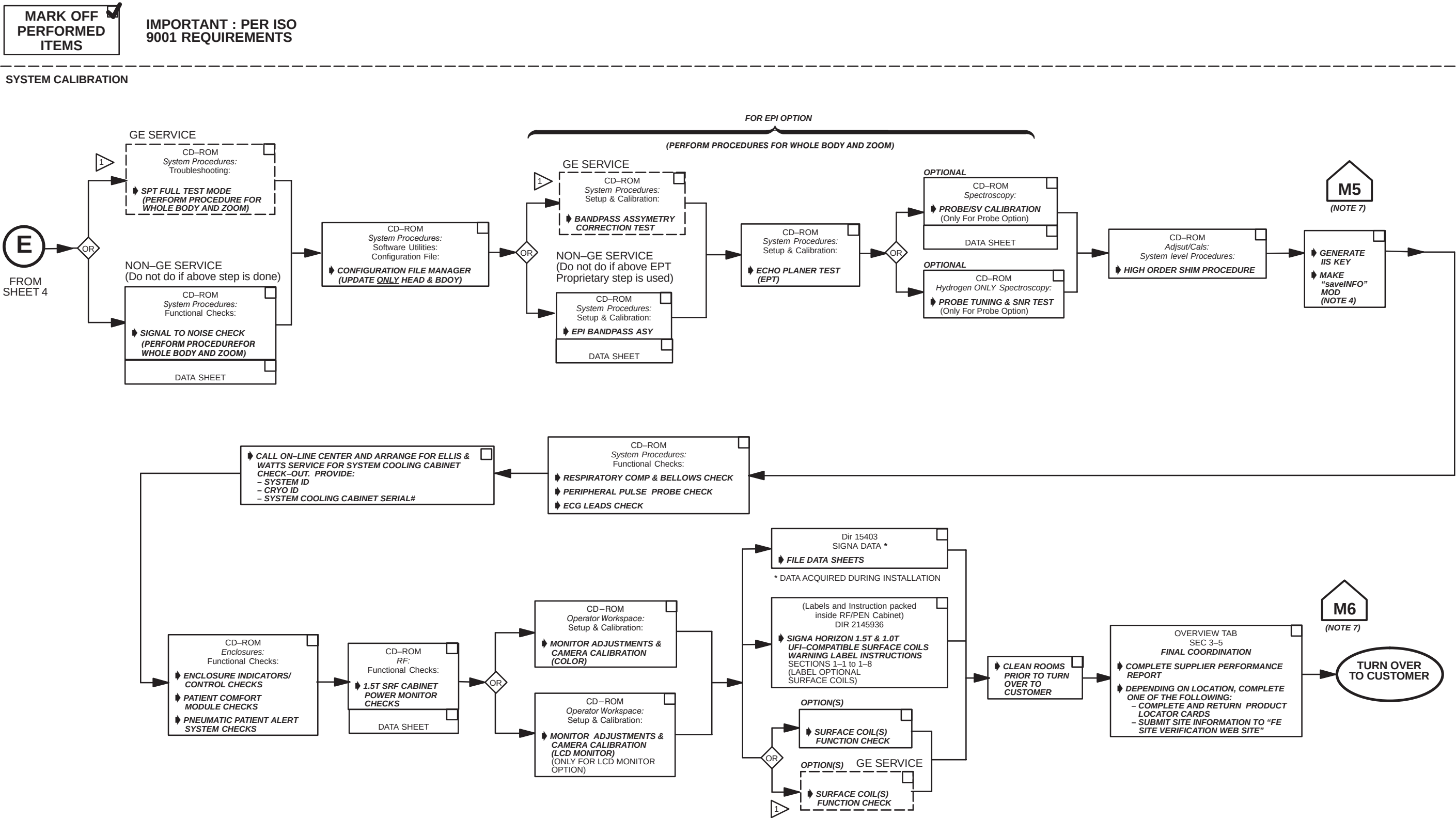
IMPORTANT: PER ISO 9001 REQUIREMENTS

SYSTEM CALIBRATION



SIGNA TwinSpeed INSTALLATION FLOW CHART ILLUSTRATION 1-2 (SHEET 4 OF 5)

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SIGNA TwinSpeed INSTALLATION FLOW CHART
ILLUSTRATION 1-2 (SHEET 5 OF 5)

SECTION 2 – SYSTEM CABLES

2-1 LOCATING AND UNPACKING CABLES

This section assumes Signa *TwinSpeed* System and options are delivered complete with all cables present.

Many of the cables are common to both the 9.x , 10.x (EXCITE I) and 11.x (EXCITE II) System. The cables that are specific to 9.x, 10.x, or 11.x installation are designated in the applicable cable maps and in the first column of the following tables as:

- * Cable used on 9.x installation only
- ** Cable used on 10.x (EXCITE I) installation only
- *** Cable used on 11.x (EXCITE II) installation only

Cables shipment containers have color coded labels which identify the cable routing areas:

- Green – Equipment Room
- Purple – Operator's Room
- Orange – Magnet Room
- Yellow – Post-installation (Phatoms, Coils)



Ty-wraps must be cut flush with no protruding sharp edges or points. Failure to do so can result in numerous laceration hazards when servicing the equipment.

The table below provides definitions for the following cable tables.

COLUMN HEADING	DESCRIPTION	SPECIAL NOTES
RUN	The run number as seen on the delivered cable	The vendor cable may not have run numbers.
"FROM" & "TO"	Labels the destination of the cable	
DESCRIPTION	Could include information on length, number of conductors, type of cable, and end plug type	Total length of cable delivered to site, type of cable specifies power from data.
CABLE TYPE	Type designation for the cable	
VOLTAGE RATING	The highest voltage that may be continuously applied to a wire	Specific to the core cable manufacturer given.
ACTUAL VOLTAGE	Nominal for continuous; maximum for variable voltage	If voltage varies during use, the maximum is given.
TEMP RATING	Maximum temperature insulating material may be used without loss of basic properties	Specific to the core cable manufacturer given.
FLAME RATING	Measure of material's ability to support combustion	Specific to the core cable manufacturer given.
REMARKS	Describe the cable's use in the MR system	

2-1 LOCATING AND UNPACKING CABLES (continued)

1. Unpack magnet room cables from the GE Magnet accessory box(s) and move them into the magnet or equipment room for routing. The shield cooler runs should have been installed previously when magnet was delivered.
 - a. For magnet and equipment room standard runs delivered with GE Magnet, refer to Table 2-3.
2. Unpack and move power and ground cable boxes to the PDU area where they will be routed. Set Run 040, 044, and 045 aside. Power and ground cables are listed in Table 2-1.

TABLE 2-1
PDU POWER AND GROUND CABLES

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
037	PD1 GND	MR2 A12 GND	36 ft Ground Wire (#10)	SO	600	0	105	VW-1	MR2 "permanent" (UL) ground
038	PD1 A2 A2	MR1 A7 GND	31 ft Ground Wire (#10)	SO	600	0	105	VW-1	SRF Cabinet ground
040	MSI	RF COM GND	69 ft Ground Wire (#1/0)	SO	600	208Y	105	VW-1	Magnet Ground Cable
044	PP1 GND	MR1 A7 GND	31 ft Ground Wire (#10)	SO	600	0	105	VW-1	RF Cabinet Separate ground
045	PP1 GND	MG6 GND	41 ft Ground Wire (#10)	SO	600	0	105	VW-1	Blower Cabinet Separate ground
048 * **	PD1 GND	OW1 A2 A5 GND	51 ft Ground Wire (#10/1)	SO	600	0	105	VW-1	Operator Workspace Cabinet ground
049 * **	OW1 A1 A1	OW1 A2 A5 J2	Power Cord	STO	600	120	90	VW-1	Operator Workspace Table to OW Cabinet
966	MR1 A7 J1	MR3 PD1	35 ft Power (5-#8)	SO	600	208Y	75	FT-4	SSM Module Power
967	MR1 CB1	MR3 PD1	35.5 ft Power (4-#8)	SO	600	208	90	FT-4	SRF Cabinet Power
968	MR1 A12 J1	MR3 PD1	35 ft Power (5-#8)	SO	600	208Y	75	FT-4	System Cabinet Power
969 * **	OW1 A2 A5 J1	MR3 PD1	65.5 ft Power (4-#10)	SO	600	120	75	FT-4	Operator Workspace Cabinet Power
988	PP1 A7 GND	TAC GND	60 ft. Ground (#8)	SO	600	0	80	VW-1	TAC to PP1 ground
989	MR3 PD1	TAC J8	25 ft. Power (3-#10)	SO	600	120	90	FT-4	TAC Power
1081 ***	OW1 A19 J1	MR3 PD1	65.5 ft Power (4-#10)	SO	600	120	75	FT-4	Operator Workspace Power
N/A	PD1 MAG/SHIM OUTLET	MAG/SHIM POWER SUPPLY	50 ft Power (5-#8) With 30A Plug and Receptacle						Cable for Compact PDU to GE Magnet Power Supply or SC Shim Supply Power

2-1 LOCATING AND UNPACKING CABLES (continued)

3. Unpack and move fixed site magnet room interconnect cables to magnet room. Refer to Table 2-2.
- Locate Body Coil RF input Cables (ship with Body Coil) and move cables to magnet room.
 - Runs 404, 414, 421, 422, 485, and 749 are shipped with the Magnet Enclosure.

TABLE 2-2
MAGNET ROOM RUNS (NOT CUT TO LENGTH)

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
262	PP1 J8	MG3 A3 A5 OUT	45 ft. RG223/U Coax	CMX	1900 VRMS	<30	80	VW-1	Body Coil receive
282	PP1 J12	MG3 A2 J6	45 ft. 25 pin Sub-D	CI3	300	<30	80	FT-4	Patient Intercom
297	PP1 J18	EO1	90 ft. 9 pin Sub-D	CI3	300	<30	80	FT-4	Emergency off switch
300	PP1 J15	MG2 A15	60 ft. 25 pin Sub-D	CI3	300	<30	80	FT-4	Bore Light power
318	PP1 J11	MG3 A7	45 ft. 9 pin Sub-D	CI3	300	38.5 VDC	80	FT-4	Long Drive Motor power
385	PP1 J20	MG2 A30	67 ft. 37 pin Sub-D	CI3	300	<30	80	FT-4	Patient Alignment Light power
387	PP1 J47	MG2 A29 J1	60 ft. 25 pin Sub-D	CI3	300	38.5 VDC	80	FT-4	Table Elevation motor power
404	MG3 A17 J1	MG2 A16 A5 J6	13.67 ft. RG58/U Coax	CI2	1900 VRMS	<30	80	VW-1	Head transmit cable take up
414	MG3 A35	MG3 A2 J9	30 ft. 9-pin Sub-D	CI3	300	<30	80	FT-4	Front Cover Microphone
421	MG2 A33 J1	MG3 A31 J1	15 ft. 15 pin Sub-D	CI3	300	<30	80	FT-4	Dock Limit
422	MG2 A33 J2	MG3 A1 J4	25 ft. 37 pin Sub-D	CI3	300	<30	80	FT-4	Long Drive Encoder/Limit Sw
458	PP1 J17	OM3	9 pin Sub-D	CI3	300	<30	80	FT-4	Optional Remote O2 Sensor
485*	MG3 A17 J2	MG2 A16 A3 OUT	13.67 ft. RG58/C Coax	CI2	1900 VRMS	<30	80	VW-1	Head receive cable take up
491	PP1-J80	MG3-A36-J7	80 ft. BNC	CMX	1900 VRMS	<30	80	VW-1	MC2 Receive
492	PP1-J81	MG3-A36-J8	80 ft. BNC	CMX	1900 VRMS	<30	80	VW-1	MC3 Receive
493	PP1-J82	MG3-A36-J9	80 ft. BNC	CMX	1900 VRMS	<30	80	VW-1	MC4 Receive
494*	PP1-A14-J87	MG3-A36-J6	80 ft. BNC	CMX	1900 VRMS	<30	80	VW-1	MC1 Receive
495*	PP1-A15-J78	MG3-A36-J10	80 ft. 15 Sub-D	CI3	300	<30	80	FT-4	Multi-coil Switching
612	PP1 SHIM	MG2 A12 A1	50 ft. Output						Shim Option
700*	PP1 A14 J86	MG2/MG3 A17 J2	45 ft. RG223/U Coax	CMX	1900 VRMS	<30	80	VW-1	Head or Surface Coil receive
715	PP1 J14	MG2 A33 J1	65 ft. 37 pin Sub-D	CI3	300	<30	80	FT-4	SRI Power
716	PP1 J89	MG2 A11 A1 J2	65 ft. 15 pin Sub-D	CI3	300	<30	80	FT-4	PAC Power
727*	PP1-A16-J77	MG3-A36-J5	80 ft. 15 Sub-D	CI3	300	<30	80	FT-4	Multi-coil Bias
728	MG3 A36 J11	MG2 A16 OUT J1	13.67 ft. BNC	CI2	500 VRMS	<30	200	VW-1	RF Cable
729	MG3 A36 J12	MG2 A16 OUT J2	13.67 ft. BNC	CI2	500 VRMS	<30	200	VW-1	RF Cable
730	MG3 A36 J13	MG2 A16 OUT J3	13.67 ft. BNC	CI2	500 VRMS	<30	200	VW-1	RF Cable
731	MG3 A36 J14	MG2 A16 OUT J4	13.67 ft. BNC	CI2	500 VRMS	<30	200	VW-1	RF Cable
732	MG3 A36 J15	MG2 A16 OUT J5	13.67 ft. BNC	CI2	500 VRMS	<30	200	VW-1	RF Cable

2-1 LOCATING AND UNPACKING CABLES (continued)

TABLE 2-2 (CONTINUED)
MAGNET ROOM RUNS (NOT CUT TO LENGTH)

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
733	MG3 A36 J16	MG2 A16 OUT J6	13.67 ft. BNC	CI2	500 VRMS	<30	200	VW-1	RF Cable
734	MG3 A36 J17	MG2 A16 OUT J7	13.67 ft. BNC	CI2	500 VRMS	<30	200	VW-1	RF Cable
735	MG3 A36 J18	MG2 A16 OUT J8	13.67 ft. BNC	CI2	500 VRMS	<30	200	VW-1	RF Cable
736*	MG3 A36 J19	MG2 A16 A8 J9	12 ft. 25 pin Sub-D	CI3	300	<30	80	FT-4	RF Cable
743	MG2 A12 A1 J2	MG3 A3 A1 J3	10 ft Coax w/M&F "N"	STO	2500 VDC	950	85	VW-1	Body Coil Drive
744	MG2 A12 A1 J1	MG3 A3 A1 J4	10 ft Coax w/M&F "N"	STO	2500 VDC	950	85	VW-1	Body Coil Drive
749	MG2 A33 J10	MG2 A36	35 ft 3 cond w/ 9-Sub D	CI3	300	<30	105	FT-4	Bore Temperature Sensor
778	PP1 A11 J72	MG2 A12 A2 J3	70 ft. MHV Coax	CI2	1900 VRMS	1000 VDC	80	VW-1	Dynamic Disable (body coil frnt)
779	PP1 A11 J73	MG2 A12 A2 J4	70 ft. MHV Coax	CI2	1900 VRMS	1000 VDC	80	VW-1	Dynamic Disable (body coil frnt)
780	PP1 A11 J74	MG3 A3 A3 J6	60 ft. MHV Coax	CI2	1900 VRMS	1000 VDC	80	VW-1	Inductive Drive
781	PP1 A11 J75	MG2 A12 A2 J5	60 ft. MHV Coax	CI2	1900 VRMS	1000 VDC	80	VW-1	Dynamic Disable (body coil rear)
782	PP1 A11 J76	MG2 A12 A2 J6	60 ft. MHV Coax	CI2	1900 VRMS	1000 VDC	80	VW-1	Dynamic Disable (body coil rear)
784	PP1 F5, F6	MG6 J1	40 ft. 2 cond power cable	SO	600	120	80	FT-4	Patient cooling blower power
848	MG2 A11 J4	MG2 A4 J5	16 ft. 25 pin Sub-D/rt. angl	CI3	300	<30	80	FT-4	Operator Display to PAC II
861 * **	MG2 A4 J4	MG2 A33J5,6,7,8,9	20 ft 5/37 pin Sub-D/80 pin	CI2	300	<30	80	VW-1	Operator Display to SRI
862	MG2 A4 J1	MG2 A5 J1	2 ft. 26 pin Rt. angle	CI2	300	<30	80	VW-1	Oper Display to Rt. Op Cntrl
863	MG2 A4 J2	MG2 A5 J2	2 ft. 26 pin Rt. angle	CI2	300	<30	80	VW-1	Oper Display to Lt. Op Cntrl
893	MG3 A3 J2	MG3 A3 A5 INPUT	6.6 Inch BNC		1900 VRMS	<30	80	VW-1	Splitter to Preamp Cable
897	MG2 A4 J3	MG2 A33 J4	20 ft. 37 pin Sub-D/26 pin	CI3	300	<30	80	FT-4	Operator Display to SRI
990	PP1 J52	Vacuum Sensor	50 ft.. 9 pin Sub-D	CI3	300	<30	80	FT-4	Vacuum Sensor
1011 ***	MG2 A4 J4	MG2 A33 J5	20 ft. Data Cable (80-#28)	CI2	300	<30	80	VW-1	SRI Splitter to Oper Display
1035 ** / ***	MG2 A16 A3/A5 J4	MG3 A36 J21	40 ft. RG58/C Coax	CI2	1900 VRMS	<30	80	VW-1	Head or Surface Coil Receive
1036 *X / ***	PP1 J84	MG3 A36 J6	BNC	CMX	1900 VRMS	<30	80	VW-1	MC4 Receive
1037 ** / ***	PP1 J6	MG3 A36 J1	BNC	CMX	1900 VRMS	<30	80	VW-1	MC4 Receive
1038 ** / ***	PP1 J7	MG3 A36 J2	BNC	CMX	1900 VRMS	<30	80	VW-1	MC4 Receive

2-1 LOCATING AND UNPACKING CABLES (continued)

TABLE 2-2 (CONTINUED)
MAGNET ROOM RUNS (NOT CUT TO LENGTH)

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
1039 ** / ***	PP1 J58	MG3 A36 J3	BNC	CMX	1900 VRMS	<30	80	VW-1	MC4 Receive
1040 ** / ***	PP1 J59	MG3 A36 J4	BNC	CMX	1900 VRMS	<30	80	VW-1	MC4 Receive
1041 ** / ***	PP1 A15 J78	MG3 A36 J10	80 ft. 15 Sub-D	CI3	300	<30	80	FT-4	SRI Power
1042 ** / ***	PP1 A16 J77	MG3 A36 J5	80 ft. 15 Sub-D	CI3	300	<30	80	FT-4	CrossPoint/8 Channel SW Control Power Cable
1043 ** / ***	MG3 A8 J11	MG3 A5 J1	0.5ft Discrete Wires (13-#22)	CI3	300	<30	60	VW-1	Head T/R Power/Coil ID LED
1049 ** / ***	MG3 A36 J19	MG2 A16 A8 J9	12 ft. 25 pin Sub-D	CI3	300	<30	80	FT-4	Multi-Coil Select Cable
1054 ***	MG3 A36 J20	MG2 A33 J15	Data Cable (9-#22)	CI3	300	<30	80	vw-1	MXA Signal Cable
1057 ***	PP1 J85	MG3 A8 J1	32 ft. R223/U Coax	CMX	1900 VRMS	<30	80	VW-1	UNTS Antenna

TABLE 2-3
GE MAGNET CABLES

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
601	MS8	MS1	Power Cable (#4/0)	SO	600	208	80	VW-1	Ramp cables, Power Assembly Positive 4/0
602	MS8	MS1	Power Cable (#4/0)	SO	600	208	80	VW-1	Ramp cables, Power Assembly Negative 4/0
603	MS7	MS1 A3 A1	Power Cable (#4/0)	SO	600	208	80	VW-1	Ramp and Shim Cable Assembly
604	MS7 or MS8	MS1 A3 A1	Data Cable (12-#22)	CI3	150	60 VDC	80	FT-4	Switch Heaters
605	MS1	MR2 A8	Data Cable (4-#22)	CI3	300	<30	105	FT-4	Instrumentation Cable, 80 feet
606	MS4	MS1 A3 A1	Data Cable (4-#22)	CI3	300	<30	105	FT-4	Emergency Run Down Unit
608	MS1	MR2 A8	Data Cable (4-#22)	CI3	300	<30	105	FT-4	Instrumentation Cable, 15 feet
609	MS7 OR MS8	PDU	Power Cable (5-#16)	SO	300	208Y	105	FT-4	Feeder Cables
621	MS5 A1	MS1 A2	7/8" Gas Flex Line	STO	N/A	N/A	85	VW-1	Supply Flexible Gas Line (PP1 J60 or J61)
622	MS5 A1	MS1 A2	7/8" Gas Flex Line	STO	N/A	N/A	85	VW-1	Return Flexible Gas Line (PP1 J60 or J61)
623	MS5 A1	PP1 F1, F2, F3, F4	Power Cable (4-#18)	SO	600	200	105	VW-1	Compressor to Pen Panel Cable (equipment room)
624	MS1 A2	PP1 F1, F2, F3, F4	Power Cable (4-#18)	SO	600	200	105	VW-1	Cold Head Control Cable

2-1 LOCATING AND UNPACKING CABLES (continued)

4. Unpack and move fiber optic cables into routing location. Refer to Table 2-4.

TABLE 2-4
FIBER OPTIC CABLES DELIVERED WITH FIXED-SITE KIT

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
708*	MR2 A11 J16	MR1 A7 TX	20 ft. 1/2 in. fiber optic conduit	N/A	N/A	0	85	N/A	Single simplex plastic fiber optic cable
709	MR3 A11 J2	MR1 A7 RX	20 ft. 3/4 in. fiber optic conduit	N/A	N/A	0	85	N/A	Single simplex plastic fiber optic cable
710*	MR2 A11 J17	MR3 A11 J1	20 ft. 3/4 in. fiber optic conduit	N/A	N/A	0	85	N/A	Four glass fiber optic and one plastic fiber optic cables
711/ 712*	MR2 A11 J13	PP1-J91	100 ft. 3/4 in. fiber optic conduit	N/A	N/A	0	85	N/A	Two duplex fiber optic and one simplex fiber optic cables
711/ 712	PP1-J91	MG2 A33 and MG2 A11 A1	100 ft. 3/4 in. fiber optic conduit	N/A	N/A	0	85	N/A	One duplex fiber optic cables and one simplex fiber optic cable and One duplex plastic fiber optic cable for PAC-II
836*	MR2 A11 J30 BIT3	OW1 A15 BIT3	72.2 ft. 3/8 in. jacketed cable	N/A	N/A	0	85	N/A	Two fiber optic cables with two spares
1033 ** / ***	MR2 A11 J16	MR1 A7 TX	20 ft. 1/2 in. fiber optic conduit	N/A	N/A	0	85	N/A	Single simplex plastic fiber optic cable
1034 ** / ***	MR2 A11 J17	MR3 A11 J1	20 ft. 3/4 in. fiber optic conduit	N/A	N/A	0	85	N/A	Four glass fiber optic and one plastic fiber optic cables
1044**	MR2 A11 J22	OW1 A15 SLOT2	72.2 ft. 3/8 in. jacketed cable	N/A	N/A	0	85	N/A	Two fiber optic cables with two spares
1045 ** / ***	MR2 A11 J13	PP1 A17 A1	3/4 in. fiber optic conduit	N/A	N/A	0	85	N/A	Two duplex fiber optic and one simplex fiber optic cables
1078 ***	MG2 A33 J30/J29	MG3 A36 J18/J19	72.2 ft. 3/8 in. jacketed cable	N/A	N/A	0	85	N/A	Two fiber optic cables, SRI to Cross Point Switch
1083 ***	MR2 A11 J22	OW1 A15 XMT1	72.2 ft. 3/8 in. jacketed cable	N/A	N/A	0	85	N/A	Two fiber optic cables with two spares

2-1 LOCATING AND UNPACKING CABLES (continued)

5. Unpack and move interconnect cables which are cut to length to the applicable room. Refer to Table 2-5.
- a. Gradient cables will be cut, re-terminated and routed according to appropriate Direction in Tab 1.

TABLE 2-5
EQUIPMENT ROOM AND MAGNET ROOM CABLES & HOSES CUT TO LENGTH

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
536	WC1	MG2 A12 A1	Flexible Tubing	N/A	N/A	N/A	N/A	N/A	Gradient Coil Cooling Line
537	WC1	MG2 A12 A1	Flexible Tubing	N/A	N/A	N/A	N/A	N/A	Gradient Coil Cooling Line
235	MR1 A7 J13	PP1 J44	7/8 in. HELIAX® Coax or	STO	6000 VDC	<30	120	VW-1	1.5T Body Coil Run created from kit
			Times Microwave LMR-900FR	CATV R	5000 VDC		85	FT-4	
236	MR1 A7 J12	PP1 J4	1/2 in. HELIAX® Coax or	STO	4000 VDC	<30	120	VW-1	1.5T Head Coil Run created from kit
			Times Microwave LMR-600FR	CATV R	4000 VDC		85	FT-4	
257	PP1 J44	MG3 A3 A1 J1	7/8 in. HELIAX® Coax or	STO	6000 VDC	<30	120	VW-1	1.5T Body Coil Transmit
			Times Microwave LMR-900FR	CATV R	5000 VDC		85	FT-4	
258	PP1 J4	MG3 A17 J1	1/2 in. HELIAX® Coax or	STO	4000 VDC	<30	120	VW-1	1.5T Head Coil Transmit
			Times Microwave LMR-600FR	CATV R	4000 VDC		85	FT-4	
976	MR3 CABI/F J3	TAC ±X	Conductor (3-#2)	STO	1000	600	90	FT-4	Twin X Gradient Power
977	MR3 CABI/F J4	TAC ±Y	Conductor (3-#2)	STO	1000	600	90	FT-4	Twin Y Gradient Power
978	MR3 CABI/F J5	TAC ±Z	Conductor (3-#2)	STO	1000	600	90	FT-4	Twin Z Gradient Power
979	TAC X+,WB-,ZM-	PP1X+,WB-,ZM-	Conductor (3-#2)	STO	1000	600	90	FT-4	Twin X Gradient Filter
980	TAC Y+,WB-,ZM-	PP1Y+,WB-,ZM-	Conductor (3-#2)	STO	1000	600	90	FT-4	Twin Y Gradient Filter
981	TAC Z+,WB-,ZM-	PP1Z+,WB-,ZM-	Conductor (3-#2)	STO	1000	600	90	FT-4	Twin Z Gradient Filter
982	PP1 ZZM+,ZM-	MG2 A12 A1 7/8	Conductor (3-#2)	STO	1000	600	90	FT-4	Twin Zzm Gradient Output
983	PP1 YZM+,ZM-	MG2 A12 A1 9/10	Conductor (3-#2)	STO	1000	600	90	FT-4	Twin Yzm Gradient Output
984	PP1 XZM+,ZM-	MG2 A12 A1 11/12	Conductor (3-#2)	STO	1000	600	90	FT-4	Twin Xzm Gradient Output
985	PP1 ZWB+,WB-	MG2 A12 A1 1/2	Conductor (3-#2)	STO	1000	600	90	FT-4	Twin ZWB Gradient Output
986	PP1 YWB+,WB-	MG2 A12 A1 3/4	Conductor (3-#2)	STO	1000	600	90	FT-4	Twin YWB Gradient Output
987	PP1 XWB+,WB-	MG2 A12 A1 5/6	Conductor (3-#2)	STO	1000	600	90	FT-4	Twin XWB Gradient Output

2-1 LOCATING AND UNPACKING CABLES (continued)

6. Verify magnet monitoring cables for GE Magnet are delivered (or if previously installed, remain in place). Refer to Table 2-6.

TABLE 2-6
MAGNET MONITORING CABLES DELIVERED WITH MAGNET

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
823	MSM1 A1 J10	MR2 A11 J24	Data Cable (15-#22)	CI3	300	<30	80	FT-4	Monitor to System Cabinet I/F
824	PP1 J10	FJ1	Data Cable (9-#22)	CI3	300	<30	80	FT-4	Monitor Sensor I/F to Penetration Panel
825	PP1 J48	MSM1 A1 J8	Data Cable (25-#22)	CI3	300	<30	80	FT-4	Magnet Monitor connector to Penetration Panel
826	MSM1 A1 J9	FJ3	Data Cable (15-#22)	CI3	300	<30	80	FT-4	Monitor to Compressor I/F
827	FJ3	MS5 A5 A1 J1 / FJ4	Data Cable (8-#22) & Data Cable (4-#22)	CI3	300	<30	80	FT-4	Compressor I/F
828	PP1 J10	MA1 A3 A1 P403	Data Cable (9-#22)	CI3	300	<30	80	FT-4	Penetration Panel to Sensor
829	PP1 J48	MS1 FJ2	Data Cable (25-#22)	CI3	300	<30	80	FT-4	Penetration Panel to Magnet I/F
830	MS1 FJ2	MS1 A2 J2/ MS1 A5 A2 J1/ MG3 TB1	3cond/3cond/25cond	CI3	300	<30	80	FT-4	Magnet I/F
831	FJ1	MSM1 A1 J7	Data Cable (8-#24)	CI3	300	<30	80	FT-4	Monitor to Sensor I/F
832	MS1 A2 J1	MS1 SLEEVE/ MS1 COLDHEAD	Data Cable (8-#22)	CM	300	<30	80	FT-4	Buffer Amp to Cooling Sleeves
833	FJ4	MS5 A1 A6 JR	Data Cable (6-#24)	CM	300	<30	80	FT-4	Compressor to Compressor I/F
939	MDP UPS	MSM4	6 ft. Power (5-#15)	SO	600	120	85	VW-1	MDP to UPS
940	MSM3 RS232	MSM1 J1	6 ft. Data Cable (9-#22)	CI3	300	<30	80	FT-4	Monitor to Modem
941	MSM1 J11	SCC j1	50 ft. 9-pin Sub-D	CI3	300	<30	80	FT-4	Monitor to Sys Cooling Cab (for TSCC & TGWC only)
942/ 1077	MSM1 J5	OW1 A2 A4 PRT7	80 ft. Cat 5 Ethernet	CI3	300	<30	80	FT-4	Monitor to OW
	MR2 A11 J24	MR2 IPG J9/ MR2 PDU J5							System Cabinet I/F

2-1 LOCATING AND UNPACKING CABLES (continued)

7. Unpack and move equipment room system cables into routing location (cabinets, Operator Workstation). Refer to Table 2-7.

TABLE 2-7
EQUIPMENT ROOM RUNS

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
229	MR2 A11 J1	MR1 A7 J3	20 ft RG223/U Coax	CMX	1900 VRMS	<30	80	VW-1	Exciter RF output
231	MR2 A11 J2	PP1 J8	40 ft RG223/U Coax	CMX	1900 VRMS	<30	80	VW-1	Body Coil receive
296	FACILITY DISCNCT	PP1 J18	45 ft 9 pin Sub-D	CI3	300	<30	80	FT-4	Emergency off switch
457	OM1	PP1 J17	9 pin Sub-D	CI3	300	<30	80	FT-4	Optional O2 Monitor cable
486*	PP1-A14-J85	MR1-A7-J22	30 ft. 9 Sub-D	CI3	300	<30	80	FT-4	Head/MC1 Receive Select Switch
487*	PP1-A15-J78	MR1-A7-J20	40 ft.15 Sub-D	CI3	300	<30	80	FT-4	Multi-coil Switching
488	PP1-J80	MR2-A11-J8	40 ft. BNC	CMX	300	<30	80	VW-1	MC2 Receive
489	PP1-J81	MR2-A11-J9	40 ft. BNC	CMX	300	<30	80	VW-1	MC3 Receive
490	PP1-J82	MR2-A11-J10	40 ft. BNC	CMX	300	<30	80	VW-1	MC4 Receive
499/233	MR2 A11 J5	PP1 J7	40 ft RG223/U Coax	CMX	300	<30	80	VW-1	Head and Surface Coil receive
611	TAC J22	PP1 SHIM							Resiisteive Shim PS Option
612	MG2 A12 A1	PP1 SHIM							Resiisteive Shim PS Option
701	MR2 A11 J15	RF Door switch	70 ft 9 pin Sub-D	CI3	300	<30	80	FT-4	RF Door switch
702	MR2 A11 J18	MR1 A7 J5	20 ft 9 pin Sub-D	CI3	300	<30	80	FT-4	Unblank/EFB
726*	PP1-A16-J77	MR1-A7-J18	30 ft. 15 Sub-D	CI3	300	<30	80	FT-4	Multi-coil Bias
768	MR1 A7 J31	PP1 J20	30 ft 37 pin Sub-D	CI3	300	<30	80	FT-4	Patient Alignment Light power
769	MR1 A7 J34	PP1 J15	30 ft 25 pin Sub-D	CI3	300	<30	80	FT-4	Bore Light power
770	MR1 A7 J41	PP1 J11	30 ft 9 pin Sub-D	CI3	300	38.5 VDC	80	FT-4	Long Drive Motor power
771	MR1 A7 J36	PP1 J47	30 ft 25 pin Sub-D	CI3	300	38.5 VDC	80	FT-4	Dock Power
772	MR1 A7 J40	PP1 J14	30 ft 37 pin Sub-D	CI3	300	<30	80	FT-4	SRI Power
773	MR1 A7 J35	PP1 A17 J1	30 ft 25 pin Sub-D	CI3	300	<30	80	FT-4	PAC & Fiber Optic I/F Power
774	MR2 A11 J25	MR1 A7 J32	20 ft 25 pin Sub-D	CI3	300	<30	80	FT-4	Hydraulic Manifold Interface
775	MR1 A7 J42	PP1 A11 J92	30 ft MHV Coax	CI2	1900 VRMS	1000 VDC	80	VW-1	100V Dynamic Disable
776	MR1 A7 J44	PP1 A11 J93	30 ft MHV Coax	CI2	1900 VRMS	1000 VDC	80	VW-1	100V Dynamic Disable
777	MR1 A7 J43	PP1 A11 J94	30 ft MHV Coax	CI2	1900 VRMS	1000 VDC	80	VW-1	100V Dynamic Disable
783	MR1 A7 J37	PP1 F5/F6	30 ft 2-#8 AWG	SO	600	120	80	FT-4	Blower Cabinet Power
788	OW1 A1 A4 J7	PP1 J12	40 ft Data (25-#22)	CI3	300	<30	80	FT-4	Intercom
789	OW1 A1 A4 J18	MR2 A11 J26	51 ft Data (9-#22)	CI3	300	<30	80	FT-4	E-stop
791	OW1 A16 PORT1	MR2 A11 J30	51 ft Data (25-#22)	CI3	300	<30	80	FT-4	TNF monitoring (Octane)
797	OW1 A2 A1 PC	OW1 A3 LCD	2 line; 3.5 mm jacks	CI3	300	<30	80	FT-4	LCD Display

2-1 LOCATING AND UNPACKING CABLES (continued)

TABLE 2-7 (CONT'D)
EQUIPMENT ROOM RUNS

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
798	OW1 A2 A3 HOST	OW1 A6	10.5 ft. Data (3-#24)	CI3	300	<30	80	FT-4	Color Monitor cable
805	OW1 A1 A4 J6	OW1 A7	3.3 ft. Data	CI3	300	<30	80	FT-4	Keyboard cable
807	OW1 A2 A3 HOST	OW1 A5	9.8 ft. Data	CI3	300	<30	80	FT-4	DASM SCSI cable
810	OW1 A1 A2 J8	OW1 A15 KBD,MS	9 ft. 25 pin Sub-D	CI3	300	<30	80	UL-1581	Keyboard/Mouse cable (Octane)
811	OW1 A2 A4	OW1 A15	12 ft. Ethernet	CI3	300	<30	80	FT-4	Communication
812	OW1 A15	OW1 A13	12 ft. 25/9 pin Sub-D	CM	300	<30	80	UL-1581	Modem
813	OW1 A2 A1 COM2	OW1 A415 PORT7	12 ft. 9 pin Sub-D	CI3	600	<30	105	VW-1	Communication
814	OW1 A2 A1 KBD,MS	OW1 A1 A4 J9	9 ft. 25 pin Sub-D	CI3	300	<30	80	FT-4	Keyboard/Mouse cable (PC)
815	OW1 A16	OW1 A15 SCSI	6 ft. 68/50 pin	CI3	300	<30	80	FT-4	SCSI Cable (Octane)
817	OW1 A2 A5 J3	OW1 A1 A2 J17	8 ft. 9 pin Sub-D	CI3	300	<30	80	FT-4	Workspace I/F Module
818	OW1 A16 J6,7,8	MR2 A11 J31	70 ft. 25/3-25 pin	CI3	300	<30	80	FT-4	TNF monitoring
819	OW1 A2 A7 SCSI	OW1 A15	10 ft. 68/50 pin	CI3	300	<30	80	FT-4	SCSI Cable
821	OW1 A15 MON-ITOR	OW1 A6 R,B,G	10.5 ft. 3 BNC/SubD	CI3	300	<30	80	FT-4	Color Monitor cable
971	MR2 A11 J23	MR3 PD1 J2	40 ft 15 pin Sub-D	CI3	300	<30	80	FT-4	PDU Status (ACGD)
972*	MR2 A11 J14	MR3 PD1 J3	40 ft 9 pin Sub-D	CI3	300	<30	80	FT-4	Remote PDU (ACGD)
974	MR3 PD1 J5	TAC J12	49 ft. 9 pin Sub-D	CI3	300	<30	80	FT-4	Twin GIP/GP
991	PP1 J52	TAC J9	50 ft. 9 pin Sub-D	CI3	300	<30	80	FT-4	TAC Communication
992	TSCC J3	TAC J10	60 ft. 9 pin/15 pin Sub-D	CI3	300	<30	80	FT-4	Twin System Cooling Cable (for TSCC & TGWC only)
1004	TAC J27	OW1 A16 PRT2	9 pin:25 pin Sub-D	CI3	300	<30	80	FT-4	MXA Signal Cable
1025 ** / ***	MR2 A11 J40	PP1 J6	40 ft. BNC	CMX	1900 VRMS	<30	80	VW-1	MC2 Receive
1026 ** / ***	MR2 A11 J41	PP1 J7	40 ft. BNC	CMX	1900 VRMS	<30	80	VW-1	MC2 Receive
1027 ** / ***	MR2 A11 J42	PP1 J58	40 ft. BNC	CMX	1900 VRMS	<30	80	VW-1	MC2 Receive
1028 ** / ***	MR2 A11 J43	PP1 J59	40 ft. BNC	CMX	1900 VRMS	<30	80	VW-1	MC2 Receive
1029 ** / ***	MR2 A11 J62	PP1 A16 J77	60 ft. 15 pin Sub-D	CI3	300	<30	80	FT-4	Monitor to Compressor I/F
1030 ** / ***	MR2 A11 J61	PP1 A16 J78	30 ft. 37 pin Sub-D	CI3	300	<30	80	FT-4	SRI Power
1031**	MR2 A11 J63	OW1 A15 SLOT1	65 ft. 25 pin Sub-D	CI3	300	<30	80	FT-4	Communication
1032	MR2-A11-J31	OW1-A16-FJ100	65 ft. 25 pin Sub-D	CI3	300	<30	80	FT-4	Serial Communication
1046	FJ100	OW1-A16-PORT 1,6,7,8	4.5 ft. 25 pin Sub-D	CI3	300	<30	80	FT-4	Serial Communication

2-1 LOCATING AND UNPACKING CABLES (continued)

TABLE 2-7 (CONT'D)
EQUIPMENT ROOM RUNS

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
1053 ***	PP1 J85	MR2 A11 J54	34.5 ft RG223/U Coax	CMX	1900 VRMS	<30	80	VW-1	UNTS Antenna
1056 ***	OW1 A17 PRT6	OW1-A12 ETHRNET	3.3 ft. Ethernet Cat 5	CI3	30	<30	80	FT-4	Laptop Access cable
1059 ***	OW1 A17 PRT2	OW1-A15 ETHRNET	5.9 ft. Ethernet Cat 5	CI3	30	<30	80	FT-4	Lan Hub cable
1060 ***	OW1 A19 J3	OW1 A21 J17	3.9 ft. 9 pin Sub-D	CI3	300	<30	80	FT-4	WIM Power cord
1061 ***	OW1 A15 KYBD/MS	OW1 A21 J8	2.9 ft. 25 pin Sub-D 2 PS/2s	CI3	300	<30	90	CL-2	Keyboard/Mouse cable
1062 ***	OW1 A20	OW1 A21 J6	13.2 ft. 50 pin SCSI, 9 pin Sub-D	CI3	300	<30	80	VW-1	SCIM cable
1063 ***	OW1 A15 Video 1	OW1 A6 HOST LCD	9.9 ft. 29 pin DVI, 15 pin VGA	CI3	300	<30	80	VW-1	Host LCD Display Video cable
1064 ***	OW1 A15 SCSI 1	OW1 A16 SCSI TOWER	9.9 ft. SCSI cable 50	CI3	300	<30	80	VW-1	SCSI Tower Data cable
1065 ***	OW1 A11 J2	OW1 A6 Pwr	9.9 ft. Power (3-#18)	SVT	300	120	80	VW-1	Host LCD Display Power cable
1066 ***	OW1 A15	OW1 A13	3.9 ft. 25/9 pn Sub-D	CM	300	<30	80	FT-4	Modem I/O cable
1067 ***	OW1 A15 SCSI 2	OW1 A5 DASM	5.9 ft. SCSI cable 50	CI3	30	<30	80	VW-1	DASM I/O cable
1068 ***	OW1 A15 Video2	OW1 A3 LCD	9.9 ft. Data (15-#28)	CI3	300	<30	80	VW-1	LCD Display Video cable
1069 ***	OW1 A11 J8	OW1 A3 Pwr	9.9 ft. Power (3-#18)	SVT	300	120	80	VW-1	LCD Display Power cable
1070 ***	OW1 A11 J11	OW1 A16 Pwr	9.9 ft. Power (3-#18)	SVT	300	120	80	VW-1	SCSI Tower Power cable
1071 ***	OW1 A11 J1	OW1 A15 Pwr	9.9 ft. Power (3-#18)	SVT	300	120	80	VW-1	Host Computer Power cable
1074 ***	OW1 A19 J2	OW1 A11	3.3 ft. Power (3-#14)	SVT	600	120	105	VW-1	Power Strip cord
1075 ***	OW1 A19 J4	OW1 A11	3.3 ft. Power (3-#14)	SVT	600	120	105	VW-1	Power Strip cord
1082 ***	OW1 A15 ETH SLOT 2	MR2 A11 J63	75 ft. Ethernet Cat 5	CI3	300	<30	80	FT-4	EXCITE Ethernet C'cable
1084 ***	OW1 A21 J6 via Run 1062	MR2 A11 J26	60 ft 9 pin Sub-D	CI3	300	<30	80	FT-4	E-stop
1085 ***	OW1 A21 J7	PP1 J12	65 ft 25 pin Sub-D	CI3	300	<30	80	FT-4	Intercom

8. Unpack additional equipment room cables for Signa TwinSpeed Options:
 - a. Refer to the service/installation manual shipped with options.
 - b. Refer to *Direction 2151387, Signa Horizon Spectroscopy Subsystem*, for 1.5T Spectroscopy (M1040JB) equipment room cables.

2-2 SORTING AND ROUTING CABLES



Ty-wraps must be cut flush with no protruding sharp edges or points. Failure to do so can result in numerous laceration hazards when servicing the equipment.

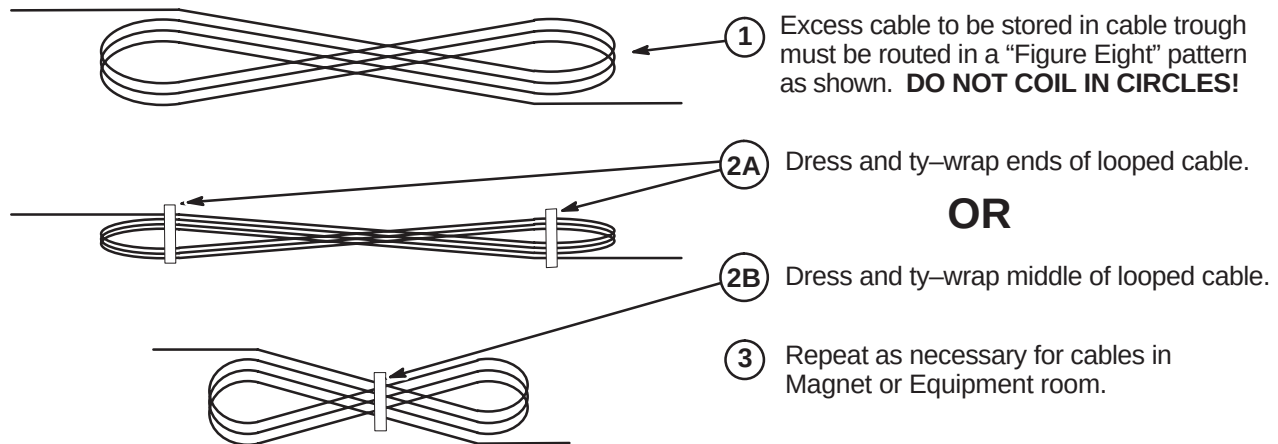
System power, ground, and data cables are color coded at each end by destination per colors shown on Cable Map.

1. Sort cables by destination. Normally, it is best to sort cables originating from the farthest cabinet or component from the PDU first. This will make it easier to pull cables through the troughs.

Note

Carefully review architectural drawings to insure that all ducts are properly installed and that all signal and power cables are accounted for before starting to route cable runs.

2. Unroll coiled cables along the intended route to insure that they will lay freely without twists or kinks. Route cables in accordance with the applicable System Interconnect Diagram.
3. Plan for storage location of coiled cable excess length if cables are to remain at delivered length. See Illustration 2-1. Some cables are usually cut to length such as gradient cables, power cables, Helix, and Fiber-optic cables.



PROPER STORAGE OF EXCESS CABLES

ILLUSTRATION 2-1

4. Place cables in provided troughs, conduit, or ducts. Leave sufficient slack for later connection when cabinets are in final position according to site architectural plans. Route cables inside exam room to Rear Pedestal area with sufficient length remaining to complete routing and connecting within Magnet Enclosure.

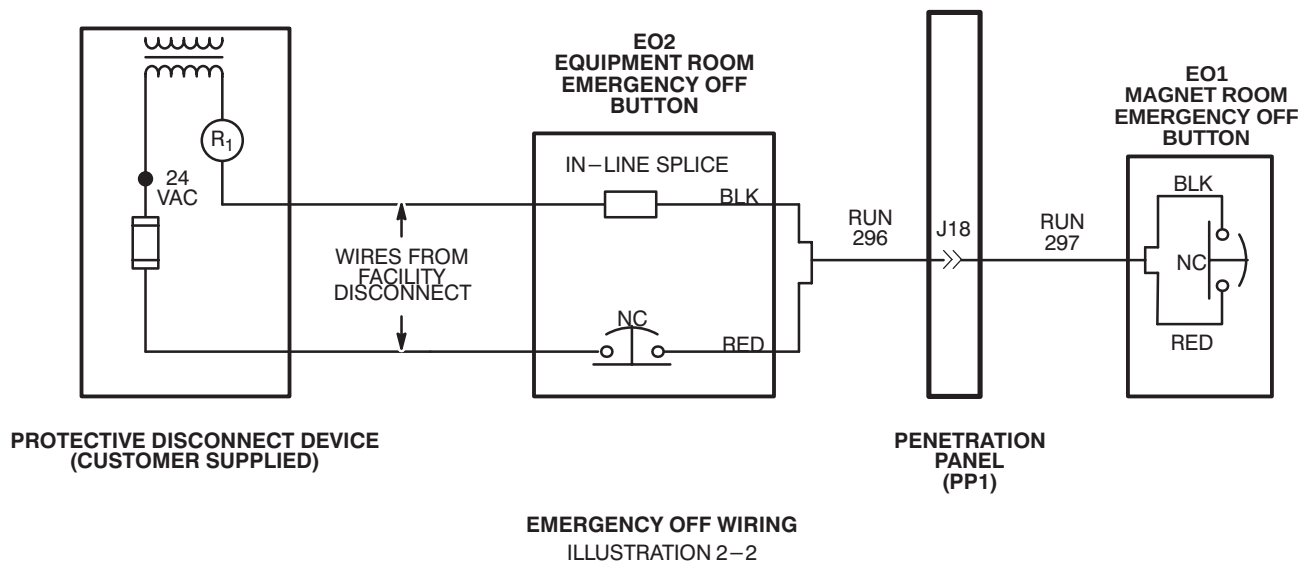
2-3 EMERGENCY OFF CONNECTIONS

Run 296 is routed to EO2 (Equipment Room Emergency Off) and connected to EO2 and wire from Facility Disconnect. Run 297 is routed from Penetration Panel to EO1 (Magnet Room Emergency Off Button). Refer to *Direction 2294003, Signa TwinSpeed Pre-Installation*, Section 5-3-1, Protective Disconnect Device, for detailed wiring diagram.

1. Complete routing of Run 296 to EO2 (Equipment Room Emergency Off) location, and trim cable to length. Locate red and black pair of wires and prepare ends for applicable terminals.

Aside: In Illustration 2-3, black and red wires are used for connections in Runs 295 and 297. Actually any pair of wires on these runs could be used so long as both ends are consistent with one another. Runs 296 and 297 are actually nine wire cables.

2. At equipment room room "Emergency Off" (EO2) location, terminate red wire at end of Run 296 with local supplied terminal and connect to customer supplied Equipment Room Emergency Off Button (EO2). See Illustration 2-3.
3. Terminate customer supplied wire (from fuse in Protective Disconnect Device) with local supplied applicable terminal and connect to customer supplied Emergency Off Button (EO2). See Illustration 2-3.
4. Terminate customer supplied wire from R1 in Protective Disconnect Device with local supplied push-on terminal.
5. Terminate black wire at end of Run 296 with local supplied "push-on" terminal.
6. Connect black wire from end of Run 296 to customer supplied wire from protective disconnect device with local supplied in-line splice. See Illustration 2-3.
7. Complete routing of Run 297 to EO1 (Magnet Room Emergency Off) location, and trim cable to length. Locate red and black pair of wires and prepare ends for applicable terminals.
8. Terminate red and black wires at end of Run 297 with local supplied terminals and connect to customer supplied Magnet Room Emergency Off Button (EO1). See Illustration 2-3.



2-4 RUN 701 (RF DOOR SWITCH) CONNECTIONS

- 1. Route Run #701 (from System Cabinet) to RF Door Switch. This 100 ft cable has a 9-pin subminiature “D” plug on one end. The other end is cut off with the black and red leads from pair #1 dressed for connecting to a switch. Should it be desired to cut off excess length, make sure that the black and red leads have continuity to pins 1 and 6 on the 9 pin subminiature D plug.
- 2. Connect black lead on Run #701 to RF Door Switch COM (common) contact.
- 3. If RF door switch is normally open, proceed to step 4. If normally closed, proceed to step 5.
- 4. Connect red lead on Run #701 to RF Door Switch N.O. (normally open) contact.
- 5. If RF door switch is not normally open, connect red lead on Run #701 to RF Door Switch N.C. (normally closed) contact.

2-5 CABLE INTERCONNECTION DOCUMENTATION

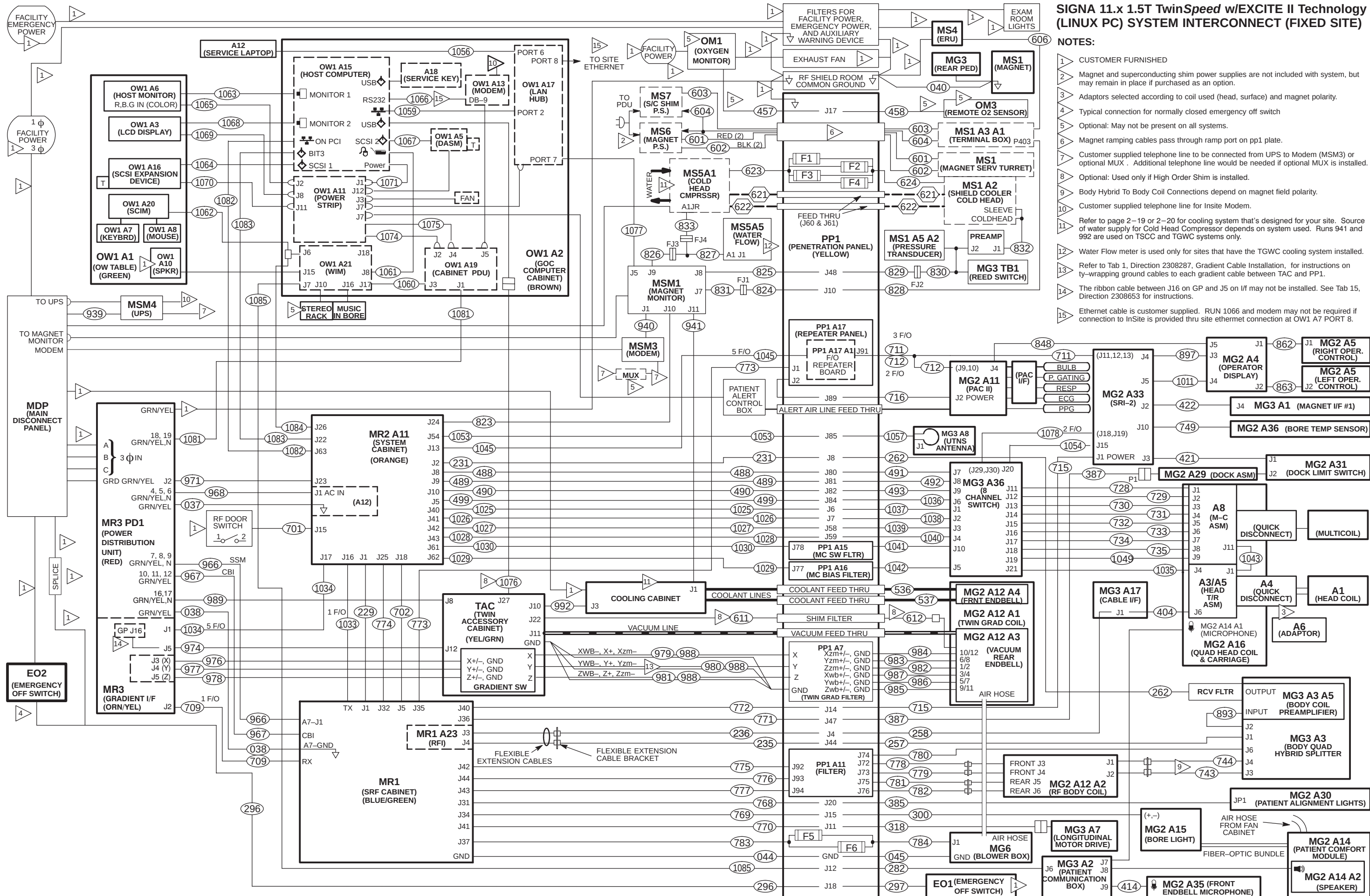
The cable maps illustrate the Signa TwinSpeed System interconnect for all supplied cables and interface with customer supplied wiring.

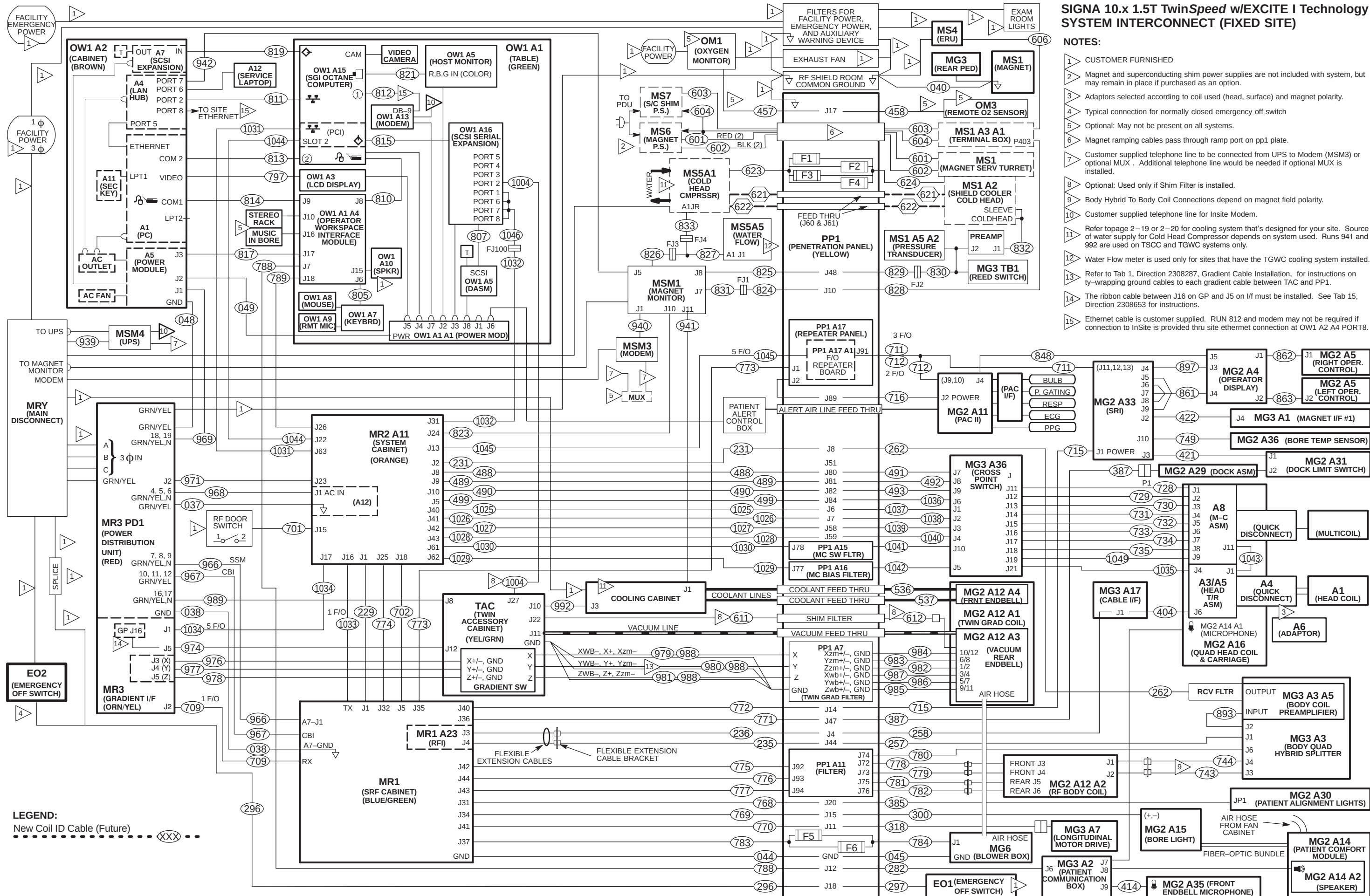
- 1. Interconnects for additional cables supplied with other options (Spectroscopy, etc.) are illustrated in the installation manual shipped with the option.
 - Connections for 1.5T Spectroscopy are in *Direction 2304882, Signa Spectroscopy Subsystem*.
 - Instructions for connections for optional laser cameras are supplied with option.
 - Connection procedures for other options are covered in the installation manual shipped with the option.
- 2. Procedures for connections made at the Magnet Enclosure, Operator Workspace, cabinets, etc. are detailed in the respective installation section within this manual.
- 3. Interconnect details for magnet subsystem are covered in the applicable magnet subsystem manual.

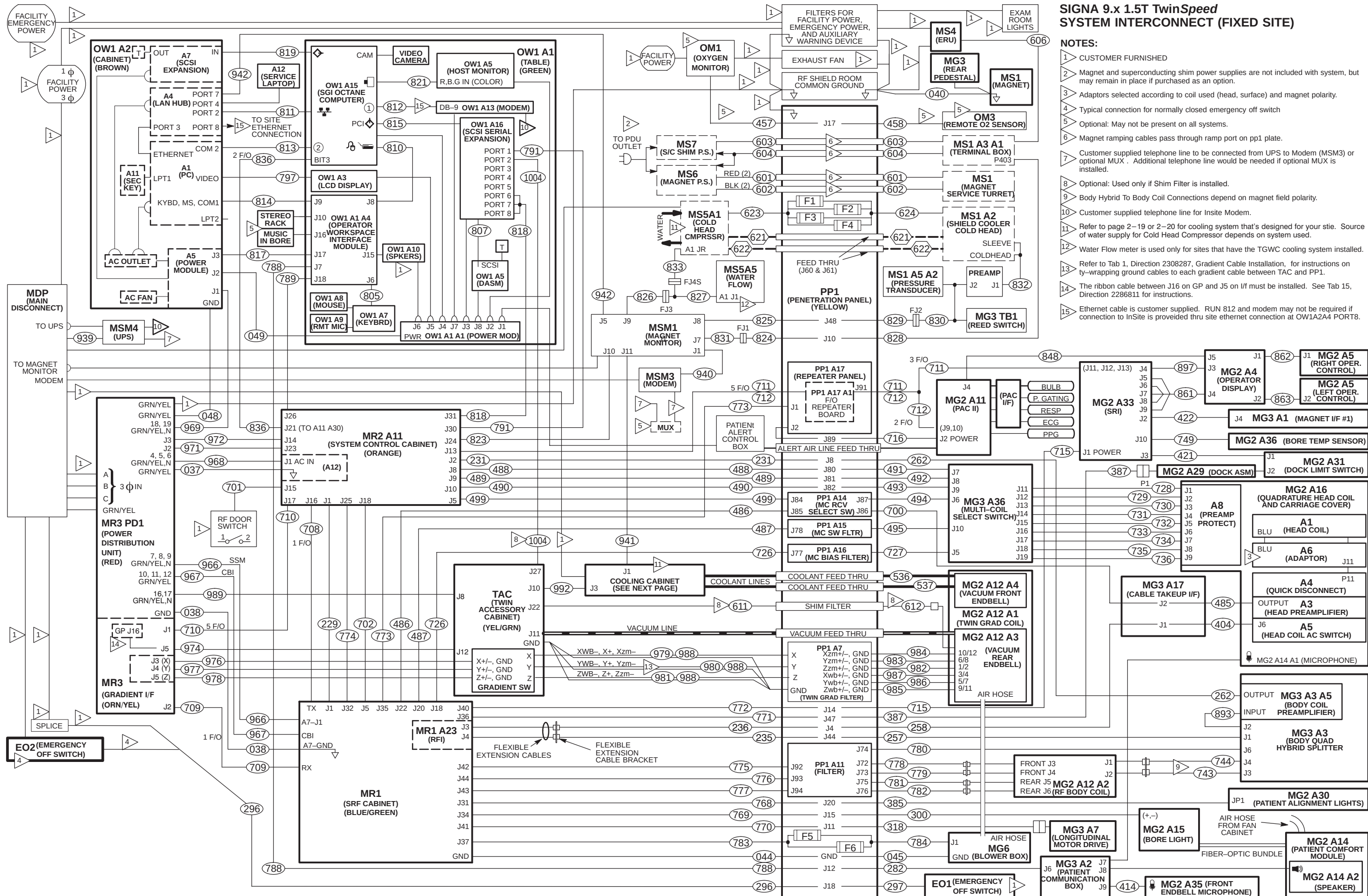
Fold-out TwinSpeed System Interconnect Cable Map is provided at the end of this section. They are:

Description	Page
Signa 11.x 1.5T TwinSpeed w/EXCITE II System Interconnect (Fixed Site)	2-15
Signa 10.x 1.5T TwinSpeed w/EXCITE I System Interconnect (Fixed Site)	2-16
Signa 9.x 1.5T TwinSpeed System Interconnect (Fixed Site)	2-17
Signa 1.5T TwinSpeed System Cooling Cabinet Interconnect Options – TSCC & TGWC	2-19
Signa 1.5T TwinSpeed System Cooling Cabinet Interconnect Options – MRCC & GWHX	2-20
Signa 11.x 1.5T TwinSpeed w/EXCITE II System Interconnect (Mobile)	2-21
Signa 10.x 1.5T TwinSpeed w/EXCITE I System Interconnect (Mobile)	2-22
Signa 9.x 1.5T TwinSpeed System Interconnect (Mobile)	2-23

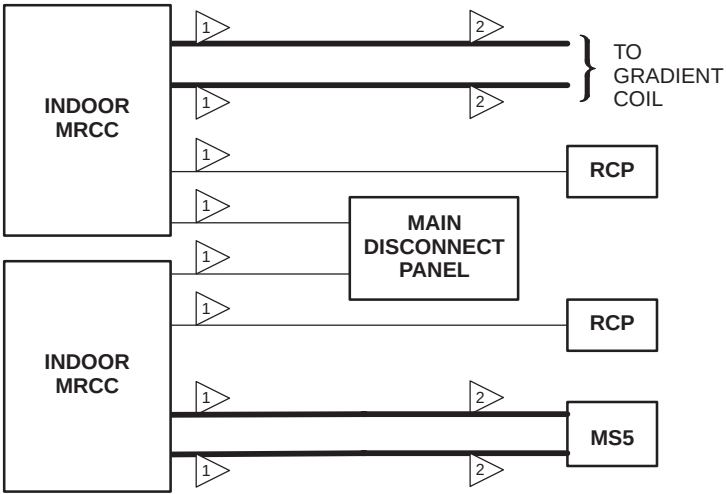
Tape map in a convenient location to check off routed cable. Upon completion of connecting cables, return fold-out map to the binder for future reference.





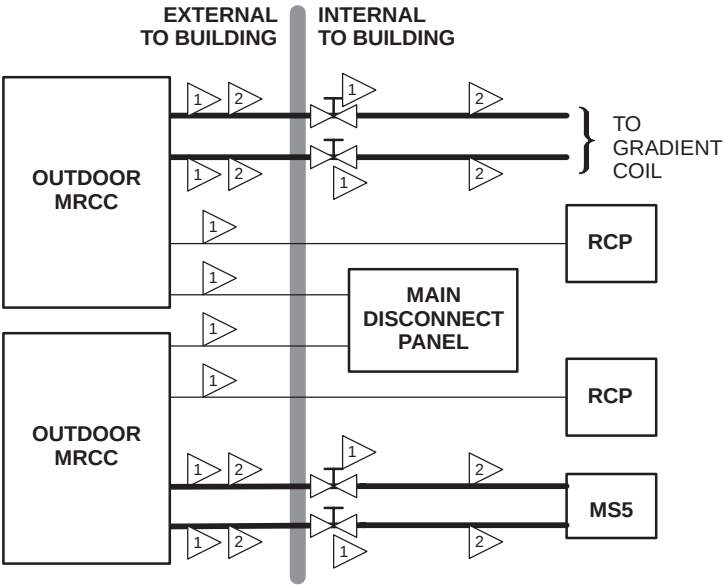


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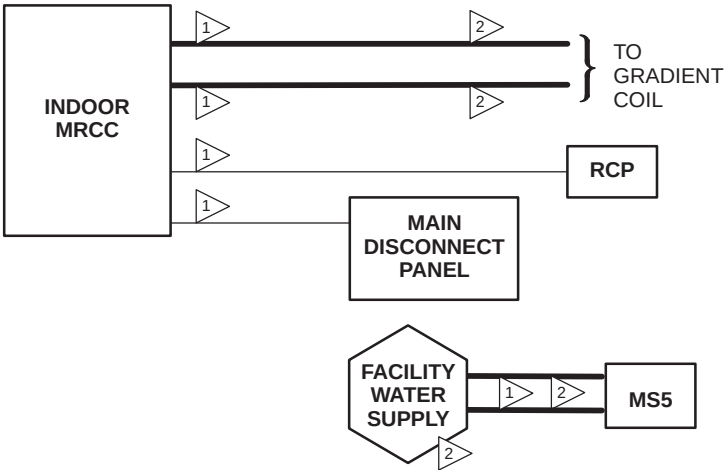
- NOTES:**
- 1 CUSTOMER FURNISHED.
 - 2 THIS GROUP CONTAINS WATER LINES WHICH SHALL BE ROUTED SEPARATE FROM ELECTRICAL LINES (I.E. POWER & SIGNAL).
- NOTE:**
- ONLY INTERCONNECTS SPECIFIC TO MRCC SUBSYSTEM EQUIPMENT SHOWN HERE.

DUAL INDOOR MRCC SUBSYSTEM GROUP INTERCONNECT DIAGRAM



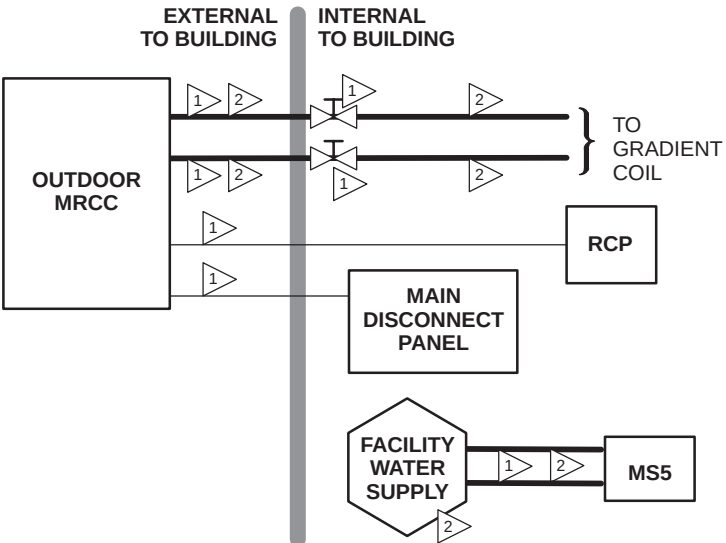
- NOTES:**
- 1 CUSTOMER FURNISHED.
 - 2 THIS GROUP CONTAINS WATER LINES WHICH SHALL BE ROUTED SEPARATE FROM ELECTRICAL LINES (I.E. POWER & SIGNAL).
- NOTE:**
- ONLY INTERCONNECTS SPECIFIC TO MRCC SUBSYSTEM EQUIPMENT SHOWN HERE.

DUAL OUTDOOR MRCC SUBSYSTEM GROUP INTERCONNECT DIAGRAM



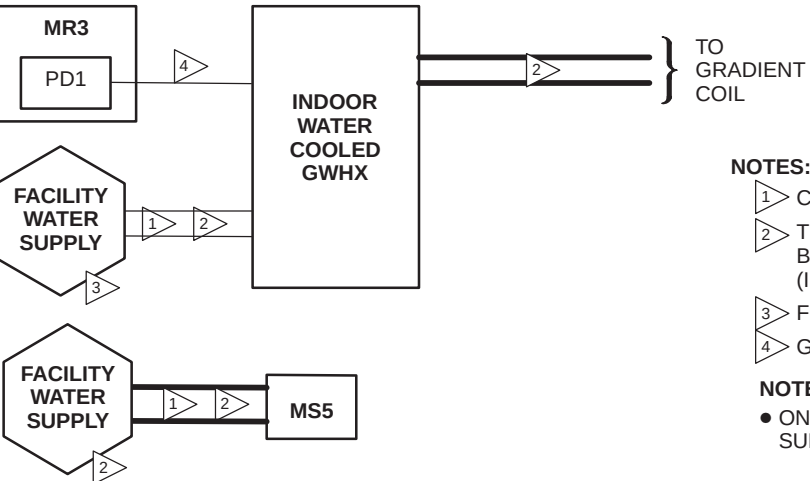
- NOTES:**
- 1 CUSTOMER FURNISHED.
 - 2 THIS GROUP CONTAINS WATER LINES WHICH SHALL BE ROUTED SEPARATE FROM ELECTRICAL LINES (I.E. POWER & SIGNAL).
- NOTE:**
- ONLY INTERCONNECTS SPECIFIC TO MRCC SUBSYSTEM EQUIPMENT SHOWN HERE.

INDOOR MRCC SUBSYSTEM GROUP INTERCONNECT DIAGRAM



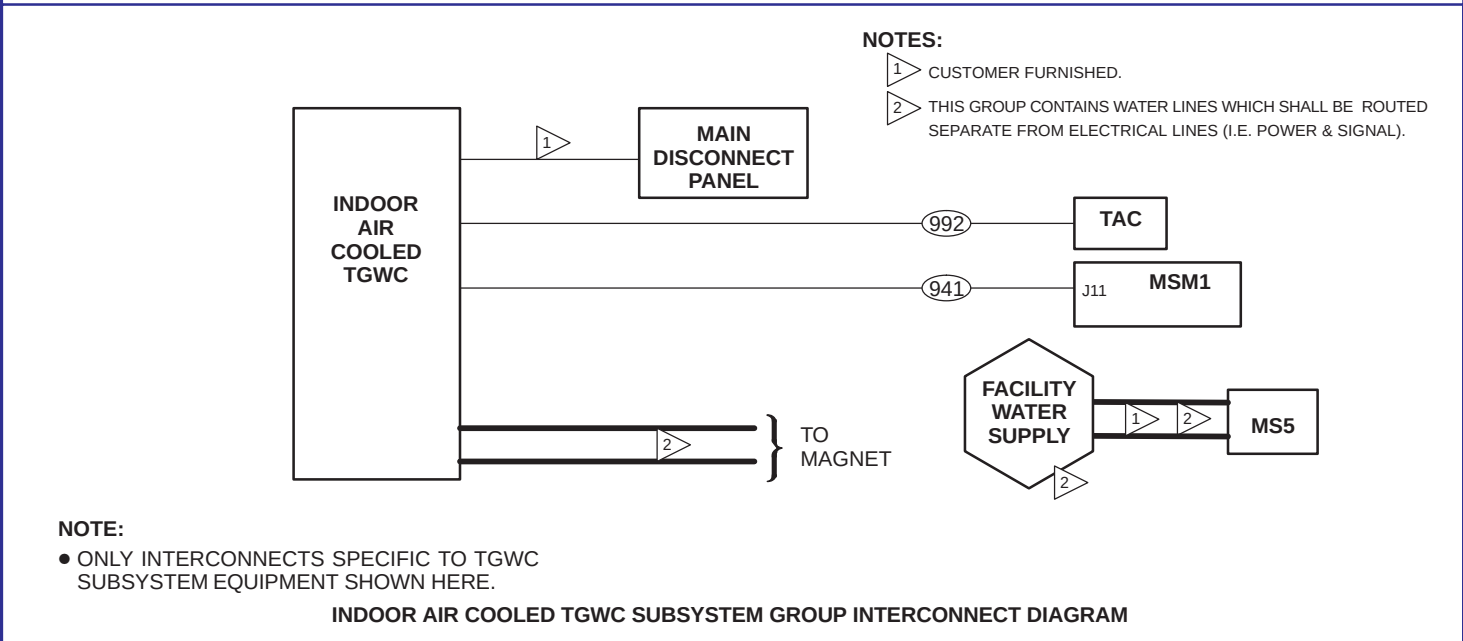
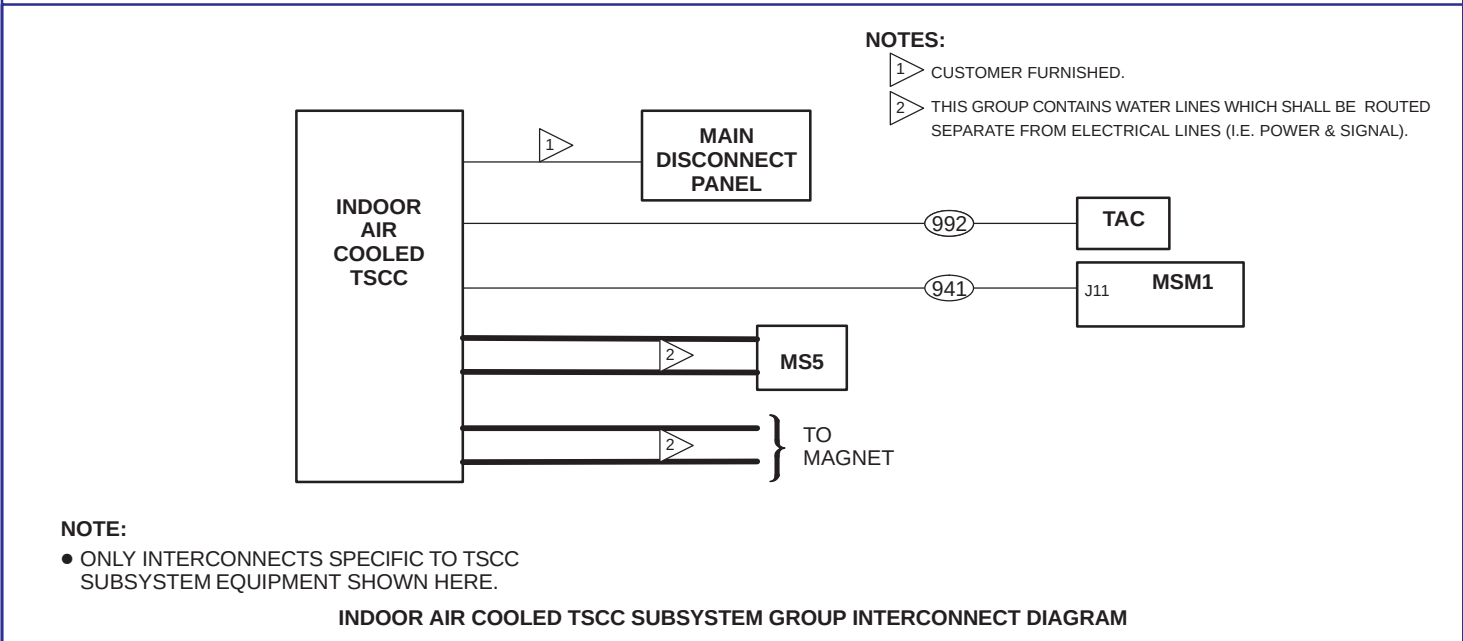
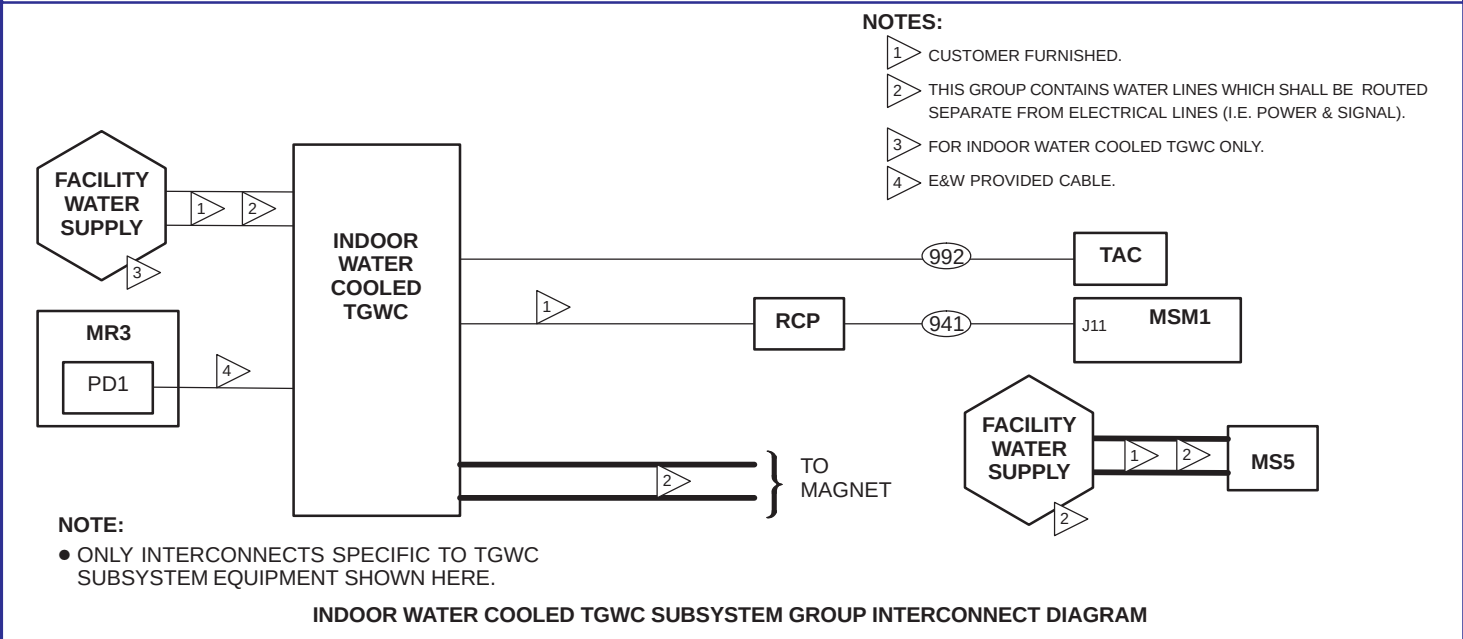
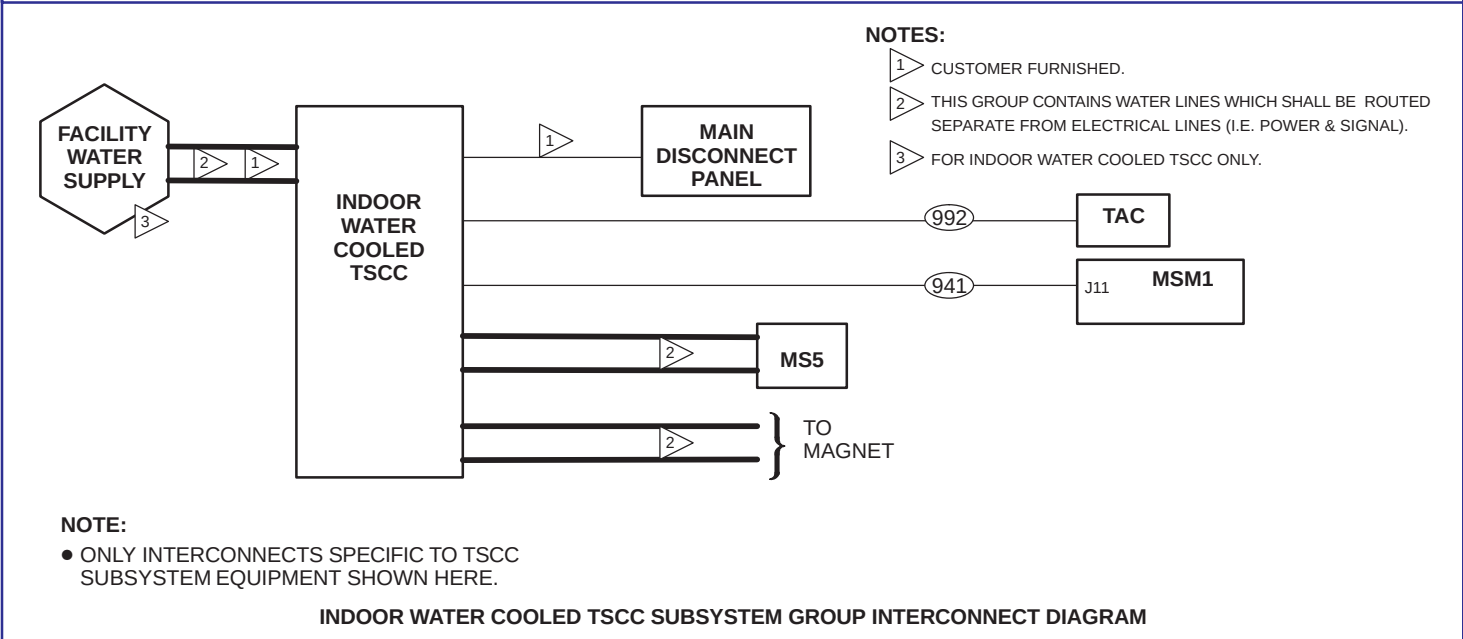
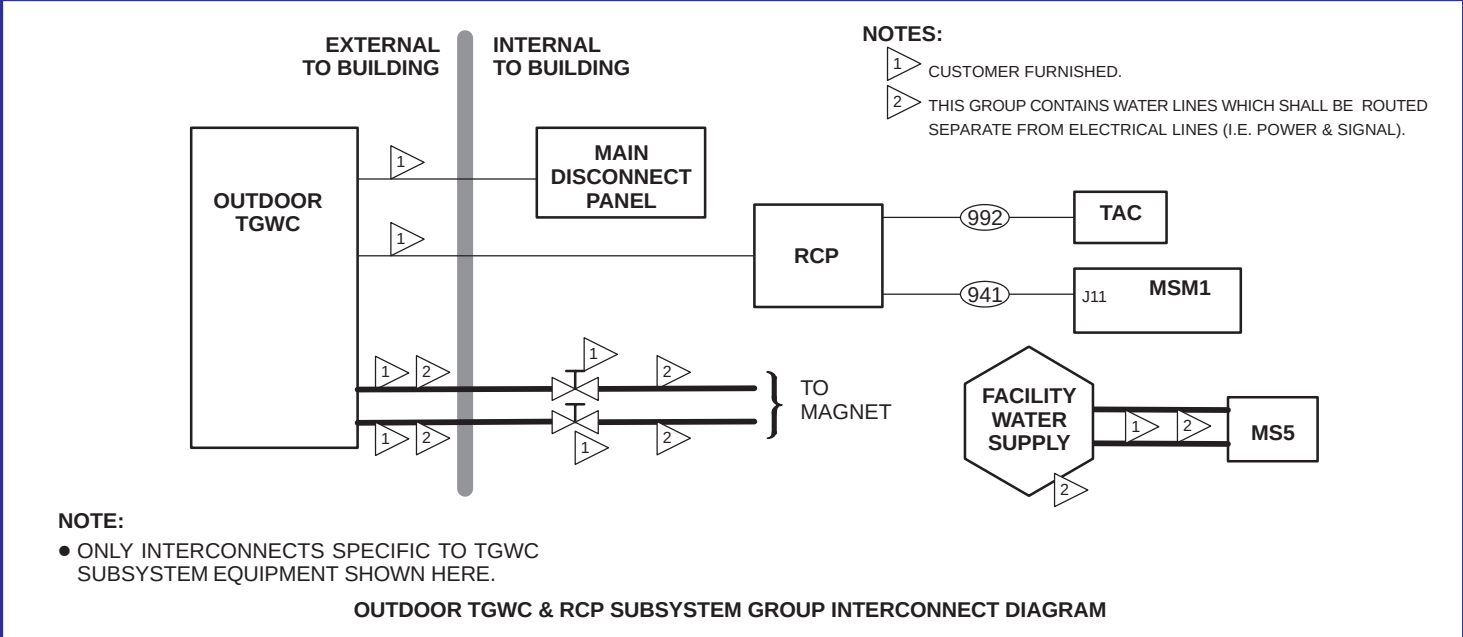
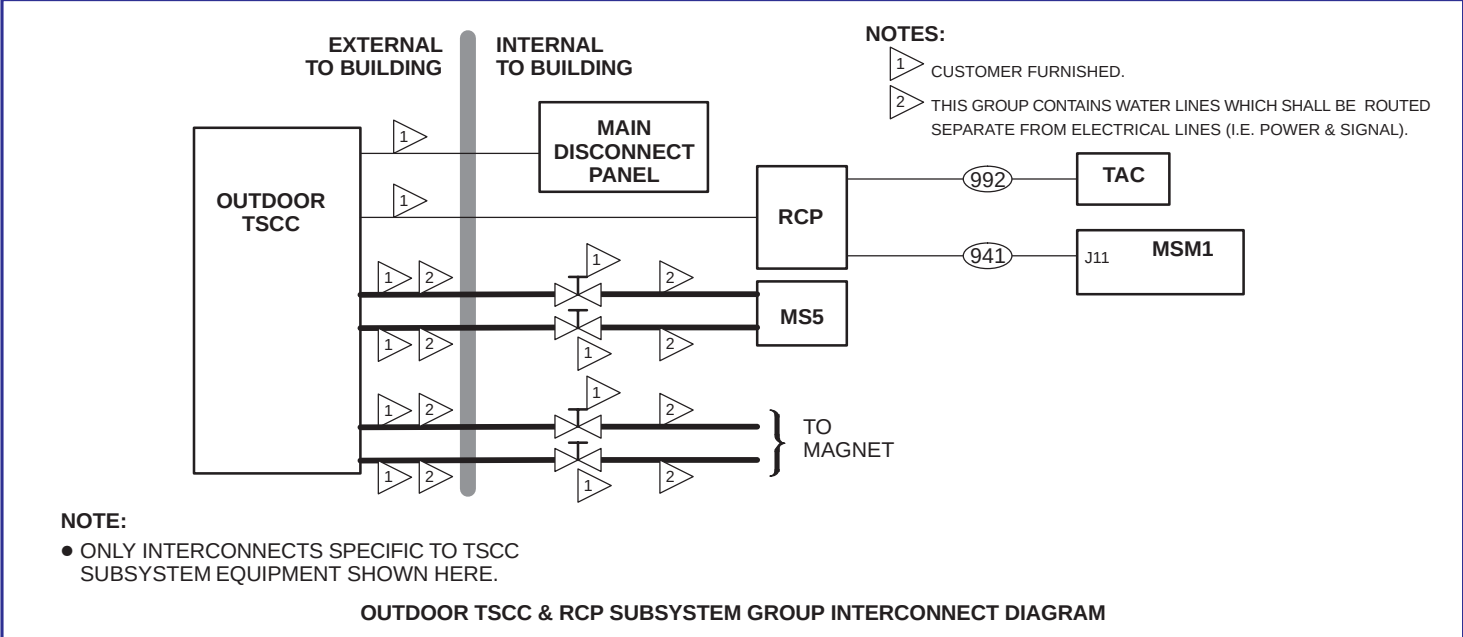
- NOTES:**
- 1 CUSTOMER FURNISHED.
 - 2 THIS GROUP CONTAINS WATER LINES WHICH SHALL BE ROUTED SEPARATE FROM ELECTRICAL LINES (I.E. POWER & SIGNAL).
- NOTE:**
- ONLY INTERCONNECTS SPECIFIC TO MRCC SUBSYSTEM EQUIPMENT SHOWN HERE.

OUTDOOR MRCC SUBSYSTEM GROUP INTERCONNECT DIAGRAM



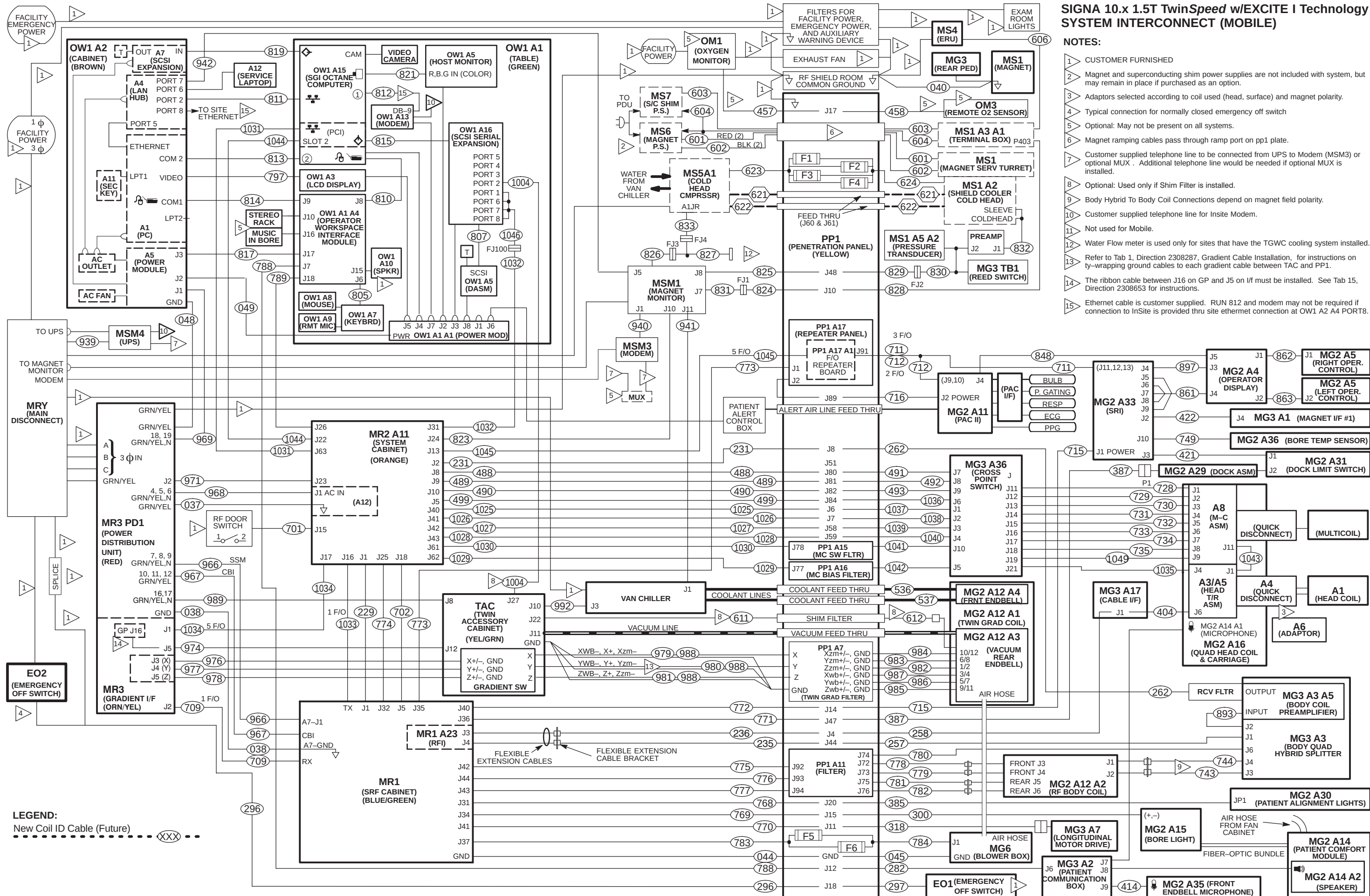
- NOTES:**
- 1 CUSTOMER FURNISHED.
 - 2 THIS GROUP CONTAINS WATER LINES WHICH SHALL BE ROUTED SEPARATE FROM ELECTRICAL LINES (I.E. POWER & SIGNAL).
 - 3 FOR INDOOR WATER COOLED GWHX ONLY.
 - 4 GWHX PROVIDED CABLE.
- NOTE:**
- ONLY INTERCONNECTS SPECIFIC TO GWHX SUBSYSTEM EQUIPMENT SHOWN HERE.

INDOOR GWHX SUBSYSTEM GROUP INTERCONNECT DIAGRAM





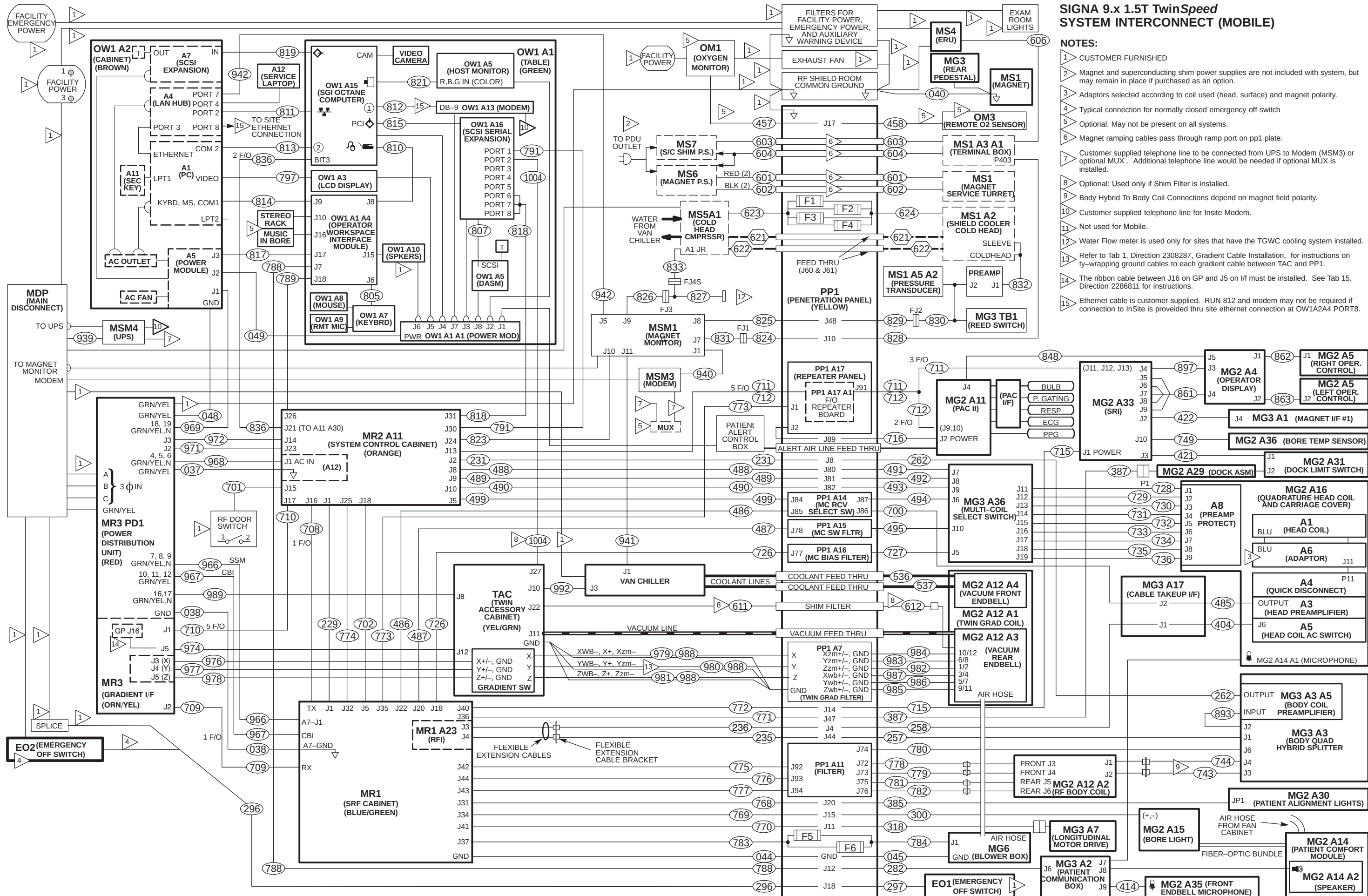
	CUSTOMER FURNISHED
2	Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
3	Adaptors selected according to coil used (head, surface) and magnet polarity.
4	Typical connection for normally closed emergency off switch
5	Optional: May not be present on all systems.
6	Magnet ramping cables pass through ramp port on pp1 plate.
7	Customer supplied telephone line to be connected from UPS to Modem (MSM3) or optional MUX . Additional telephone line would be needed if optional MUX is installed.
8	Optional: Used only if High Order Shim is installed.
9	Body Hybrid To Body Coil Connections depend on magnet field polarity.
10	Customer supplied telephone line for Insite Modem.
11	Not used for Mobile.
12	Water Flow meter is used only for sites that have the TGWC cooling system installed
13	Refer to Tab 1, Direction 2308287, Gradient Cable Installation, for instructions on ty-wrapping ground cables to each gradient cable between TAC and PP1.
14	The ribbon cable between J16 on GP and J5 on I/f must be installed. See Tab 15, Direction 2308653 for instructions.
15	Ethernet cable is customer supplied. RUN 1066 and modem may not be required if connection to InSite is provided thru site ethernet connection at OW1 A7 PORT 8.



**SIGNA 10.x 1.5T TwinSpeed w/EXCITE I Technology
SYSTEM INTERCONNECT (MOBILE)**

NOTES:

- 1 CUSTOMER FURNISHED
- 2 Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
- 3 Adaptors selected according to coil used (head, surface) and magnet polarity.
- 4 Typical connection for normally closed emergency off switch
- 5 Optional: May not be present on all systems.
- 6 Magnet ramping cables pass through ramp port on pp1 plate.
- 7 Customer supplied telephone line to be connected from UPS to Modem (MSM3) or optional MUX . Additional telephone line would be needed if optional MUX is installed.
- 8 Optional: Used only if Shim Filter is installed.
- 9 Body Hybrid To Body Coil Connections depend on magnet field polarity.
- 10 Customer supplied telephone line for Insite Modem.
- 11 Not used for Mobile.
- 12 Water Flow meter is used only for sites that have the TGWC cooling system installed.
- 13 Refer to Tab 1, Direction 2308287, Gradient Cable Installation, for instructions on ty-wrapping ground cables to each gradient cable between TAC and PP1.
- 14 The ribbon cable between J16 on GP and J5 on I/f must be installed. See Tab 15, Direction 2308653 for instructions.
- 15 Ethernet cable is customer supplied. RUN 812 and modem may not be required if connection to InSite is provided thru site ethernet connection at OW1 A2 A4 PORT8.



SIGNA 9.x 1.5T TwinSpeed SYSTEM INTERCONNECT (MOBILE)

NOTES:

- 1 CUSTOMER FURNISHED
- 2 Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
- 3 Adaptors selected according to coil used (head, surface) and magnet polarity.
- 4 Typical connection for normally closed emergency off switch
- 5 Optional: May not be present on all systems.
- 6 Magnet ramping cables pass through ramp port on pp1 plate.
- 7 Customer supplied telephone line to be connected from UPS to Modem (MSM3) or optional MUX. Additional telephone line would be needed if optional MUX is installed.
- 8 Optional: Used only if Shim Filter is installed.
- 9 Body Hybrid To Body Coil Connections depend on magnet field polarity.
- 10 Customer supplied telephone line for Insite Modem.
- 11 Not used for Mobile.
- 12 Water Flow meter is used only for sites that have the TGWC cooling system installed.
- 13 Refer to Tab 1, Direction 2308287, Gradient Cable Installation, for instructions on ty-wrapping ground cables to each gradient cable between TAC and PP1.
- 14 The ribbon cable between J16 on GP and J5 on I/F must be installed. See Tab 15, Direction 2286811 for instructions.
- 15 Ethernet cable is customer supplied. RUN 812 and modem may not be required if connection to InSite is provided thru site ethernet connection at OW1A2A4 PORT8.

SECTION 3 – SYSTEM START UP

3-1 RETURNING SHIPPING MATERIAL

The shipping material must be returned **as soon as possible**. Prompt return is required so that production shipments can continue with these labor saving dollies and carts. These can be included with the “Return pile”, Recycling Operation will route them to appropriate location.

3-2 PHANTOMS AND SERVICE KITS

1. Locate Phantom Kit (Loader with Head TLT sphere, 4.38 inch Surface Coil) and SPT Phantom Cart Kit.
2. Unpack the phantoms carefully to reduce risk of damage, breakage, or chemical spill.
3. Locate the multilingual (provided in eight languages) phantom attention labels shipped with each of the phantoms.



Three types of multilingual phantom attention labels are provided and each type must be applied to the correct phantom. The directions at the top of each label sheet describe which phantoms that can receive that label.

4. Read the directions provided at the top of each of the multilingual attention label sheets and apply the appropriate language attention label for the site to each of the phantoms that should receive that type of label.
 - a. Each phantom will receive the correct type of attention label written in the appropriate language for that site.
 - b. Note that the LVshim phantom assembly will receive 3 labels, one for each phantom component.
5. Place service phantoms into SPT Phantom Cart or service storage cabinet to be organized and ready to be used during applicable checks and calibrations. The DQA Phantom is to be stored in a customer designated location.
6. Locate TPS RF I/F Kit (if applicable) and Signa Spares Kit. Place kits into service storage cabinet for ready use.

3-3 POWER ON

1. Notify field service and other installation personnel that are working at the site that the PDU main disconnect locks and tags will be removed so PDU power can be turned on.
2. Restore power to the PDU from main disconnect.
3. Refer to the CD-ROM, Set-Up & Calibration, and perform the following:
 - PDU Voltage Checks Compact
 - Startup Checks Compact
 - Transformer Taps Compact
4. Press the FULL ON button on the PDU front control panel.
5. Except for the ACGD Gradient Cabinet power modules, switch all cabinet circuit breakers to ON. The ACGD power module circuit breakers should remain OFF until room humidity stabilizes under 60%.

3-4 CALIBRATION AND SYSTEM CHECKS

Refer to Installation Flow Chart to perform:

- Software load and Configuration file edit.
- Functional checks
- Set-up and calibration
- System performance checks

3-5 FINAL COORDINATION

1. Confirm that the blower box has been anchored properly and installed outside the minimum service area.
2. Complete assembly of PPG cable and probe head.
3. Verify that Operator Alert squeeze bulb will operate alarm.
4. Replace cabinet covers that have not been previously replaced.
5. Make sure all Magnet Enclosure and Rear Pedestal covers that have not been installed.
6. Dock Patient Transport in position at front of bore.
7. Record and enter applicable data into applicable site configuration files and records.
8. Complete Product Locator information for all serialized components, new or updated, that were installed via either:
 - **(U.S. Only)** *FE Site Verification Web Site*
or
 - Process and return product locator installation cards for all serialized components to:
Product Locator File, P.O. Box 414, W-523, Milwaukee, WI 53201-0414

Refer to Section 1 for details on submitting Product Locator information.

Note

Failure to fill out and return Product Locator Cards may result in failure of your site to receive future FMI's.

9. Process the **Supplier Performance Report** located in Common Forms. A reference copy of the form is provided at the end of this section.
10. Store the delivered site's set of service tools and spares kit in service cabinet at site.
11. Locate surface coil ("quick disconnect") adapters and any optional surface coils and positioning accessories and place on Patient Transport. Be sure only the correct polarity (normal or reverse **but not both**) head coil adapter is left at site.
12. Locate Box "To be opened by Applications Specialist" and set aside for later visit by Applications Specialist.
13. Leave the site's set of Manuals and CD-ROM at site. Organize and set up reference cabinet for the Manuals.
14. Locate Material Safety Data Sheets (MSDS) packed with phantoms and gradient coil coolant. They must be retained on site. Customer is to be informed that material with MSDS was brought into site and customer should know/decide where MSDS should be retained at site.
15. Verify that coordination of application support for instruction of site users on the appropriate Software Release has been made before returning system to the users. Turn site over to applications who will instruct users.

GE Medical Systems Field Service

SUPPLIER PERFORMANCE REPORT

Send completed forms to the 4 addresses at the end of this form.

Supplier Name: _____ Date: _____

Work Performed / Evaluated: _____

Commodity / Item Purchased: _____

GE Customer Name: _____

FDO Number: _____ Test Equipment Bar-code No; _

GEMS Evaluator Name: _____

Please rate suppliers as 1 to 5 on each of the 15 questions below:

1 = Unacceptable 2 = Poor 3 = Average 4 = Excellent 5 = Outstanding

There are 5 groups with three related questions in each group. Groups can be rated either as a group or 3 separate questions.

Any rating below 3 triggers a discrepancy review with the supplier.

Corrective actions will be faster and more effective when you provide examples and/or detail information.

Supplier schedules well and:

1	starts on time?	1	2	3	4	5
2	finishes on time?	1	2	3	4	5
3	avoids disruptions?	1	2	3	4	5

Supplier is flexible and:

4	responds to GEMS/customer requests?	1	2	3	4	5
5	cooperates with changes in schedule?	1	2	3	4	5
6	deals with unexpected problems?	1	2	3	4	5

Supplier tools and equipment are:

7	appropriate for contracted work?	1	2	3	4	5
8	adequate for contracted work?	1	2	3	4	5
9	calibrated and in good working order?	1	2	3	4	5

Supplier meets GEMS quality standards with employees who:

10	are trained, skilled and technically competent?	1	2	3	4	5
11	always keep job sites clean and free of debris?	1	2	3	4	5
12	ensure both function and appearance are as expected?	1	2	3	4	5

Supplier employees enhance GEMS image by their:

13	appearance?	1	2	3	4	5
14	behavior?	1	2	3	4	5
15	attitude?	1	2	3	4	5

Examples / Details: _____

Any rating below 3 triggers a discrepancy review with the supplier.

SEND COMPLETED FORMS TO:

1. Senior Operations Specialist for your LCT
2. The contractor who performed the work
3. ISS (already in default distribution)
4. Sourcing Supplier Quality (already in default distribution)

SYSTEM INSTALLATION INSTRUCTIONS

1-1 **Tab 1, System, contains installation instructions for the following:**

Use only the applicable System Installation Instruction Directions

- *DIRECTION 2147593* – 1.5T Heliax Cable Installation Instructions
- *DIRECTION 2389819* – 1.5T Times Microwave Installation Instructions
- *DIRECTION 2308287* – Signa TwinSpeed ACGD Gradient Cable Installation Instructions
- *DIRECTION 2287106* – Signa ACGD Fiber Optic Cable Installation Instructions
- *DIRECTION 2338872* – Signa EXCITE I Fiber Optic Cable Installation Instructions
- *DIRECTION 2398342* – Signa EXCITE II Fiber Optic Cable Installation Instructions
- *DIRECTION 2287109* – Signa ACGD System Power Cable Installation Instructions

1.5T HELIAX CABLE INSTALLATION

INSTALLATION STEPS SUMMARY

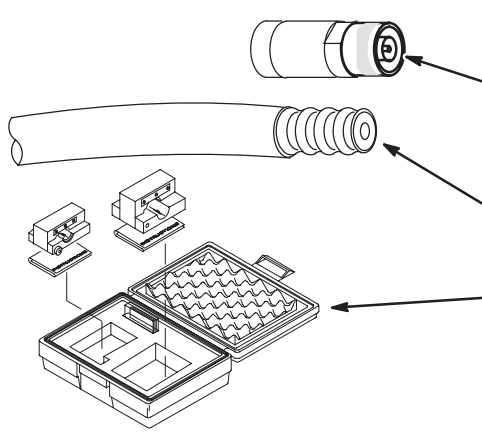
- ☐ A – Routing, Cutting, Terminating Heliax Cables
- ☐ B – Connecting Runs 235, 236, 257, and 258

CAUTION

Do not subject 1/2 inch diameter heliax cable to a bend radius less than 5 inches (127mm). Do not subject 7/8 inch diameter heliax cable to a bend radius less than 10 inches (254mm). This cable will be permanently damaged if it is kinked by a bend radius less than the specified minimum.

☐ A – ROUTING, CUTTING, TERMINATING HELIAX CABLES

Runs 236 and 258 are created during installation from 50 foot and/or 55 foot cables along with four pre-marked labels. The installer may use both lengths of cables as provided, or if storage of excess length is a problem, cut one length (If runs are short, it may be feasible to cut one cable into two lengths) or both lengths and re-terminate each cut end. Either way, labels are attached after length is decided and cables are installed. Runs 235 and 257 are created during installation from a 100 foot cable along with four pre-marked labels.

- 
- ① Decide if cables are to be cut and re-terminated. If yes, continue to Step 2. If no, proceed to Step B.
 - ② Procure a new connector (46-230011P1) for each cable that needs to be shortened and re-terminated.
 - ③ Route cables and mark at length for cut-off.
 - ④ Cut cables to length with a fine-tooth hack saw. Remove about one inch of jacket to expose outer conductor.
 - ⑤ Obtain the Heliax Termination Kit (46-320362G1) field service tool and follow instructions supplied with the kit. This will enable precise cut-off of cable.
 - ⑥ Attach new connector(s) to cable(s) according to instruction sheets supplied with connector.

☐ B – CONNECTING RUNS 235, 236, 257, AND 258

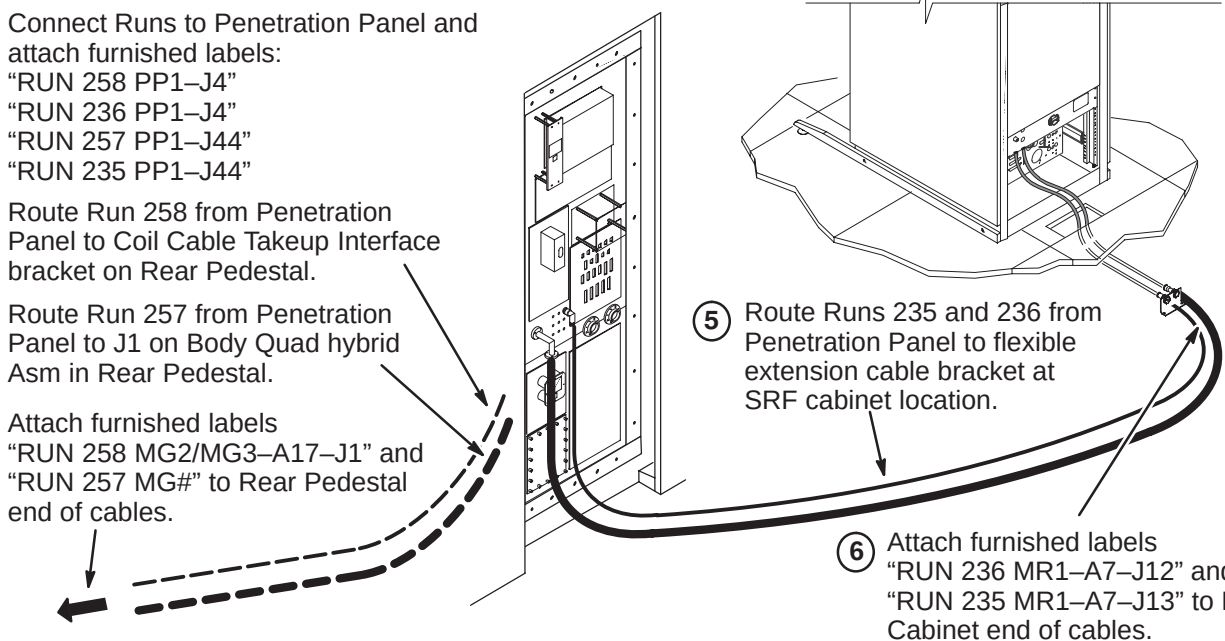
- 
- ① Connect Runs to Penetration Panel and attach furnished labels:
"RUN 258 PP1-J4"
"RUN 236 PP1-J4"
"RUN 257 PP1-J44"
"RUN 235 PP1-J44"
 - ② Route Run 258 from Penetration Panel to Coil Cable Takeup Interface bracket on Rear Pedestal.
 - ③ Route Run 257 from Penetration Panel to J1 on Body Quad hybrid Asm in Rear Pedestal.
 - ④ Attach furnished labels "RUN 258 MG2/MG3-A17-J1" and "RUN 257 MG#" to Rear Pedestal end of cables.
 - ⑤ Route Runs 235 and 236 from Penetration Panel to flexible extension cable bracket at SRF cabinet location.
 - ⑥ Attach furnished labels "RUN 236 MR1-A7-J12" and "RUN 235 MR1-A7-J13" to RF Cabinet end of cables.

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<u>SECTION</u>	<u>PAGE</u>
1-1 TIMES MICROWAVE RF CABLES INFORMATION	1-2
1-2 LAYOUT OF CABLES	1-3
1-3 TERMINATE AND CONNECT RF BODY CABLES AT PENETRATION PANEL	1-4
1-4 TERMINATE AND CONNECT RF HEAD COIL CABLES	1-5

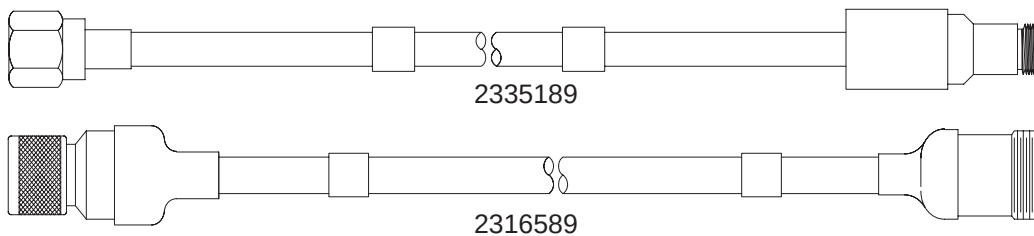
1-1 TIMES MICROWAVE RF CABLES INFORMATION

Four cables and other associated parts to be installed are provided for this procedure.

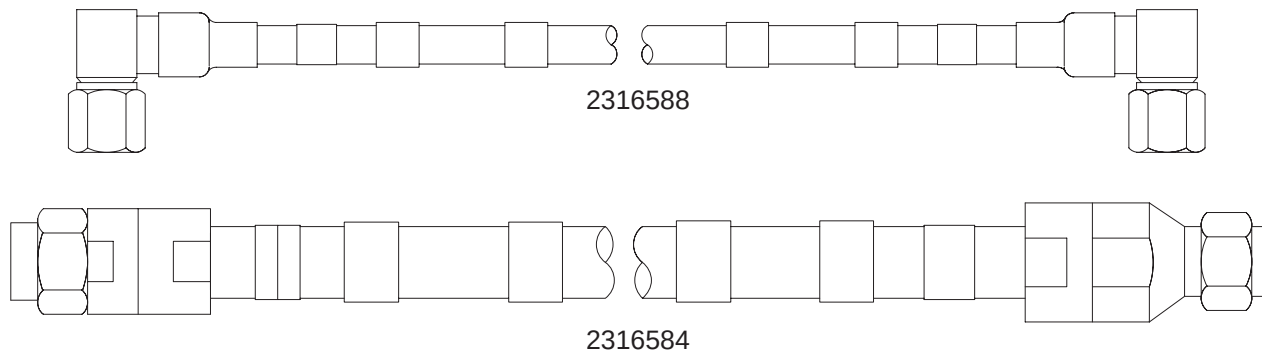


Do not subject LMR-600FR cable (Head) to a bend radius less than 1.5 inches (138mm). Do not subject LMR-900FR cable (Body) to a bend radius less than 3 inches (75mm). The cables will be permanently damaged if it is kinked by a bend radius less than the specified minimum.

The two short cables, 2335189 (Head) and 2316589 (Body) are connected to the SRFD2 cabinet.



The two long cables, 2316588 (Head, Runs 236 & 258) and 2316584 (Body, Runs 235 & 257), will be cut to length and terminated for routing in the equipment and scan rooms.

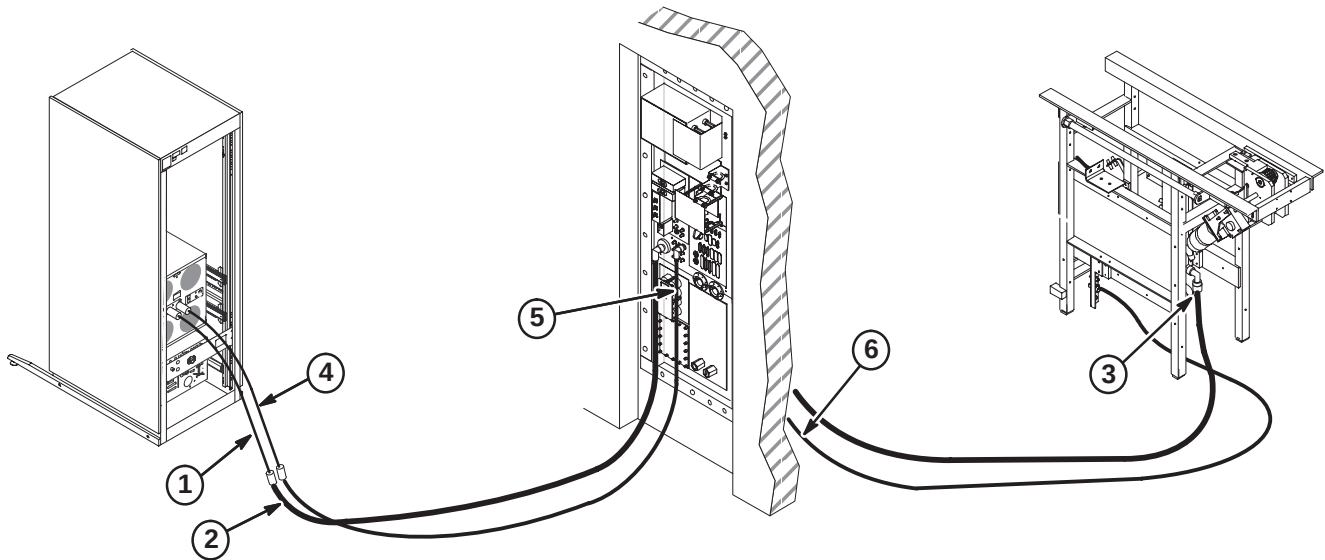


The remaining connectors, fasteners, and labels will be installed as directed in the following steps.

1-2 LAYOUT OF CABLES

RF cables will be routed and connected as indicated in the following steps.

1. Connect extension cable 2316589 to J4 Body Output at rear of RF module in RF cabinet.
2. Connect end of Times RF Body cable (2316584) marked "(RUN 235) FJ5" to Body extension cable. Route to J44 at Penetration Panel. Determine if any take-up slack is needed and cut to length. Refer to *Times Connector Installation Procedures* provided with cables for cutting information. Refer to Section 1-3 for termination procedure of cut end.
3. Move remainder of RF Body cable to Scan Room and connect other end of cable (2316584) marked "(RUN 257) MG3A3-J1" to Rear Pedestal as shown below. Route to J44 at Penetration Panel. Determine if any take-up slack is needed and cut to length. Refer to *Times Connector Installation Procedures* provided with cables for cutting information. Refer to Section 1-3 for termination procedure of cut end.
4. Connect extension cable 2335189 to J3 Head Output at rear of RF module in RF cabinet.
5. Connect end of Times RF Head cable (2316588) marked "(RUN 236) PP1-J4" to J4 at Penetration Panel. Route to RF Head extension cable at rear of RF cabinet. Determine if any take-up slack is needed and cut to length. Refer to *Times Connector Installation Procedures* provided with cables for cutting information. Refer to Section 1-4 for termination procedure of cut end.
6. Move remainder of RF Head cable to Scan Room and connect other end of cable marked (RUN 258) PP1-J4" to Penetration Panel at J4. Route to Rear Pedestal Takeup bracket J1. Determine if any take-up slack is needed and cut to length. Refer to *Times Connector Installation Procedures* provided with cables for cutting information. Refer to Section 1-4 for termination procedure of cut end.



1-3 TERMINATE AND CONNECT RF BODY CABLES AT PENETRATION PANEL

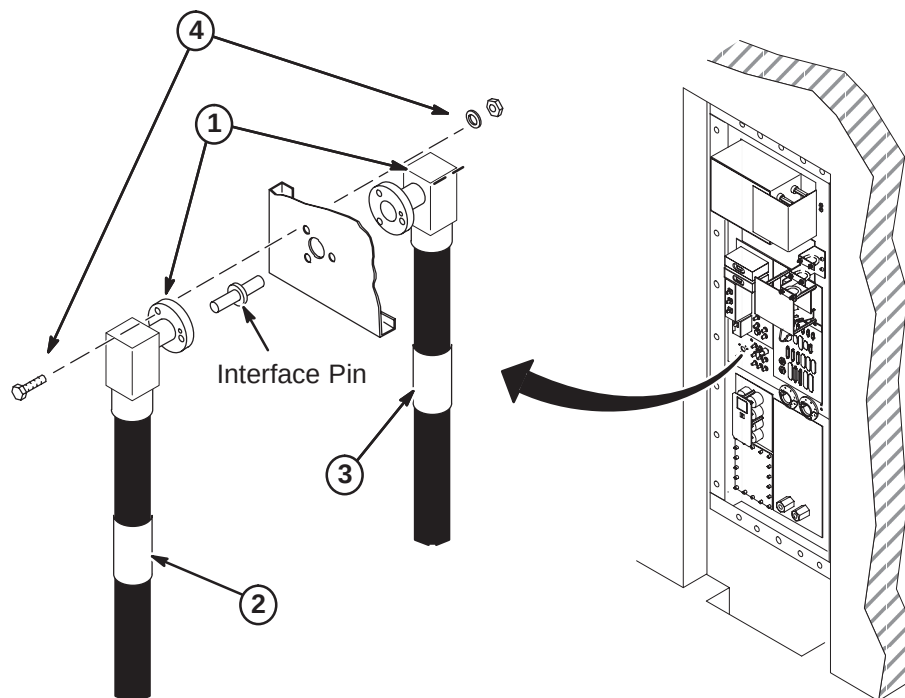
Perform the following steps for terminating and connecting the RF Body cables at the Penetration Panel J44.

1. Refer to *Times Connector Installation Procedures* provided with cables for information on installing an EIA Right Angle Connector (2319986) on each end of the RF cables. Use provided LMR900 Stripping Tool (2335190).

Note

Make sure the fastener holes in the connector flanges align with the holes in the Penetration Panel so the cables will route toward floor when installed.

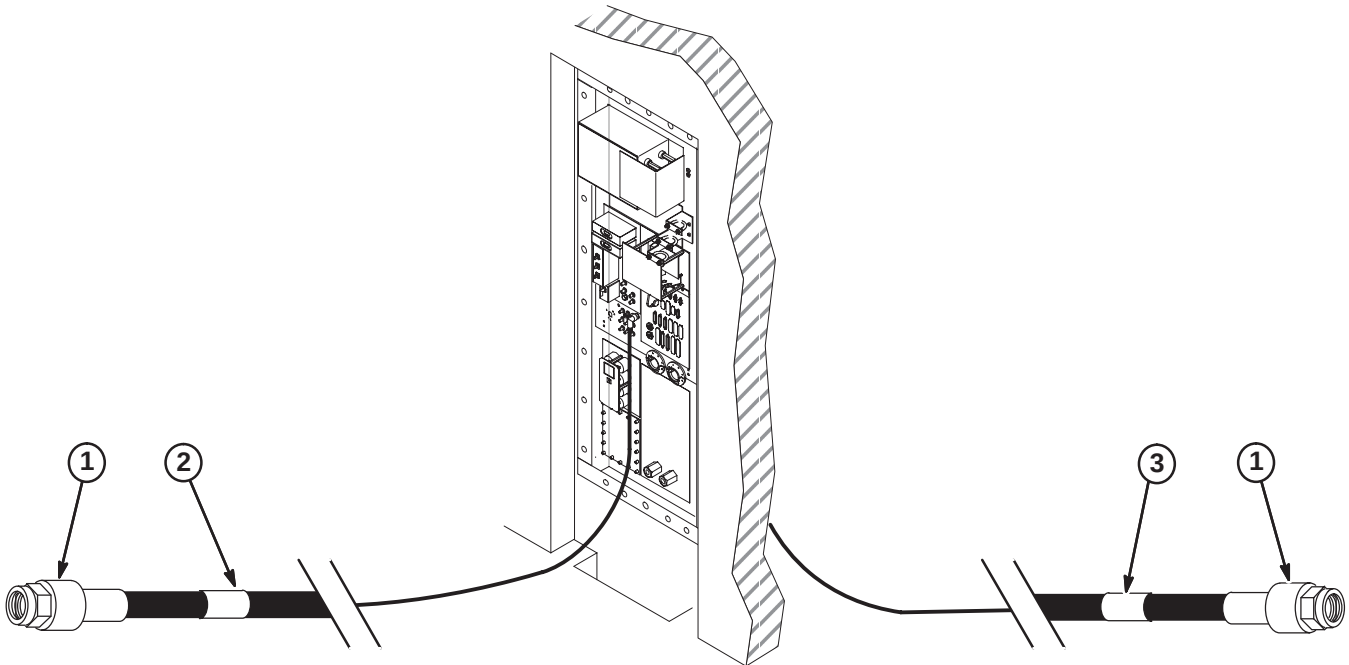
2. Attach label "RUN 235, PP1-J44" (2357230) to cable as shown on Equipment room side of Penetration Panel.
3. Attach label "RUN 257, PP1-J44" (2357230-2) to cable as shown on Scan room side of Penetration Panel.
4. Insert Interface Pin into one of the connectors. At J44, align connectors on each side of Penetration Panel and push together. Fasten together using three of each of the following:
 - 1/4-20 x 1.25 Hex Head Bolt (brass)
 - 1/4 Screw Size Washer (phos bronze)
 - 1/4-20, 7/16" Hex x 3/16" thk Nut (brass)



1-4 TERMINATE AND CONNECT RF HEAD COIL CABLES

Perform the following steps for terminating and connecting the RF Head cables in the Equipment and Scan rooms.

1. Refer to *Times Connector Installation Procedures* provided with cables for information on installing an Straight Male N Connector (2352195) on each end of the RF cable. Use provided LMR600 Stripping Tool (2352193).
 - Connect Equipment room end to Extension cable at FJ6.
 - Connect Scan room end to MG3A17-J1 at Interface bracket on Rear Pedestal.
2. Attach label "RUN 236 FJ6" (2357230-3) to cable as shown in Equipment room.
3. Attach label "RUN 258 MG3A17-J1" (2357230-4) to cable as shown in Scan room.



INSTALLATION STEPS SUMMARY

- ☐ A1 – Gradient Cable Information and Supplied Labels
- ☐ A2 – Instructions for Magnet Room Cable Installation

☐ A1 – GRADIENT CABLE INFORMATION AND SUPPLIED LABELS

Nine cables have been delivered to site for either non–EXCITE, EXCITE I or EXCITE II. The difference between the cables is the length of six of the nine cables. All cables have three lead wires with terminals and attached cable labels at one end. The leads on this cable end are connected to the Twin Gradient Filter on the Penetration Panel. The other end is cut flush with no labels or terminals.

The nine cables with supplied labels are as follows for each group:

NON–EXCITE & EXCITE I GRADIENT CABLES:

RUN	PART #	LABEL	LENGTH	SUPPLIED LABELS
979	2100948–34	PP1–A7–X	75.5 ft.	(1) RUN 976, (1) RUN 979, (2) X+, (1) X–, (1) GND, (1) XZM–, (1) XWB–
980	2100948–35	PP1–A7–Y	75.5 ft.	(1) RUN 977, (1) RUN 980, (2) Y+, (1) Y–, (1) GND, (1) YZM– (1) YWB–
981	2100948–36	PP1–A7–Z	75.5 ft.	(1) RUN 978, (1) RUN 981, (2) Z+, (1) Z–, (1) GND, (1) ZZM– (1) ZWB–
982	2100948–42	PP1–A7–Z _{ZM}	50.5 ft.	(1) RUN 982, (1) ZZM– (1) ZZM+
983	2100948–41	PP1–A7–Y _{ZM}	50.5 ft.	(1) RUN 983, (1) YZM– (1) YZM+
984	2100948–40	PP1–A7–X _{ZM}	50.5 ft.	(1) RUN 984, (1) XZM– (1) XZM+
985	2100948–45	PP1–A7–Z _{WB}	50.5 ft.	(1) RUN 985, (1) ZWB– (1) ZWB+
986	2100948–44	PP1–A7–Y _{WB}	50.5 ft.	(1) RUN 986, (1) YWB– (1) YWB+
987	2100948–43	PP1–A7–X _{WB}	50.5 ft.	(1) RUN 987, (1) XWB– (1) XWB+

Note: Verify that 50.5 ft. (15.4M) length of supplied Runs 982, 983, 984, 985, 986, and 987 cable is sufficient to route from the Penetration Panel to Body Coil Gradient Cables at rear of magnet.

EXCITE II – CONFIG A GRADIENT CABLES:

RUN	PART #	LABEL	LENGTH	SUPPLIED LABELS
979	2345400–10	PP1–A7–X	59.5 ft.	(1) RUN 976, (1) RUN 979, (2) X+, (1) X–, (1) GND, (1) XZM–, (1) XWB–
980	2345400–11	PP1–A7–Y	59.5 ft.	(1) RUN 977, (1) RUN 980, (2) Y+, (1) Y–, (1) GND, (1) YZM– (1) YWB–
981	2345400–12	PP1–A7–Z	59.5 ft.	(1) RUN 978, (1) RUN 981, (2) Z+, (1) Z–, (1) GND, (1) ZZM– (1) ZWB–
982	2345400–18	PP1–A7–Z _{ZM}	26.5 ft.	(1) RUN 982, (1) ZZM– (1) ZZM+
983	2345400–17	PP1–A7–Y _{ZM}	26.5 ft.	(1) RUN 983, (1) YZM– (1) YZM+
984	2345400–16	PP1–A7–X _{ZM}	26.5 ft.	(1) RUN 984, (1) XZM– (1) XZM+
985	2345400–24	PP1–A7–Z _{WB}	26.5 ft.	(1) RUN 985, (1) ZWB– (1) ZWB+
986	2345400–23	PP1–A7–Y _{WB}	26.5 ft.	(1) RUN 986, (1) YWB– (1) YWB+
987	2345400–22	PP1–A7–X _{WB}	26.5 ft.	(1) RUN 987, (1) XWB– (1) XWB+

Note: Verify that 26.5 ft. (8.1M) length of supplied Runs 982, 983, 984, 985, 986, and 987 cable is sufficient to route from the Penetration Panel to Body Coil Gradient Cables at rear of magnet.

EXCITE II – CONFIG C GRADIENT CABLES:

RUN	PART #	LABEL	LENGTH	SUPPLIED LABELS
979	2345400–10	PP1–A7–X	59.5 ft.	(1) RUN 976, (1) RUN 979, (2) X+, (1) X–, (1) GND, (1) XZM–, (1) XWB–
980	2345400–11	PP1–A7–Y	59.5 ft.	(1) RUN 977, (1) RUN 980, (2) Y+, (1) Y–, (1) GND, (1) YZM– (1) YWB–
981	2345400–12	PP1–A7–Z	59.5 ft.	(1) RUN 978, (1) RUN 981, (2) Z+, (1) Z–, (1) GND, (1) ZZM– (1) ZWB–
982	2345400–21	PP1–A7–Z _{ZM}	47.5 ft.	(1) RUN 982, (1) ZZM– (1) ZZM+
983	2345400–20	PP1–A7–Y _{ZM}	47.5 ft.	(1) RUN 983, (1) YZM– (1) YZM+
984	2345400–19	PP1–A7–X _{ZM}	47.5 ft.	(1) RUN 984, (1) XZM– (1) XZM+
985	2345400–27	PP1–A7–Z _{WB}	47.5 ft.	(1) RUN 985, (1) ZWB– (1) ZWB+
986	2345400–26	PP1–A7–Y _{WB}	47.5 ft.	(1) RUN 986, (1) YWB– (1) YWB+
987	2345400–25	PP1–A7–X _{WB}	47.5 ft.	(1) RUN 987, (1) XWB– (1) XWB+

Note: Verify that 39.5 ft. (12M) length of supplied Runs 982, 983, 984, 985, 986, and 987 cable is sufficient to route from the Penetration Panel to Body Coil Gradient Cables at rear of magnet.

WARNING!

FERROUS MATERIAL HAZARD! THE RATCHETING CRIMP TOOL REQUIRED FOR THIS PROCEDURE CONTAINS FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME A DANGEROUS PROJECTILE. KEEP ALL FERROUS TOOLS AS FAR FROM THE MAGNET AS POSSIBLE.

IF MAGNET IS AT FULL FIELD – CUT GRADIENT CABLE TO LENGTH AND REMOVE FROM MAGNET ROOM TO PERFORM CABLE TERMINATION PROCEDURES.

TOOLS REQUIRED: 2134776 – Cable Crimper/Stripper Kit

☐ A2 – INSTRUCTIONS FOR MAGNET ROOM CABLE INSTALLATION

- 1

Move Run 982, 983, 984, 985, 986, and 987 cables to magnet room. The end of cable with terminals will be connected to Penetration Panel. See Procedures **B1** and **B2**.
- 2

Route cables from Gradient Filter of Penetration Panel to Body Coil gradient cables at rear of magnet.
- 3

Verify sufficient slack of at least two feet (610mm) is allowed for termination and connection. Mark location for cutting. Cut cables to length and label cut ends with applicable run number. **Do not cut Body Coil gradient cables.**
- 4

Locate furnished bag of parts and attach terminal and labels to each cable which was cut to length. See Procedures **B3** and **B4**.
- 5

See Procedure **C1** and **C2** for connection of Runs 982, 983, 984, 985, 986, and 987 to Body Coil gradient cables.

INSTALLATION STEPS SUMMARY

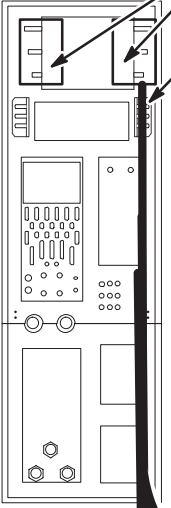
☐ B1 – Connect ZM (Zoom) Cables to Penetration Panel

☐ B2 – Connect WB (Whole Body) Cables to Penetration Panel

☐ B3 – Terminate Runs 982, 983, and 984 for Magnet Room

☐ B4 – Terminate Runs 985, 986, and 987 for Magnet Room

MAGNET ROOM VIEW



B1 – CONNECT ZM (ZOOM) CABLES TO PENETRATION PANEL

1 Remove gradient filter terminal covers.

2 Route Run 982, 983, and 984 to right side of gradient filter on Penetration Panel.

3 Connect six ZM– and ZM+ wires to gradient filter terminals as marked.

4 Align cables to obtain maximum spacing from other cables and conductive metal objects.

5 Tighten nuts to a torque of 35 ft–lbs (67–74Nm) or after tightening, verify that nothing is loose.

6 Route and secure each cable Run to strain-relief bracket with U–bolts.

7 Secure GND lead wire on Ground Stud for each cable Run.

Note: Red tape indicates positive (+) wires.

Note: When tightening nuts, it is important to hold screw body.

Note: Final check is by hand. Make sure screw is fixed, nut is tight, and cable is not detached from terminal lug.

MAGNET ROOM

EQUIPMENT ROOM

MAGNET ROOM

ZZM–

ZZM+

ZWB–

Z+

ZZM–

YZM–

YZM+

XZM–

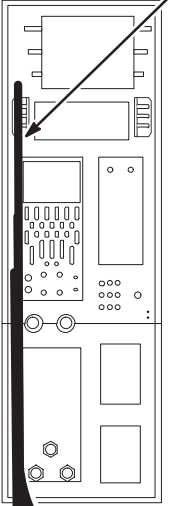
XZM+

RUN 982

RUN 983

RUN 984

MAGNET ROOM VIEW



B2 – CONNECT WB (WHOLE BODY) CABLES TO PENETRATION PANEL

1 Route Run 985, 986, and 987 to left side of gradient filter on Penetration Panel.

2 Connect six WB– and WB+ wires to gradient filter terminals as marked.

3 Align cables to obtain maximum spacing from other cables and conductive metal objects.

4 Tighten nuts to a torque of 35 ft–lbs (67–74Nm) or after tightening, verify that nothing is loose.

5 Secure GND lead wire on Ground Stud for each cable Run.

6 Route and secure each cable Run to strain-relief bracket with U–bolts.

7 Replace gradient filter terminal covers.

Note: Red tape indicates positive (+) wires.

Note: When tightening nuts, it is important to hold screw body.

Note: Final check is by hand. Make sure screw is fixed, nut is tight, and cable is not detached from terminal lug.

EQUIPMENT ROOM

MAGNET ROOM

Y+

YZM

ZWB+

ZWB–

YWB–

XZM

YWB+

YWB–

XWB–

X+

XWB+

XWB–

RUN 987

RUN 986

RUN 985

B3 – TERMINATE RUNS 982, 983, AND 984 FOR MAGNET ROOM

Note: The following steps are required to terminate the end of the cable that was cut in Step A2. The other end of the cable is connected to the Penetration Panel.

1 Remove 24 in. (288 mm) of jacket from end of cable Run 982.

2 Trim rubber cable filler strands flush with end of jacket.

3 Cut and remove GROUND wire.

4 Strip 0.75 in. (19 mm) of insulation from end of each wire.

5 Wrap red tape (from Pen Panel end of cable) around wire "ONE".

6 Place furnished "ZZM+" label on wire "ONE". Place furnished "ZZM–" label on wire "TWO".

7 Place furnished "Run 982" label on cable.

8 Repeat above process for Runs 983 and 984.

Note: Red tape, Yzm+ (RUN 983) and Xzm+ (RUN 984) labels are attached to wire "ONE".

B4 – TERMINATE RUNS 985, 986, AND 987 FOR MAGNET ROOM

Note: The following steps are required to terminate the end of the cable that was cut in Step A2. The other end of the cable is connected to the Penetration Panel.

1 Remove 24 in. (288 mm) of jacket from end of cable Run 985.

2 Trim rubber cable filler strands flush with end of jacket.

3 Cut and remove GROUND wire.

4 Strip 0.75 in. (19 mm) of insulation from end of each wire.

5 Wrap red tape (from Pen Panel end of cable) around wire "ONE".

6 Place furnished "ZWB+" label on wire "ONE". Place furnished "ZWB–" label on wire "TWO".

7 Place furnished "Run 985" label on cable.

8 Repeat above process for Runs 986 and 987.

Note: Red tape, Ywb+ (RUN 986) and Xwb+ (RUN 987) labels are attached to wire "ONE".

2 of 5

INSTALLATION STEPS SUMMARY

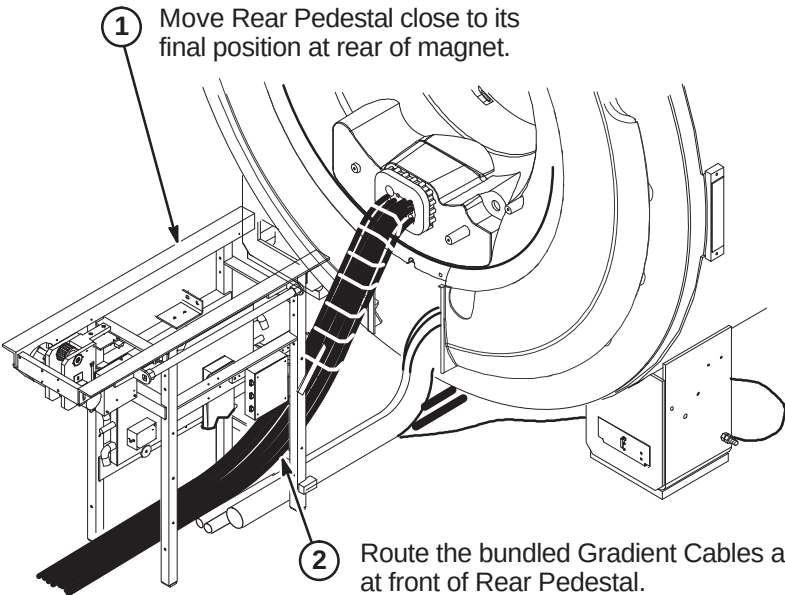
- C1 – Routing of Body Coil Gradient Cables
- C2 – Connection of Body Coil Gradient Cables

CAUTION

Rear End Bell **MUST** be installed on magnet with gradient coil cables routed thru block before connecting gradient coil cables to gradient cables routed from Penetration Panel with splices in cable trough. Failure to do so will result in inability to install Rear End Bell.

C1 – ROUTING OF BODY COIL GRADIENT CABLES

Note: Cables and hoses can be routed to exit out of rear or side of Rear Pedestal.

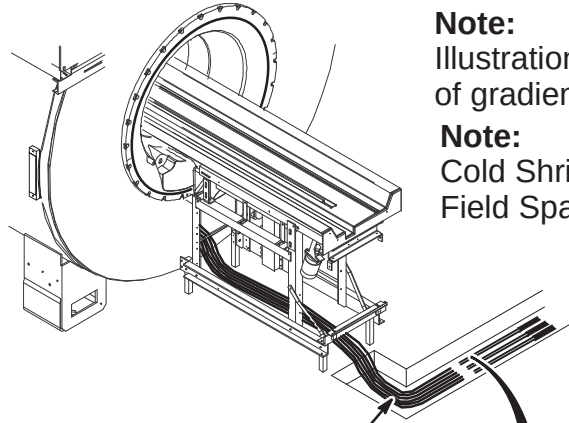


Note: Routing the Gradient Cables above rather than below the cross brace will reduce white pixels

CAUTION

Do not cut TRM Body Coil gradient cables. Only Gradient Output cables from Penetration Panel are to be cut to length.

C2 – CONNECTION OF BODY COIL GRADIENT CABLES



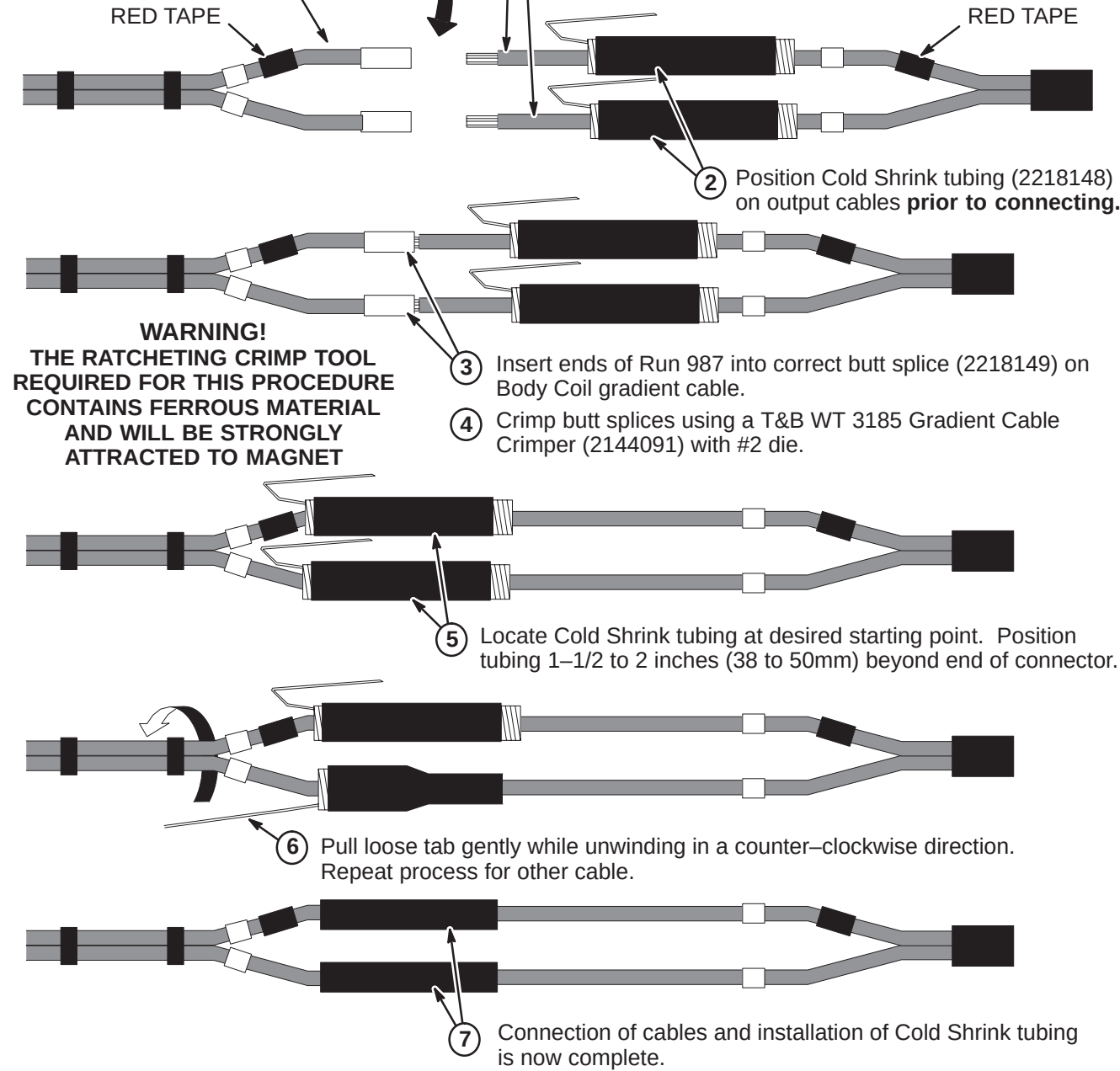
Note: Illustration below shows a typical cable routing for connection of gradient output cables within an enclosed cable trough.

Note: Cold Shrink Tubing (2218148) can be found in the Magnet Field Spares Kit delivered with the Magnet

Gradient Output Cables from Penetration Panel

- 1 Cable connection process shown below is for Run 987. Repeat process for Runs 982, 983, 984, 985 and 986.

Note: Red tape, XWB+ (RUN 987), YWB+ (RUN 986), ZWB+ (RUN 985), XzM+ (RUN 984), YzM+ (RUN 983), and ZzM+ (RUN 982) labels are attached to wire "ONE".



WARNING!
THE RATCHETING CRIMP TOOL
REQUIRED FOR THIS PROCEDURE
CONTAINS FERROUS MATERIAL
AND WILL BE STRONGLY
ATTRACTED TO MAGNET

INSTALLATION STEPS SUMMARY

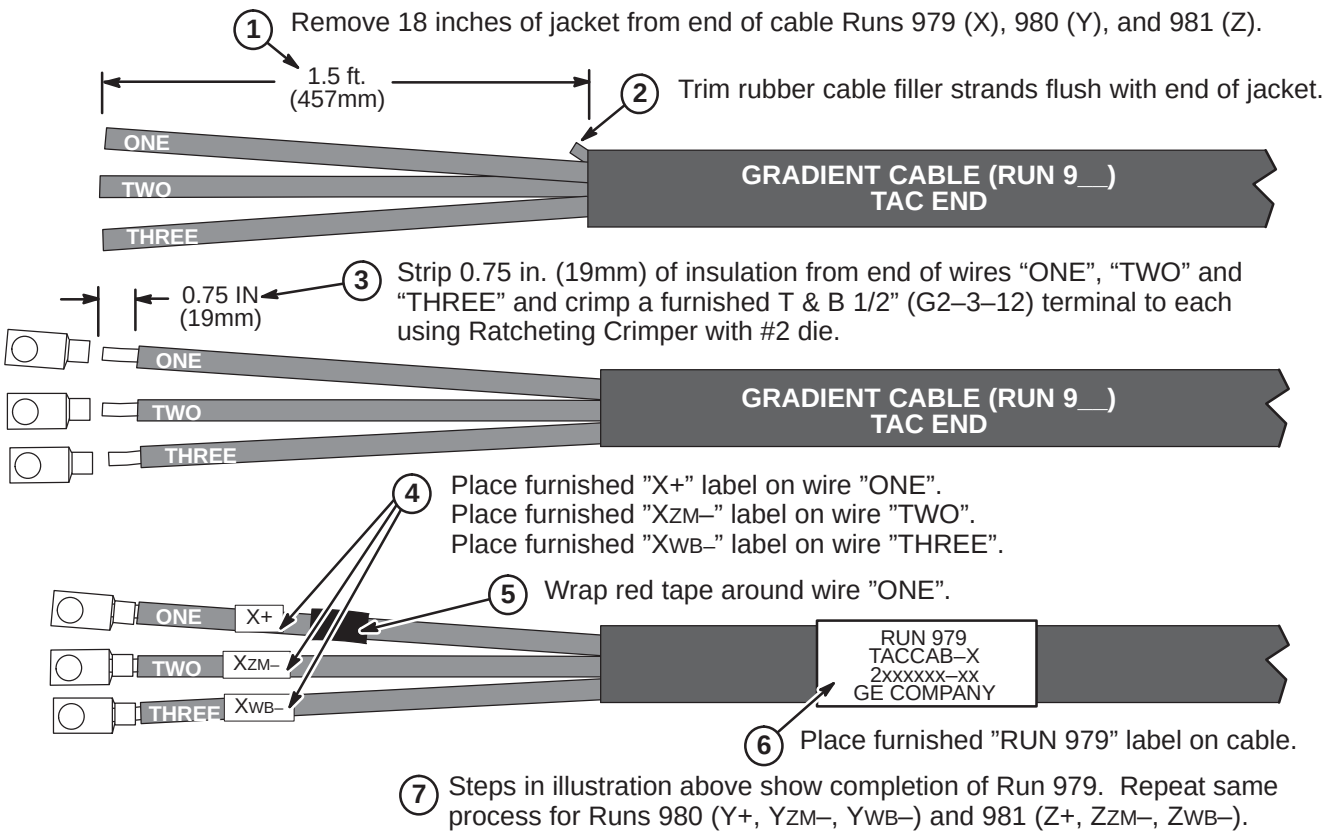
- ❑ D1 – Cut cables to length and route in Equipment Room
- ❑ D2 – Terminate Runs 979, 980, and 981 For TAC Cabinet from Penetration Panel
- ❑ D3 – Route Ground Cables to Runs 979, 980, and 981
- ❑ D4 – Connection Ground Cables and Runs 979, 980, and 981 to Penetration Panel

❑ D1 – CUT CABLES TO LENGTH AND ROUTE IN EQUIPMENT ROOM

- 1 Move Run 979, 980, and 981 cables to equipment room. Route cables from Gradient Filter of Penetration Panel to right side of interface panel at rear of Twin Accessory Cabinet (TAC). The end of cable with terminals will be connected to Penetration Panel.
- 2 Verify sufficient slack of at least two feet (610mm) is provided for termination and connection. Mark location for cutting. Cut cables to length and label cut ends with applicable run number.
- 3 Route remainder of cables from rear of TAC to rear of ACGD/PDU cabinet. Make sure enough slack is provided for termination and connection to cabinet. Mark location for cutting. Cut cables to length.
- 4 Locate furnished bag of parts and attach terminals and labels to each cable which was cut to length. See Procedures C2 and D2a or D2b for TAC Cabinet terminations and D1 for ACGD/PDU cabinet terminations.
- 5 See Procedure C3 for routing and terminating of Run 988 ground wires.
- 6 See Procedure C4 for connection to Equipment room side of Penetration Panel.
- 7 See TAC installation instructions (DIRECTION 2308633 IN TAB 17) for connection of Runs 976, 977, 978, 979, 980, and 981 to TAC cabinet.
- 8 See ACGD/PDU Cabinet installation instructions (DIRECTION 2308653 IN TAB 15) for connection of Runs 976, 977, and 978 to ACGD/PDU cabinet.

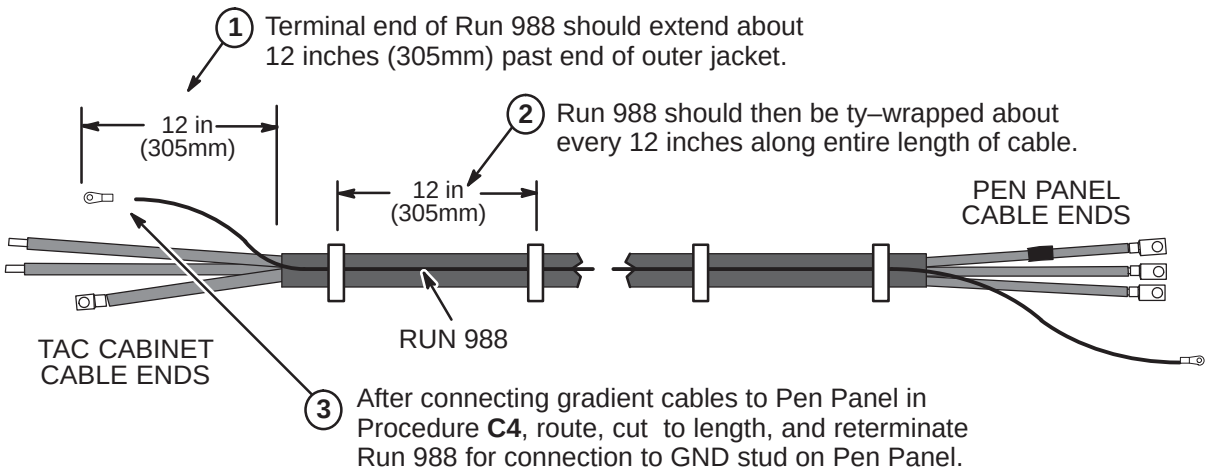
❑ D2 – TERMINATE RUNS 979, 980, & 981 FOR TAC CABINET FROM PEN PANEL

Note: The following steps are required to terminate the end of the cable that was cut in Procedure C1, Step 2. The other end of the cable is connected to the Penetration Panel.



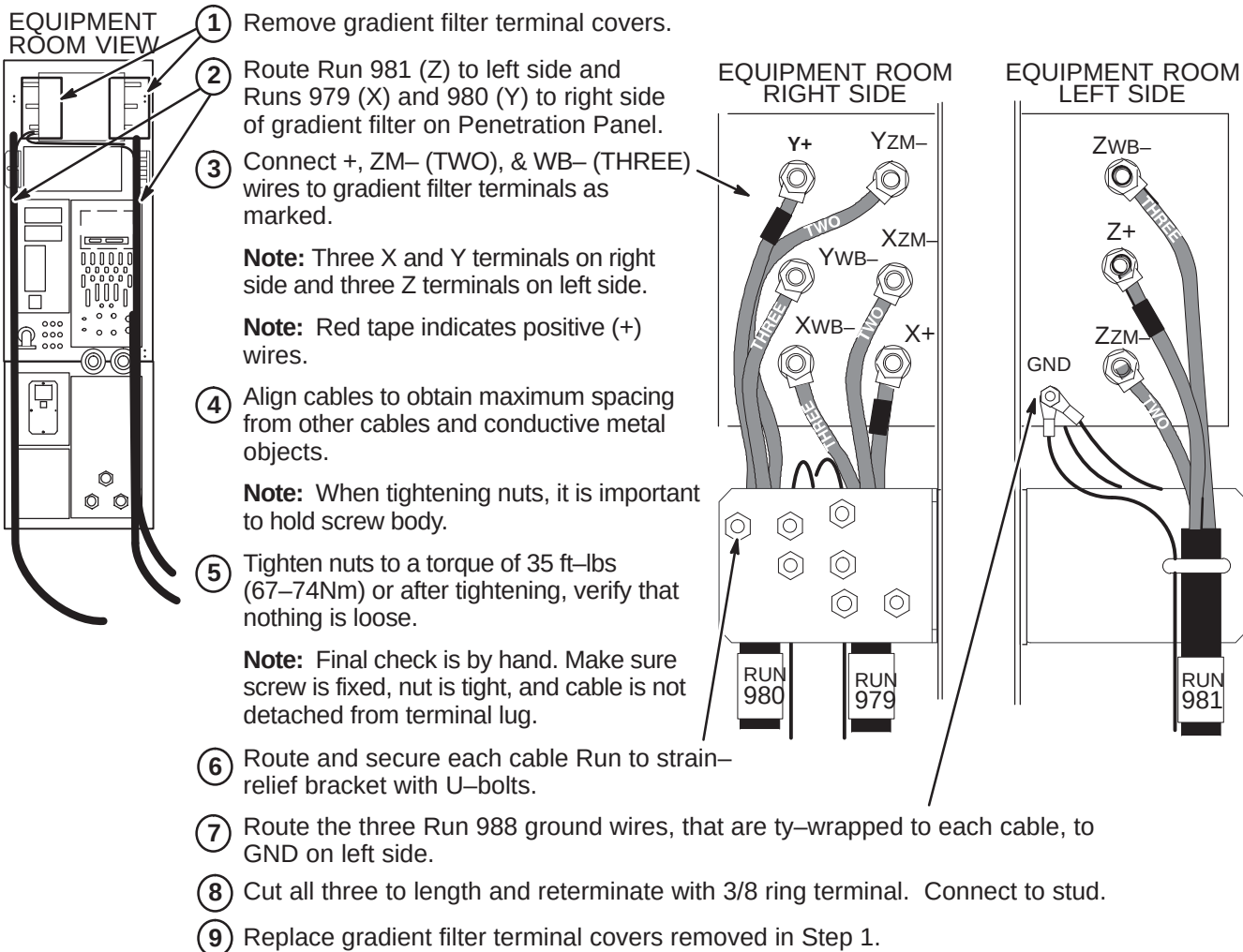
❑ D3 – ROUTE GROUND CABLES TO RUNS 979, 980, AND 981

Note: For each gradient cable routed between the Pen Panel and TAC, a ground wire, Run 988 (2283297), has to be ty-wrapped to each cable.



Excess ground cable can not be coiled. It must be cut to length and reterminated.

❑ D4 – CONNECT GROUND CABLES AND RUNS 979, 980, AND 981 TO PEN PANEL



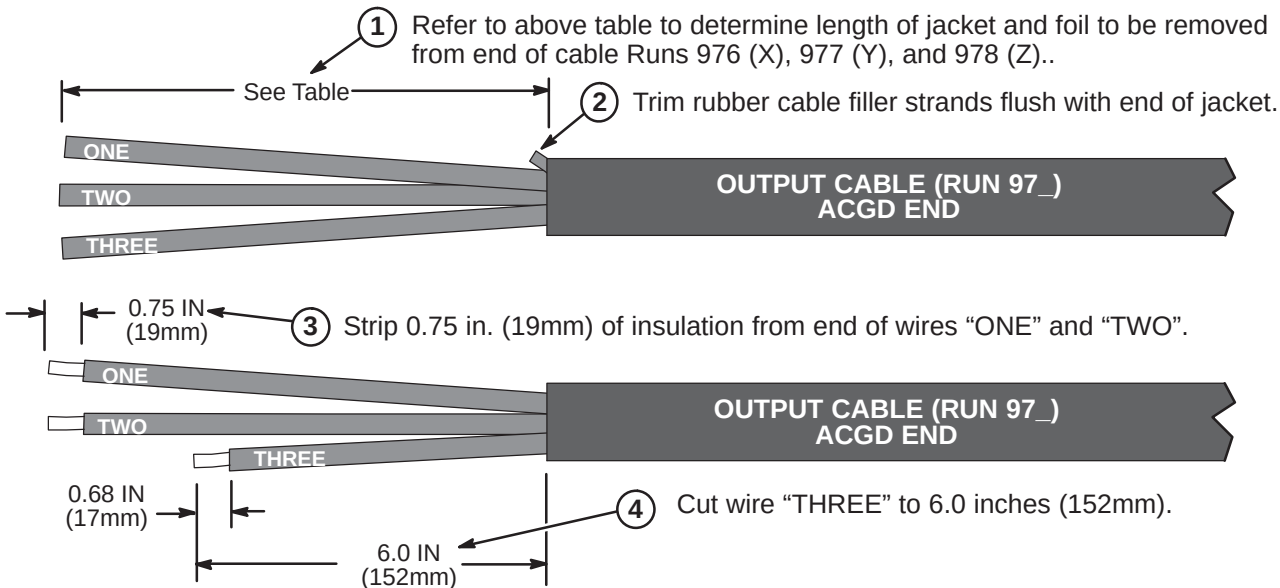
INSTALLATION STEPS SUMMARY

- ☐ E1 – Terminate Runs 976, 977, and 978 for Connection to ACGD/PDU Cabinet
- ☐ E2a – Connect Runs 976, 977, and 978 to TAC Cabinet from ACGD/PDU OR
- ☐ E2b – Terminate Runs 976, 977, and 978 for TAC Cabinet from ACGD/PDU

E1 – TERMINATE RUN 976, 977, & 978 FOR CONNECTION TO ACGD/PDU CABINET

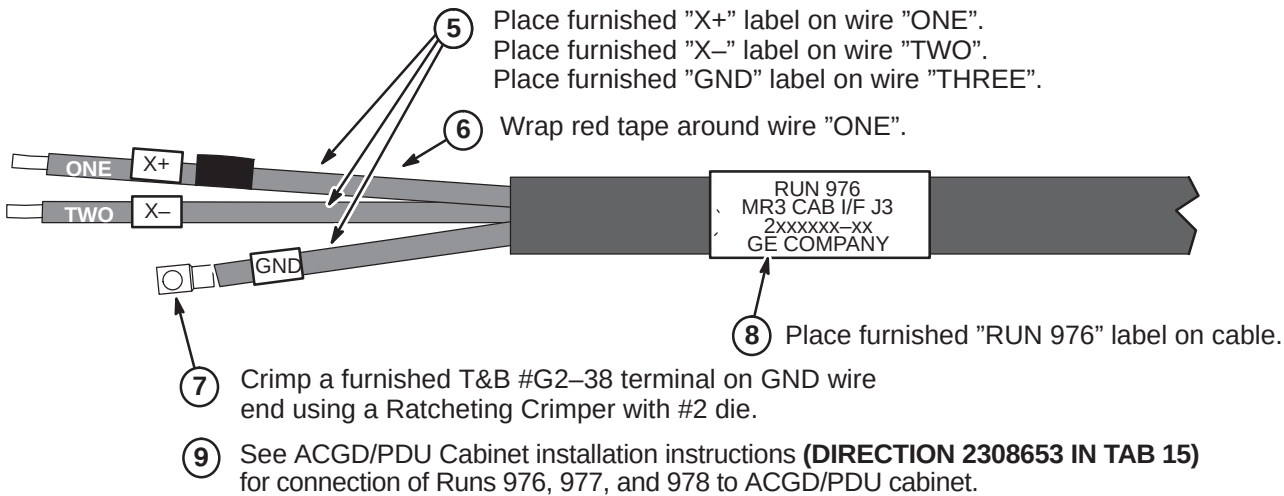
Note: The following steps are required to terminate the end of the cable that was cut in Procedure **C1, Step 3**. The other end of the cable is connected to the TAC Cabinet.

	Run 976	Run 977	Run 978
Jacket Length to be Removed	8.0 IN (203mm)	11.0 IN (280mm)	14.0 IN (356mm)



Note: Steps in illustration below show completion of Run 976. Repeat same process for Runs 977 (Y+, Y-) and 978 (Z+, Z-).

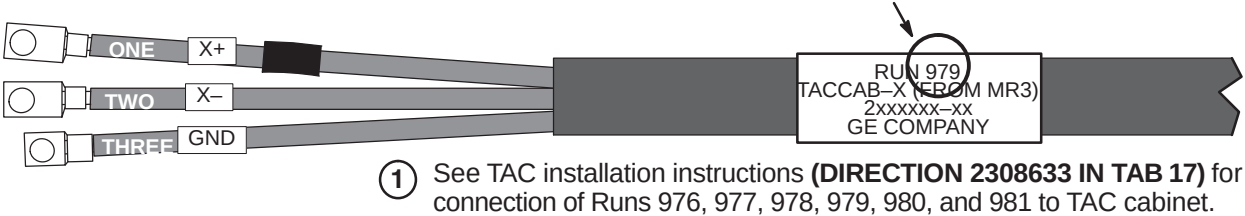
Red tape, Y+ (RUN 977) and Z+ (RUN 978) labels are attached to wire "ONE".



E2a – CONNECT RUNS 976, 977, AND 978 TO TAC CABINET FROM ACGD/PDU

A change is in process where cables delivered to site will be preterminated for the TAC Cabinet connection of Runs 976, 977, and 978. (The remainder of the cable cut in Procedure **C1**. If this is the cable delivered to your site, then proceed to Step 1 below.

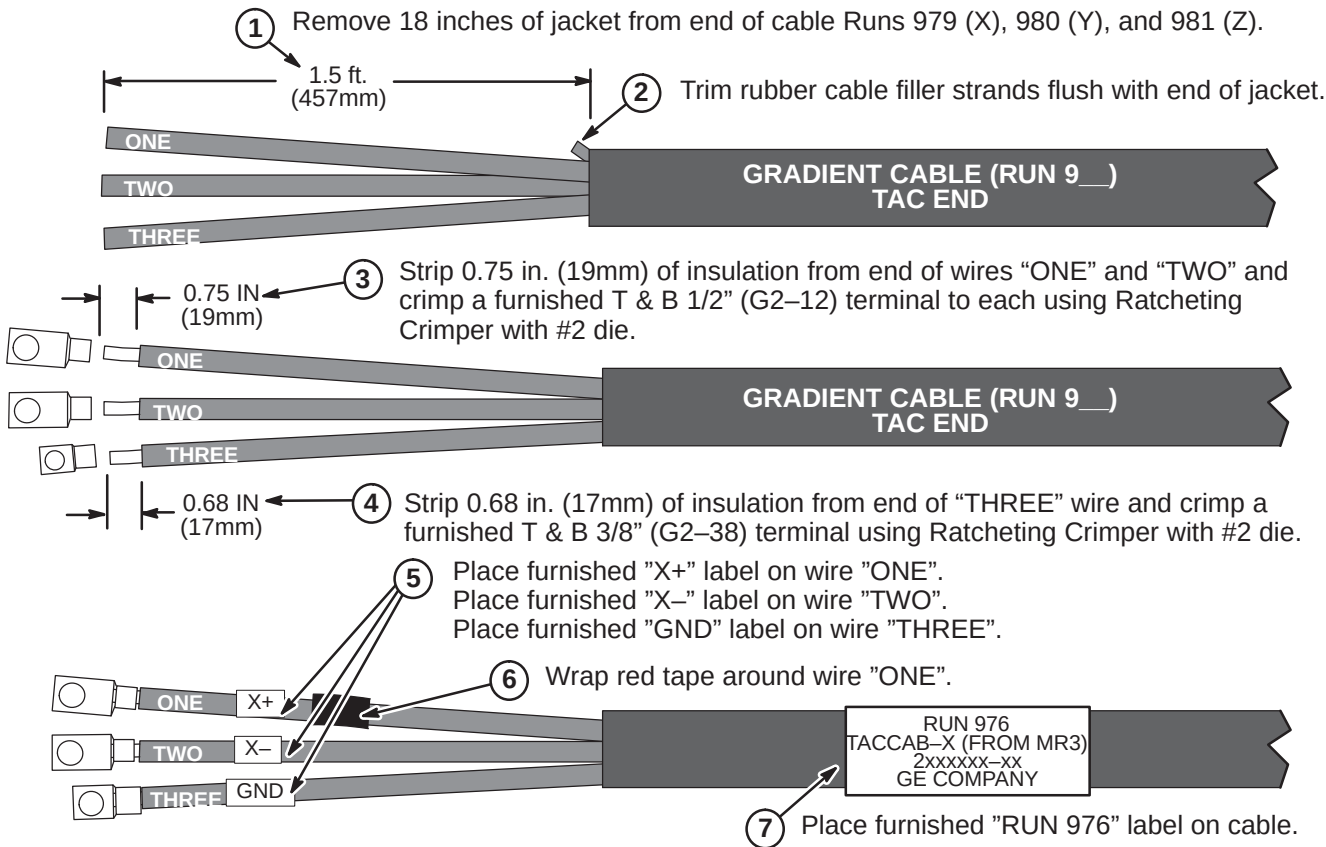
Note: Early production cables contain an error on the label. Correct the label(s) as follows: Change Run 979 to 976, 980 to 977, and 981 to 978.



Note: Early versions of the gradient cables delivered to site required the cable wires to be terminated for connection to the TAC Cabinet for Runs 976, 977, and 978 as in Procedure **D2b** below. Complete steps 1 thru 9 in Procedure **D2b** below is these are the style of cables delivered to your site.

E2b – TERMINATE RUNS 976, 977, AND 978 FOR TAC CABINET FROM ACGD/PDU

Note: The following steps are required to terminate the end of the cable that was cut in Procedure **C1, Step 3**. The other end of the cable is connected to the ACGD/PDU Cabinet.



8 Steps in illustration above show completion of Run 976. Repeat same process for Runs 977 (Y+, Y-, GND) and 978 (Z+, Z-, GND).

9 See TAC installation instructions (DIRECTION 2308633 IN TAB 17) for connection of Runs 976, 977, 978, 979, 980, and 981 to TAC cabinet.

INSTALLATION STEPS SUMMARY

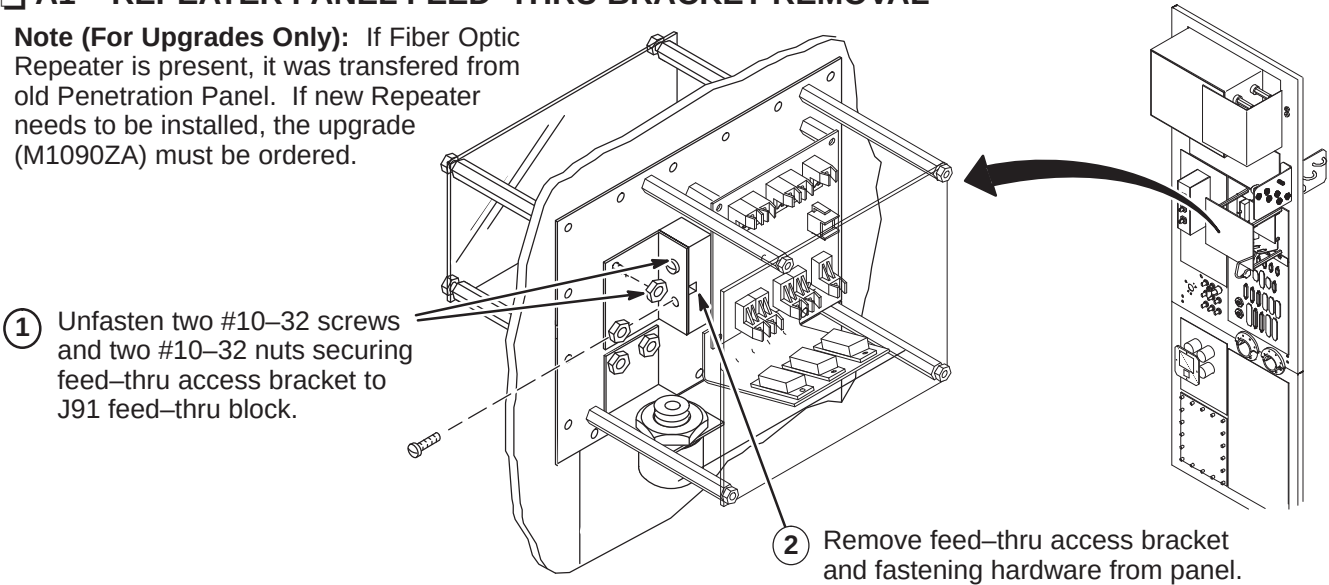
- A1 – Repeater Panel Feed-thru Bracket Removal
- A2 – Routing Run 711/712 in Magnet Room
- A3 – Connecting Magnet Room Run 711/712 to Repeater Panel
- A4 – Route/Connect Equipment Room Run 711/712 to Repeater Panel
- A5 – Route/Connect Fiber Optic Equipment Room Runs 708, 709, 710, 711/712, and 836

CAUTION

Fiber optic cables are easily damaged! Handle fiber optic cables very carefully. Failure to do so may cause intermittent problems difficult to isolate. Do not bend fiber optic cables to radius smaller than two inches.

A1 – REPEATER PANEL FEED-THRU BRACKET REMOVAL

Note (For Upgrades Only): If Fiber Optic Repeater is present, it was transferred from old Penetration Panel. If new Repeater needs to be installed, the upgrade (M1090ZA) must be ordered.



A2 – ROUTING RUN 711/712 IN MAGNET ROOM

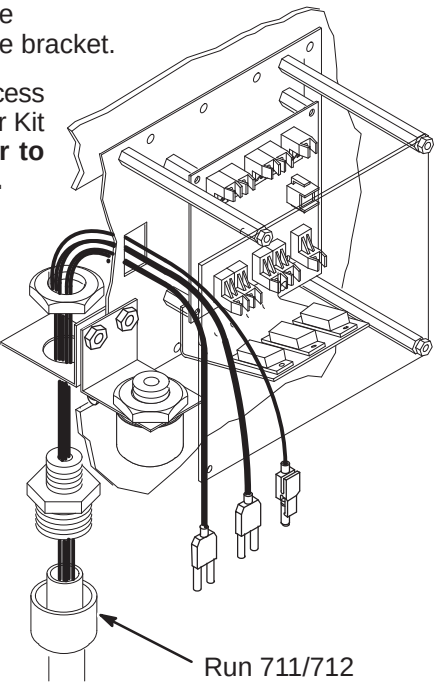
- 1 Move Run 711/712 (46-328079G9) to the Magnet Room side of the Penetration Panel. The “PP1-J91” end will be attached to interface bracket.

Note: If Run 711/712 excess conduit is a problem in the magnet room, excess may be cutoff and ends terminated with the Fiber Optic Connector Repair Kit per page 2. **If cables need to be shortened, mark each cable prior to cutting. This will aid in identifying the cables when reterminating.**

- 2 Attach Run 711/712 to access bracket. Refer to Direction 2123149 (packed with bag of FO Conduit connectors) for connection installation.
- 3 Route ends of individual fiber-optic Runs through J91 opening to the Repeater Board (PP1 A17 A1)
- 4 Route Run 711/712 to Magnet Enclosure Rear Pedestal and underneath magnet.

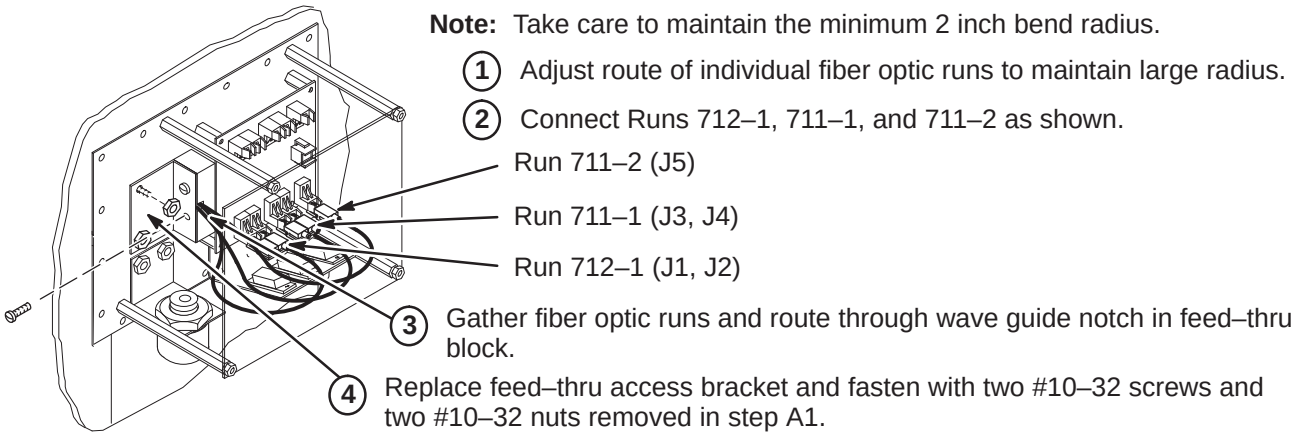
Note: For the next two steps, refer to Magnet Enclosure Installation (Tab 2) for connecting Run 711 to SRI Module and Run 712 to PAC-II.

- 5 Route Run 711/712 to the front right side of the magnet (SRI location).
- 6 Route Run 712 into the magnet enclosure front cover, across the top and down the right side to the PAC-II.



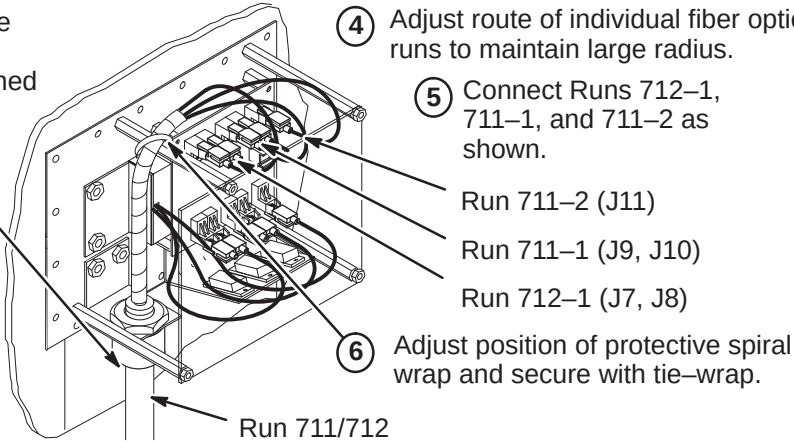
A3 – CONNECTING MAGNET ROOM RUN 711/712 TO REPEATER PANEL

- Note:** Take care to maintain the minimum 2 inch bend radius.
- 1 Adjust route of individual fiber optic runs to maintain large radius.
 - 2 Connect Runs 712-1, 711-1, and 711-2 as shown.



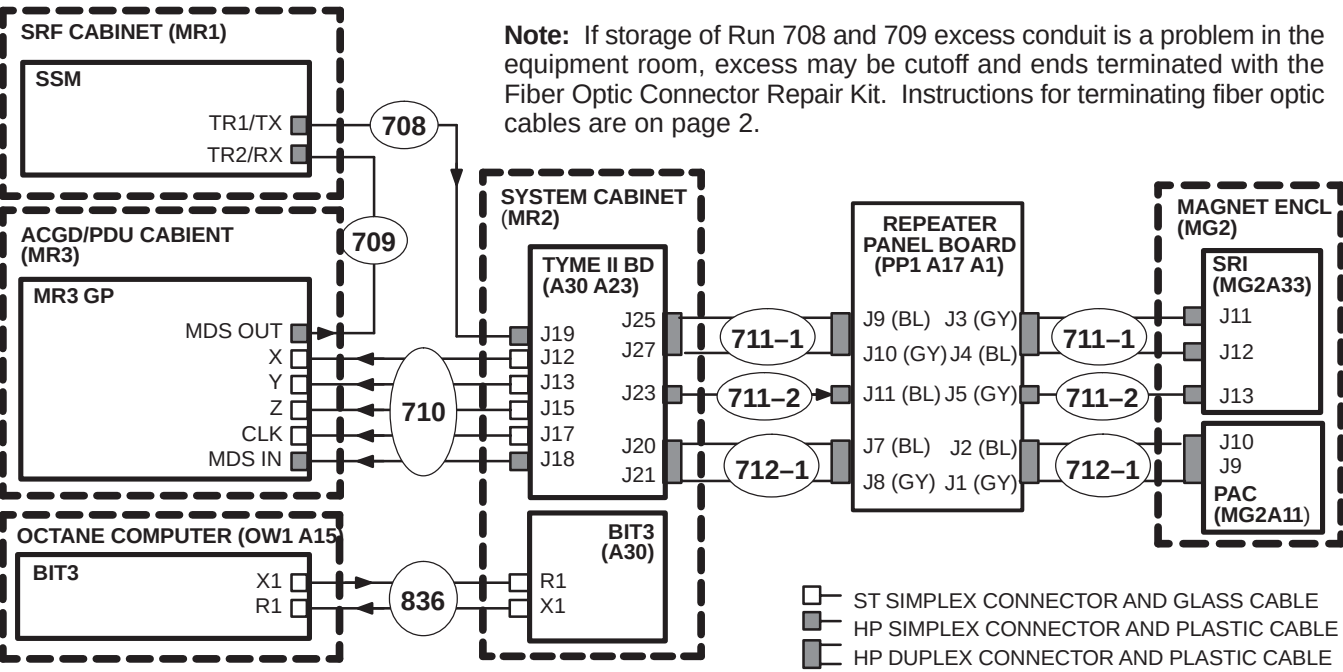
A4 – ROUTE/CONNECT EQUIPMENT ROOM RUN 711/712 TO REPEATER PANEL

- 1 Move Run 711/712 (46-328078G5) to the Equipment Room side of the Penetration Panel. The “PP1-A17” end will be attached to interface bracket.
- 2 Attach Run 711/712 to access bracket. Refer to Direction 2123149 (packed with bag of FO Conduit connectors) for connector installation instructions.
- 3 Route other end of Run 711/712 to System Cabinet. Refer to Direction 2170004 (Tab 8) for connecting fiber optics in System Cabinet.
- 4 Adjust route of individual fiber optic runs to maintain large radius.
- 5 Connect Runs 712-1, 711-1, and 711-2 as shown.
- 6 Adjust position of protective spiral wrap and secure with tie-wrap.



A5 – ROUTE/CONNECT F/O EQUIPMENT ROOM RUNS 708, 709, 710, 711/712 & 836

- 1 Unroll coiled fiber optic conduits for Runs 708, 709, and 710 along intended route. Connection of Runs is shown in diagram below. Plan for storage of excess Run 710 conduit of as it cannot be reterminated.



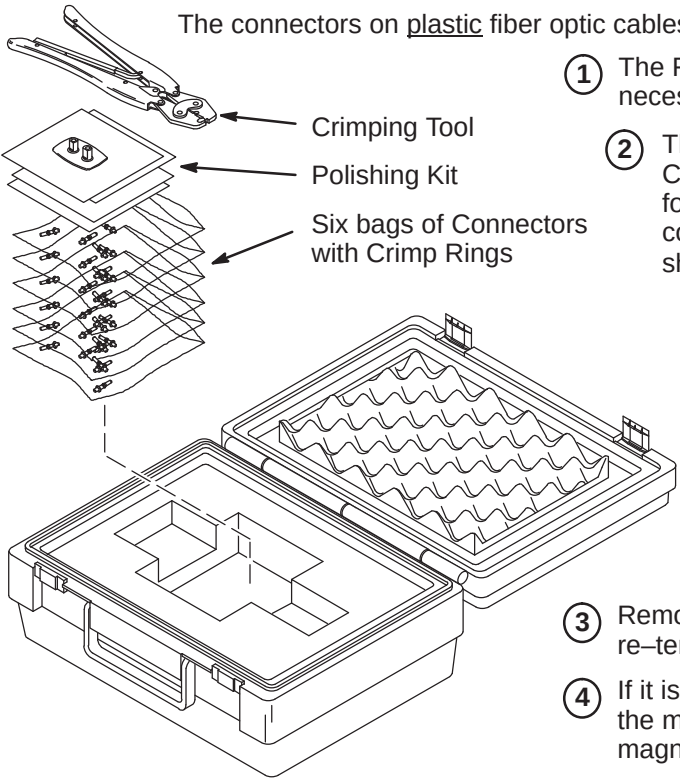
INSTALLATION STEPS SUMMARY

- ❑ B1 – Fiber Optic Cable Termination Kit
- ❑ B2 – Stripping Fiber Optic Outer Jacket
- ❑ B3 – Crimping Connector to Fiber Optic Cable
- ❑ B4 – Trimming Excess Fiber Optic from Connector
- ❑ B5 – Polishing Fiber Optic End

WARNING!

FERROUS MATERIAL HAZARD! CRIMP TOOL AND OTHER TOOLS REQUIRED FOR THIS PROCEDURE CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. KEEP ALL FERROUS TOOLS AS FAR FROM THE MAGNET AS POSSIBLE.

❑ B1 – FIBER OPTIC CABLE TERMINATION KIT



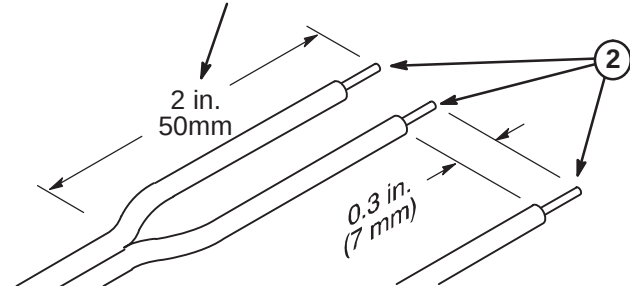
- 1 The Fiber Optic Repair Kit, 46-301450G1, contains all the necessary materials to replace a damaged connector.
- 2 The "Spares Kit", delivered with the Shipping Collector, contains a 46-320119P1 Termination Kit for Fiber Optic Cables. This kit consists of consumables such as terminals and polishing sheets used to terminate cables during installation.

Note: The connectors on glass fiber optic cables **cannot** be replaced using the F/O Repair Kit. At this time, there is no kit available to repair glass fiber optic cable connectors.

- 3 Remove the end of fiber optic cable to be re-terminated from magnet room if possible.
- 4 If it is not possible to remove the end to be repaired from the magnet room, the cable must be as far from the magnet as possible before it can be worked on.
- 5 Cut cable to desired length.

❑ B2 – STRIPPING FIBER OPTIC OUTER JACKET

- 1 For Duplex Cables, separate the two fibers approximately 2 in. (50mm) back from the ends.



Note: The separated duplex cable must be stripped to equal lengths on each side of the cable. This allows proper seating of the cable into the connector.

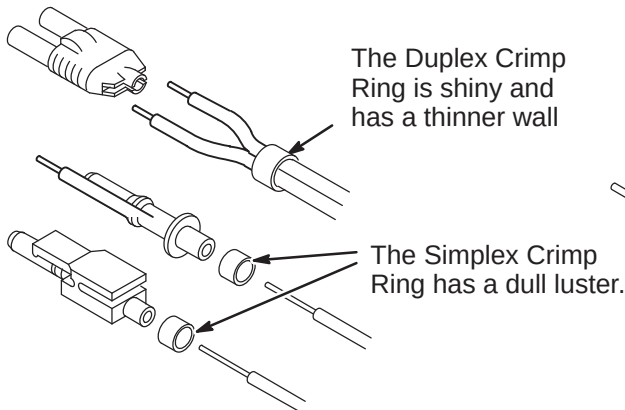
- 2 Strip off approximately 0.3 in. (7 mm) of the outer jacket with the 16 gauge wire strippers. Excess webbing on duplex cable may have to be trimmed to allow the connector to slide over the cable.

❑ B3 – CRIMPING CONNECTOR TO FIBER OPTIC CABLE

- 1 Place the Crimp Ring and Connector over the end of the cable.

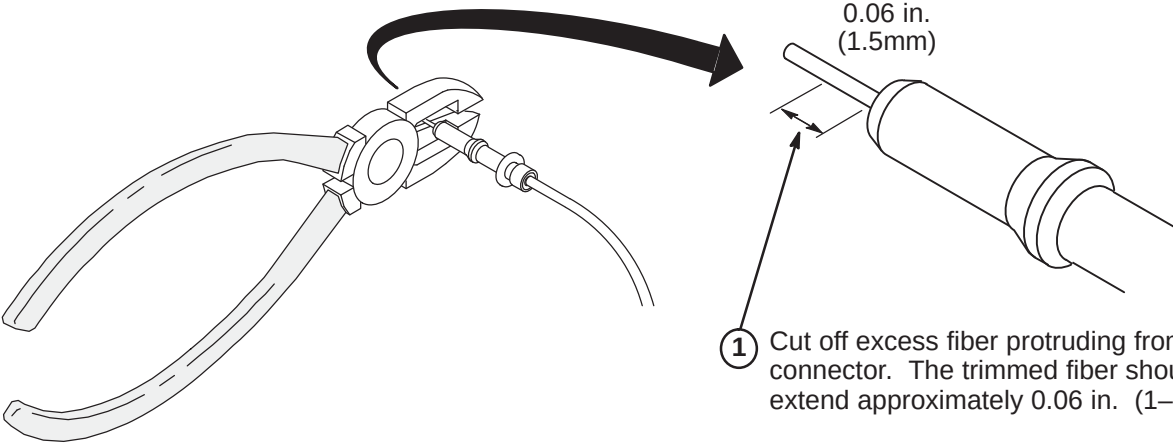
Note: The Connector Crimp Rings cannot be interchanged between simplex and duplex connectors.

- 2 The fiber should extend approximately 0.12 in. (3mm) through the end of the connector.



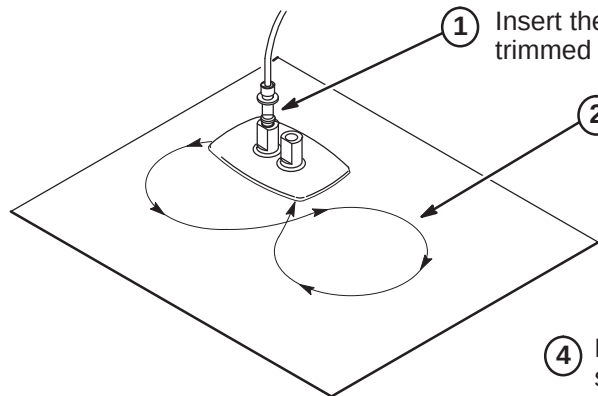
- 3 Carefully position the ring so that it is entirely on the connector and then crimp the ring in place with the crimping tool.

❑ B4 – TRIMMING EXCESS FIBER FROM CONNECTOR



- 1 Cut off excess fiber protruding from the connector. The trimmed fiber should extend approximately 0.06 in. (1-1/2mm).

❑ B5 – POLISHING FIBER OPTIC END



- 1 Insert the connector fully into the polishing fixture with the trimmed fiber end protruding from the bottom of the fixture.

- 2 Place the 600 grit abrasive paper on a flat smooth surface. Pressing down on the indicator, polish the fiber and the connector using a figure eight motion until the connector is flush with the end of the polishing fixture.

- 3 Wipe the connector and fixture with a clean cloth or tissue.

- 4 Place the flush connector and polishing fixture on the dull side of the 3 micron pink lapping film and continue to polish the fiber and connector for approximately 25 strokes.

Note: The four dots on the bottom of the polishing fixture are wear indicators. Replace the polishing fixture when any dot is no longer visible.

- 5 The fiber end should be flat, smooth and clean. The cable end can now be connected.

INSTALLATION STEPS SUMMARY

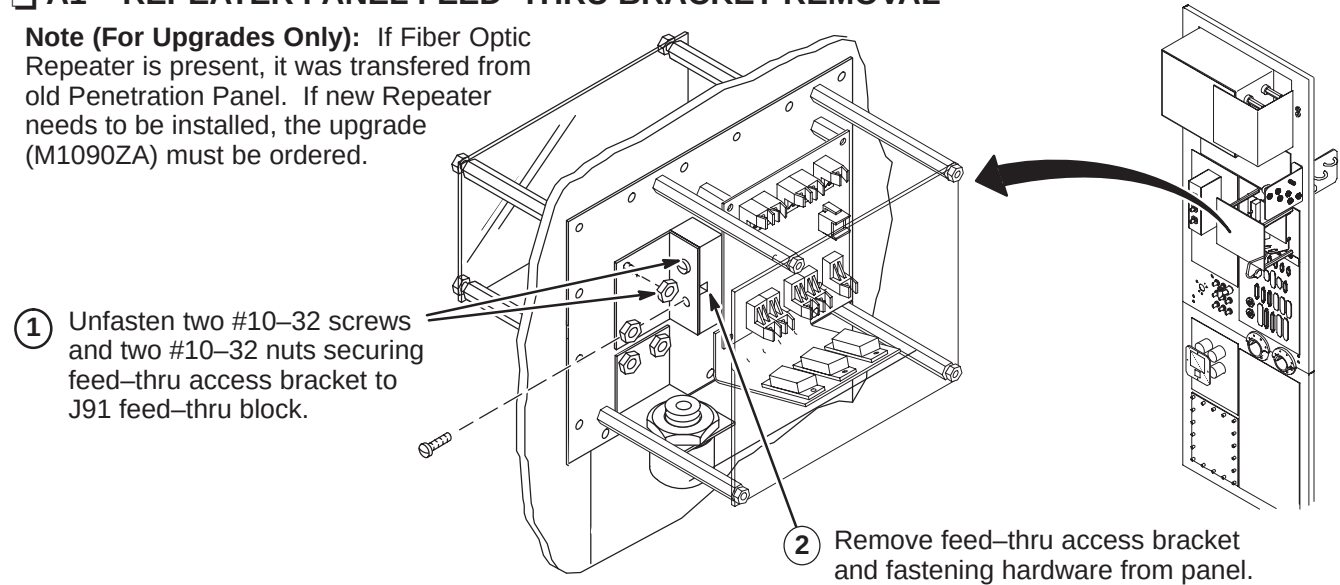
- A1 – Repeater Panel Feed-thru Bracket Removal
- A2 – Routing Run 711/712 in Magnet Room
- A3 – Connecting Magnet Room Run 711/712 to Repeater Panel
- A4 – Route/Connect Equipment Room Run 1045 to Repeater Panel
- A5 – Cable Map of Fiber Optic Runs 709, 711/712, 1032, 1033, 1034, 1044, and 1045

CAUTION

Fiber optic cables are easily damaged! Handle fiber optic cables very carefully. Failure to do so may cause intermittent problems difficult to isolate. Do not bend fiber optic cables to radius smaller than two inches.

A1 – REPEATER PANEL FEED-THRU BRACKET REMOVAL

Note (For Upgrades Only): If Fiber Optic Repeater is present, it was transferred from old Penetration Panel. If new Repeater needs to be installed, the upgrade (M1090ZA) must be ordered.



A2 – ROUTING RUN 711/712 IN MAGNET ROOM

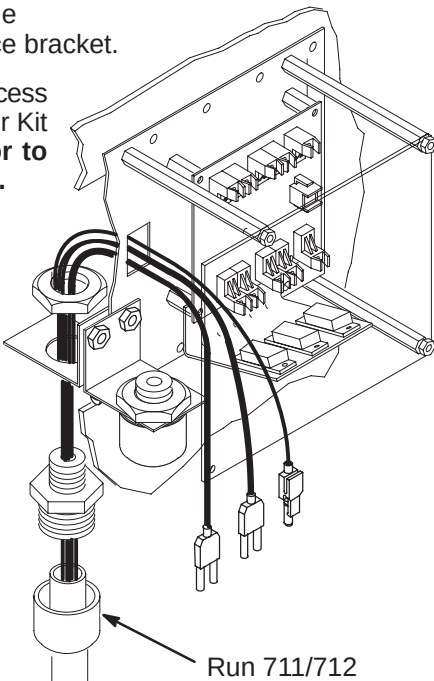
- 1 Move Run 711/712 (46-328079G9) to the Magnet Room side of the Penetration Panel. The “PP1-J91” end will be attached to interface bracket.

Note: If Run 711/712 excess conduit is a problem in the magnet room, excess may be cutoff and ends terminated with the Fiber Optic Connector Repair Kit per page 2. **If cables need to be shortened, mark each cable prior to cutting. This will aid in identifying the cables when reterminating.**

- 2 Attach Run 711/712 to access bracket. Refer to Direction 2123149 (packed with bag of FO Conduit connectors) for connection installation.
- 3 Route ends of individual fiber-optic Runs through J91 opening to the Repeater Board (PP1 A17 A1)
- 4 Route Run 711/712 to Magnet Enclosure Rear Pedestal and underneath magnet.

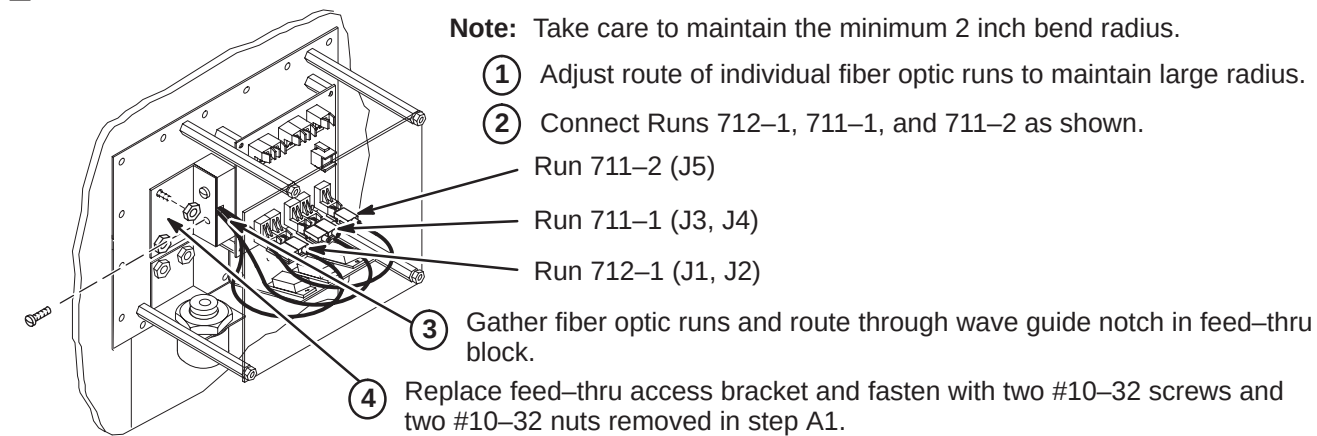
Note: For the next two steps, refer to Magnet Enclosure Installation (Tab 2) for connecting Run 711 to SRI Module and Run 712 to PAC-II.

- 5 Route Run 711/712 to the front right side of the magnet (SRI location).
- 6 Route Run 712 into the magnet enclosure front cover, across the top and down the right side to the PAC-II.



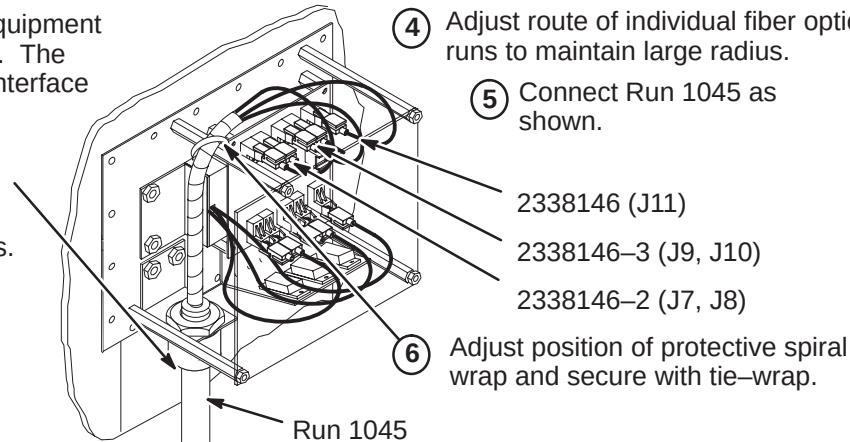
A3 – CONNECTING MAGNET ROOM RUN 711/712 TO REPEATER PANEL

Note: Take care to maintain the minimum 2 inch bend radius.



A4 – ROUTE/CONNECT EQUIPMENT ROOM RUN 1045 TO REPEATER PANEL

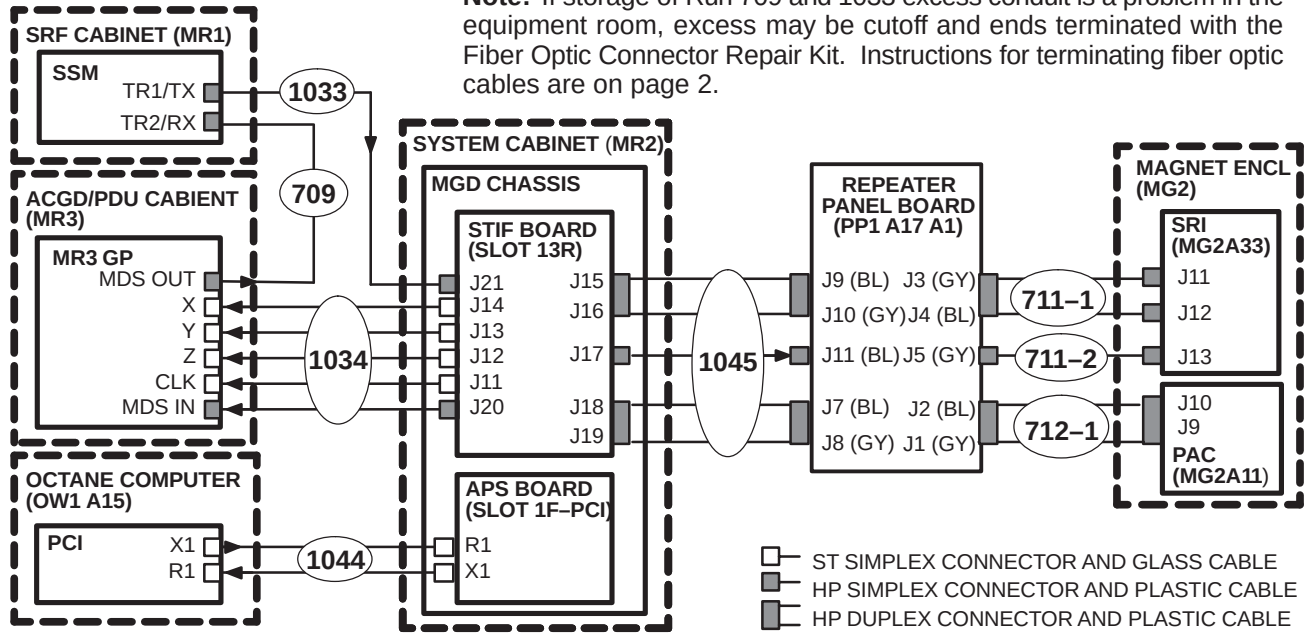
- 1 Move Run 1045 (2331846) to the Equipment Room side of the Penetration Panel. The “PP1-A17” end will be attached to interface bracket.
- 2 Attach Run 1045 to access bracket. Refer to Direction 2123149 (packed with bag of FO Conduit connectors) for connector installation instructions.
- 3 Route other end of Run 1045 to System Cabinet. Refer to Direction 2338502 (Tab 8) for connecting fiber optics in System Cabinet.
- 4 Adjust route of individual fiber optic runs to maintain large radius.
- 5 Connect Run 1045 as shown.
- 6 Adjust position of protective spiral wrap and secure with tie-wrap.



A5 – CABLE MAP OF F/O RUNS 709, 711/712, 1032, 1033, 1034, 1044, AND 1045

- 1 Unroll coiled fiber optic conduits for Runs 709, 1033, and 1034 along intended route. Connection of Runs is shown in diagram below. Plan for storage of excess Run 1034 conduit of as it cannot be reterminated.

Note: If storage of Run 709 and 1033 excess conduit is a problem in the equipment room, excess may be cutoff and ends terminated with the Fiber Optic Connector Repair Kit. Instructions for terminating fiber optic cables are on page 2.



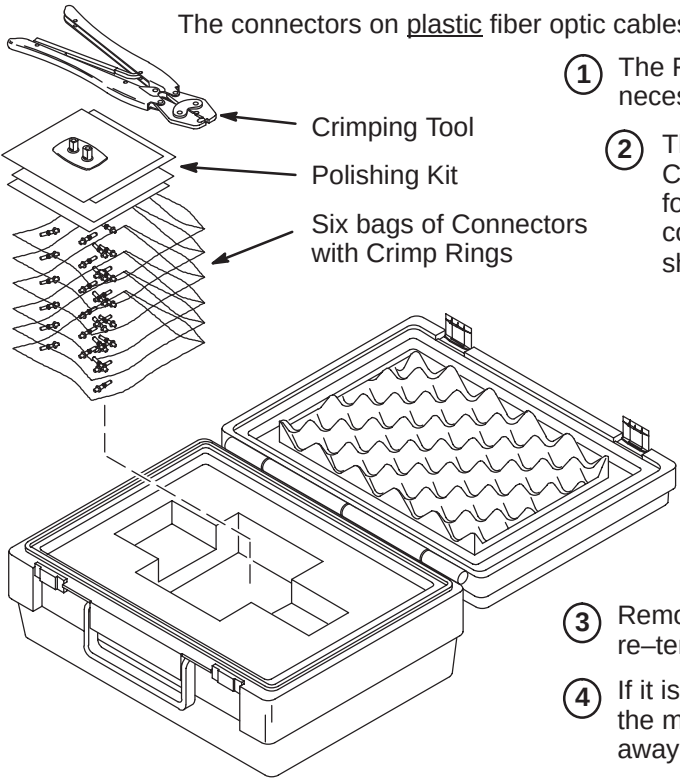
INSTALLATION STEPS SUMMARY

- ❑ B1 – Fiber Optic Cable Termination Kit
- ❑ B2 – Stripping Fiber Optic Outer Jacket
- ❑ B3 – Crimping Connector to Fiber Optic Cable
- ❑ B4 – Trimming Excess Fiber Optic from Connector
- ❑ B5 – Polishing Fiber Optic End

WARNING!

FERROUS MATERIAL HAZARD! CRIMP TOOL AND OTHER TOOLS REQUIRED FOR THIS PROCEDURE CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. KEEP ALL FERROUS TOOLS AT LEAST 10 FEET AWAY FROM THE MAGNET.

❑ B1 – FIBER OPTIC CABLE TERMINATION KIT



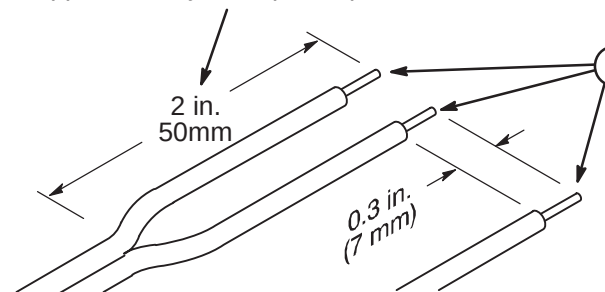
- 1 The Fiber Optic Repair Kit, 46-301450G1, contains all the necessary materials to replace a damaged connector.
- 2 The "Spares Kit", delivered with the Shipping Collector, contains a 46-320119P1 Termination Kit for Fiber Optic Cables. This kit consists of consumables such as terminals and polishing sheets used to terminate cables during installation.

Note: The connectors on glass fiber optic cables **cannot** be replaced using the F/O Repair Kit. At this time, there is no kit available to repair glass fiber optic cable connectors.

- 3 Remove the end of fiber optic cable to be re-terminated from magnet room if possible.
- 4 If it is not possible to remove the end to be repaired from the magnet room, the cable must be at least 10 feet away from the magnet before it can be worked on.
- 5 Cut cable to desired length.

❑ B2 – STRIPPING FIBER OPTIC OUTER JACKET

- 1 For Duplex Cables, separate the two fibers approximately 2 in. (50mm) back from the ends.



Note: The separated duplex cable must be stripped to equal lengths on each side of the cable. This allows proper seating of the cable into the connector.

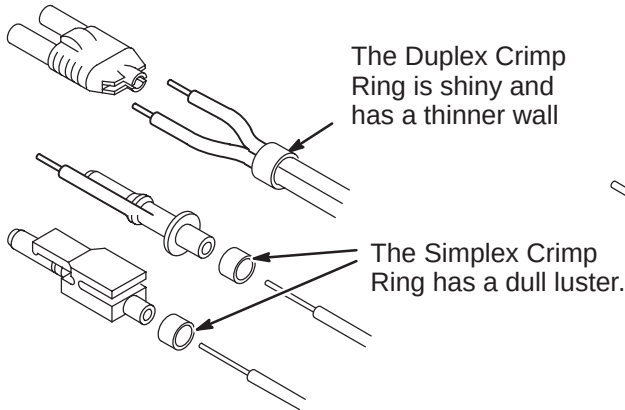
- 2 Strip off approximately 0.3 in. (7 mm) of the outer jacket with the 16 gauge wire strippers. Excess webbing on duplex cable may have to be trimmed to allow the connector to slide over the cable.

❑ B3 – CRIMPING CONNECTOR TO FIBER OPTIC CABLE

- 1 Place the Crimp Ring and Connector over the end of the cable.

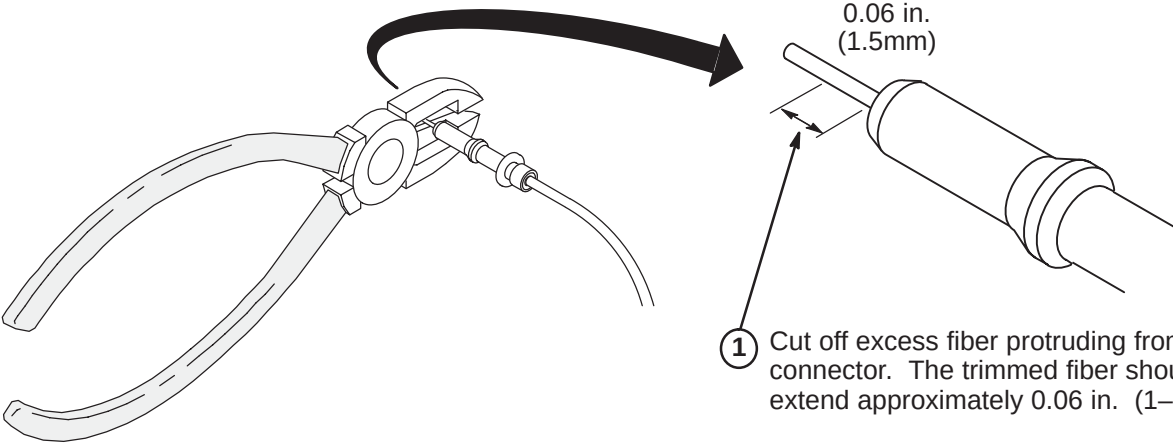
Note: The Connector Crimp Rings cannot be interchanged between simplex and duplex connectors.

- 2 The fiber should extend approximately 0.12 in. (3mm) through the end of the connector.



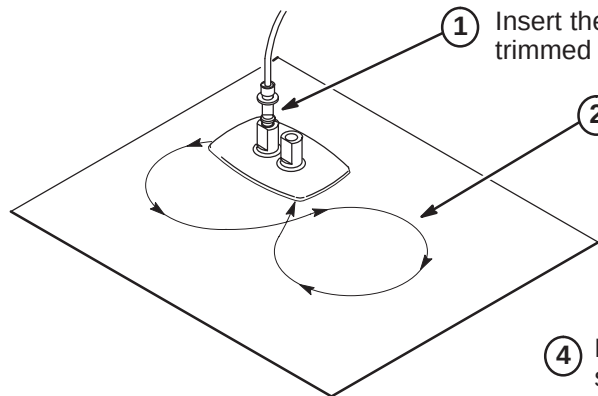
- 3 Carefully position the ring so that it is entirely on the connector and then crimp the ring in place with the crimping tool.

❑ B4 – TRIMMING EXCESS FIBER FROM CONNECTOR



- 1 Cut off excess fiber protruding from the connector. The trimmed fiber should extend approximately 0.06 in. (1-1/2mm).

❑ B5 – POLISHING FIBER OPTIC END



- 1 Insert the connector fully into the polishing fixture with the trimmed fiber end protruding from the bottom of the fixture.
- 2 Place the 600 grit abrasive paper on a flat smooth surface. Pressing down on the indicator, polish the fiber and the connector using a figure eight motion until the connector is flush with the end of the polishing fixture.
- 3 Wipe the connector and fixture with a clean cloth or tissue.
- 4 Place the flush connector and polishing fixture on the dull side of the 3 micron pink lapping film and continue to polish the fiber and connector for approximately 25 strokes.
- 5 The fiber end should be flat, smooth and clean. The cable end can now be connected.

Note: The four dots on the bottom of the polishing fixture are wear indicators. Replace the polishing fixture when any dot is no longer visible.

INSTALLATION STEPS SUMMARY

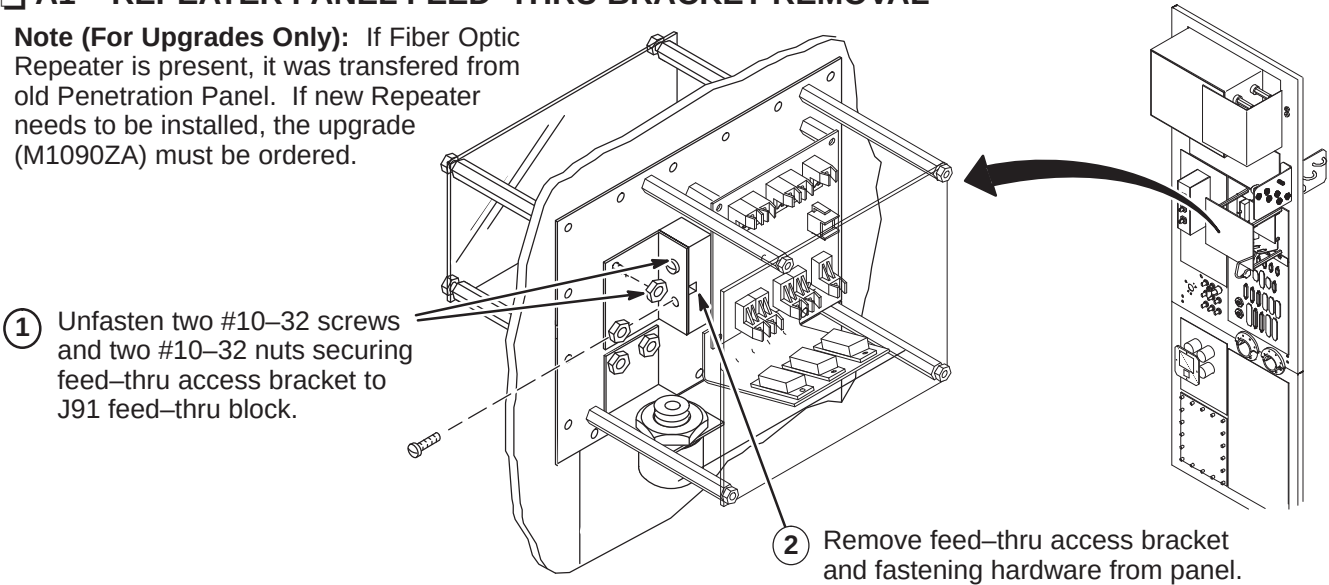
- A1 – Repeater Panel Feed-thru Bracket Removal
- A2 – Routing Run 711/712 in Magnet Room
- A3 – Connecting Magnet Room Run 711/712 to Repeater Panel
- A4 – Route/Connect Equipment Room Run 1045 to Repeater Panel
- A5 – Cable Map of Fiber Optic Runs 709, 711/712, 1032, 1033, 1034, 1044, and 1045

CAUTION

Fiber optic cables are easily damaged! Handle fiber optic cables very carefully. Failure to do so may cause intermittent problems difficult to isolate. Do not bend fiber optic cables to radius smaller than two inches.

A1 – REPEATER PANEL FEED-THRU BRACKET REMOVAL

Note (For Upgrades Only): If Fiber Optic Repeater is present, it was transferred from old Penetration Panel. If new Repeater needs to be installed, the upgrade (M1090ZA) must be ordered.



A2 – ROUTING RUN 711/712 IN MAGNET ROOM

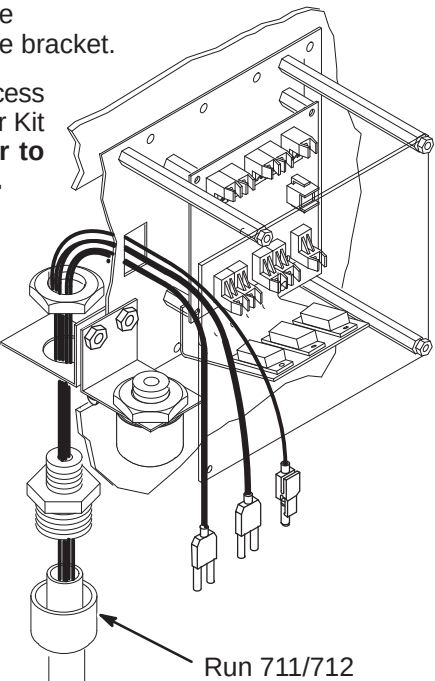
- 1 Move Run 711/712 (46-328079G9) to the Magnet Room side of the Penetration Panel. The “PP1-J91” end will be attached to interface bracket.

Note: If Run 711/712 excess conduit is a problem in the magnet room, excess may be cutoff and ends terminated with the Fiber Optic Connector Repair Kit per page 2. **If cables need to be shortened, mark each cable prior to cutting. This will aid in identifying the cables when reterminating.**

- 2 Attach Run 711/712 to access bracket. Refer to Direction 2123149 (packed with bag of FO Conduit connectors) for connection installation.
- 3 Route ends of individual fiber-optic Runs through J91 opening to the Repeater Board (PP1 A17 A1)
- 4 Route Run 711/712 to Magnet Enclosure Rear Pedestal and underneath magnet.

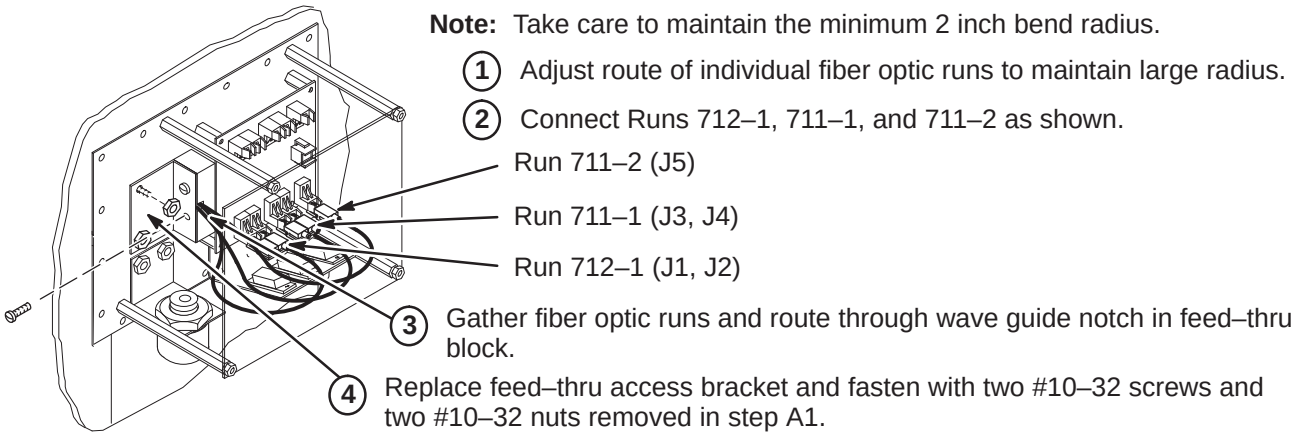
Note: For the next two steps, refer to Magnet Enclosure Installation (Tab 2) for connecting Run 711 to SRI Module and Run 712 to PAC-II.

- 5 Route Run 711/712 to the front right side of the magnet (SRI location).
- 6 Route Run 712 into the magnet enclosure front cover, across the top and down the right side to the PAC-II.

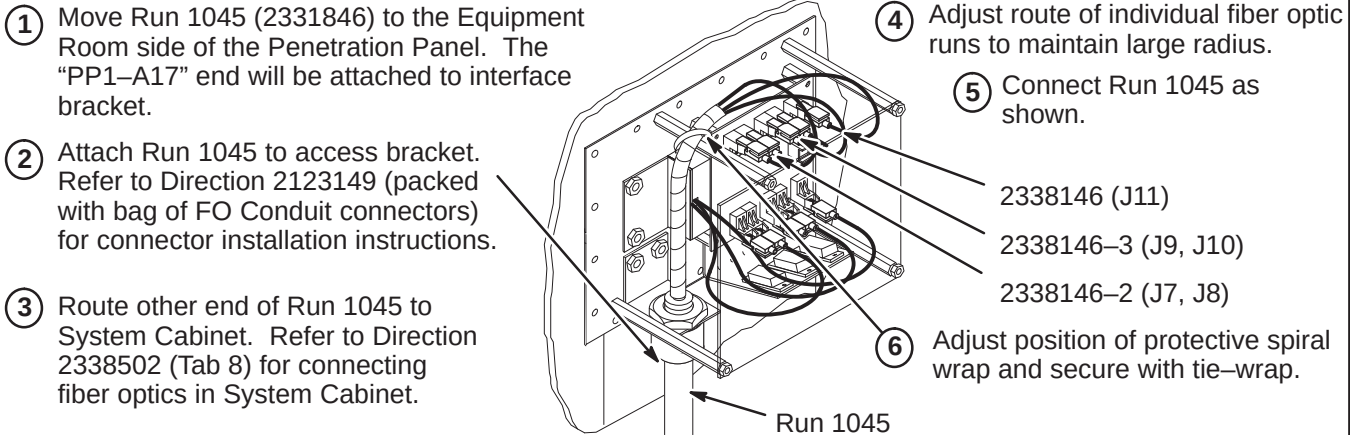


A3 – CONNECTING MAGNET ROOM RUN 711/712 TO REPEATER PANEL

Note: Take care to maintain the minimum 2 inch bend radius.

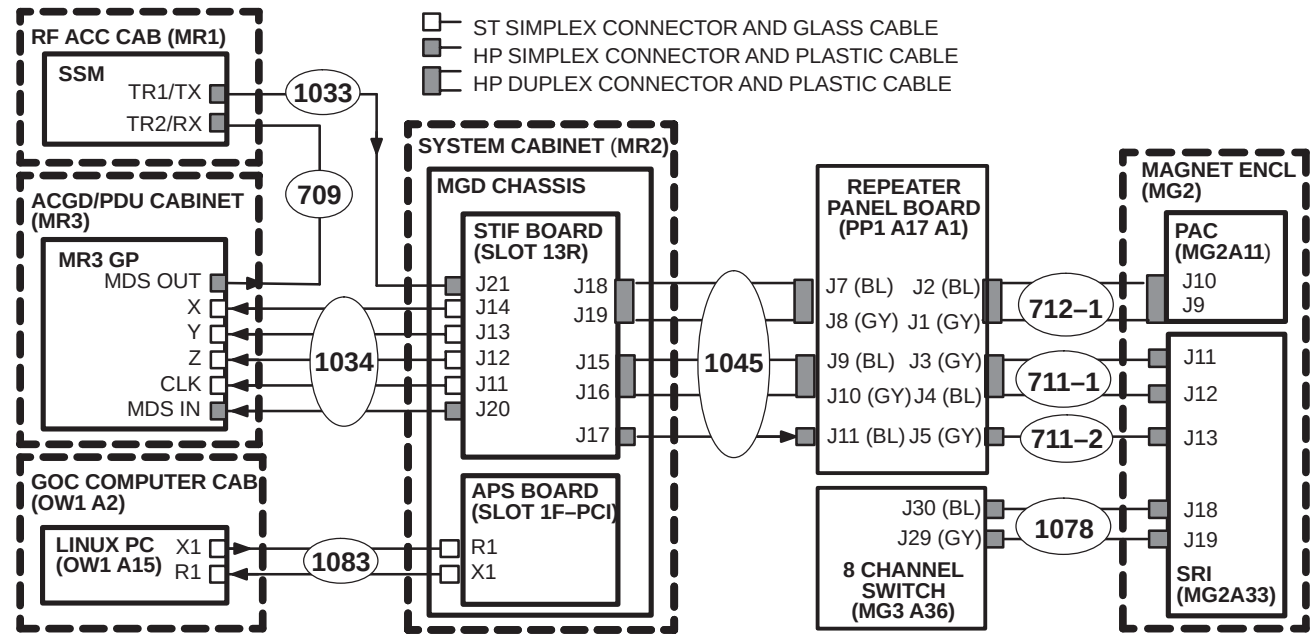


A4 – ROUTE/CONNECT EQUIPMENT ROOM RUN 1045 TO REPEATER PANEL



A5 – CABLE MAP OF F/O RUNS 709, 1033, 1034, 1045, 1078, AND 1083

- 1 Unroll coiled fiber optic conduits for Runs 709, 1033, 1034, 1045, 1078, and 1083 along intended routes. Connection of Runs is shown in diagram below. Plan for storage of excess Run 1034 conduit of as it cannot be reterminated.



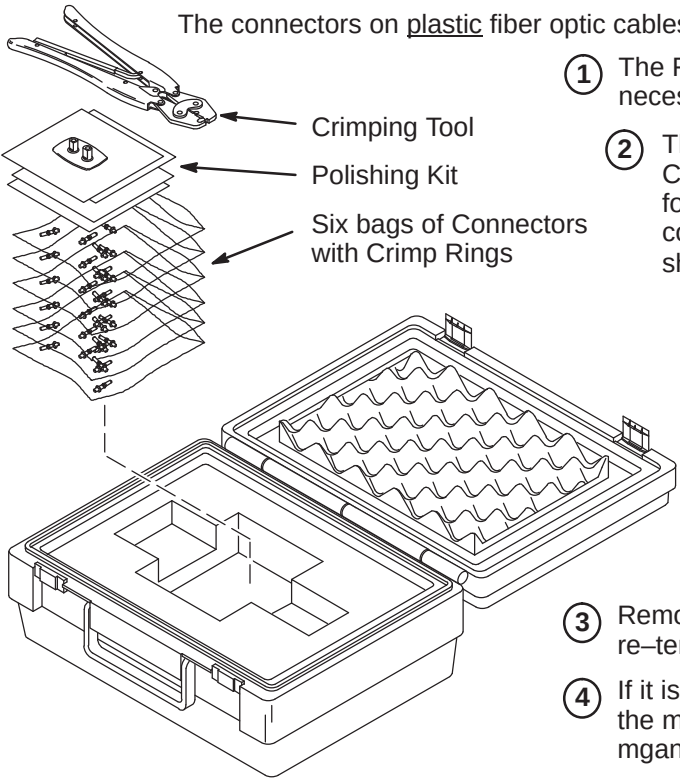
INSTALLATION STEPS SUMMARY

- ❑ B1 – Fiber Optic Cable Termination Kit
- ❑ B2 – Stripping Fiber Optic Outer Jacket
- ❑ B3 – Crimping Connector to Fiber Optic Cable
- ❑ B4 – Trimming Excess Fiber Optic from Connector
- ❑ B5 – Polishing Fiber Optic End

WARNING!

FERROUS MATERIAL HAZARD! CRIMP TOOL AND OTHER TOOLS REQUIRED FOR THIS PROCEDURE CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. KEEP ALL FERROUS TOOLS AS FAR FROM THE MAGNET AS POSSIBLE.

❑ B1 – FIBER OPTIC CABLE TERMINATION KIT



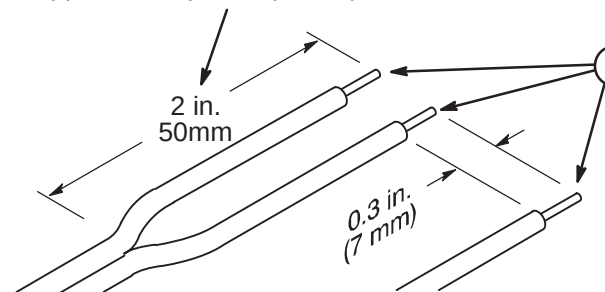
- The connectors on plastic fiber optic cables can be replaced if damaged or cut off to shorten.
- 1 The Fiber Optic Repair Kit, 46-301450G1, contains all the necessary materials to replace a damaged connector.
 - 2 The "Spares Kit", delivered with the Shipping Collector, contains a 46-320119P1 Termination Kit for Fiber Optic Cables. This kit consists of consumables such as terminals and polishing sheets used to terminate cables during installation.

Note: The connectors on glass fiber optic cables **cannot** be replaced using the F/O Repair Kit. At this time, there is no kit available to repair glass fiber optic cable connectors.

- 3 Remove the end of fiber optic cable to be re-terminated from magnet room if possible.
- 4 If it is not possible to remove the end to be repaired from the magnet room, the cable must be as far from the magnet as possible before it can be worked on.
- 5 Cut cable to desired length.

❑ B2 – STRIPPING FIBER OPTIC OUTER JACKET

- 1 For Duplex Cables, separate the two fibers approximately 2 in. (50mm) back from the ends.



Note: The separated duplex cable must be stripped to equal lengths on each side of the cable. This allows proper seating of the cable into the connector.

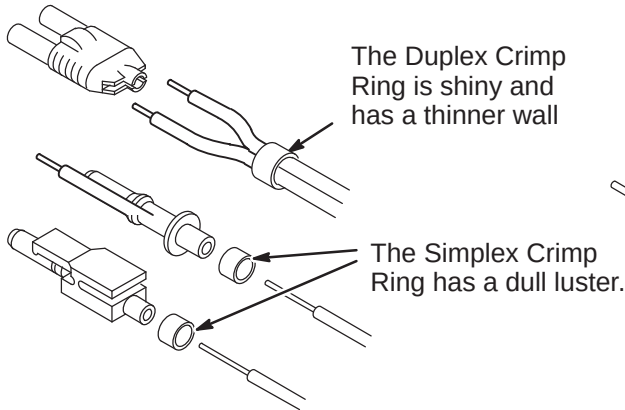
- 2 Strip off approximately 0.3 in. (7 mm) of the outer jacket with the 16 gauge wire strippers. Excess webbing on duplex cable may have to be trimmed to allow the connector to slide over the cable.

❑ B3 – CRIMPING CONNECTOR TO FIBER OPTIC CABLE

- 1 Place the Crimp Ring and Connector over the end of the cable.

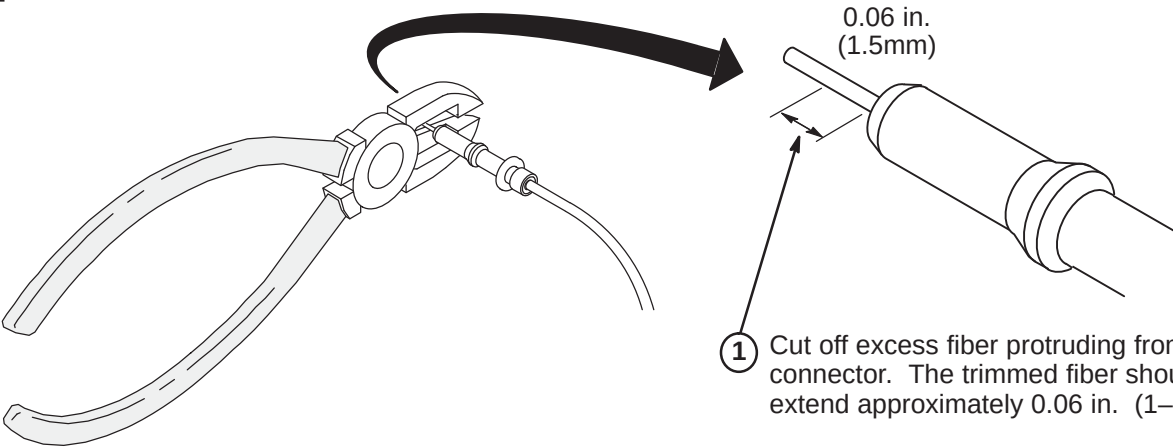
Note: The Connector Crimp Rings cannot be interchanged between simplex and duplex connectors.

- 2 The fiber should extend approximately 0.12 in. (3mm) through the end of the connector.



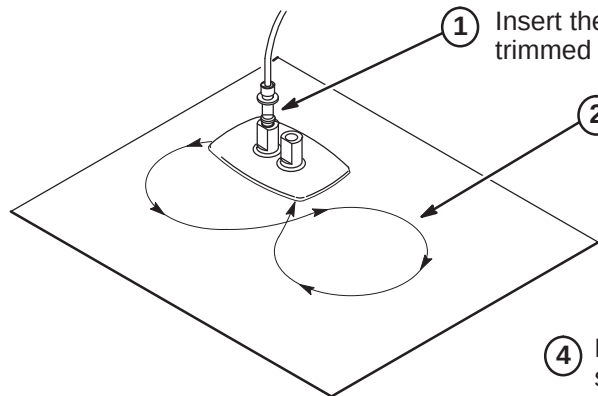
- 3 Carefully position the ring so that it is entirely on the connector and then crimp the ring in place with the crimping tool.

❑ B4 – TRIMMING EXCESS FIBER FROM CONNECTOR



- 1 Cut off excess fiber protruding from the connector. The trimmed fiber should extend approximately 0.06 in. (1-1/2mm).

❑ B5 – POLISHING FIBER OPTIC END



Note: The four dots on the bottom of the polishing fixture are wear indicators. Replace the polishing fixture when any dot is no longer visible.

- 3 Wipe the connector and fixture with a clean cloth or tissue.
- 4 Place the flush connector and polishing fixture on the dull side of the 3 micron pink lapping film and continue to polish the fiber and connector for approximately 25 strokes.
- 5 The fiber end should be flat, smooth and clean. The cable end can now be connected.

INSTALLATION STEPS SUMMARY

- A – System Power Cable Installation
- B – Power Cable Re-termination
- C – Power Cable Pre-connect checks
- D – Power Cable Connections Details

A – SYSTEM POWER CABLE INSTALLATION

1. Verify that all power and ground cables for the Signa System with ACGD are present. Refer to Section 2–1, LOCATING AND UNPACKING CABLES.
2. Lengths of PDU Power cables for:

	FIXED SITE	MOBILE	T/R
Run 966, SSM	35.0 ft. (10.7M)	35.0 ft. (10.7M)	48.0 ft. (14.6M)
Run 967, SRF Cabinet	35.5 ft. (11.1M)	29.5 ft. (9.0M)	49.0 ft. (15.0M)
Run 968, System Cabinet	35.0 ft. (10.7M)	25.0 ft. (7.6M)	45.0 ft. (13.7M)
Run 969, Operator Workspace Computer	65.5 ft. (20.0M)	33.0 ft. (10.0M)	33.0 ft. (10.0M)
Run 970, Water Chiller (Fixed–Site) (Note 1)	82.0 ft. (25.0M)		
Run 973, Heat Exchanger (Mob & T/R) (Note 1)		82.0 ft. (25.0M)	82.0 ft. (25.0M)
Run 989, Twin Accessory Cabinet (Note 1)	25.0 ft. (7.62M)	25.0 ft. (7.62M)	25.0 ft. (7.62M)

Note 1: Runs 970 and 973 are for Non–TwinSpeed systems, Run 989 is only used on TwinSpeed.

If insufficient room exists to store excess cables:

Plan for storage location if cables are not being cut to length. Proceed to Step C.

If insufficient room exists to store excess cables:

They can be cut shorter and re-terminated at the PDU. Complete Step B. Then proceed to Step C.

3. Route all power cables per applicable System Interconnect Diagram. Unroll cables to lay without twists or kinks from cabinet to PDU through provided troughs, conduit, or ducts.

B – POWER CABLE RE-TERMINATION

1. Cut power and ground cables to length as necessary. Leave sufficient slack for later connection when cabinets are in final position according to site architectural plans.
2. Cut and trim insulation for connection of power and ground cable conductors to connectors at PDU.
3. Transfer color code tape, run–number labels, and applicable wire marking labels as each cable is cut to size.

C – POWER CABLE PRE-CONNECT CHECKS

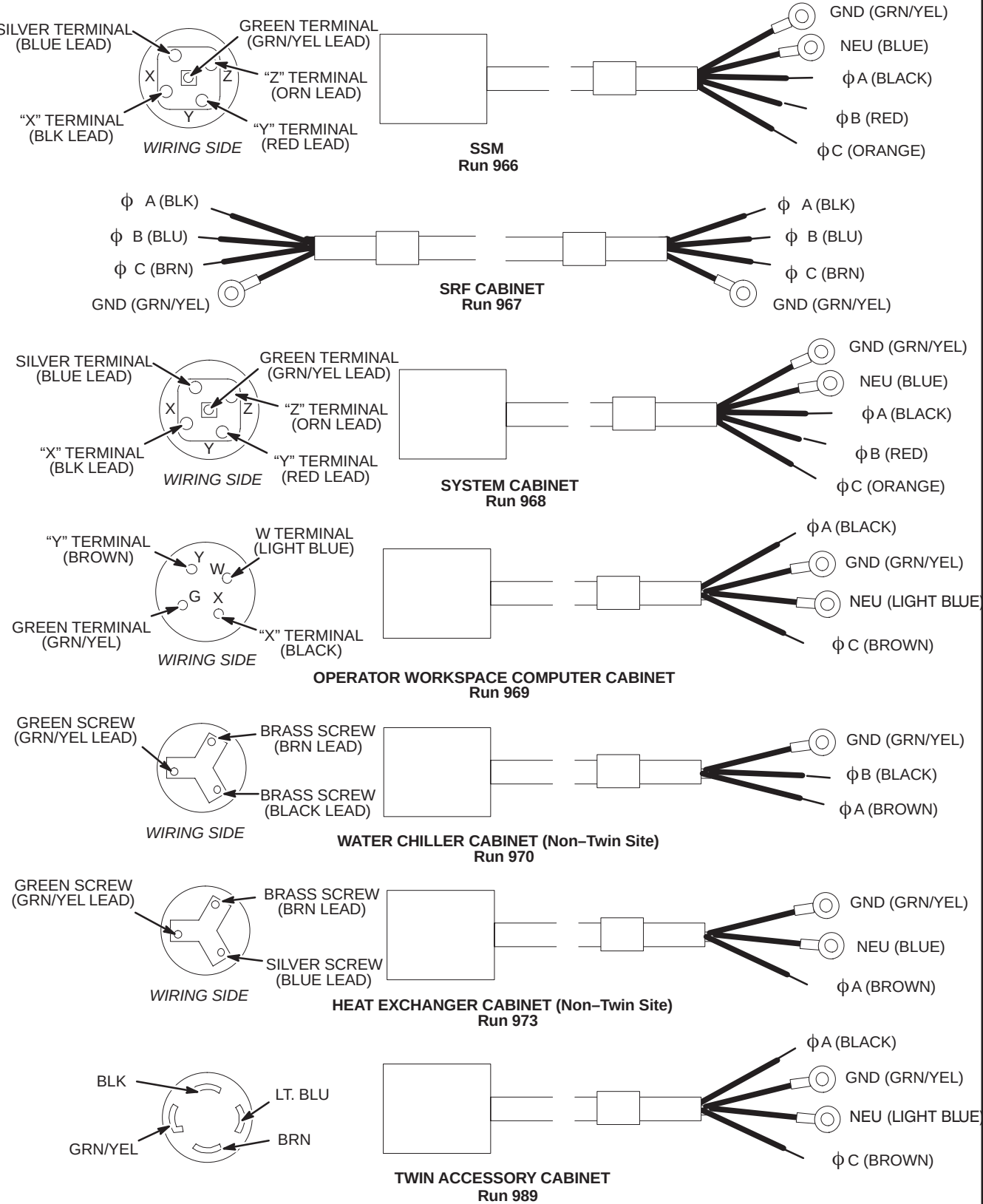
Checks for opens, shorts, and crossed connections are made for field re-terminated power cables prior to connection to PDU and cabinets. This does not provide ground resistance checks for cabinets. Ground resistance checks are in MR CD–ROM, Set Up and Calibration: System, GROUNDING AND LEAKAGE CURRENT CHECKS.

1. Use the ohmmeter and the cable plug connector Illustration to check each power cable for the following:
 - Open circuit between cable neutral and cable ground.
 - Open circuit between cable neutral and each line on the cable.
 - Open circuit between cable ground and each line on the cable.
 - Less than 1 ohm between cable neutral and the light blue (or white) conductor.
 - Less than 1 ohm between cable ground and the green/yellow conductor.
2. Any out-of-spec conditions must be immediately corrected. See Procedure D for plug connection details. After all power cables have been satisfactorily checked out, continue installation.

D – POWER CABLE CONNECTIONS DETAILS

Note

After performing cable pre-connect checks, avoid potential problems by taping up exposed ACGD Gradient Cabinet cables or immediately install ACGD/PDU Gradient Cabinet and connect terminals to power modules before PDU power is turned on.



MAGNET ENCLOSURE INSTALLATION INSTRUCTIONS

1-1 **Tab 2, Magnet Enclosure, contains installation instructions for the following:**

- *DIRECTION 2309132* – Fixed & T/R TwinSpeed Magnet Enclosure Installation Instructions
- *DIRECTION 2398593* – EXCITE Fixed & T/R TwinSpeed Magnet Enclosure Installation Instructions

LIST OF EFFECTIVE PAGES

PAGE	REV	PAGE	REV
1	6	7	6
2	4	8	1
3	1	9	1
4	3	10	1
5	2	11	5
6	1	12	5

WARNING!

FERROUS MATERIAL HAZARD! PARTS REQUIRED FOR THIS UPGRADE INSTALLATION CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES.

IF MAGNET IS AT FULL FIELD – KEEP ALL FERROUS TOOLS AND MATERIALS AS FAR FROM THE MAGNET AS POSSIBLE.

GENERAL ENCLOSURE NOTES:

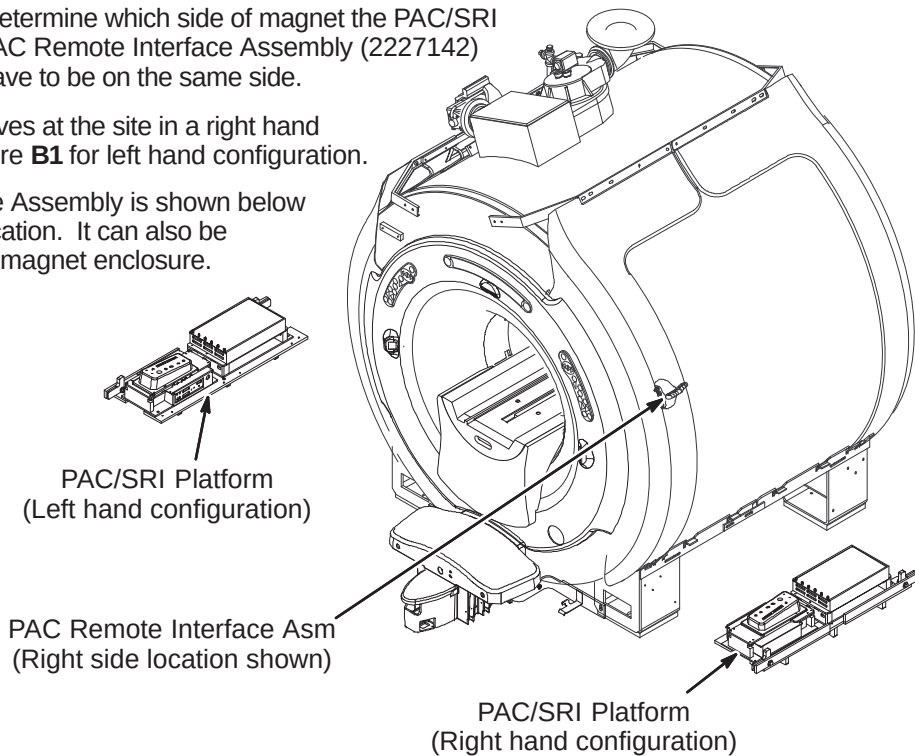
- Front, rear, left, and right, as used in this section, are defined as follows: the patient entrance end of the magnet is the front; the opposite end is the rear; left and right are in respect to a person's left and right while facing the patient entrance end of the magnet.
- A 1-ounce tube of Permatex "Anti-Seize" lubricant (2119594) is provided. Follow instructions on back of tube. "Anti-Seize" lubricant is to be applied to threads of stainless-steel screws before installing into tapped holes of magnet to prevent galling and seizing of cap screw.

INSTALLATION PROCEDURES SUMMARY

- ☐ A1 – Determine PAC/SRI Platform and PAC Remote Interface Location
- ☐ A2 – Route/Connect Coolant Hoses
- ☐ A3 – Coolant Valve Installation

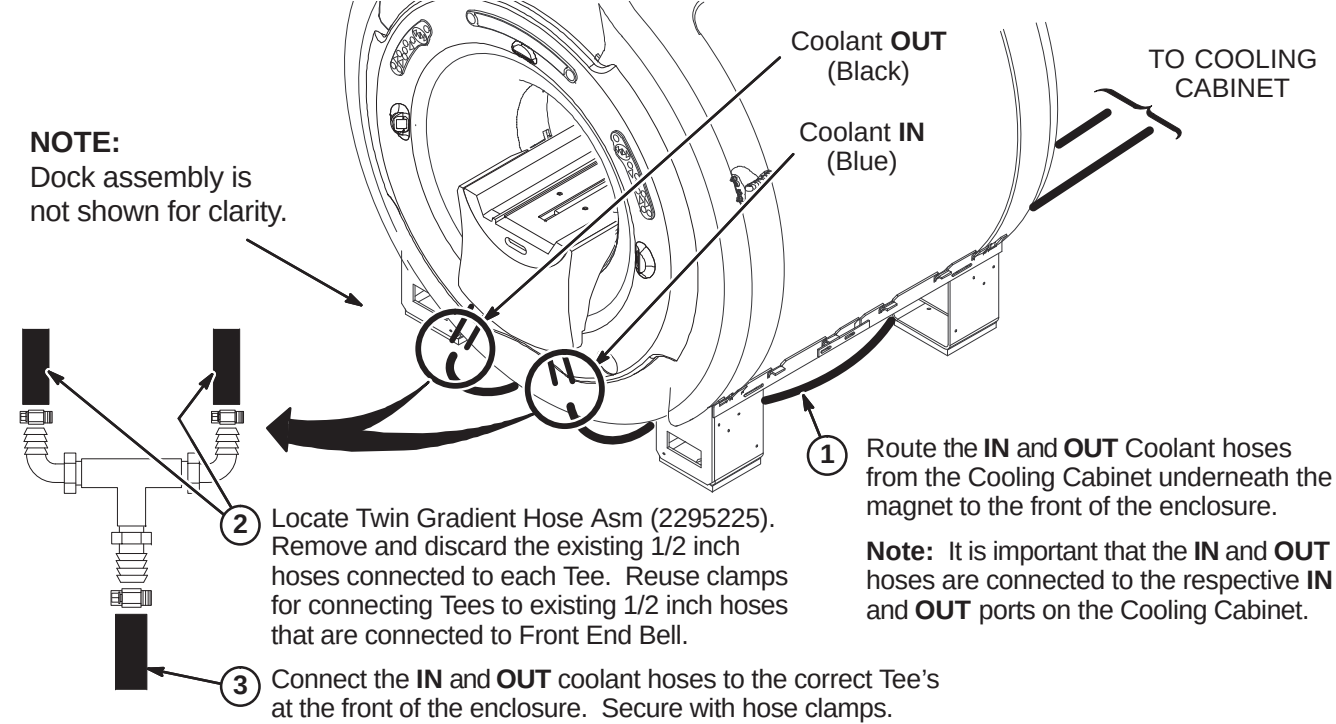
A1 – DETERMINE PAC/SRI PLATFORM AND PAC REMOTE INTERFACE LOCATION

- 1 Consult with customer to determine which side of magnet the PAC/SRI Platform (2223440) and PAC Remote Interface Assembly (2227142) should be located. They have to be on the same side.
- 2 The PAC/SRI Platform arrives at the site in a right hand configuration. See Procedure B1 for left hand configuration.
- 3 The PAC Remote Interface Assembly is shown below installed in the right side location. It can also be mounted on the left side of magnet enclosure.

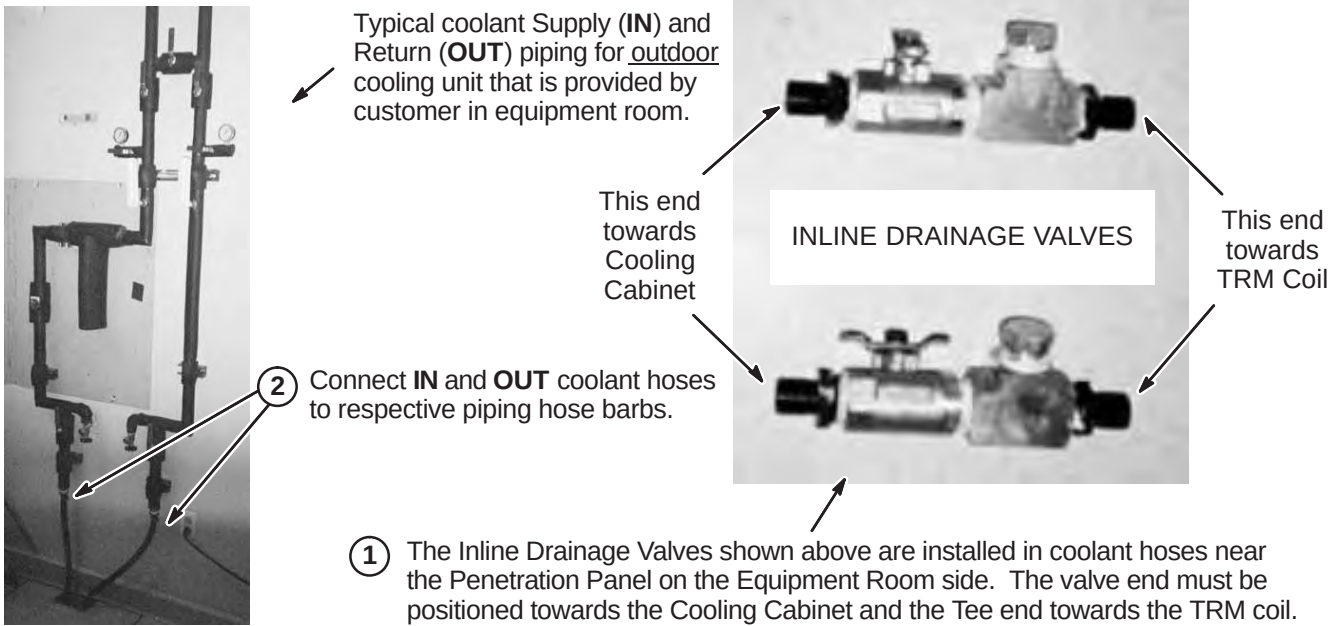


A2 – ROUTE/CONNECT COOLANT HOSES

NOTE: Magnet is delivered to site with four 1/2 inch hoses connected to the Front End Bell. The Twin Gradient Hose Assembly (2295225) is delivered to site and also has four duplicate 1/2" hoses . **Discard the 1/2 inch black (2293610-2) and blue (2293610-3) hoses that are part of 2295225.** Connect the IN and OUT coolant hoses from the Cooling Cabinet to the 1/2 inch hoses attached to the Front End Bell using the Tees and clamps shown below.



A3 – COOLANT VALVE INSTALLATION

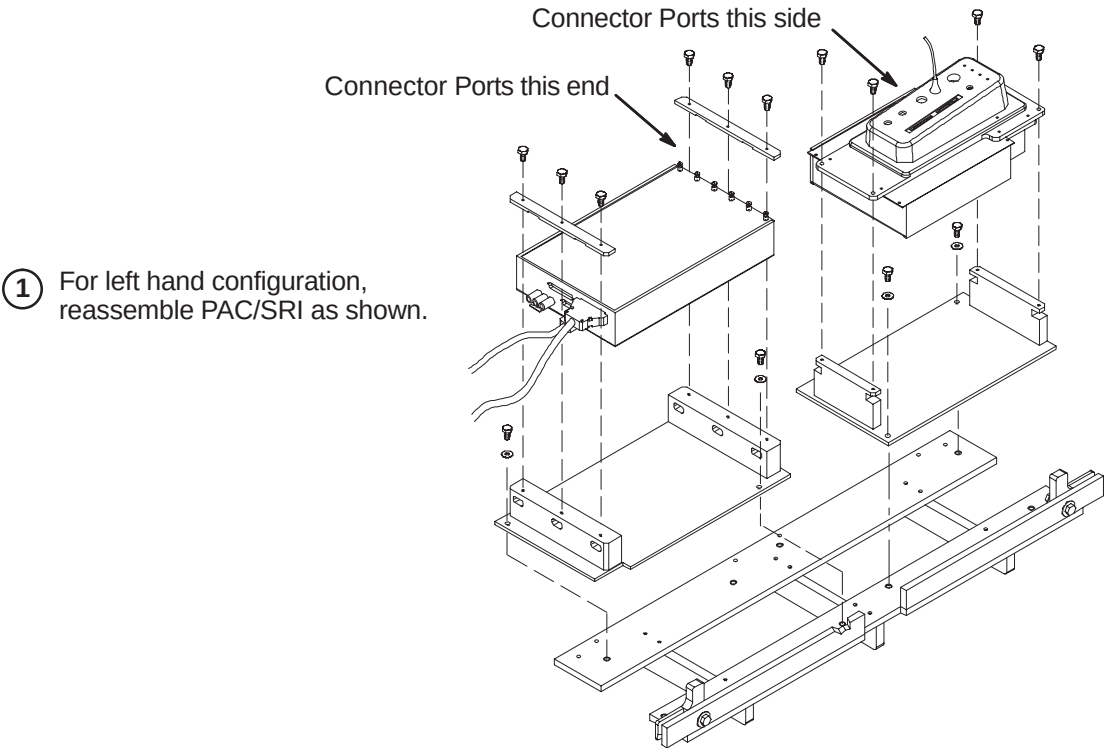


COOLANT LEAK TEST:
For Fixed sites only; refer to applicable vendor supplied Cooling Cabinet Manual. Manual can also be found on the MR Service Methods CD – ROM: Blowers & Chillers/System Operation
If leak is found, determine source of leak and repair it. **DO NOT ADD MORE COOLANT UNTIL ALL LEAKS ARE FOUND AND REPAIRED.**

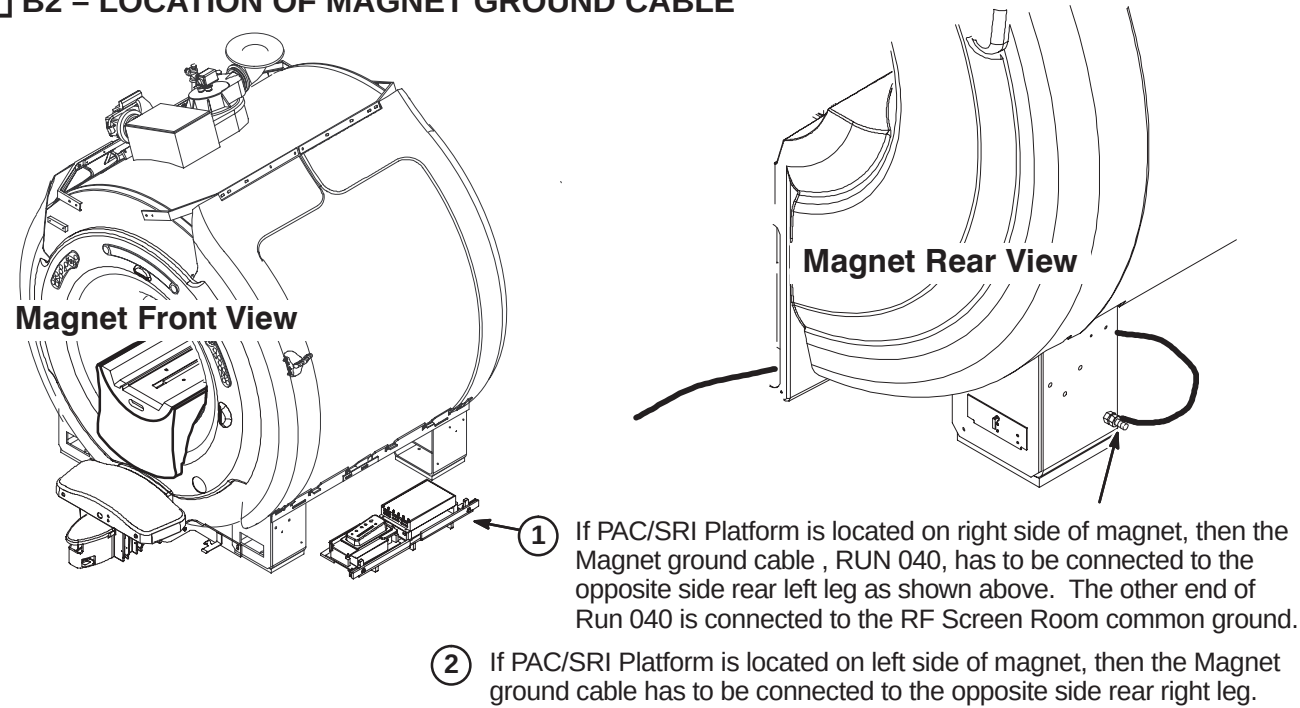
INSTALLATION PROCEDURES SUMMARY

- ☐ B1 – PAC/SRI Platform Left Hand Configuration (If Required)
- ☐ B2 – Location of Magnet Ground Cable
- ☐ B3 – Preparation of Hoses and Cables Under Rear Pedestal
- ☐ B4 – Location of Rear Pedestal

☐ B1 – PAC/SRI PLATFORM LEFT HAND CONFIGURATION (IF REQUIRED)

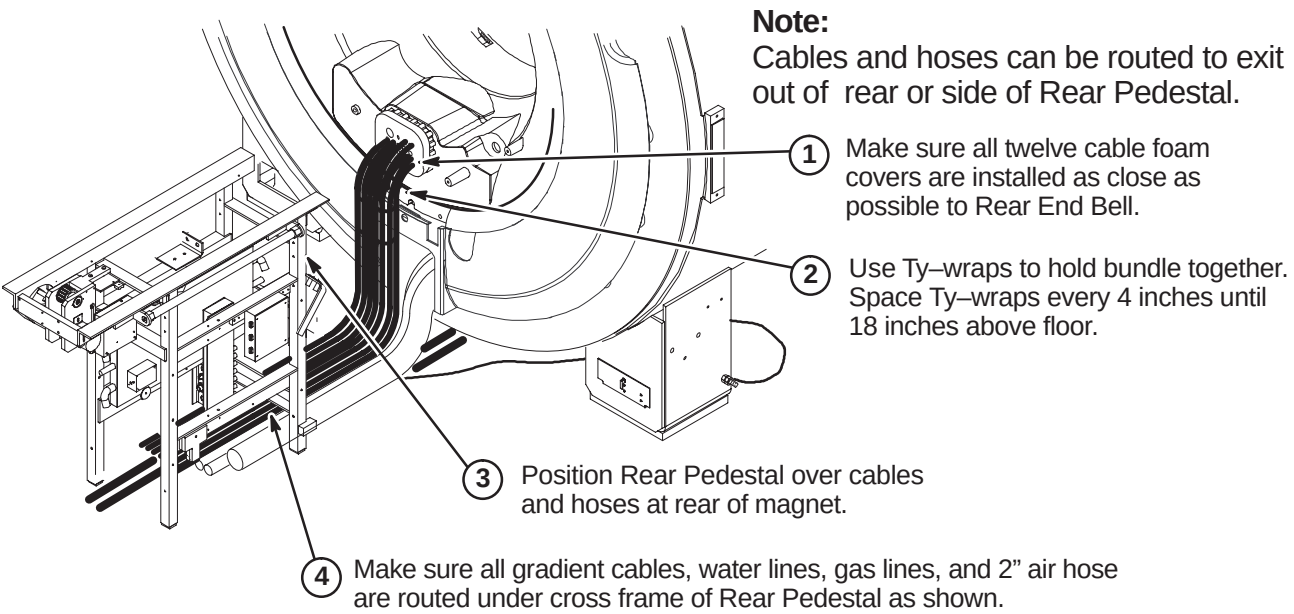


☐ B2 – LOCATION OF MAGNET GROUND CABLE

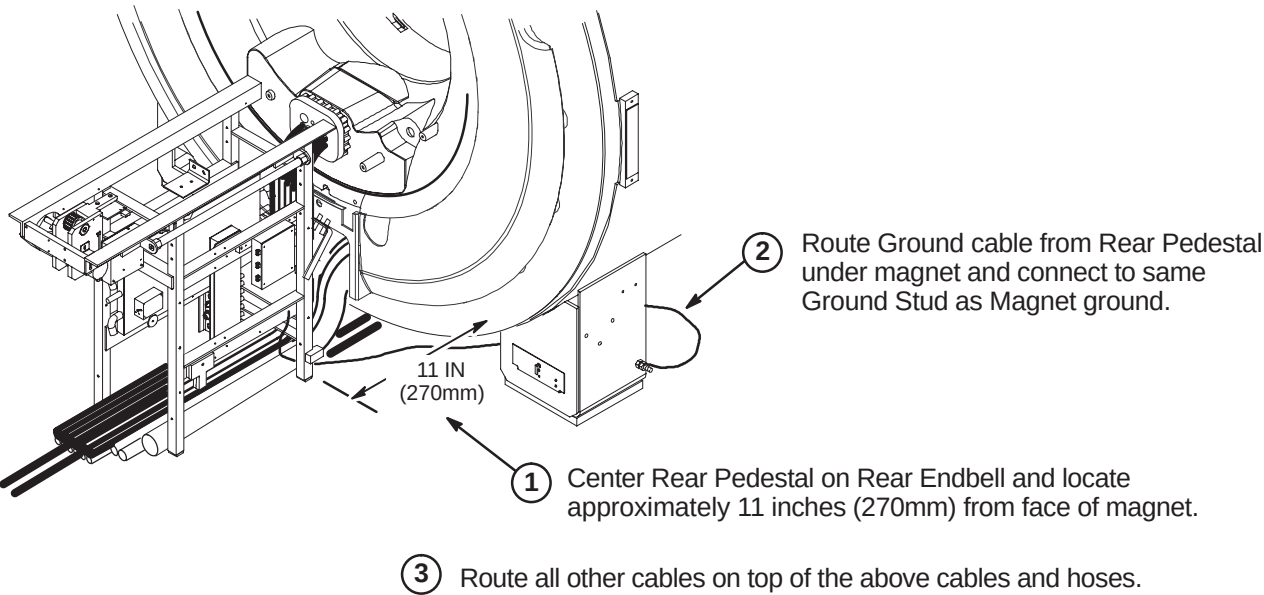


☐ B3 – PREPARATION OF HOSES AND CABLES UNDER REAR PEDESTAL

Note:
The Bridge sections mounted on the Rear Pedestal and in the bore of the magnet are not shown for clarity.



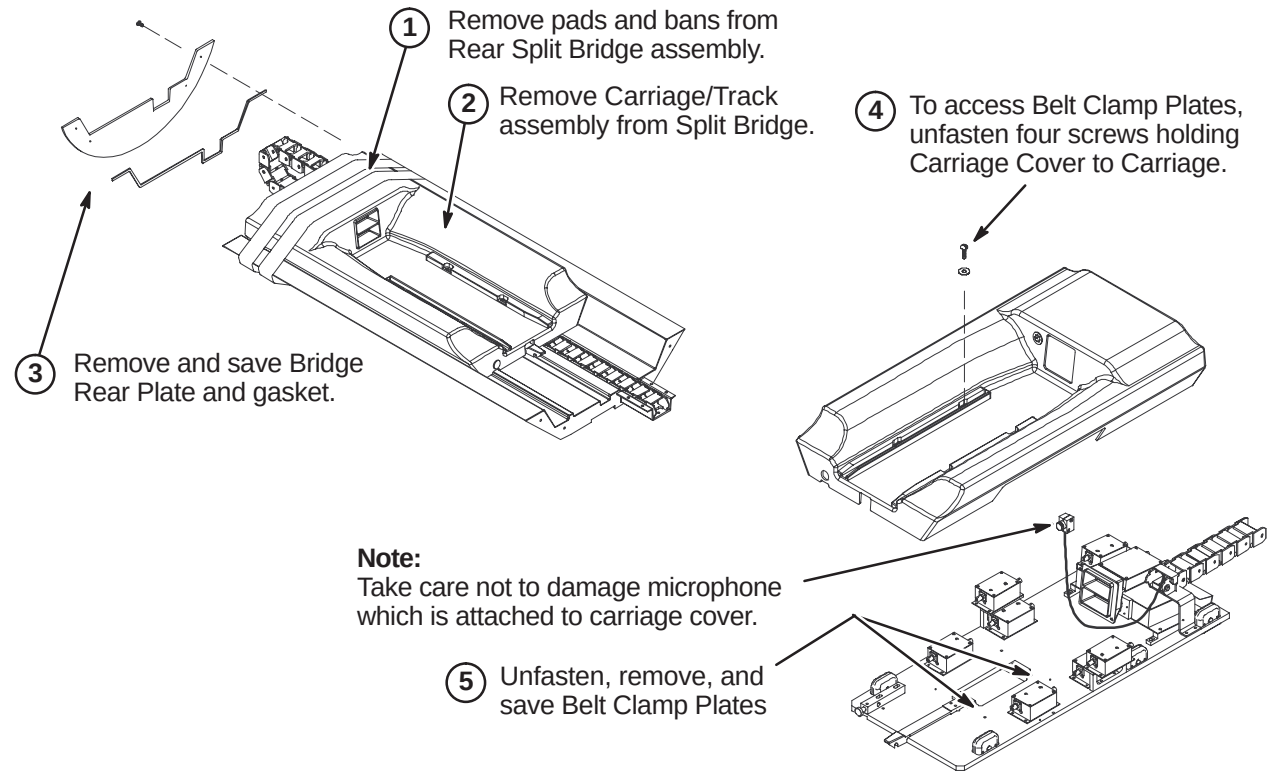
☐ B4 – LOCATION OF REAR PEDESTAL



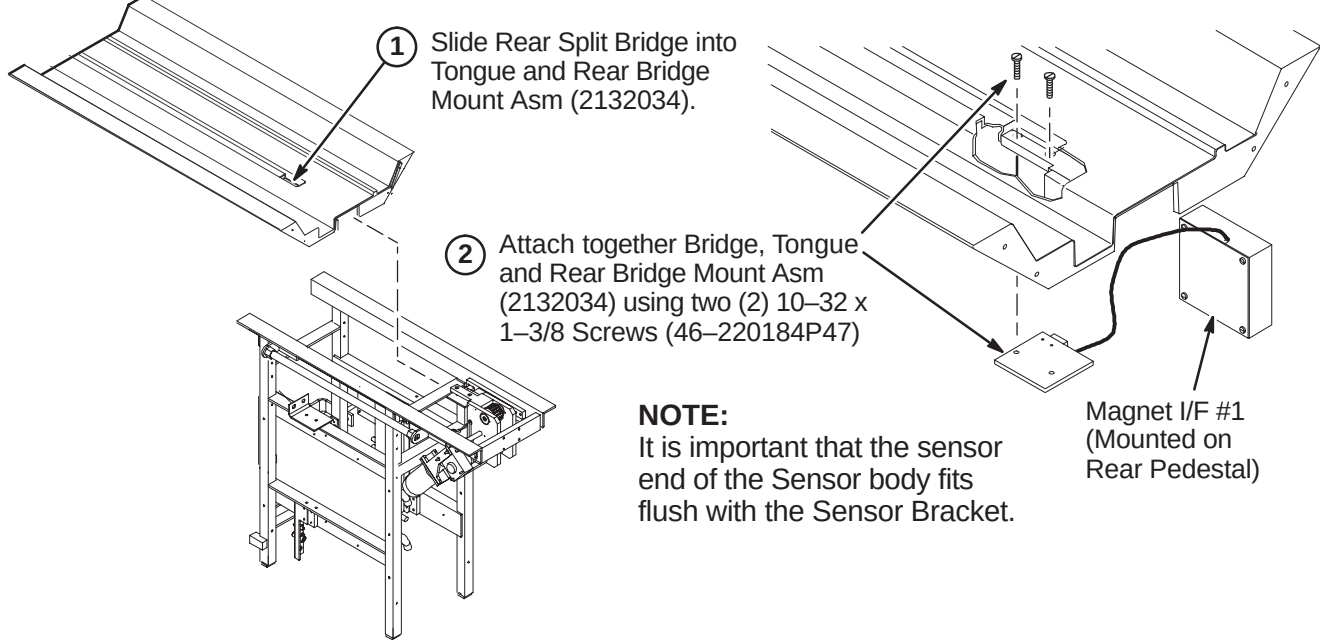
INSTALLATION PROCEDURES SUMMARY (for new style Split Bridge)

- C1 – Preparing Rear Split Bridge for Installation
- C2 – Attaching Rear Split Bridge to Rear Pedestal
- C3 – Attaching Rear Split Bridge to Front Split Bridge
- C4 – Aligning Bridge on Rear Pedestal
- C5 – Installing Bridge Locators

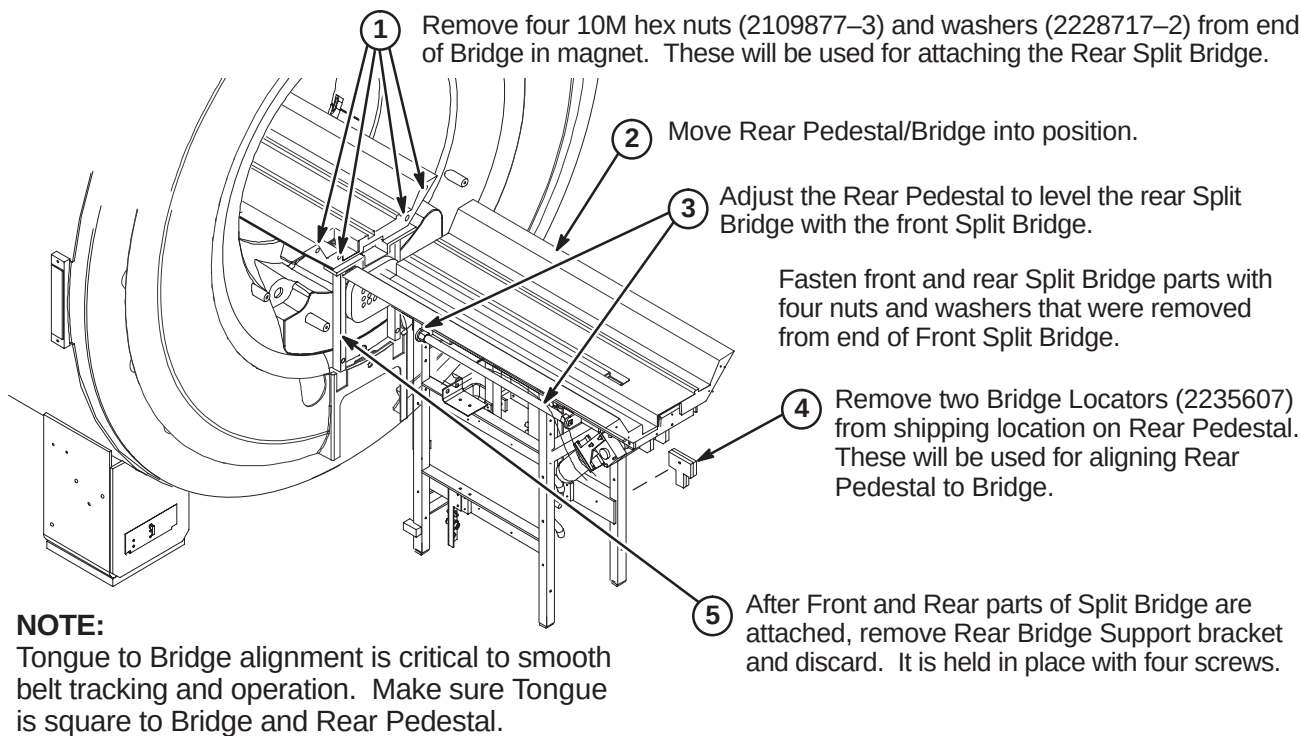
C1 – PREPARING REAR SPLIT BRIDGE FOR INSTALLATION



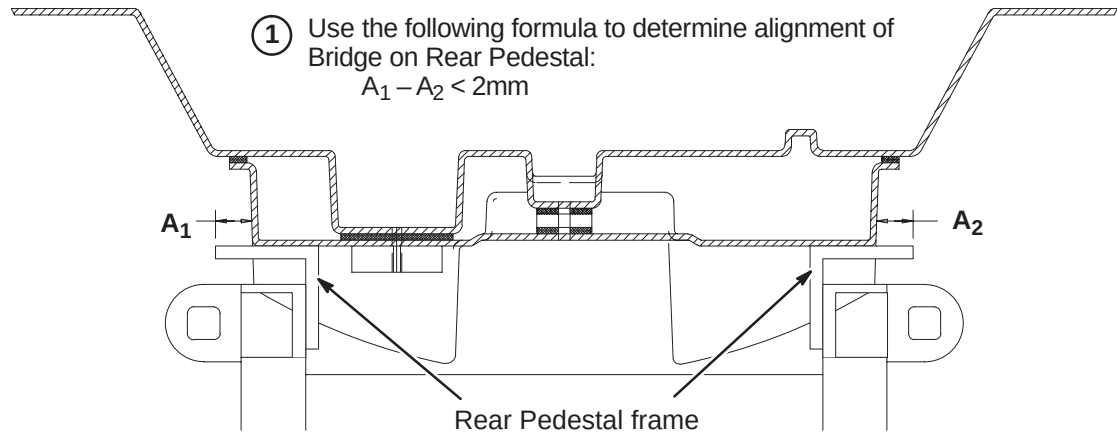
C2 – ATTACHING REAR SPLIT BRIDGE TO REAR PEDESTAL



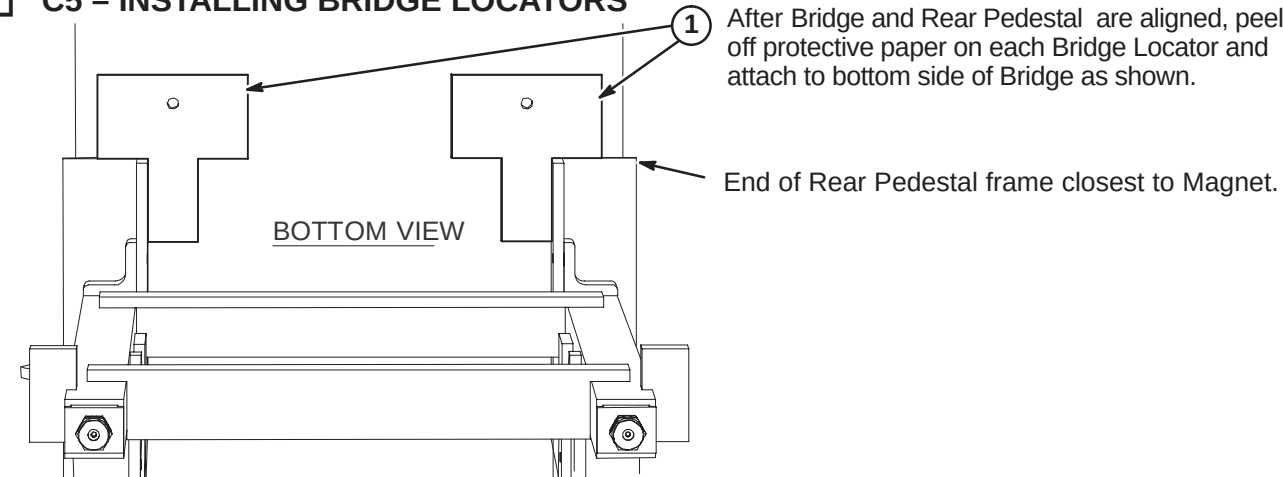
C3 – PREPARATION FOR FRONT AND REAR BRIDGE CONNECTION



C4 – ALIGNING BRIDGE ON REAR PEDESTAL



C5 – INSTALLING BRIDGE LOCATORS

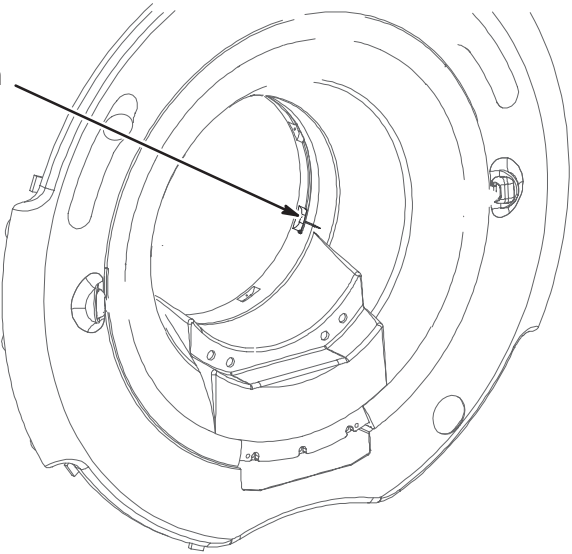


INSTALLATION PROCEDURES SUMMARY

- ❑ D1 – Leveling Front of Bridge
- ❑ D2 – Leveling Bridge
- ❑ D3 – Installing Carriage on Bridge
- ❑ D4 – Routing Drive Belt

❑ D1 – LEVELING FRONT OF BRIDGE

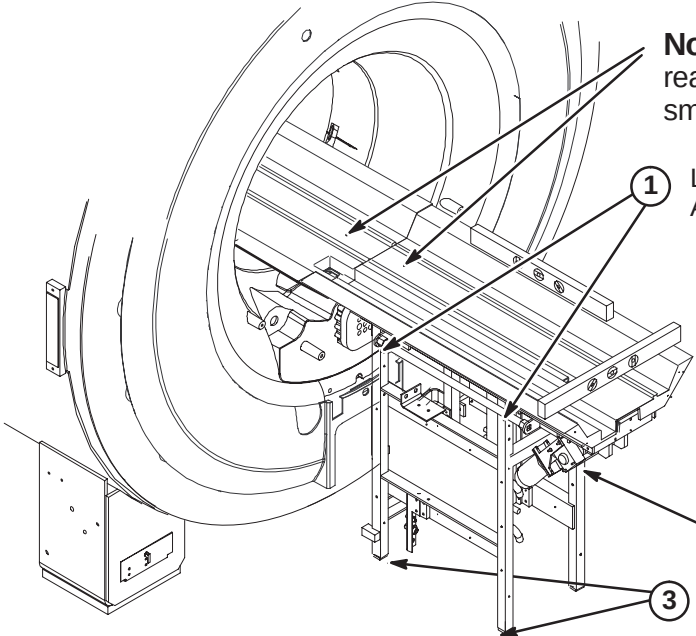
① Top of front end of Bridge should line up with alignment marks on side of Front End Bell



② Put a level across front of Bridge.

③ Level front of Bridge by adjusting right or left slider screws to center the bubble.

❑ D2 – LEVELING BRIDGE



Note: Make sure that surfaces between front and rear Bridge surfaces are even. This will provide smooth travel for the Carriage Cover wheels.

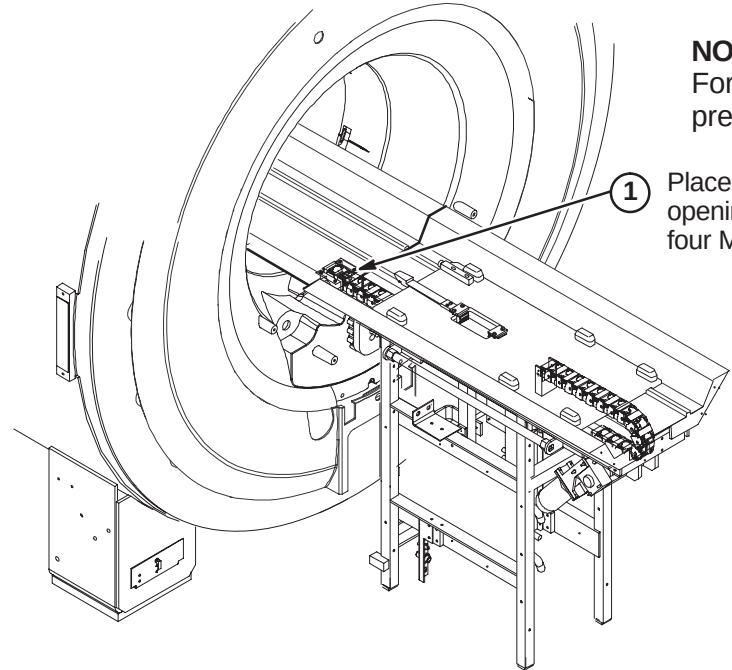
① Level Rear Pedestal Bridge and tighten Vertical Adjustment Fasteners.

NOTE: Rear Pedestal must be adjusted to maintain full contact between bottom of Bridge and top of Rear Pedestal.

② Make sure that Longitudinal Drive Asm is level.

③ Adjust leg screws on both sides as needed to level Rear Pedestal.

❑ D3 – INSTALLING CARRIAGE ON BRIDGE

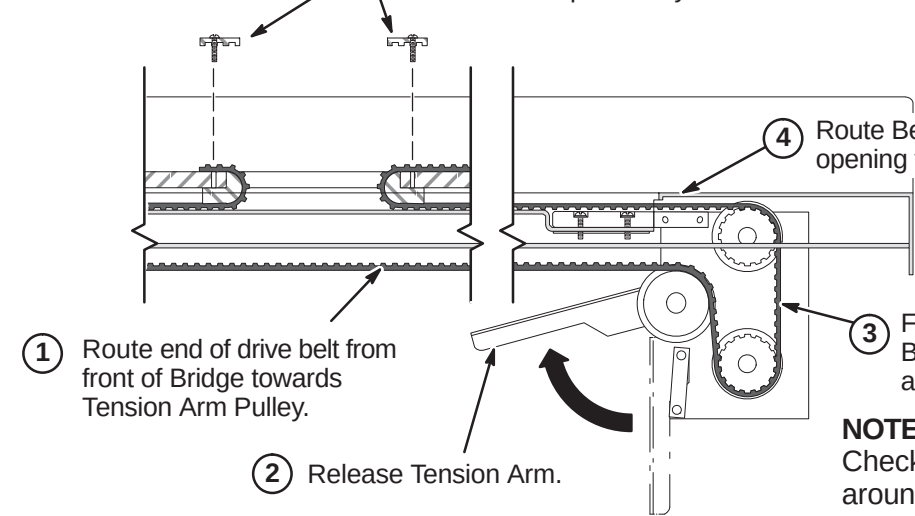


NOTE: For clarity, the carriage is shown without preamps, preamp protect module, and cables.

① Place Carriage on Bridge and route all cables thru opening in Bridge and fasten plate to Front Bridge with four M6 x 12mm nylon screws (2296958) provided.

❑ D4 – ROUTING OF DRIVE BELT

⑤ Position Belt under Clamp Plates and re-attach with previously removed screws.



① Route end of drive belt from front of Bridge towards Tension Arm Pulley.

② Release Tension Arm.

④ Route Belt under and through opening to top of Carriage.

③ Feed back end of Belt underneath Bridge, around Tension Arm Pulley and rear pulleys into Bridge.

NOTE: Check to verify that belt is not caught around spacers near rear pulley.

INSTALLATION PROCEDURES SUMMARY (for new style Split Bridge)

- ☐ M1 – Final Adjustment of Drive Belt
- ☐ M2 – Installation of Sensor Flag
- ☐ M3A – Route Non-MGD Runs 404 and 485 (and Multi-coil Cables if Required)
OR
- ☐ M3B – Route/Connect MGD Cables from LPCA to Rear Pedestal
- ☐ M4 – Installing Carriage Cover to Trolley and Bridge Cover

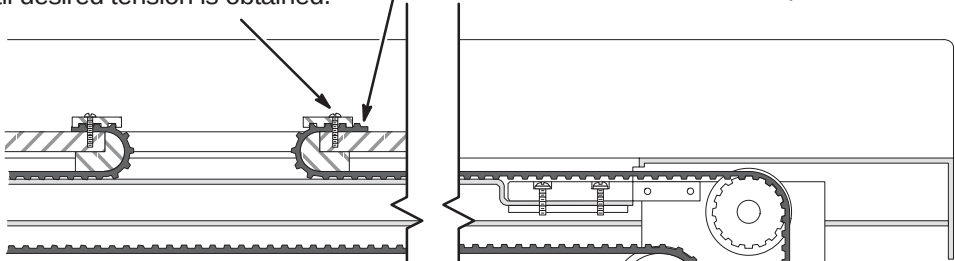
E1 – FINAL ADJUSTMENT OF DRIVE BELT

2 (IF NECESSARY):

Loosen Clamp and adjust position of Belt one tooth at a time until desired tension is obtained.

3 Tighten Clamps and trim excess Belt length.

4 Fasten Carriage Cover to Carriage with four screws removed in procedure E1.



1 Check Belt tension by closing Tension Arm. (See Note)

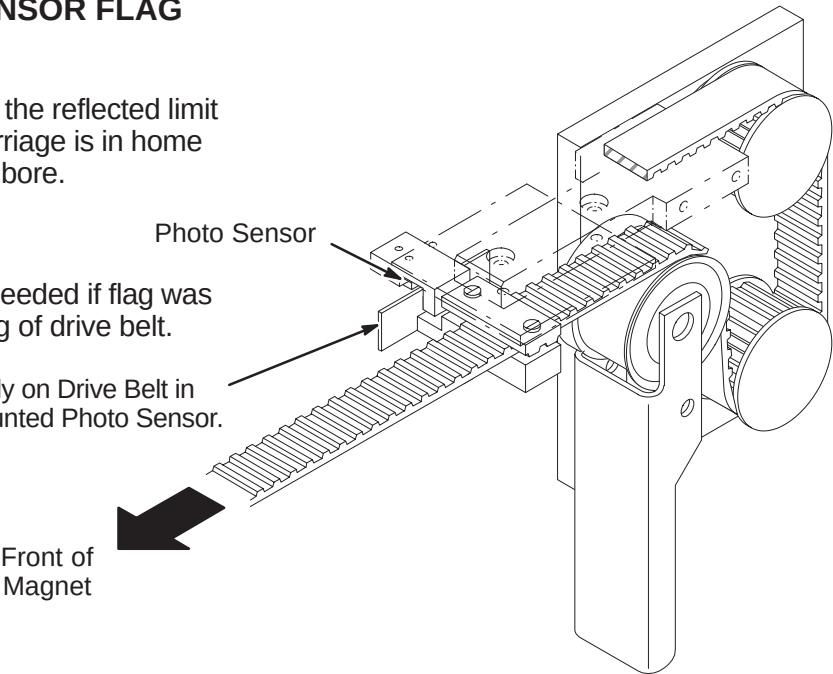
NOTE:
Tension Arm should lock crisply into position without excessive force (which could cause later problems). If adjustment is necessary go to step 2. If not, go to step 3.

E2 – INSTALLATION OF SENSOR FLAG

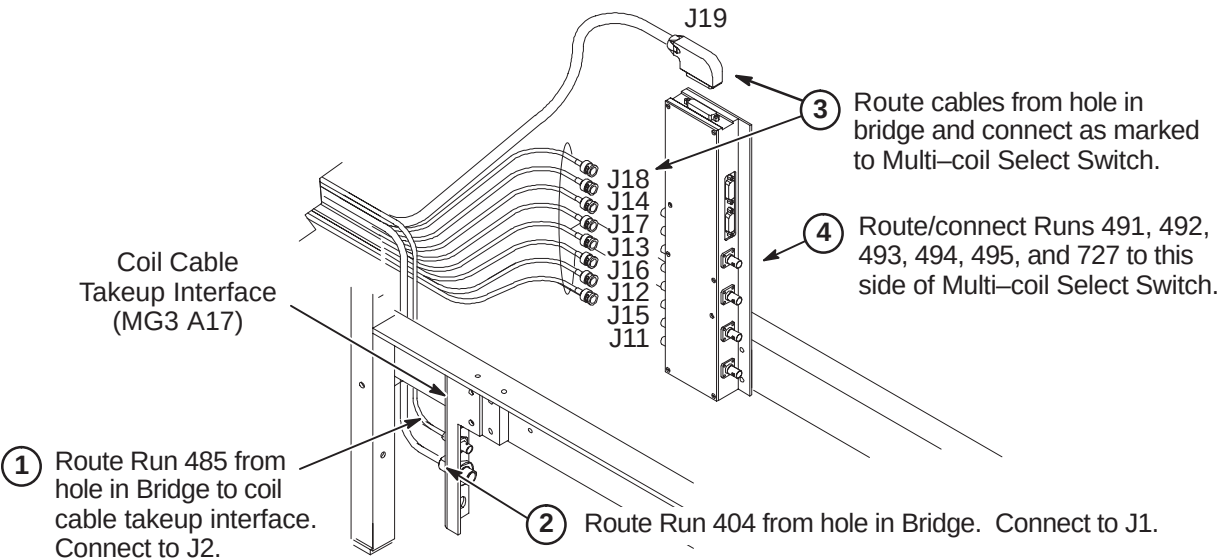
NOTE:
Function of the Flag is to interrupt the reflected limit switch sensor beam when the Carriage is in home position all the way forward in the bore.

NOTE:
The following Step is needed if flag was removed during routing of drive belt.

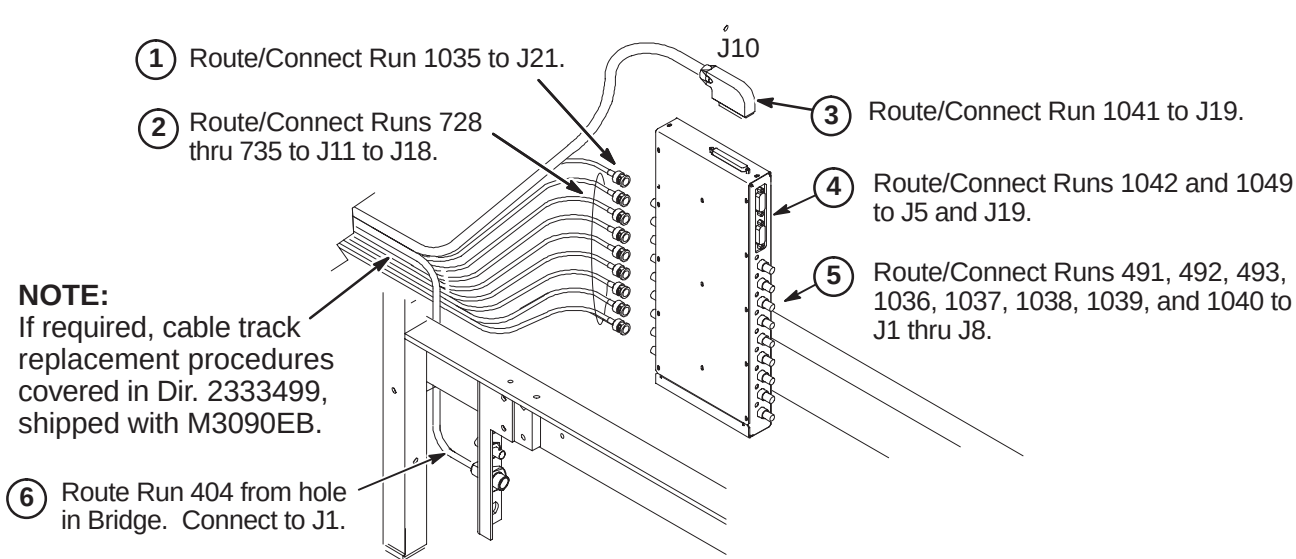
1 Install Flag Assembly on Drive Belt in line with Bridge mounted Photo Sensor.



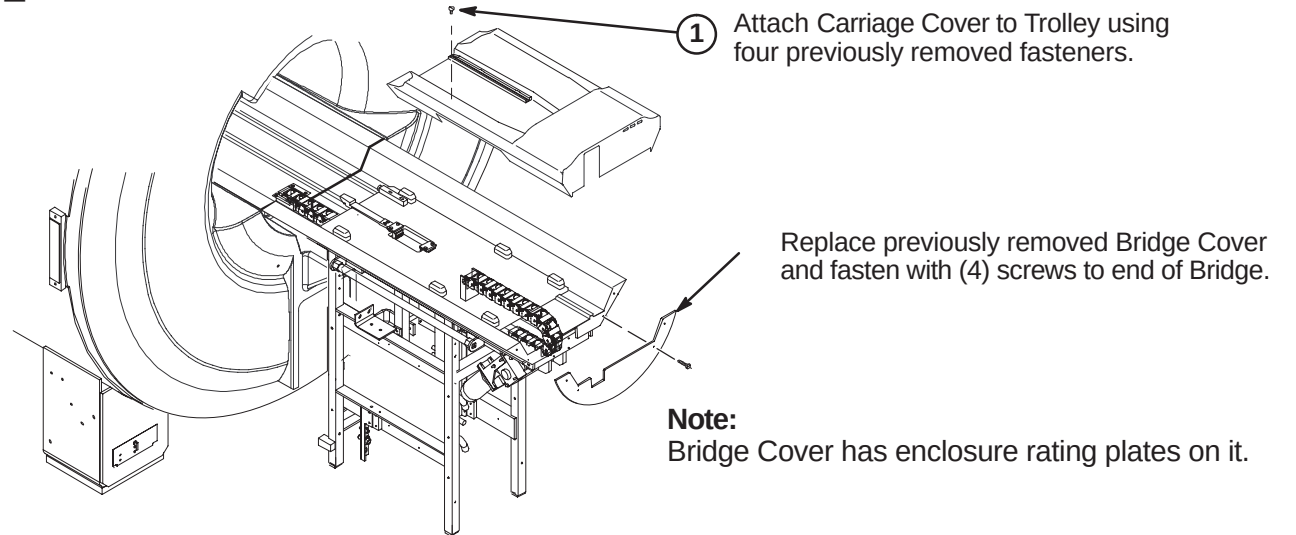
E3A – ROUTE NON-MGD RUNS 404, 485, AND MULTI-COIL CABLES



E3B – ROUTE/CONNECT MGD CABLES FROM LPCA TO REAR PEDESTAL



E4 – INSTALLING CARRIAGE COVER TO TROLLEY AND BRIDGE COVER

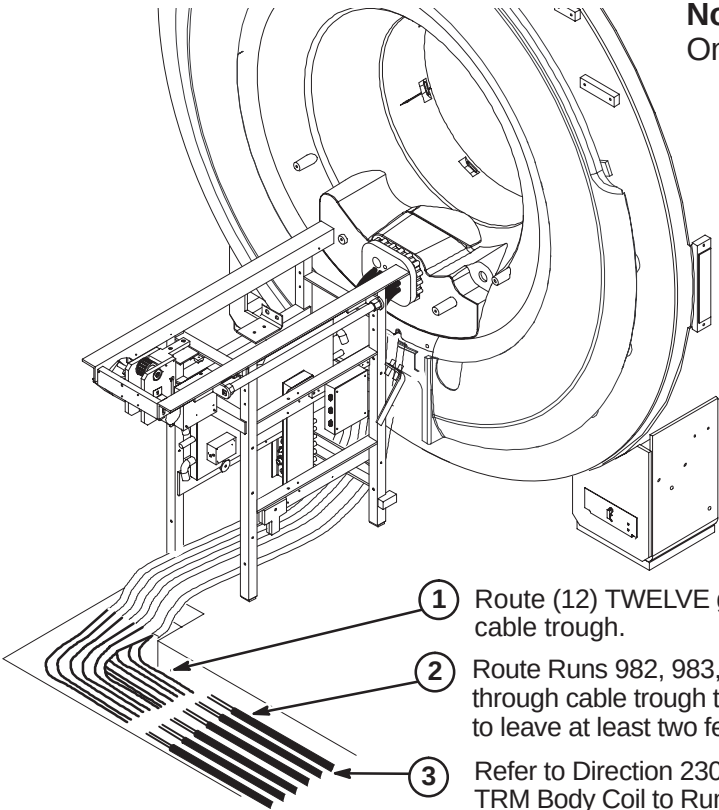


INSTALLATION PROCEDURES SUMMARY

- E1 – Routing and Connecting Gradient Output Cables
- E2 – Route/Connect Runs 282, 414, 422, 781, 782, & Front End Bell Speaker Cable
- E3 – Route/Connect Runs 257, 262, 743, 744, and 780
- E4 – Route/Connect Runs 258, 318, 700, and Drive Cables

E1 – ROUTING AND CONNECTING GRADIENT OUTPUT CABLES

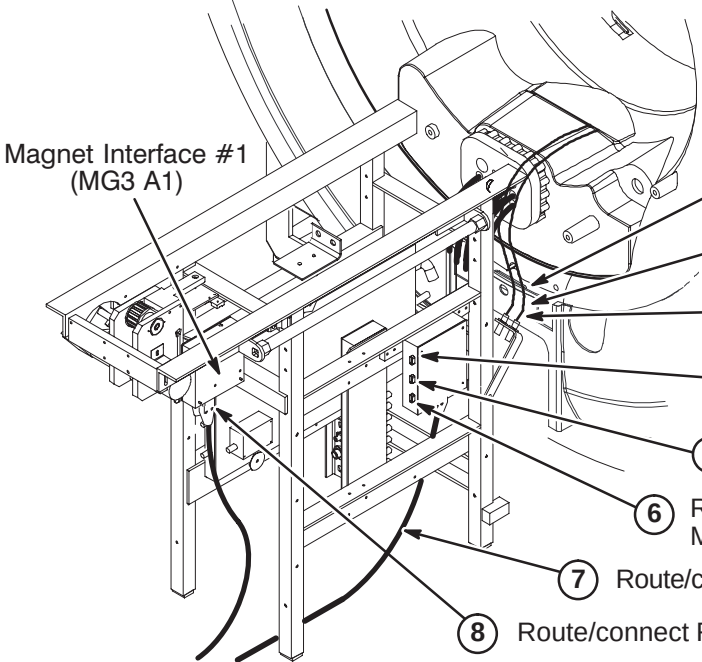
Note:
Only Gradient Cables are shown for clarity.



TRM COIL CABLE LABELS		OUTPUT CABLE LABELS		
WIRE #	LABEL	RUN #	LABEL	WIRE #
1	Z ZM–	982	ZM–	2
2	Z ZM+	982	ZM+	1
3	X WB–	987	WB–	2
4	X WB+	987	WB+	1
5	Y WB+	986	WB+	1
6	Y ZM–	983	ZM–	2
7	Y WB–	986	WB–	2
8	Y ZM+	983	ZM+	1
9	Z WB+	985	WB+	1
10	X ZM–	984	ZM–	2
11	Z WB–	985	WB–	2
12	X ZM+	984	ZM+	1

- 1 Route (12) TWELVE gradient Output Cables from TRM Body Coil to cable trough.
- 2 Route Runs 982, 983, 984, 985, 986, and 987 from Penetration Panel through cable trough to ends of Body Coil gradient cables. Make sure to leave at least two feet (610mm) of slack before cutting to length.
- 3 Refer to Direction 2308287 in Tab 1 for connecting Output cables from TRM Body Coil to Runs 982, 983, 984, 985, 986, and 987.

E2 – ROUTE/CONNECT RUNS 282, 414, 422, 781, 782, & FRONT END BELL CABLE



- 1 Route and connect RF Bias cables from RF Coil to bracket as marked.
- 2 Route and connect Run 781 from Bias cable J5 to J75 on the Penetration Panel.
- 3 Route and connect Run 782 from Bias cable J6 to J76 on the Penetration Panel.
- 4 Route/connect microphone cable from Carriage Cover to J7.
- 5 Route/connect Front End Bell microphone cable Run 414 to J9.
- 6 Route/connect speaker cable from Patient Comfort Module to J8.
- 7 Route/connect Run 282 from J12 on Pen Panel to J6.
- 8 Route/connect Run 422 from J2 on SRI to J4 on Magnet Interface #1.

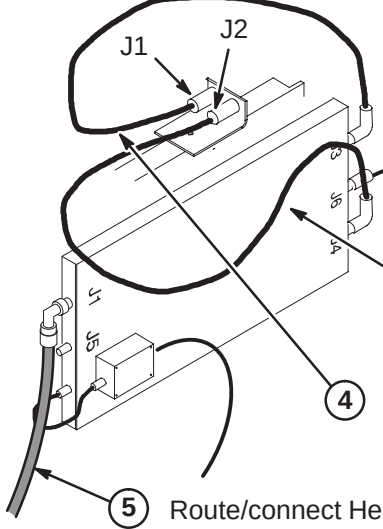
E3 – ROUTE/CONNECT RUNS 257, 262, 743, 744, AND 780

CAUTION

The RF Drive Input cables, Runs 743 and 744, utilize a corrugated solid RF shield. If mishandled, the cable can be kinked and subsequently broken.

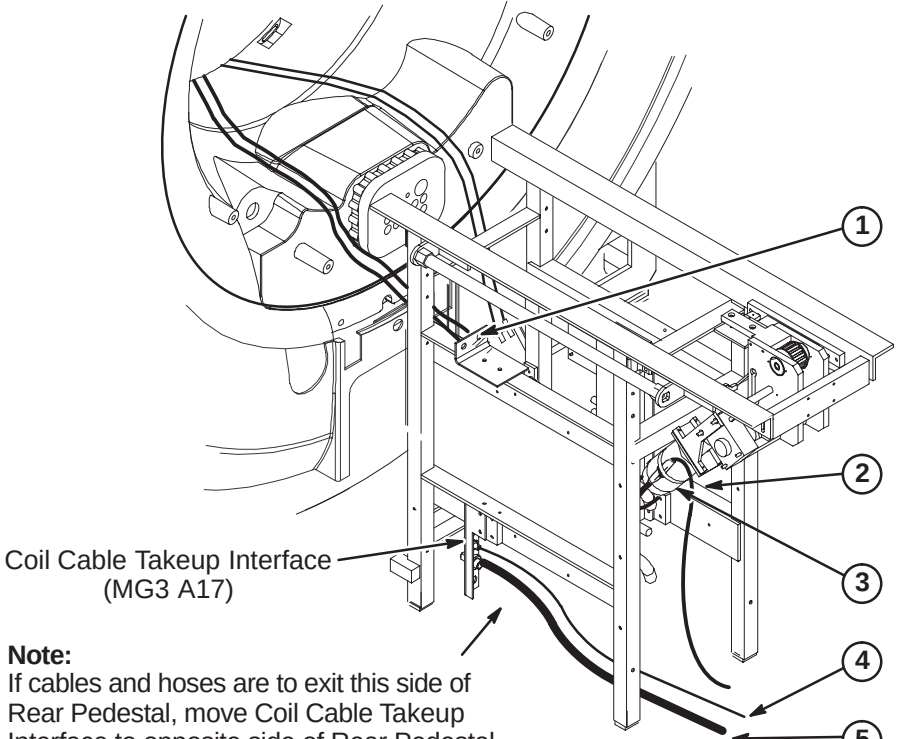
Also, use care when connecting these cable to the bulkhead connectors J1 and J2 and RF Splitter connectors J3 and J4. Cross threading or inadequate engagement of the center pin can cause arcing and damage the cables.

Note: Runs 743 and 744, Body Coil to RF Splitter cable connections may be reversed depending on magnet field polarity. Consult local Field Engineer for proper routing.



- 1 Route/connect Run 780 from J74 on Pen Panel to J6 on RF Splitter.
- 2 Route and connect Run 744 from J1 on bracket to J4 on the RF Splitter.
- 3 Route and connect Run 743 from J2 on the bracket to J3 on the RF Splitter.
- 4 Route/connect Run 262 from J8 on Pen Panel to Body Coil Pre-amplifier.
- 5 Route/connect Heliax® Run 257 to J1 on RF Splitter.

E4 – ROUTE/CONNECT RUNS 258, 318, 700, AND DRIVE CABLES



- 1 Route I & Q RF Drive Input cables from RF Coil and connect as marked to J1 and J2 on bracket.
- 2 Route/connect Run 318 to short cable on Longitudinal Motor Drive (MG3 A7).
- 3 Ty-wrap cable to motor to prevent movement.
- 4 Route/connect Run 700 to J2 on Coil Cable Takeup Interface.
- 5 Route/connect Run 258 to J1 on Coil Cable Takeup Interface.

Note:
If cables and hoses are to exit this side of Rear Pedestal, move Coil Cable Takeup Interface to opposite side of Rear Pedestal.
If Interface bracket is moved to other side, connectors on bracket need to be removed, turned end for end, and reinstalled.

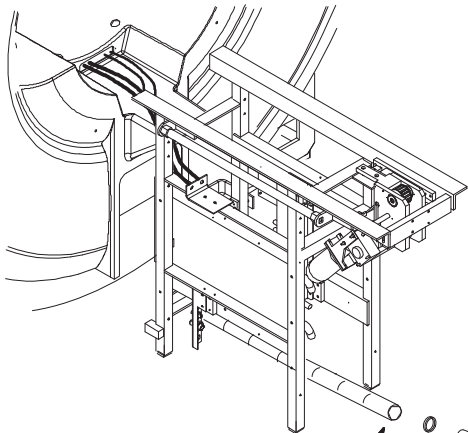
INSTALLATION PROCEDURES SUMMARY

- F1 – Install Blower Box and Route/Connect Air Hose
- F2 – Install Bore Lamp Assembly
- F3 – Route/Connect Vacuum Hose and Elbow

F1 – INSTALL BLOWER BOX AND ROUTE/CONNECT AIR HOSE

WARNING!

RF SHIELD INTEGRITY MUST BE MAINTAINED FOR MOUNTING BLOWER BOX WITHIN THE MAGNET ROOM



- Refer to site design (architectural)/customer construction drawings or contact PMI for locating and securing Blower Box (2121199).
- Check for correct installation of the anchors:
 - Confirm that the anchors are physically secured to the building structure, not to a removable computer floor, raised floor, or false floor.
 - Confirm there are at least four (4) anchors.
 - If any or the anchors appears wobbly or loose, stop the installation and contact the PMI to fix anchoring.
- Remove any mounting hardware from the anchors. Hoses should face the back of the room
- Position the blower box on top of 4 or more anchors. Hoses should face the back of the room.
- Secure the blower box using the hardware provided by the RF shield room vendor to attach.

Note:
The following items are part of Blower Hose Kit, 2129412–5.

- Connect 2 inch air hose from Rear End Bell to Reducer (2175143) with 2 inch clamp (46–208765P28). Do not cut or reduce the length of the 2 inch hose.
- Connect 4 inch air hose (46–282666P11) to reducer with 4 inch clamp (46–208765P34). If necessary, the 4 inch hose may need to be cut to reduce length.
- Route 4 inch hose to Blower Cabinet. Trim to appropriate length and attach to vent opening with 4 inch Hose Clamp (46–208765P34).

CAUTION

THE BLOWER BOX CONTAINS FERROUS MATERIAL. The proper mounting of the Blower Box is critical to safety. It MUST be mounted per these instructions

IMPORTANT!

Location of cabinet and type of fastener used for securing Blower Box outside of minimum service area should be specified in the Site Design (architectural/Construction) drawings. If proper mounting area is not specified, contact the Project Manager of Installation (PMI).

NOTE: If needed, Blower Hose Cover can be cut and modified.

F2 – INSTALL BORE LAMP ASSEMBLY

Note: Cables and hoses routed in Procedure below are not shown for clarity.

-
- Move Bore light Assembly (2222547) to area of left rear magnet leg.
 - Connect the (+) and (–) wires of Run 300 to the terminal strip as shown.
 - Route Run 300 thru opening in rear leg, under Magnet and Rear Pedestal and connect to J15 on Penetration Panel.
 - Route fiber bundle from Rear End Bell thru leg opening and insert into bushing. Fasten to bushing with set screws.
 - Insert into clip on Bore light Assembly. Bushing flanges to be inside clip.
 - Peel off protective paper on velcro and install assembly in opening in rear leg.

F3 – ROUTE/CONNECT VACUUM HOSE AND ELBOW

Note:
Vacuum Grease **must** be applied to all Centering Rings (2293608) and outside edge of vacuum connector flanges.



- Install Elbow (2293609) to End Bell using centering ring (2293608) and chain clamp (2293618–2). Tighten chain clamp to “finger tight”.
- Route vacuum hose (2293607) from Elbow on End Bell, thru Penetration Panel, to connection at rear of TAC Cabinet in Equipment Room. Cut to length.
- At each end of vacuum hose insert Hose Connector (2293616), apply vacuum grease to outside edge of connector and fasten with two (2) Clamps (2294010–3).
- Apply vacuum grease to outside edge of Hose Connector flanges and respective connector flanges on TAC and End Bell.
- Connect each end of Vacuum Hose using centering ring (2293608) and chain clamp (2293618–2). Tighten chain clamp to “finger tight”.

INSTALLATION PROCEDURES SUMMARY

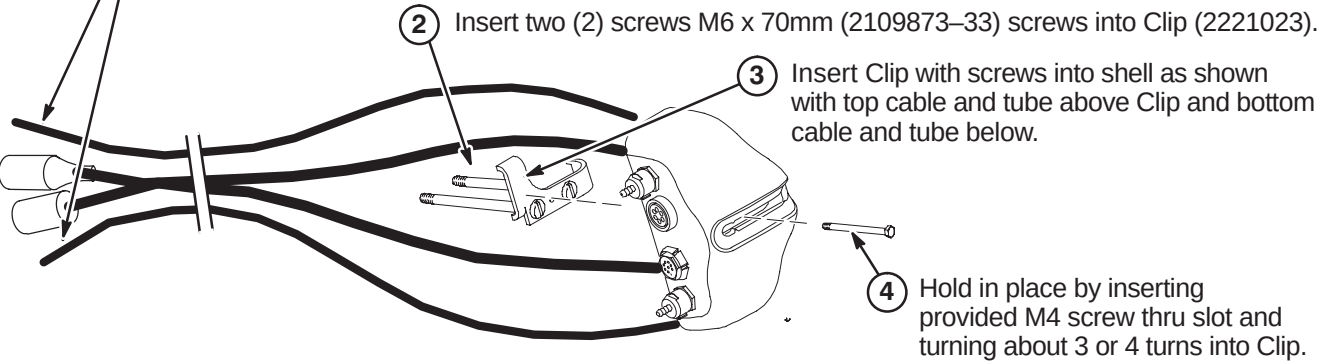
- H1 – Prepare Cables and Tubes for Routing
- H2 – Position Remote PAC Interface Mounting Clip
- H3 – Route Remote PAC Interface Cables and Tubes
- H4 – Installing Remote PAC Interface to Arc Cover
- H5 – Install PAC Hanger Bar

□ H1 – PREPARE CABLES AND TUBES FOR ROUTING

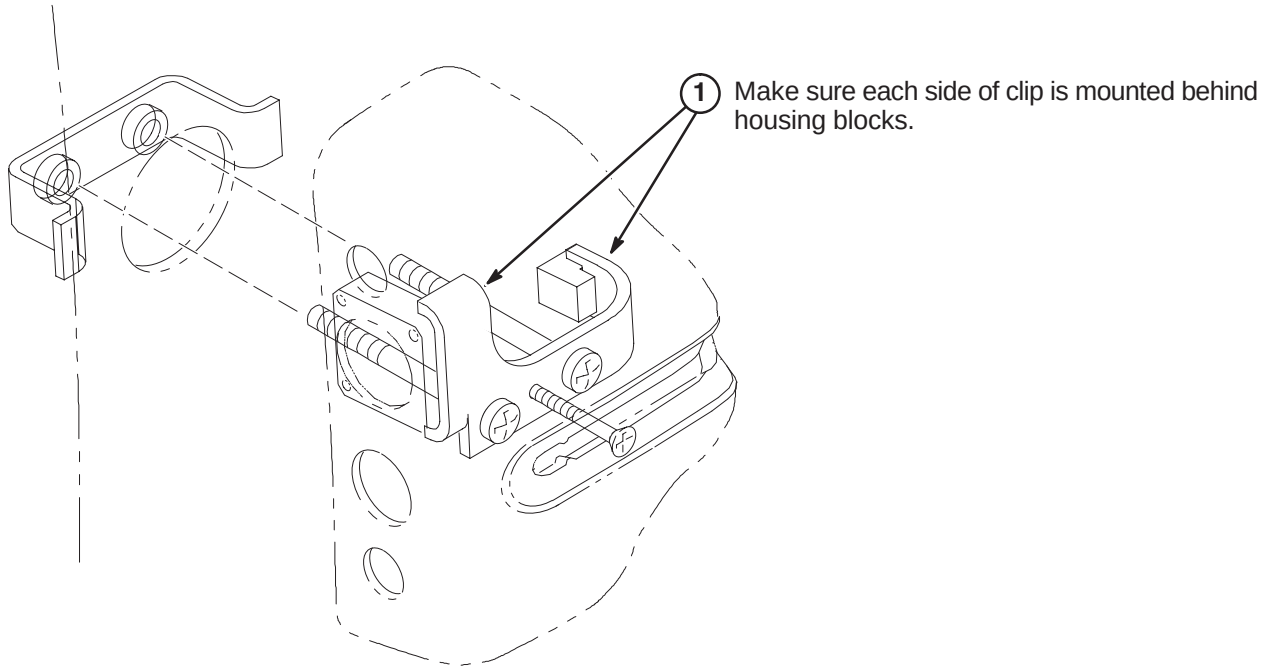
Note:
Consult with customer for correct location of Remote PAC Interface assembly. It must be on the same side as the PAC/SRI Platform, as positioned in Procedure **A1**

- ① Apply tape around end of pneumatic tubes and mark “TOP” to the opposite end of the tube connected to the top connector in the Remote PAC Interface Assembly and “BOTTOM” to the tube connected to the lower connector.

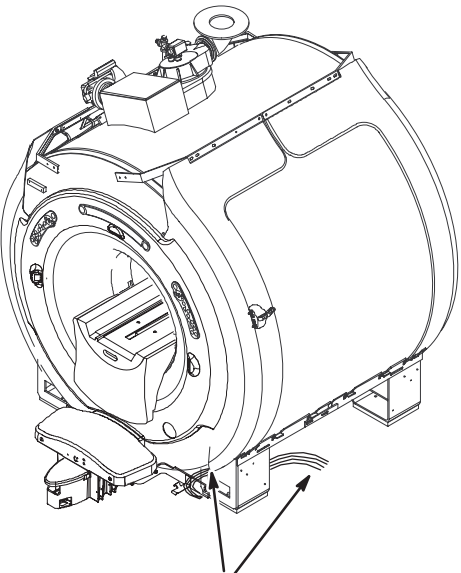
Note: Top two cables will be routed over top of bracket and bottom two cables will be routed under bracket.



□ H2 – POSITION REMOTE PAC INTERFACE MOUNTING CLIP

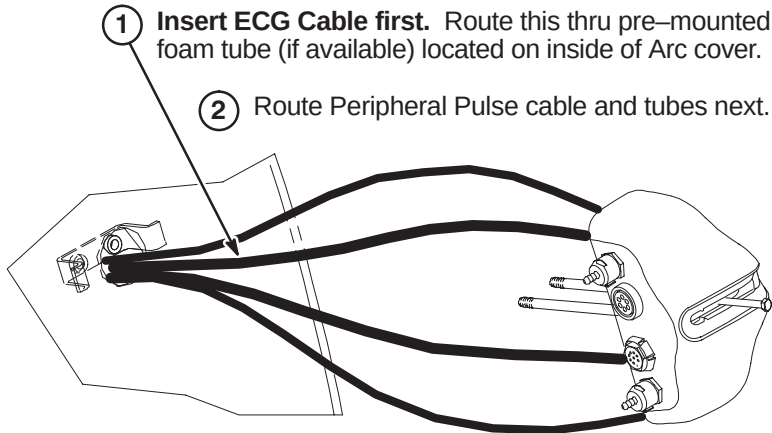


□ H3 – ROUTE REMOTE PAC INTERFACE CABLES AND TUBES



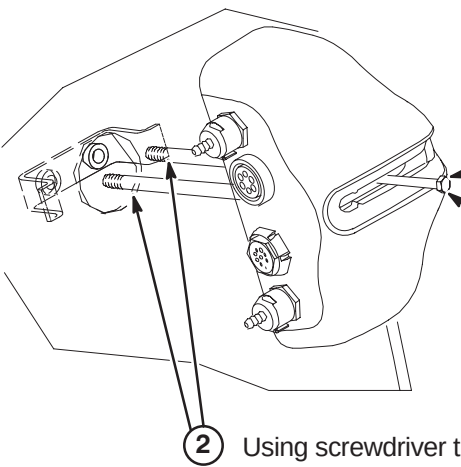
Note: Top two cables will be routed over top of bracket and bottom two cables will be routed under bracket.

Note: Insert cables/tubes separately thru hole and make sure they come out at bottom of arc cover.



- ③ Route around foot of magnet to PAC/SRI Platform location.

□ H4 – INSTALLING REMOTE PAC INTERFACE TO ARC COVER



Note:
Routed cables and tubes not shown for clarity.

① Use M4 screw thru slot to lift and guide small end of Clamp over block inside housing.

③ Remove M4 screw and save for use at later time since it would be required if Remote PAC Interface Assembly would need to be removed at any time in the future.

OR
Turn the M4 screw inward until there is clearance for sliding the PAC Hanger Bar in **T5** into place.

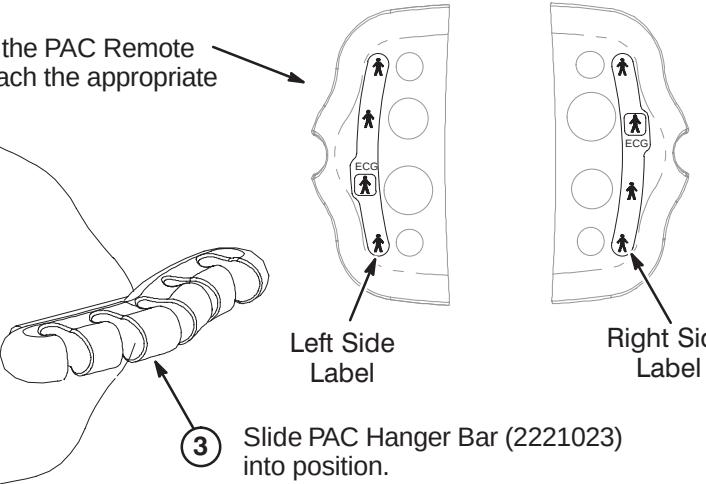
- ② Using screwdriver thru housing slot, tighten down the M6 screws to lock housing in place.

□ H5 – INSTALL PAC HANGER BAR

- ① Depending on left or right side installation of the PAC Remote Interface Asm on the magnet enclosure, attach the appropriate label that is provided with the kit.

- ② Attach cables as indicated below.

Respiration
ECG
Peripheral Gating
Patient Call



INSTALLATION PROCEDURES SUMMARY

- J1 – Dock Adjustment for Patient Transport Gap
- J2 – Route and Connect Magnet Enclosure Front Area Cables
- J3 – Cable Connection to PAC Module

J1 – DOCK ADJUSTMENT FOR PATIENT TRANSPORT GAP

Note:
Gap between Patient Transport front rubber inlay and front of End Bell must be 0.75 Inches $\pm$ 0.25 (19.0 $\pm$ 6.4mm). See Step 1 below.

WARNING!

DOCK CONTAINS FERROUS MATERIAL. HANDLE WITH CARE DURING INSTALLATION.

1 Start with two (2) 11mm washers on each side between Dock and Dock Support. If adjustment is required, add or remove washers until proper gap is achieved.

Note:
Floor stud installed by RF room vendor/mechanical contractor.

J2 – ROUTE AND CONNECT MAGNET ENCLOSURE FRONT AREA CABLES

Note:
Dock Assembly not shown for clarity.

CAUTION
No metal to metal contact is allowed for connectors to other areas. i.e. connector touching the magnet.

1 Route Runs 778 and 779 to front RF Bias cables J3 & J4 and connect as marked.

2 Route Run 749, Bore Temperature Sensor, from Front End Bell and connect at J10 on SRI.

3 Route Front End Bell Microphone and Speaker cables underneath magnet to ReaPedestal and connect to Patient Communication Box as marked.

4 Connect Run 387 to Dock Cable (MG2 A29 P1). If needed to prevent contact with other metal parts, cover connectors with provided velcro jacket or tape.

5 Connect Dock Cable to J2 on Dock Limit Box

6 Route/Connect Run 421 from SRI J3 and connect to J1 on Dock Limit Box.

7 Route cables from Operator Display to area of PAC/SRI Platform and connect as marked to SRI and PAC.

J3 – CABLE CONNECTION TO PAC MODULE

Note:
Refer to Tab 1, System, for instructions on routing and installing fiber optic cables.

1 Connect from the Remote PAC Interface to the PAC Module the following below:

2 Remove shield cover by removing one screw and unscrew shield from module.

Note: Loosen screws and slide module cover open if access to PAC-II board is required to determine correct fiber optic connection.

3 Route Run 712 fiber optic cables through shield into PAC module and connect to J9 and J10 on circuit board.

4 Screw shield in place by hand. Shield can be rotated slightly to ease fiber optic connection. Reinstall shield cover.

5 Connect Run 716

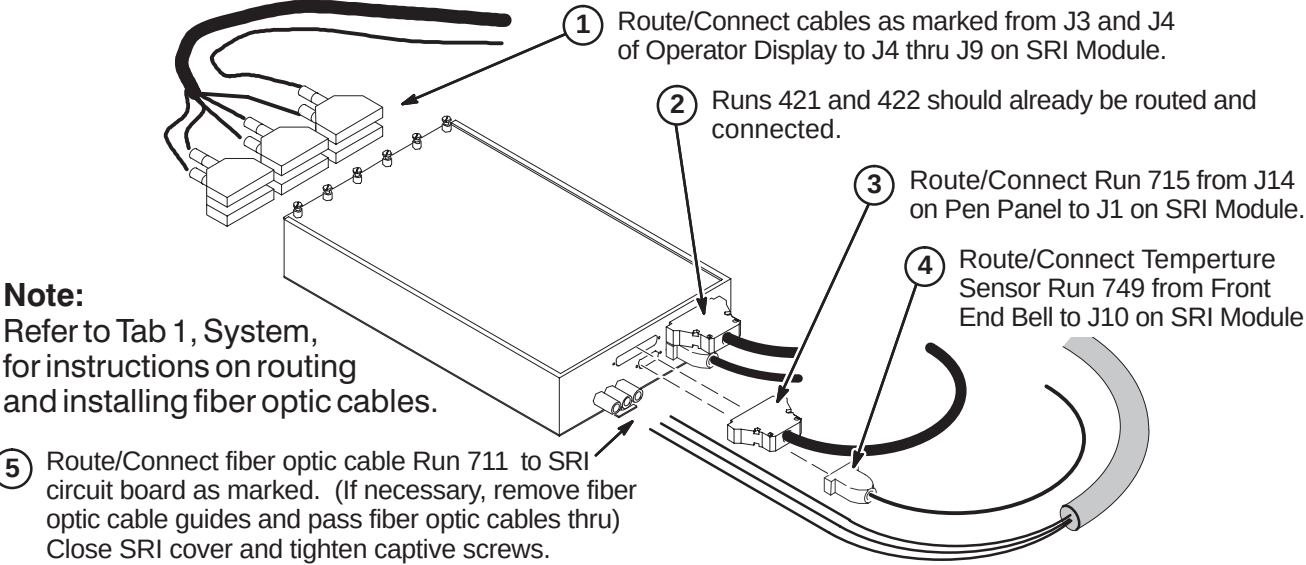
6 Connect Pneumatic Tubing from Pen Panel to tubing extension on PAC Module.

7 Connect cable from J5 on Operator Display to J4 on PAC Module.

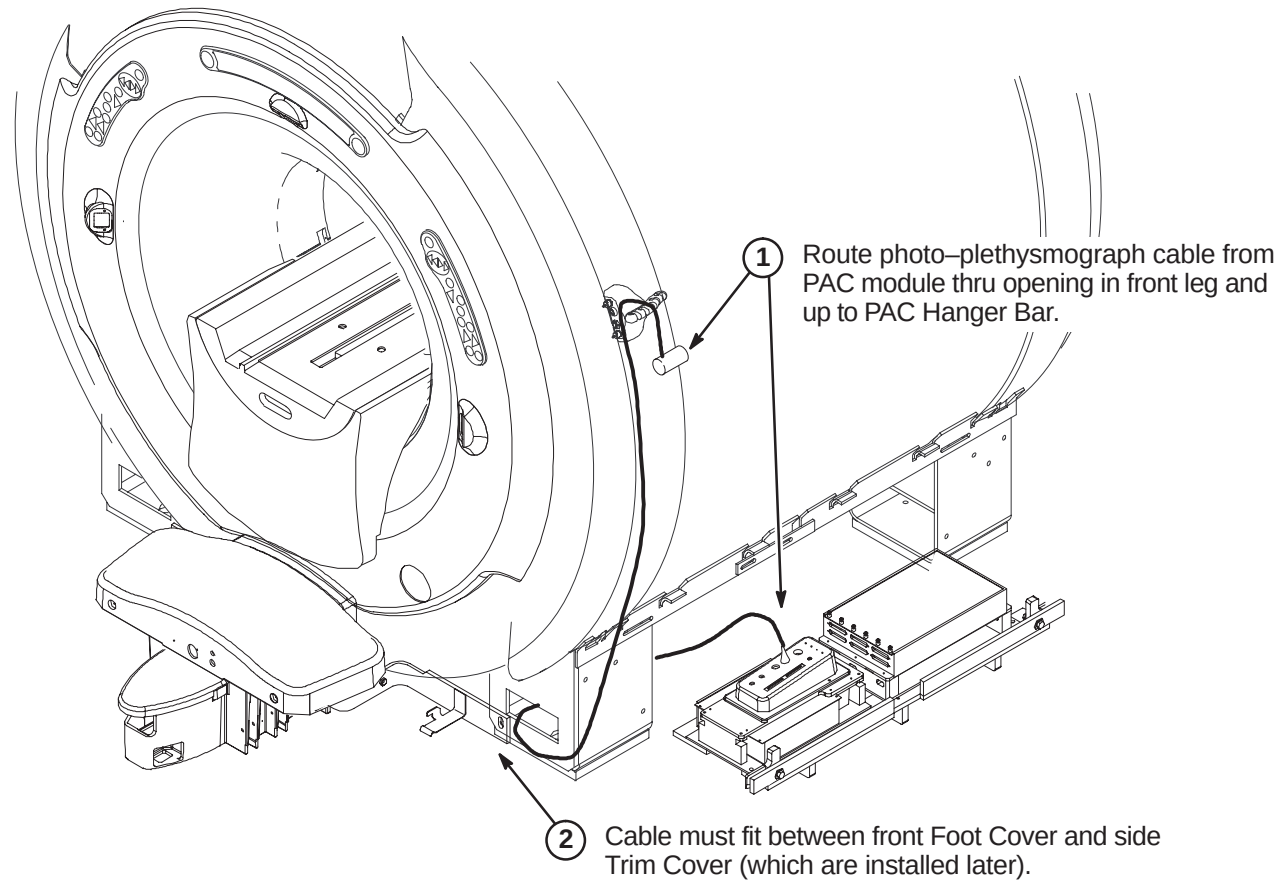
INSTALLATION PROCEDURES SUMMARY

- ❑ K1 – Connect Cables to SRI Module
- ❑ K2 – Routing of Photo-Plethysmograph Cable
- ❑ K3 – Route and Tie-wrap Cables to PAC/SRI Platform
- ❑ K4 – Install PAC/SRI Platform under Magnet

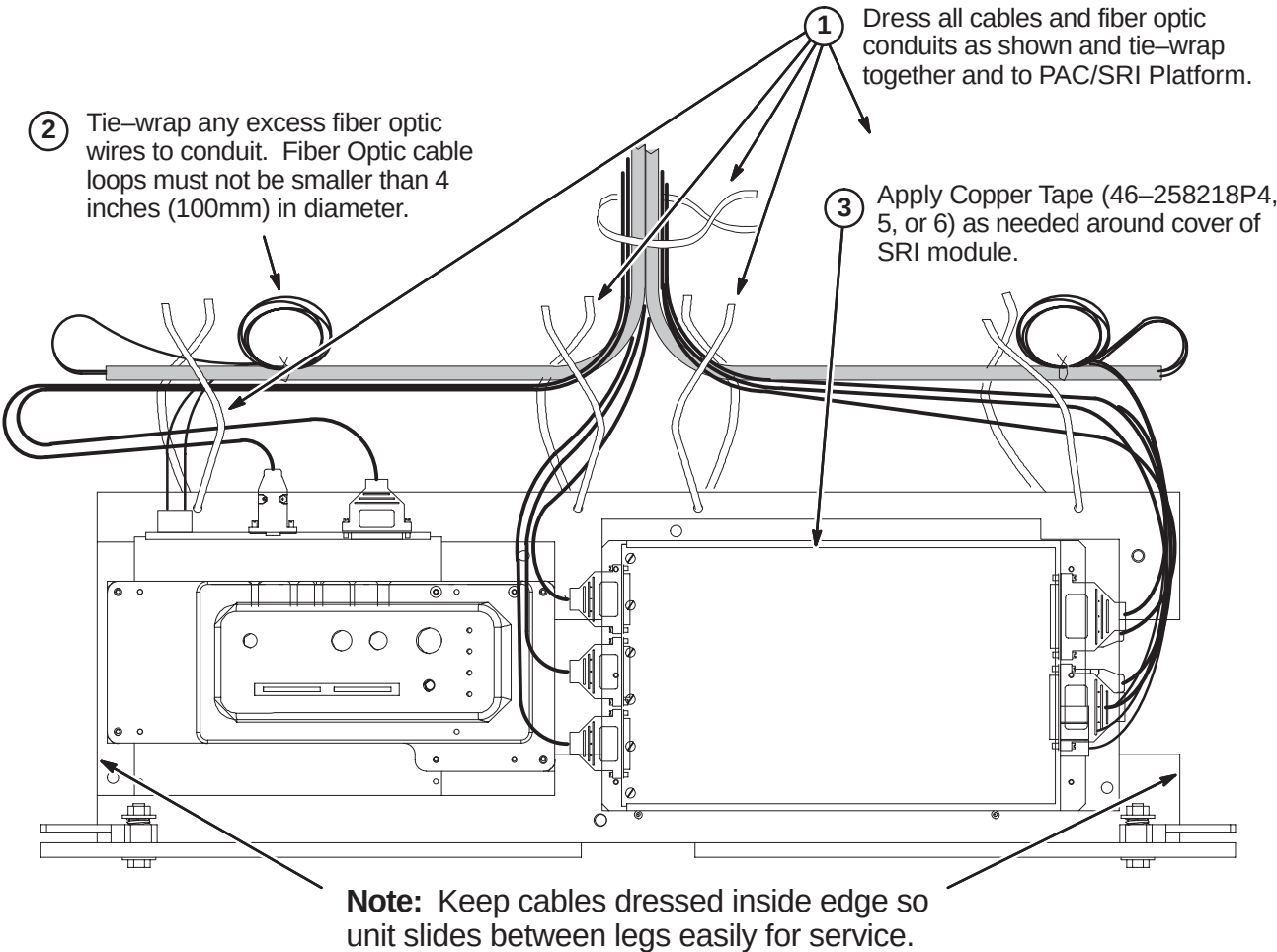
❑ K1 – CONNECT CABLES TO SRI MODULE



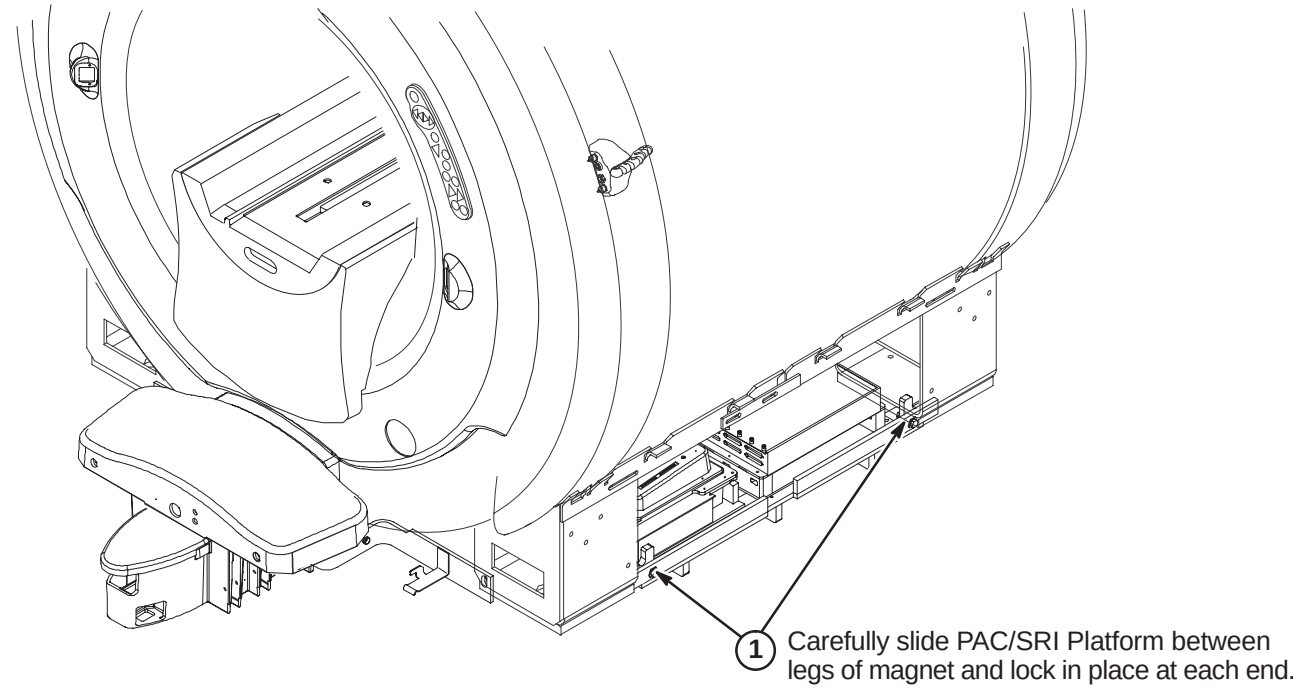
❑ K2 – ROUTING OF PHOTO-PLETHYSMOGRAPH CABLE



❑ K3 – ROUTE AND TIE-WRAP CABLES TO PAC/SRI PLATFORM



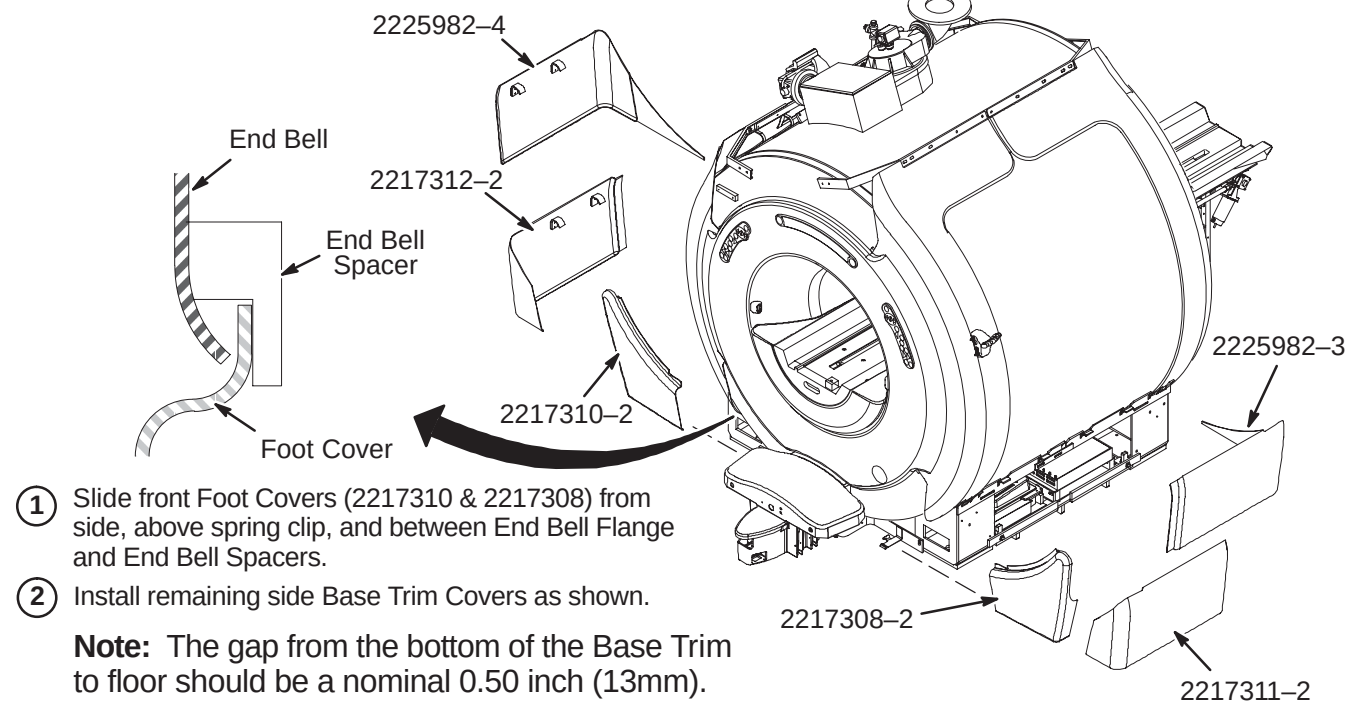
❑ K4 – INSTALL PAC/SRI PLATFORM UNDER MAGNET



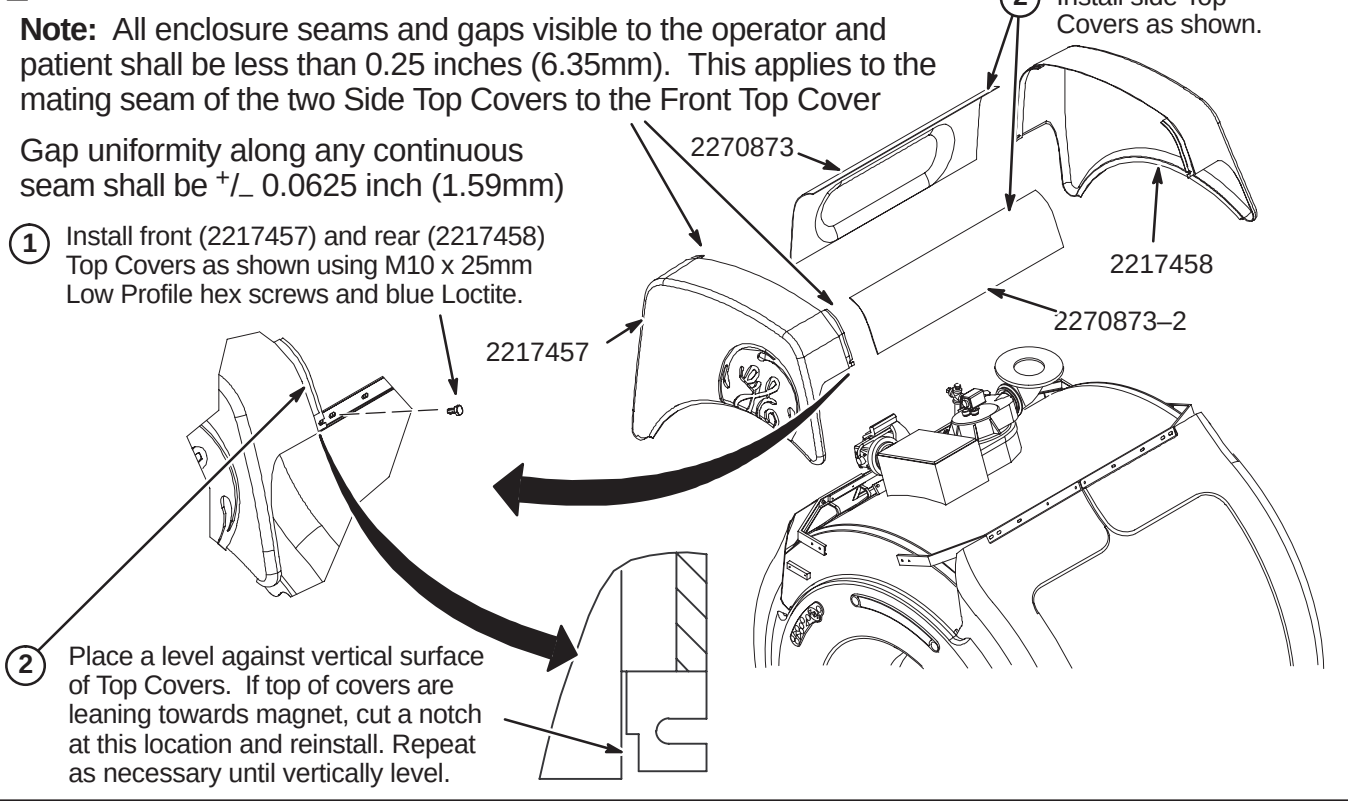
INSTALLATION PROCEDURES SUMMARY

- ❑ L1 – Install Enclosure Base Trim Covers
- ❑ L2 – Install Enclosure Top Covers
- ❑ L3 – Install Rear Pedestal Covers

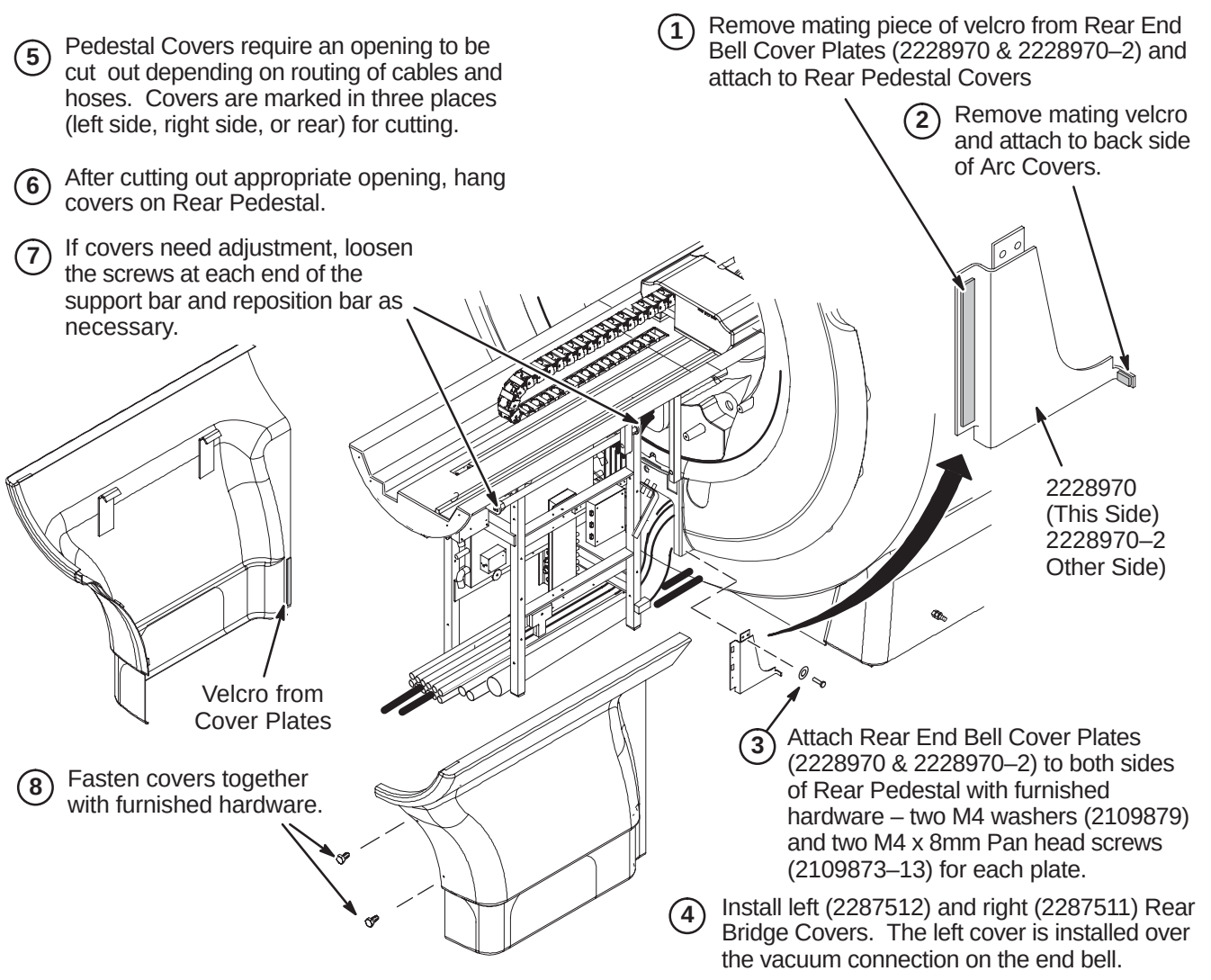
❑ L1 – INSTALL ENCLOSURE BASE TRIM COVERS



❑ L2 – INSTALL ENCLOSURE TOP COVERS



❑ L3 – INSTALL REAR PEDESTAL COVERS



INSTALLATION PROCEDURES SUMMARY

- M1 – Install Laser Alignment Light Warning Labels

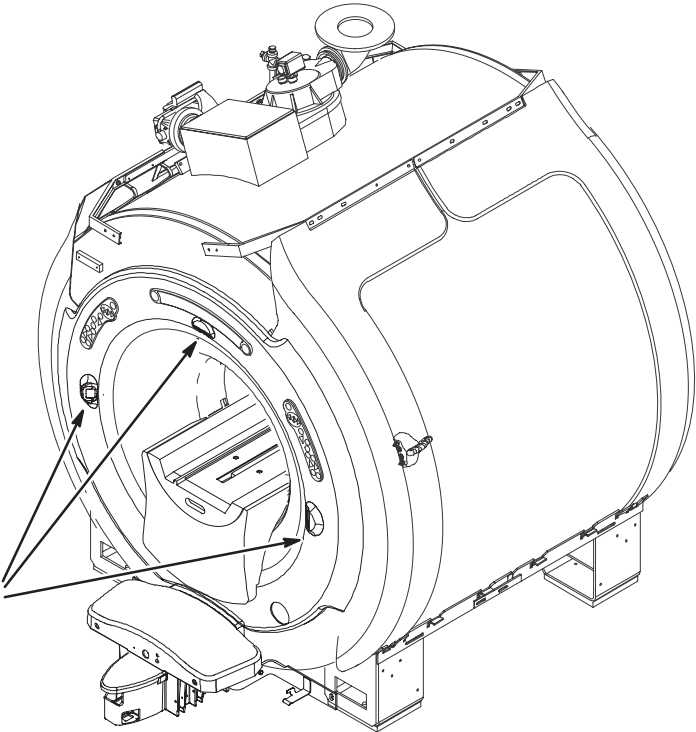
M1 – INSTALL LASER ALIGNMENT LIGHT WARINING LABELS

A sheet of Laser Alignment Light Warning Labels has been supplied with the magnet, attached to the magnet bridge.

The magnet enclosure is shipped with a label in English attached below each of the three laser lights.



English



- 1 For those sites which require a label other than English, peel off the appropriate label and affix over the English labels at the three laser light locations.



French



German



Portuguese



Spanish



Italian



Swedish

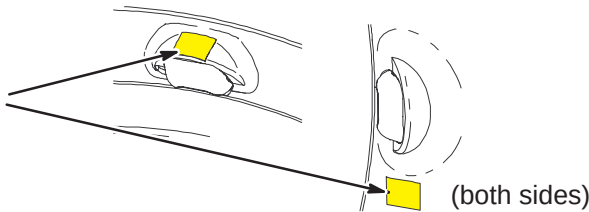


Japanese



Chinese

- 2 If enclosure does not have labels attached, attach new labels as indicated.



LIST OF EFFECTIVE PAGES

PAGE	REV	PAGE	REV
1	3	8	3
2	1	9	1
3	1	10	1
4	1	11	1
5	1	12	2
6	1	13	2
7	1		

WARNING!

FERROUS MATERIAL HAZARD! PARTS REQUIRED FOR THIS UPGRADE INSTALLATION CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES.

IF MAGNET IS AT FULL FIELD – KEEP ALL FERROUS TOOLS AND MATERIALS AS FAR FROM THE MAGNET AS POSSIBLE.

GENERAL ENCLOSURE NOTES:

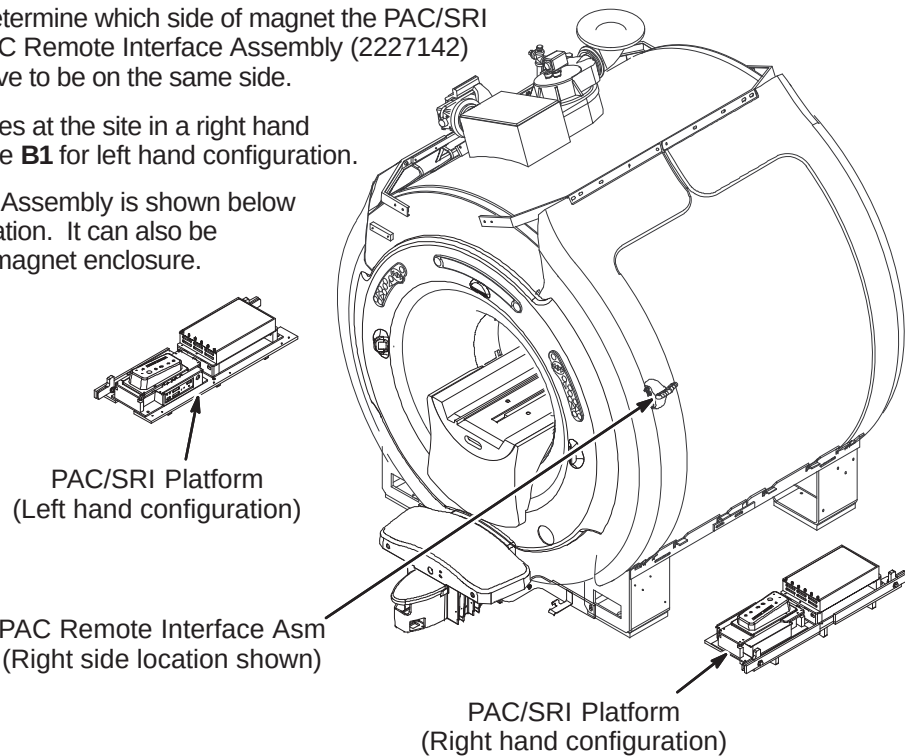
- Front, rear, left, and right, as used in this section, are defined as follows: the patient entrance end of the magnet is the front; the opposite end is the rear; left and right are in respect to a person’s left and right while facing the patient entrance end of the magnet.
- A 1-ounce tube of Permatex “Anti-Seize” lubricant (2119594) is provided. Follow instructions on back of tube. “Anti-Seize” lubricant is to be applied to threads of stainless-steel screws before installing into tapped holes of magnet to prevent galling and seizing of cap screw.

INSTALLATION PROCEDURES SUMMARY

- ☐ A1 – Determine PAC/SRI Platform and PAC Remote Interface Location
- ☐ A2 – Cooling Cabinet Requirements and Start Up
- ☐ A3 – Route/Connect Coolant Hoses to Tees
- ☐ A4 – Install Ground Clamps on Tees

A1 – DETERMINE PAC/SRI PLATFORM AND PAC REMOTE INTERFACE LOCATION

- 1 Consult with customer to determine which side of magnet the PAC/SRI Platform (2223440) and PAC Remote Interface Assembly (2227142) should be located. They have to be on the same side.
- 2 The PAC/SRI Platform arrives at the site in a right hand configuration. See Procedure B1 for left hand configuration.
- 3 The PAC Remote Interface Assembly is shown below installed in the right side location. It can also be mounted on the left side of magnet enclosure.



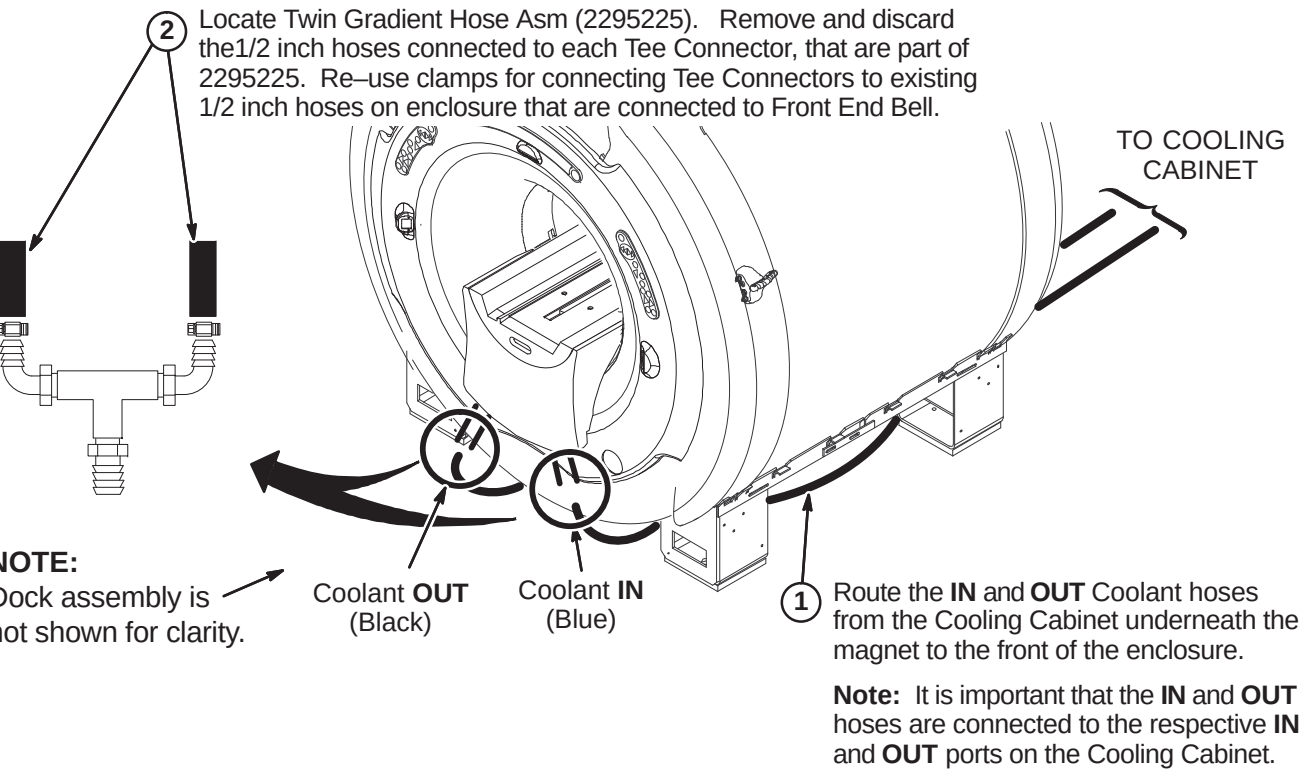
A2 – COOLING CABINET REQUIREMENTS AND START UP

The Cooling Cabinet must be installed and electrically powered prior to the start of this procedure.

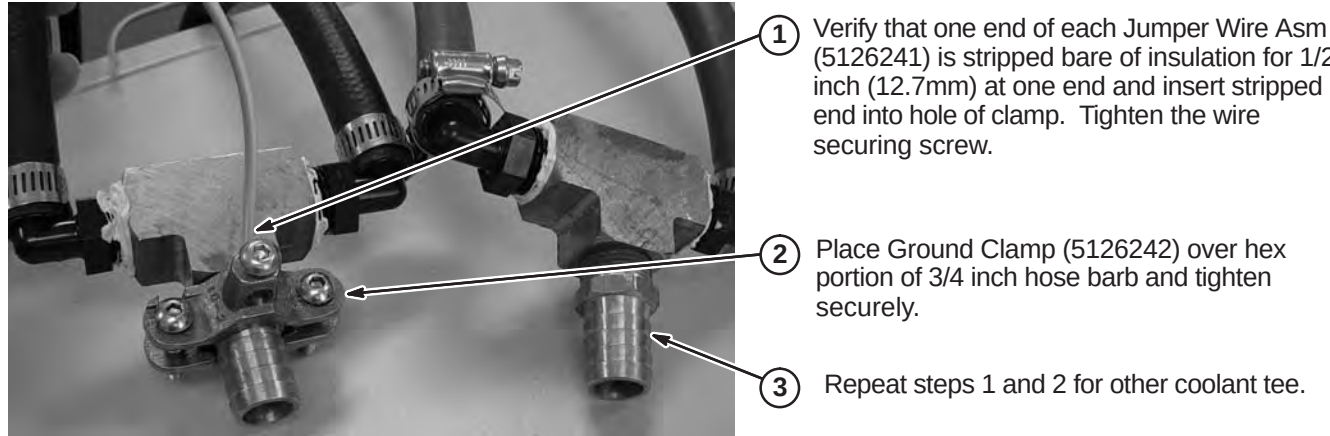
Consult vendor supplied manual that was delivered with Cooling Cabinet for installation, start-up, and operating instructions and procedures.

A3 – ROUTE/CONNECT COOLANT HOSES TO TEES

NOTE: Magnet is delivered to site with four 1/2 inch hoses connected to the Front End Bell. The Twin Gradient Hose Assembly (2295225) is delivered to site and also has four duplicate 1/2” hoses . **Discard the 1/2 inch black (2293610–2) and blue (2293610–3) hoses that are part of 2295225.** Connect the IN and OUT coolant hoses from the Cooling Cabinet to the 1/2 inch hoses attached to the Front End Bell using the Tees and clamps shown below.



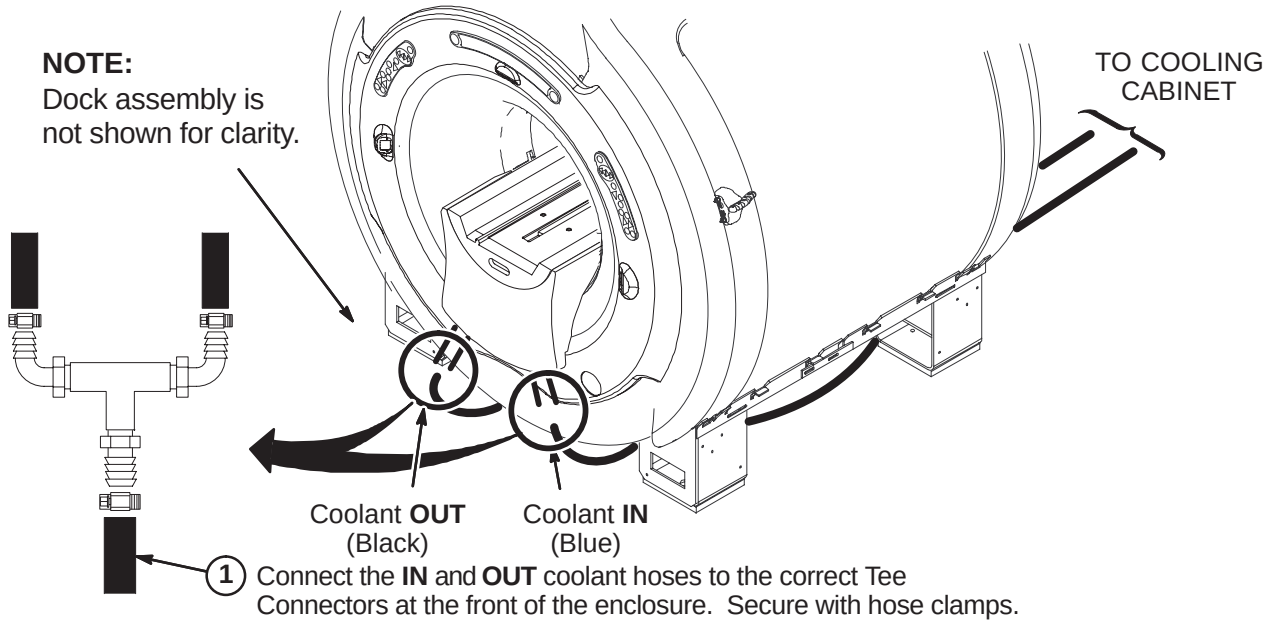
A4 – INSTALL GROUND CLAMPS ON TEES



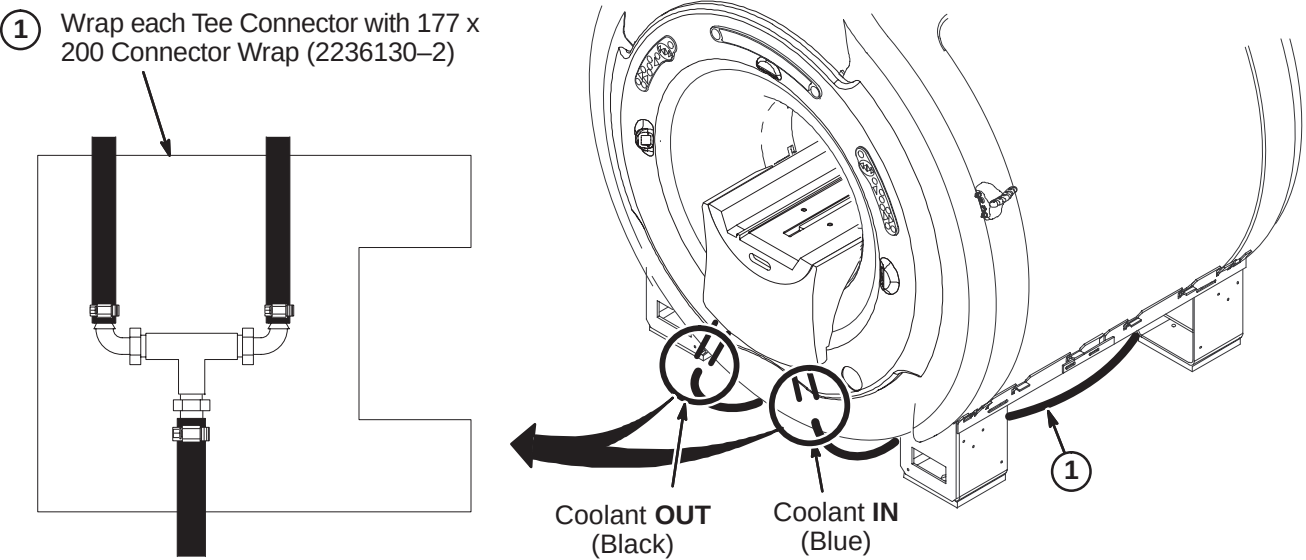
INSTALLATION PROCEDURES SUMMARY

- A5 – Connect Coolant Hoses to Tees
- A6 – Install Connector Wraps
- A7 – Route/Connect Ground Wires to Magnet
- A8 – Coolant Valve Installation

A5 – CONNECT COOLANT HOSES TO TEES



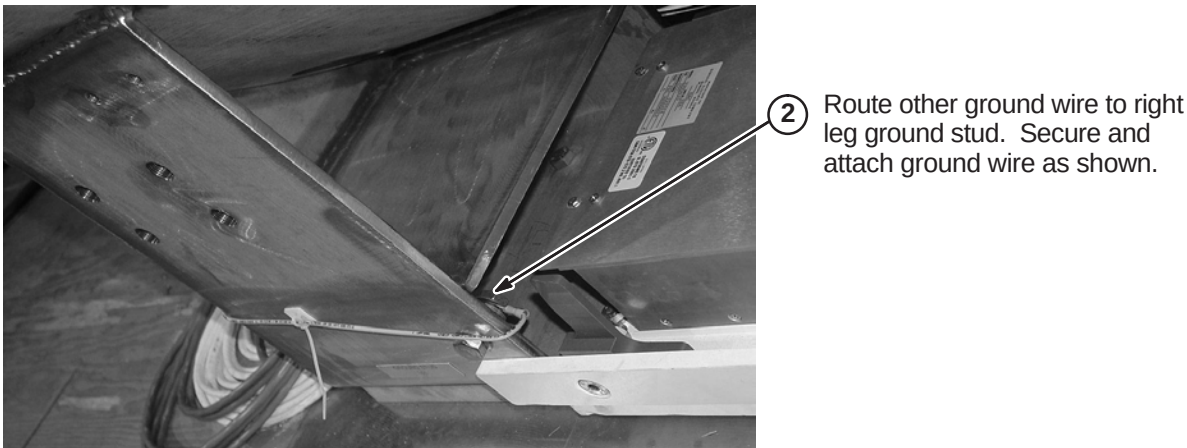
A6 – INSTALL CONNECTOR WRAPS



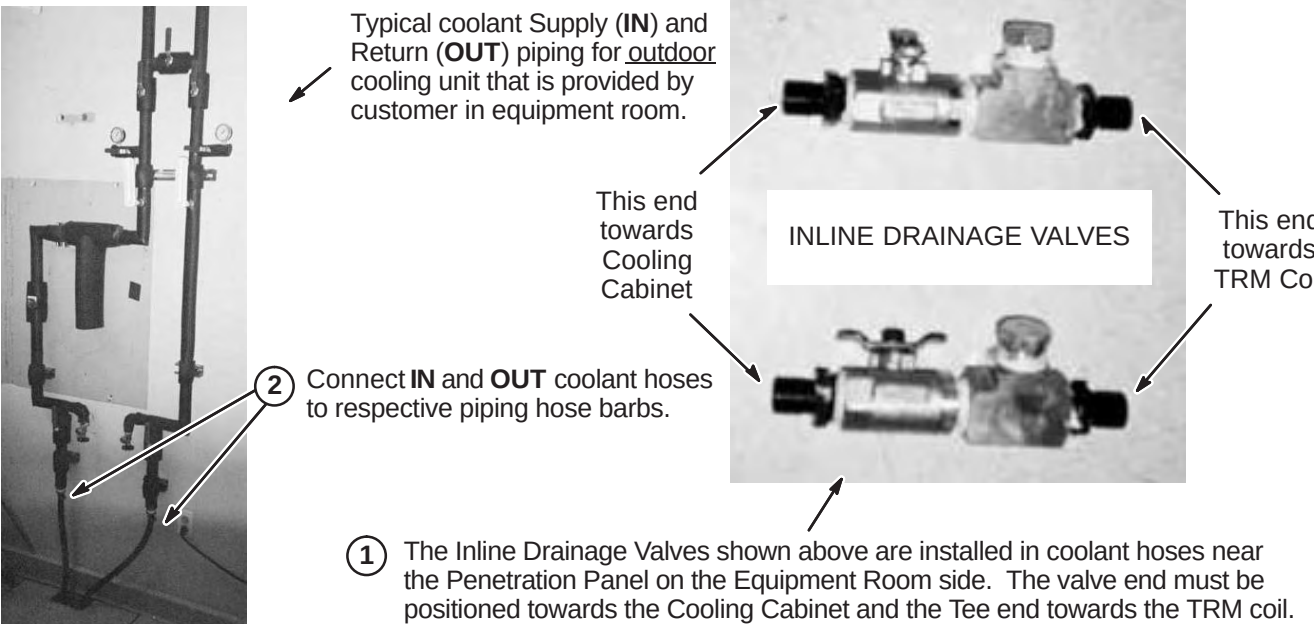
A7 – ROUTE/CONNECT GROUND WIRES TO MAGNET

Note:
If Run 040, magnet ground cable to RF Room, is connected to one of the front legs, the attaching nut can not be used for attaching any of the ground wires routed in this procedure.

If needed, remove nut from one of the unused ground studs on another leg and use to attach ground wire to above stud.



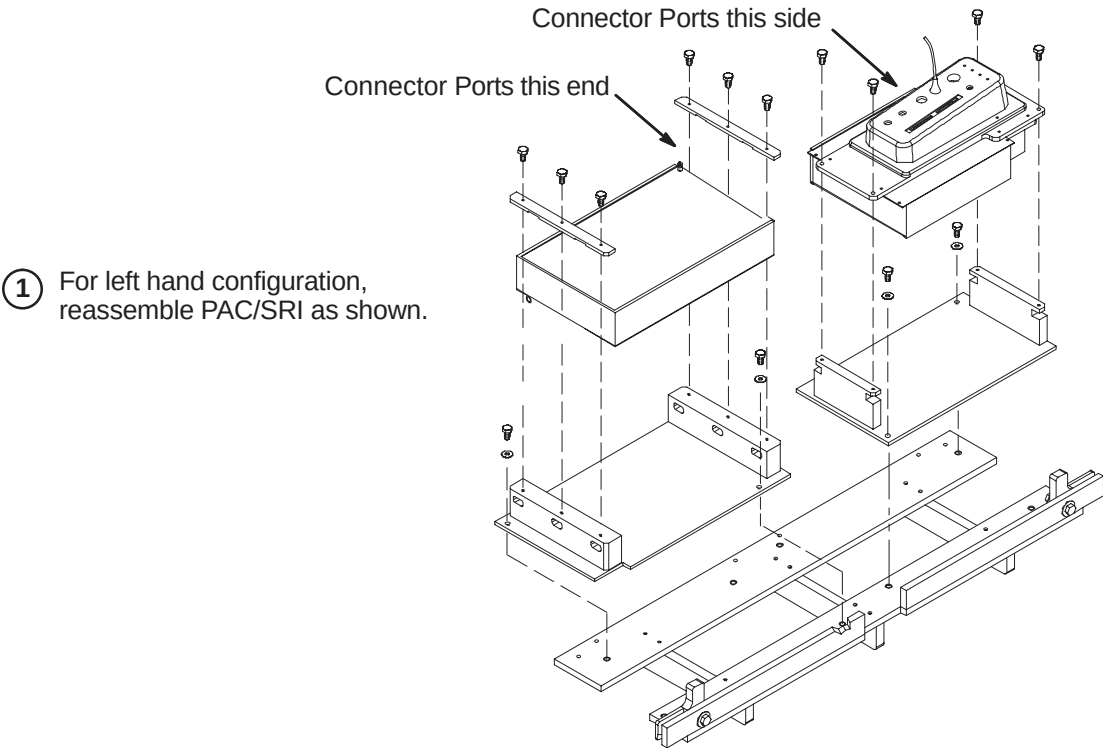
A8 – COOLANT VALVE INSTALLATION



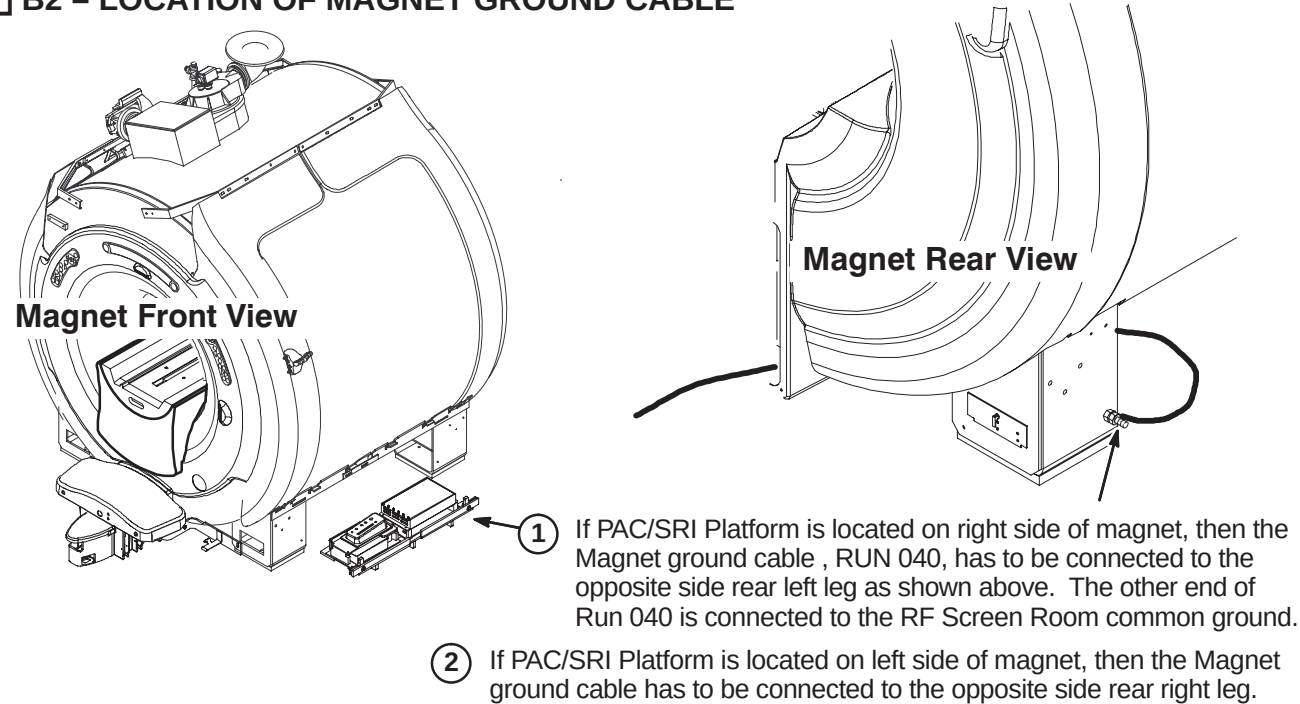
INSTALLATION PROCEDURES SUMMARY

- ☐ B1 – PAC/SRI Platform Left Hand Configuration (If Required)
- ☐ B2 – Location of Magnet Ground Cable
- ☐ B3 – Preparation of Hoses and Cables Under Rear Pedestal
- ☐ B4 – Location of Rear Pedestal

☐ B1 – PAC/SRI PLATFORM LEFT HAND CONFIGURATION (IF REQUIRED)



☐ B2 – LOCATION OF MAGNET GROUND CABLE

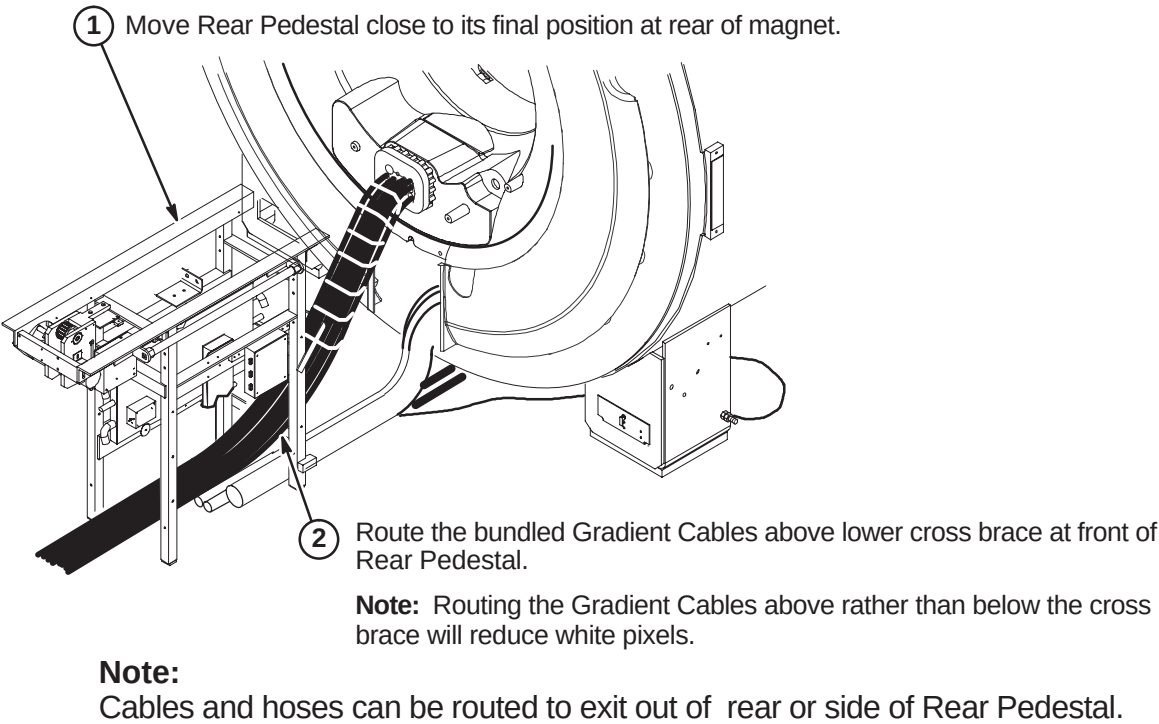


☐ B3 – PREPARATION OF HOSES AND CABLES UNDER REAR PEDESTAL

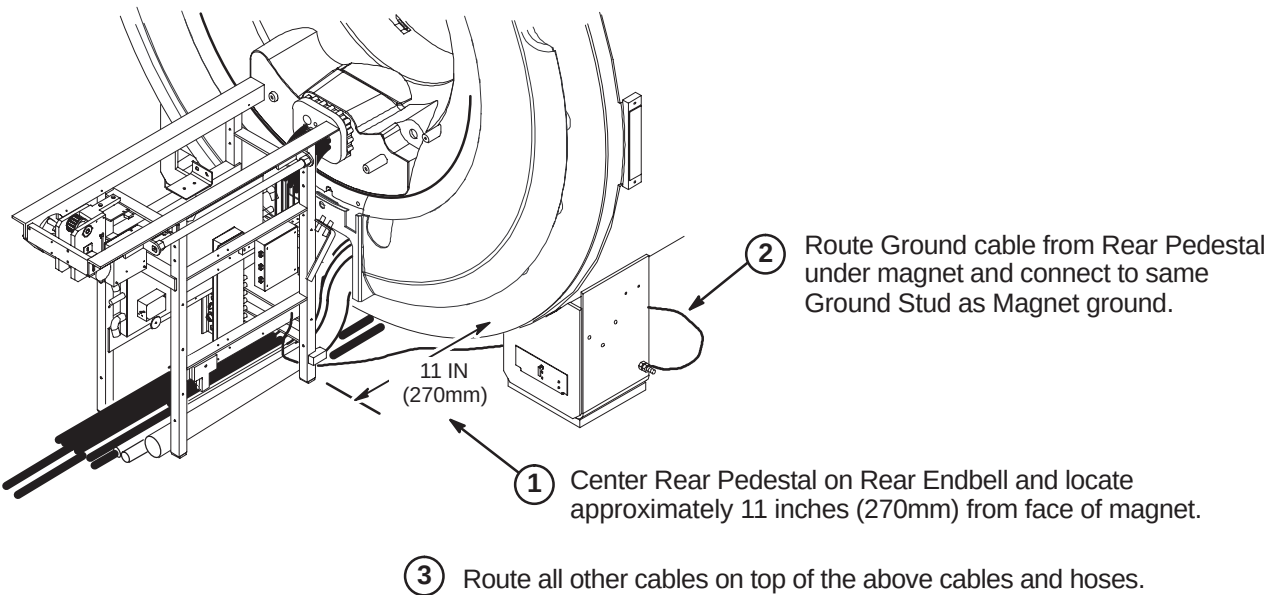
CAUTION

Rear End Bell **MUST** be installed on magnet with gradient coil cables routed thru block before connecting gradient coil cables to gradient cables routed from Penetration Panel with splices in cable trough. Failure to do so will result in inability to install Rear End Bell.

NOTE: Gradient cables will arrive bundled with tie-wraps installed.



☐ B4 – LOCATION OF REAR PEDESTAL



INSTALLATION PROCEDURES SUMMARY (for new style Split Bridge)

- C1 – Preparing Rear Split Bridge for Installation
- C2 – Attaching Rear Split Bridge to Rear Pedestal
- C3 – Attaching Rear Split Bridge to Front Split Bridge
- C4 – Aligning Bridge on Rear Pedestal
- C5 – Installing Bridge Locators

C1 – PREPARING REAR SPLIT BRIDGE FOR INSTALLATION

1 Remove pads and bans from Rear Split Bridge assembly.

2 Remove Carriage/Track assembly from Split Bridge.

3 Remove and save Bridge Rear Plate and gasket.

4 To access Belt Clamp Plates, unfasten four screws holding Carriage Cover to Carriage.

5 Unfasten, remove, and save Belt Clamp Plates

Note:
Take care not to damage microphone which is attached to carriage cover.

C2 – ATTACHING REAR SPLIT BRIDGE TO REAR PEDESTAL

1 Slide Rear Split Bridge into Tongue and Rear Bridge Mount Asm (2132034).

2 Attach together Bridge, Tongue and Rear Bridge Mount Asm (2132034) using two (2) 10–32 x 1–3/8 Screws (46–220184P47)

NOTE:
It is important that the sensor end of the Sensor body fits flush with the Sensor Bracket.

Magnet I/F #1 (Mounted on Rear Pedestal)

C3 – PREPARATION FOR FRONT AND REAR BRIDGE CONNECTION

1 Remove four 10M hex nuts (2109877–3) and washers (2228717–2) from end of Bridge in magnet. These will be used for attaching the Rear Split Bridge.

2 Move Rear Pedestal/Bridge into position.

3 Adjust the Rear Pedestal to level the rear Split Bridge with the front Split Bridge.

4 Remove two Bridge Locators (2235607) from shipping location on Rear Pedestal. These will be used for aligning Rear Pedestal to Bridge.

5 After Front and Rear parts of Split Bridge are attached, remove Rear Bridge Support bracket and discard. It is held in place with four screws.

NOTE:
Tongue to Bridge alignment is critical to smooth belt tracking and operation. Make sure Tongue is square to Bridge and Rear Pedestal.

C4 – ALIGNING BRIDGE ON REAR PEDESTAL

1 Use the following formula to determine alignment of Bridge on Rear Pedestal:
 $A_1 - A_2 < 2\text{mm}$

A₁ A₂

Rear Pedestal frame

C5 – INSTALLING BRIDGE LOCATORS

1 After Bridge and Rear Pedestal are aligned, peel off protective paper on each Bridge Locator and attach to bottom side of Bridge as shown.

End of Rear Pedestal frame closest to Magnet.

BOTTOM VIEW

INSTALLATION PROCEDURES SUMMARY

- D1 – Leveling Bridge
- D2 – Leveling Front of Bridge
- D3 – Installing Carriage on Bridge
- D4 – Routing Drive Belt

□ D2 – LEVELING BRIDGE

NOTE:
Rear Pedestal must be adjusted to maintain full contact between bottom of Bridge and top of Rear Pedestal.

Rear End Bell notch.

NOTE: Make sure that surfaces between front and rear Bridge surfaces are even. This will provide smooth travel for the Carriage wheels.

① Level Bridge and make sure that Bridge surface is no less than 2mm and no more than 4mm above the notches on the Front and Rear End Bells.

NOTE:
LPCA and Cable Track not shown for clarity.

② Tighten Vertical Adjustment Fasteners.

③ Adjust leg screws on both sides as needed to level Rear Pedestal.

□ D1 – LEVELING FRONT OF BRIDGE

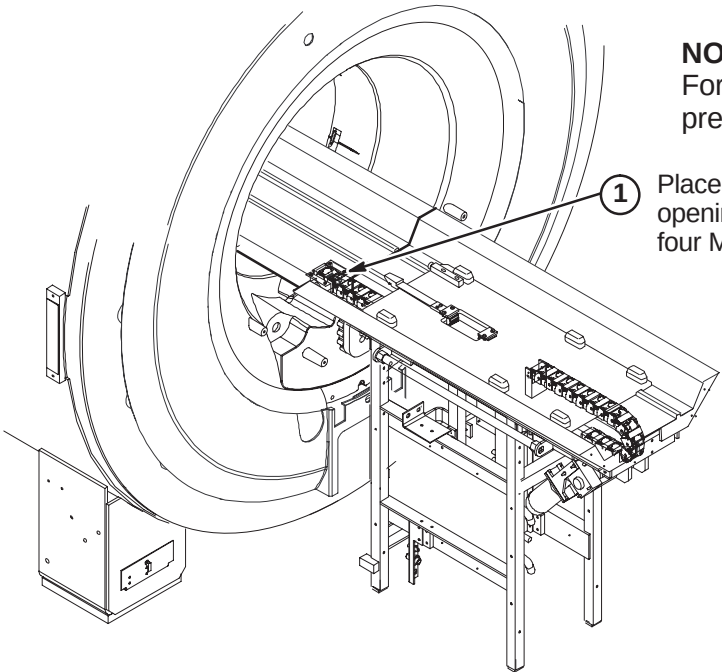
① Level Bridge and make sure that Bridge surface is no less than 2mm and no more than 4mm above the notches on the Front End Bell.



② Put a level across front of Bridge.

③ Level front of Bridge by adjusting right or left slider screws to center the bubble.

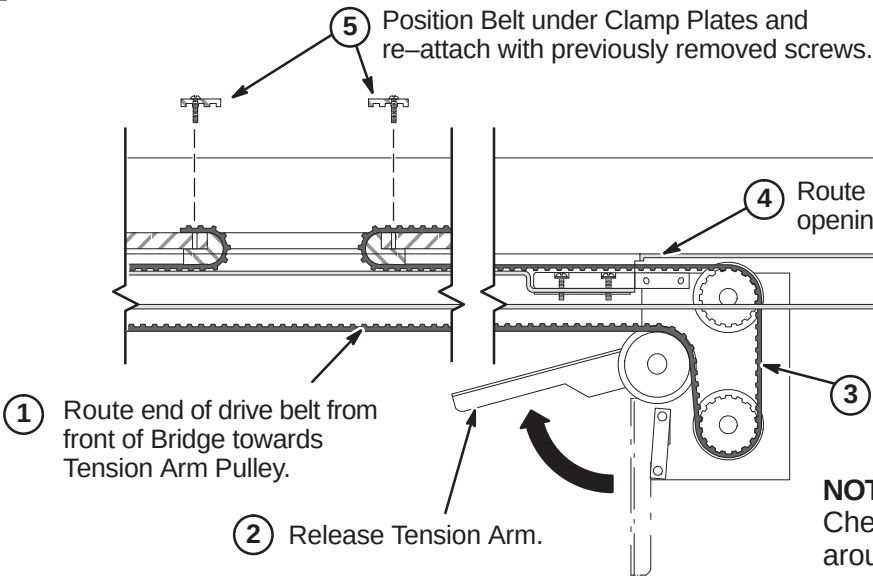
□ D3 – INSTALLING CARRIAGE ON BRIDGE



NOTE:
For clarity, the carriage is shown without preamps, preamp protect module, and cables.

① Place Carriage on Bridge and route all cables thru opening in Bridge and fasten plate to Front Bridge with four M6 x 12mm nylon screws (2296958) provided.

□ D4 – ROUTING OF DRIVE BELT



① Route end of drive belt from front of Bridge towards Tension Arm Pulley.

② Release Tension Arm.

④ Route Belt under and through opening to top of Carriage.

③ Feed back end of Belt underneath Bridge, around Tension Arm Pulley and rear pulleys into Bridge.

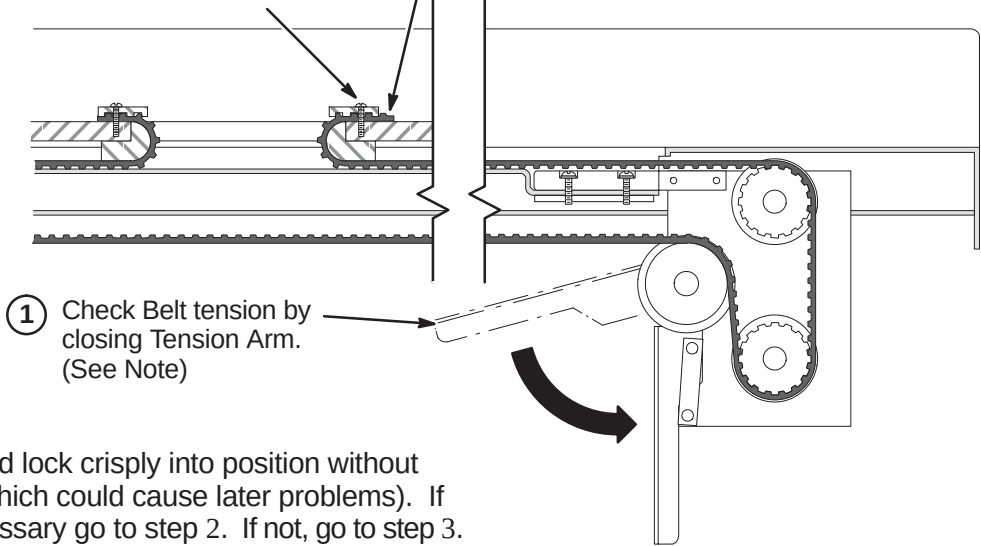
NOTE:
Check to verify that belt is not caught around spacers near rear pulley.

INSTALLATION PROCEDURES SUMMARY (for new style Split Bridge)

- M1 – Final Adjustment of Drive Belt
- M2 – Installation of Sensor Flag
- M3 – Route and Connect Runs from LPCA and SRI to Rear Pedestal
- M4 – Installing Carriage Cover to Trolley and Bridge Cover

E1 – FINAL ADJUSTMENT OF DRIVE BELT

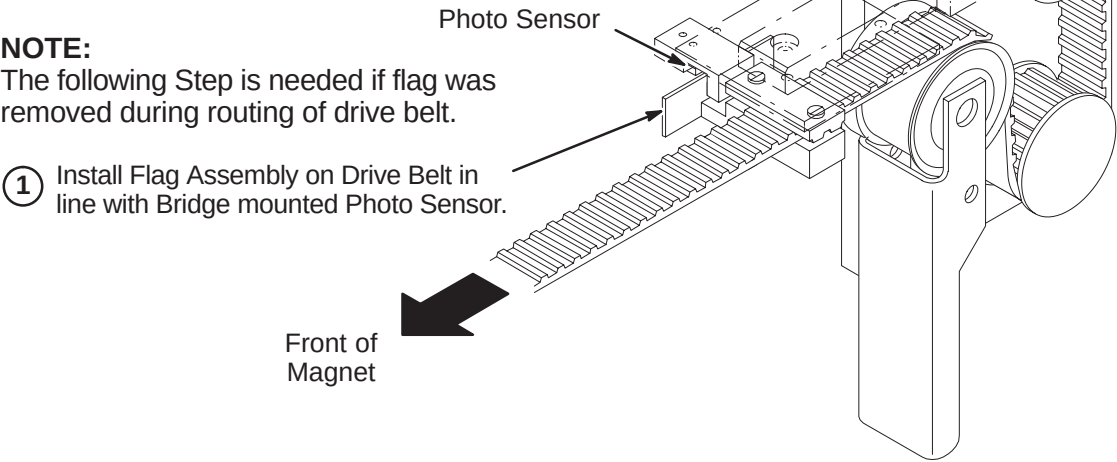
- (IF NECESSARY): Loosen Clamp and adjust position of Belt one tooth at a time until desired tension is obtained.
- Tighten Clamps and trim excess Belt length.
- Fasten Carriage Cover to Carriage with four screws removed in procedure E1.



NOTE:
Tension Arm should lock crisply into position without excessive force (which could cause later problems). If adjustment is necessary go to step 2. If not, go to step 3.

E2 – INSTALLATION OF SENSOR FLAG

NOTE:
Function of the Flag is to interrupt the reflected limit switch sensor beam when the Carriage is in home position all the way forward in the bore.

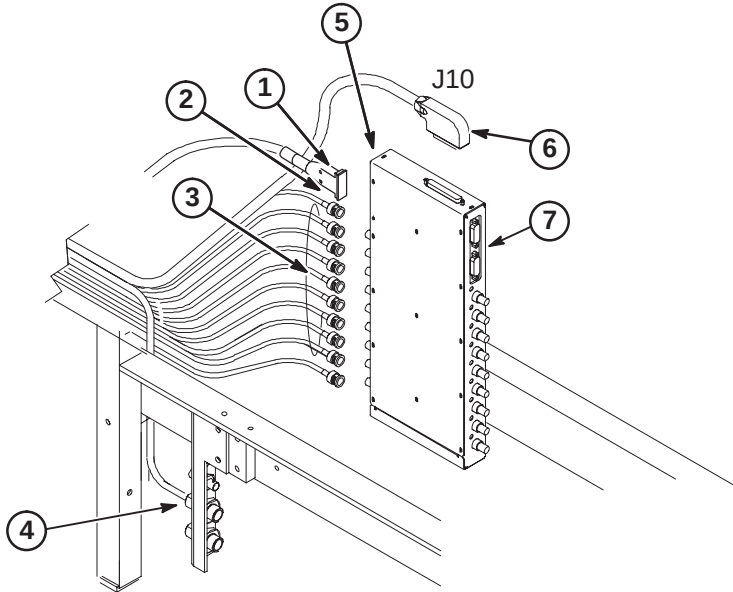


NOTE:
The following Step is needed if flag was removed during routing of drive belt.

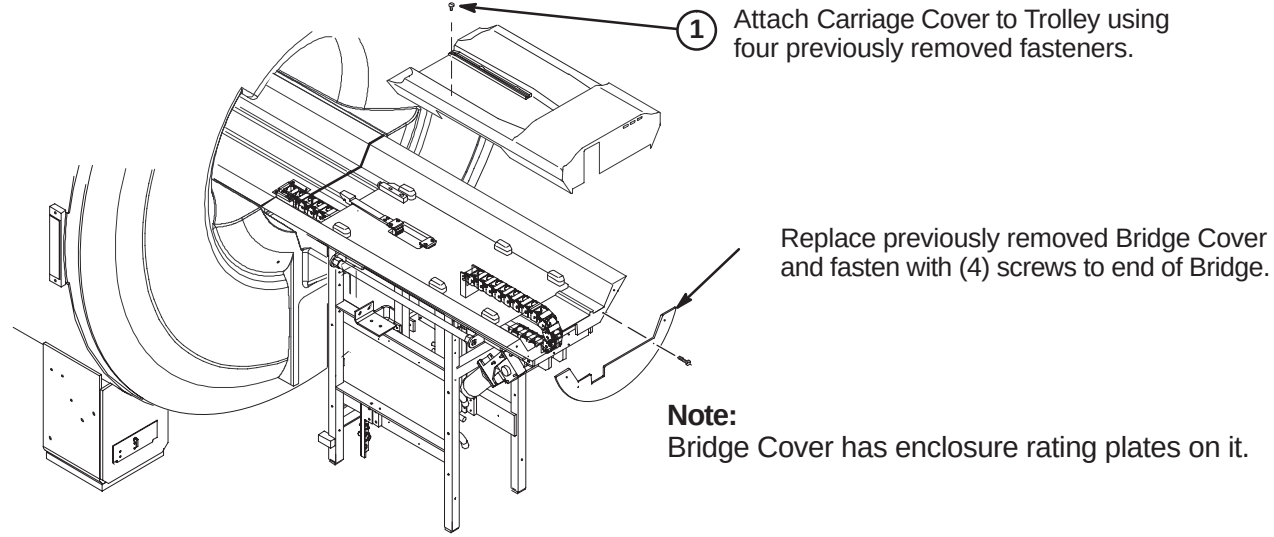
- Install Flag Assembly on Drive Belt in line with Bridge mounted Photo Sensor.

E3 – ROUTE AND CONNECT RUNS FROM LPCA AND SRI TO REAR PEDESTAL

- Route/Connect Run 1054 to J20.
- Route/Connect Run 1035 to J21.
- Route/Connect Runs 728 thru 735 to J11 to J18 and Run 1049 to J19.
- Route Run 404 and 485 from Carriage and connect to J1 and J3 respectively.
- Route/Connect Run 1078 from SRI to J29 and J30.
- Route/Connect Run 1041 from J78 on Penetration Panel to J19.
- Route/Connect Run 1042 from J77 on Penetration Panel to J5.



E4 – INSTALLING CARRIAGE COVER TO TROLLEY AND BRIDGE COVER

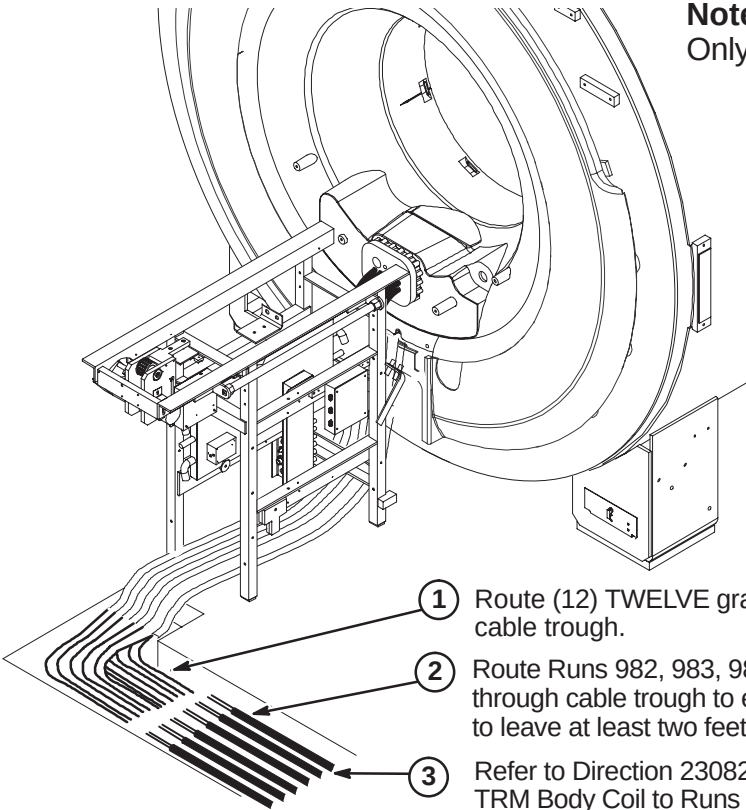


INSTALLATION PROCEDURES SUMMARY

- E1 – Routing and Connecting Gradient Output Cables
- E2 – Route/Connect Connect Cables at Rear Pedestal
- E3 – Route/Connect Runs 257, 262, 743, 744, and 780
- E4 – Route/Connect Runs 258, 318, and Drive Cables

E1 – ROUTING AND CONNECTING GRADIENT OUTPUT CABLES

Note:
Only Gradient Cables are shown for clarity.

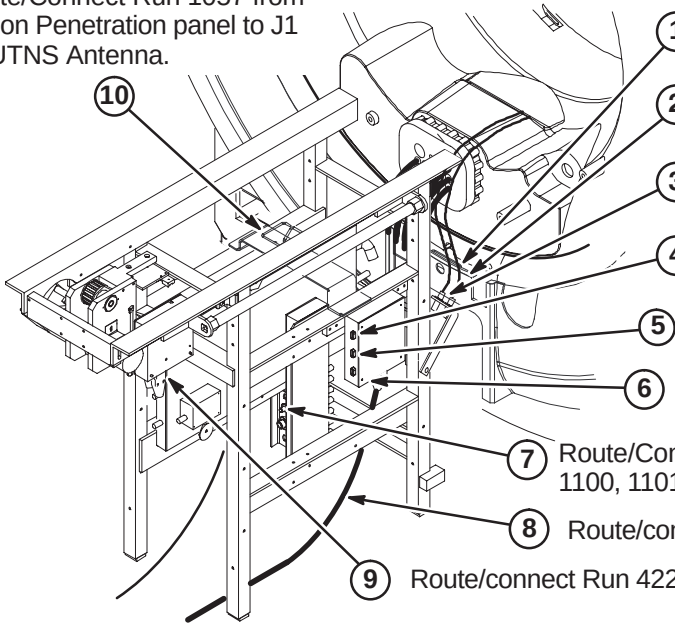


TRM COIL CABLE LABELS		OUTPUT CABLE LABELS		
WIRE #	LABEL	RUN #	LABEL	WIRE #
1	Z ZM–	982	ZM–	2
2	Z ZM+	982	ZM+	1
3	X WB–	987	WB–	2
4	X WB+	987	WB+	1
5	Y WB+	986	WB+	1
6	Y ZM–	983	ZM–	2
7	Y WB–	986	WB–	2
8	Y ZM+	983	ZM+	1
9	Z WB+	985	WB+	1
10	X ZM–	984	ZM–	2
11	Z WB–	985	WB–	2
12	X ZM+	984	ZM+	1

- 1 Route (12) TWELVE gradient Output Cables from TRM Body Coil to cable trough.
- 2 Route Runs 982, 983, 984, 985, 986, and 987 from Penetration Panel through cable trough to ends of Body Coil gradient cables. Make sure to leave at least two feet (610mm) of slack before cutting to length.
- 3 Refer to Direction 2308287 in Tab 1 for connecting Output cables from TRM Body Coil to Runs 982, 983, 984, 985, 986, and 987.

E2 – ROUTE/CONNECT CABLES AT REAR PEDESTAL

Route/Connect Run 1057 from J85 on Penetration panel to J1 on UTNS Antenna.



- 1 Route and connect RF Bias cables from RF Coil to bracket as marked.
- 2 Route and connect Run 781 from Bias cable J5 to J75 on the Penetration Panel.
- 3 Route and connect Run 782 from Bias cable J6 to J76 on the Penetration Panel.
- 4 Route/connect microphone cable from Carriage Cover to J7.
- 5 Route/connect Front End Bell microphone cable Run 414 to J9.
- 6 Route/connect speaker cable from Patient Comfort Module to J8.
- 7 Route/Connect Runs 1095, 1096, 1097, 1098, 1099, 1100, 1101, and 1102 to J1 thru J9 on 8 Channel Switch.
- 8 Route/connect Run 282 from J12 on Pen Panel to J6.
- 9 Route/connect Run 422 from J2 on SRI to J4 on Magnet Interface #1.

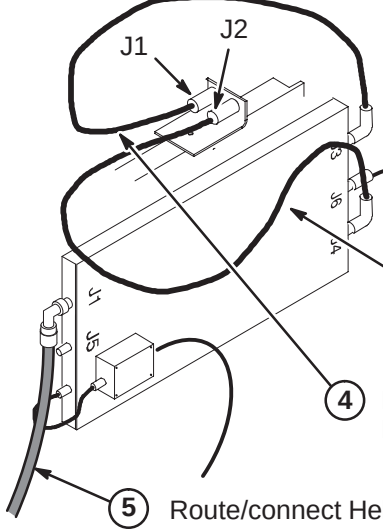
E3 – ROUTE/CONNECT RUNS 257, 262, 743, 744, AND 780

CAUTION

The RF Drive Input cables, Runs 743 and 744, utilize a corrugated solid RF shield. If mishandled, the cable can be kinked and subsequently broken.

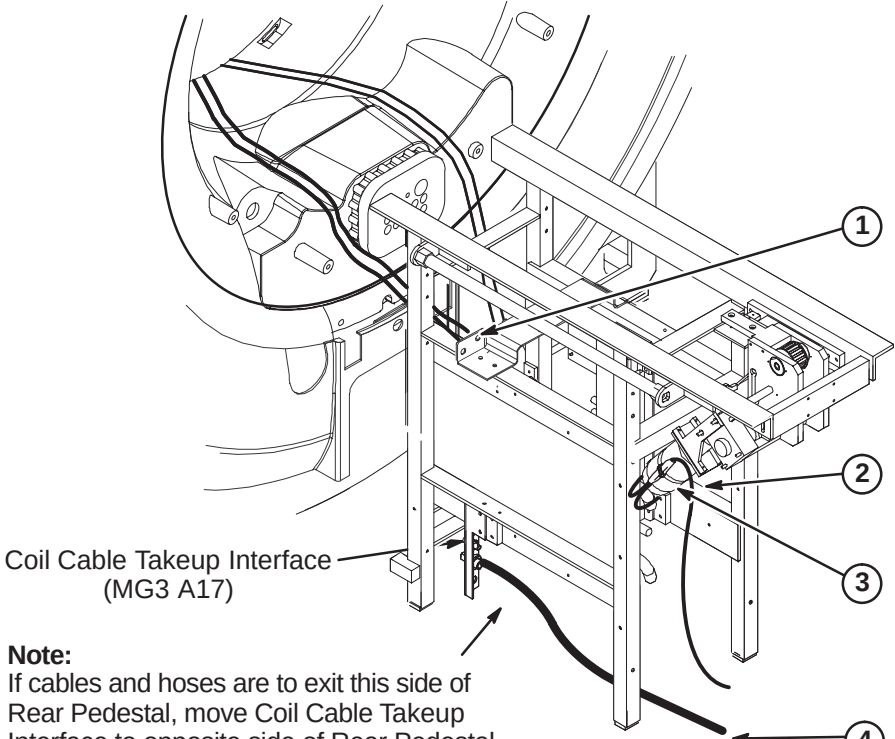
Also, use care when connecting these cable to the bulkhead connectors J1 and J2 and RF Splitter connectors J3 and J4. Cross threading or inadequate engagement of the center pin can cause arcing and damage the cables.

Note: Runs 743 and 744, Body Coil to RF Splitter cable connections may be reversed depending on magnet field polarity. Consult local Field Engineer for proper routing.



- 1 Route/connect Run 780 from J74 on Pen Panel to J6 on RF Splitter.
- 2 Route and connect Run 744 from J1 on bracket to J4 on the RF Splitter.
- 3 Route and connect Run 743 from J2 on the bracket to J3 on the RF Splitter.
- 4 Route/connect Run 262 from J8 on Pen Panel to Body Coil Preampifier.
- 5 Route/connect Heliax® Run 257 to J1 on RF Splitter.

E4 – ROUTE/CONNECT RUNS 258, 318, AND DRIVE CABLES



- 1 Route I & Q RF Drive Input cables from RF Coil and connect as marked to J1 and J2 on bracket.
- 2 Route/connect Run 318 to short cable on Longitudinal Motor Drive (MG3 A7).
- 3 Ty-wrap cable to motor to prevent movement.
- 4 Route/connect Run 258 to J1 on Coil Cable Takeup Interface.

Note:
If cables and hoses are to exit this side of Rear Pedestal, move Coil Cable Takeup Interface to opposite side of Rear Pedestal.
If Interface bracket is moved to other side, connectors on bracket need to be removed, turned end for end, and reinstalled.

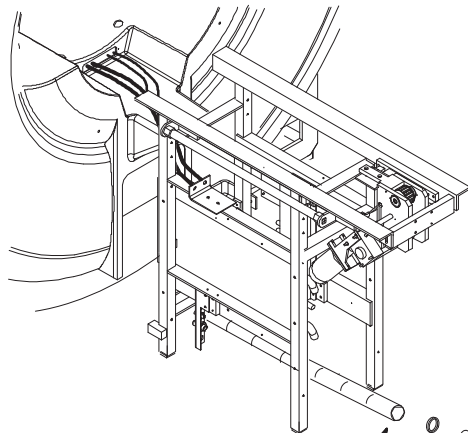
INSTALLATION PROCEDURES SUMMARY

- F1 – Install Blower box and Route/Connect Air Hose
- F2 – Install Bore Lamp Assembly
- F3 – Route/Connect Vacuum Hose and Elbow

F1 – INSTALL BLOWER BOX AND ROUTE/CONNECT AIR HOSE

WARNING!

RF SHIELD INTEGRITY MUST BE MAINTAINED FOR MOUNTING BLOWER BOX WITHIN THE MAGNET ROOM



- Refer to site design (architectural)/customer construction drawings or contact PMI for locating and securing Blower Box (2121199).
- Check for correct installation of the anchors:
 - Confirm that the anchors are physically secured to the building structure, not to a removable computer floor, raised floor, or false floor.
 - Confirm there are at least four (4) anchors.
 - If any or the anchors appears wobbly or loose, stop the installation and contact the PMI to fix anchoring.
- Remove any mounting hardware from the anchors. Hoses should face the back of the room
- Position the blower box on top of 4 or more anchors. Hoses should face the back of the room.
- Secure the blower box using the hardware provided by the RF shield room vendor to attach.

Note:
The following items are part of Blower Hose Kit, 2129412–5.

- Connect 2 inch air hose from Rear End Bell to Reducer (2175143) with 2 inch clamp (46–208765P28). Do not cut or reduce the length of the 2 inch hose.
- Connect 4 inch air hose (46–282666P11) to reducer with 4 inch clamp (46–208765P34). If necessary, the 4 inch hose may need to be cut to reduce length.
- Route 4 inch hose to Blower Cabinet. Trim to appropriate length and attach to vent opening with 4 inch Hose Clamp (46–208765P34).

CAUTION

THE BLOWER BOX CONTAINS FERROUS MATERIAL. The proper mounting of the Blower Box is critical to safety. It MUST be mounted per these instructions

IMPORTANT!

Location of cabinet and type of fastener used for securing Blower Box outside of minimum service area should be specified in the Site Design (architectural/Construction) drawings. If proper mounting area is not specified, contact the Project Manager of Installation (PMI).

NOTE: If needed, Blower Hose Cover can be cut and modified.

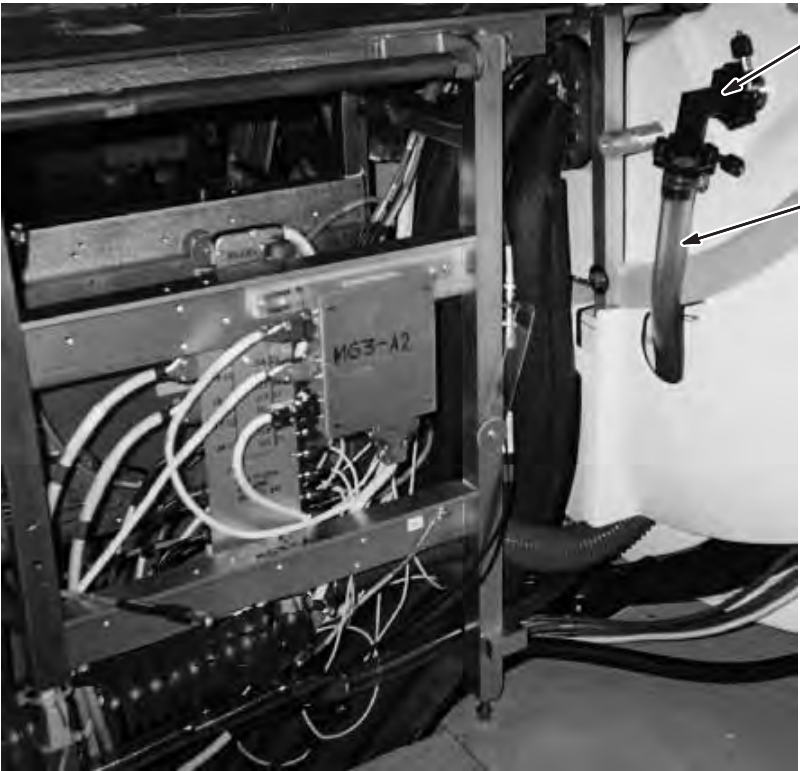
F2 – INSTALL BORE LAMP ASSEMBLY

Note: Cables and hoses routed in Procedure below are not shown for clarity.

-
- Move Bore light Assembly (2222547) to area of left rear magnet leg.
 - Connect the (+) and (–) wires of Run 300 to the terminal strip as shown.
 - Route Run 300 thru opening in rear leg, under Magnet and Rear Pedestal and connect to J15 on Penetration Panel.
 - Route fiber bundle from Rear End Bell thru leg opening and insert into bushing. Fasten to bushing with set screws.
 - Insert into clip on Bore light Assembly. Bushing flanges to be inside clip.
 - Peel off protective paper on velcro and install assembly in opening in rear leg.

F3 – ROUTE/CONNECT VACUUM HOSE AND ELBOW

Note:
Vacuum Grease **must** be applied to all Centering Rings (2293608) and outside edge of vacuum connector flanges.



- Install Elbow (2293609) to End Bell using centering ring (2293608) and chain clamp (2293618–2). Tighten chain clamp to “finger tight”.
- Route vacuum hose (2293607) from Elbow on End Bell, thru Penetration Panel, to connection at rear of TAC Cabinet in Equipment Room. Cut to length.
- At each end of vacuum hose insert Hose Connector (2293616), apply vacuum grease to outside edge of connector and fasten with two (2) Clamps (2294010–3).
- Apply vacuum grease to outside edge of Hose Connector flanges and respective connector flanges on TAC and End Bell.
- Connect each end of Vacuum Hose using centering ring (2293608) and chain clamp (2293618–2). Tighten chain clamp to “finger tight”.

INSTALLATION PROCEDURES SUMMARY

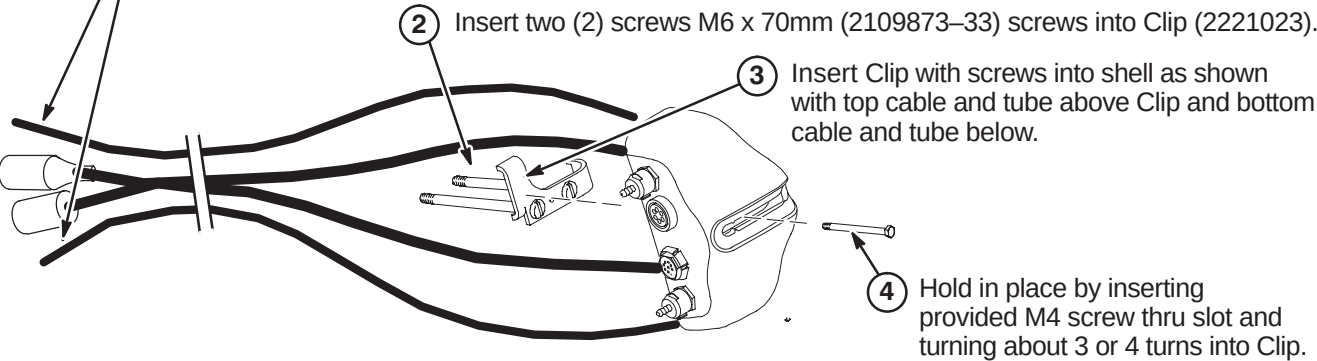
- H1 – Prepare Cables and Tubes for Routing
- H2 – Position Remote PAC Interface Mounting Clip
- H3 – Route Remote PAC Interface Cables and Tubes
- H4 – Installing Remote PAC Interface to Arc Cover
- H5 – Install PAC Hanger Bar

□ H1 – PREPARE CABLES AND TUBES FOR ROUTING

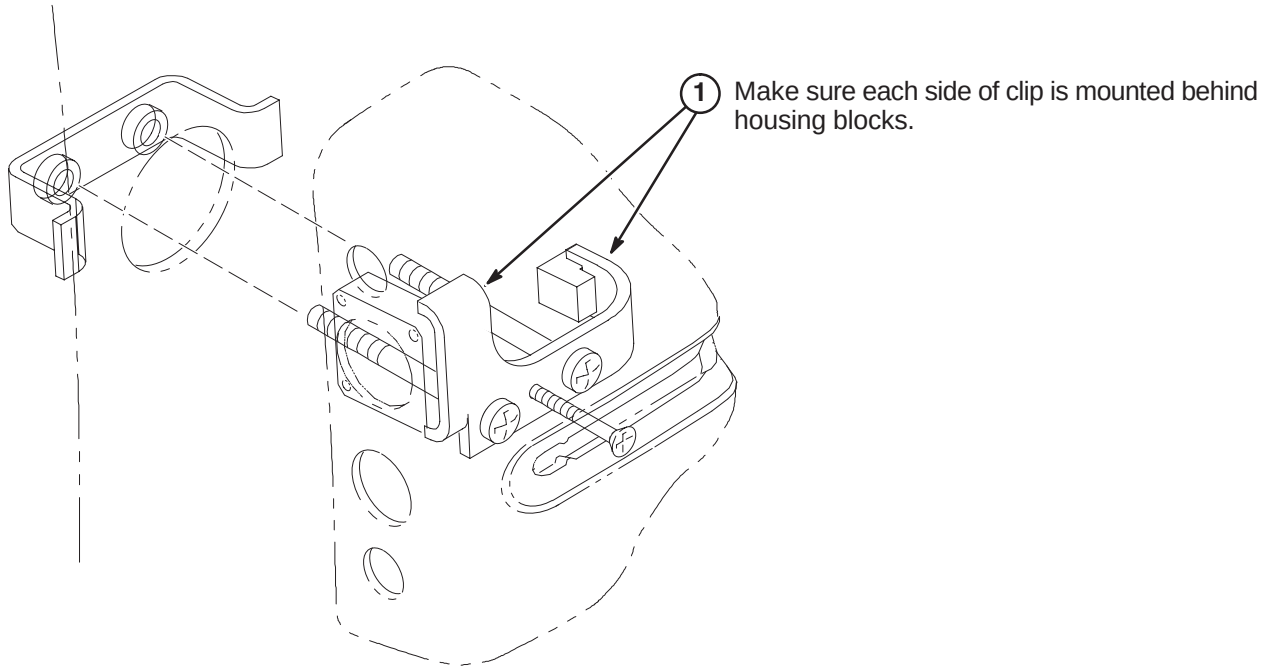
Note:
Consult with customer for correct location of Remote PAC Interface assembly. It must be on the same side as the PAC/SRI Platform, as positioned in Procedure A1

- ① Apply tape around end of pneumatic tubes and mark "TOP" to the opposite end of the tube connected to the top connector in the Remote PAC Interface Assembly and "BOTTOM" to the tube connected to the lower connector.

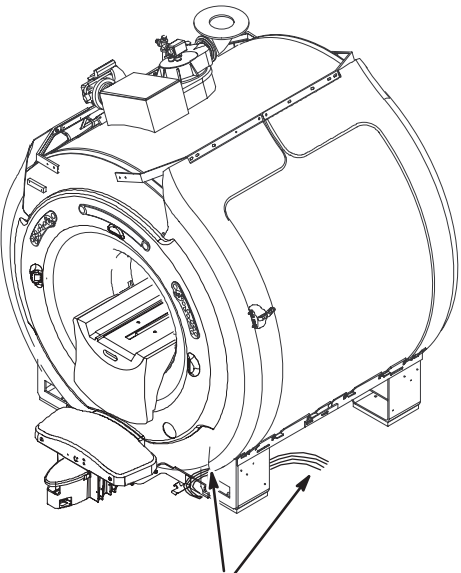
Note: Top two cables will be routed over top of bracket and bottom two cables will be routed under bracket.



□ H2 – POSITION REMOTE PAC INTERFACE MOUNTING CLIP

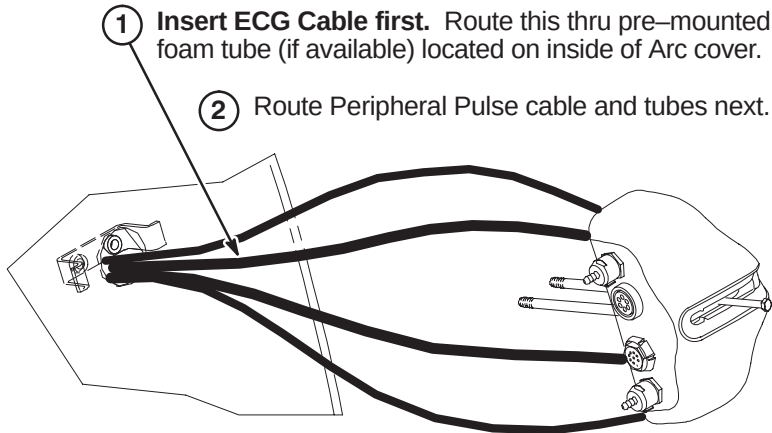


□ H3 – ROUTE REMOTE PAC INTERFACE CABLES AND TUBES



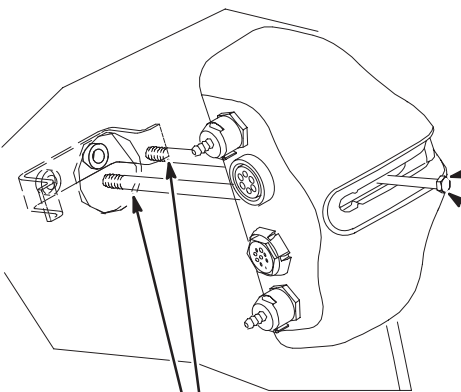
Note: Top two cables will be routed over top of bracket and bottom two cables will be routed under bracket.

Note: Insert cables/tubes separately thru hole and make sure they come out at bottom of arc cover.



- ③ Route around foot of magnet to PAC/SRI Platform location.

□ H4 – INSTALLING REMOTE PAC INTERFACE TO ARC COVER



Note:
Routed cables and tubes not shown for clarity.

- ① Use M4 screw thru slot to lift and guide small end of Clamp over block inside housing.

③ Remove M4 screw and save for use at later time since it would be required if Remote PAC Interface Assembly would need to be removed at any time in the future.

OR
Turn the M4 screw inward until there is clearance for sliding the PAC Hanger Bar in T5 into place.

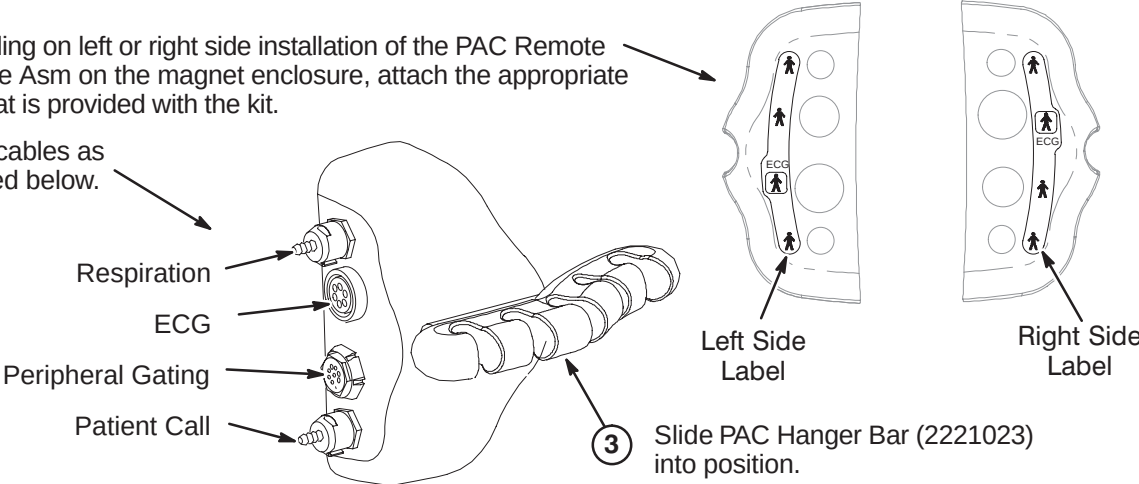
- ② Using screwdriver thru housing slot, tighten down the M6 screws to lock housing in place.

□ H5 – INSTALL PAC HANGER BAR

- ① Depending on left or right side installation of the PAC Remote Interface Asm on the magnet enclosure, attach the appropriate label that is provided with the kit.

- ② Attach cables as indicated below.

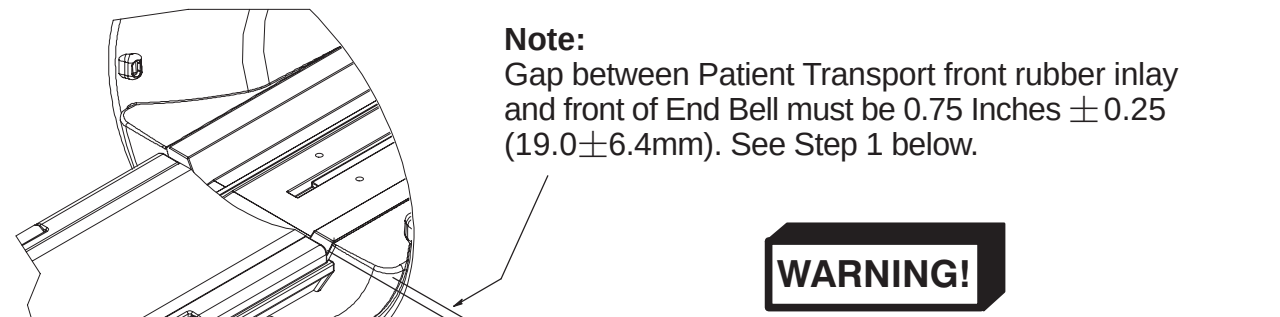
Respiration
ECG
Peripheral Gating
Patient Call



INSTALLATION PROCEDURES SUMMARY

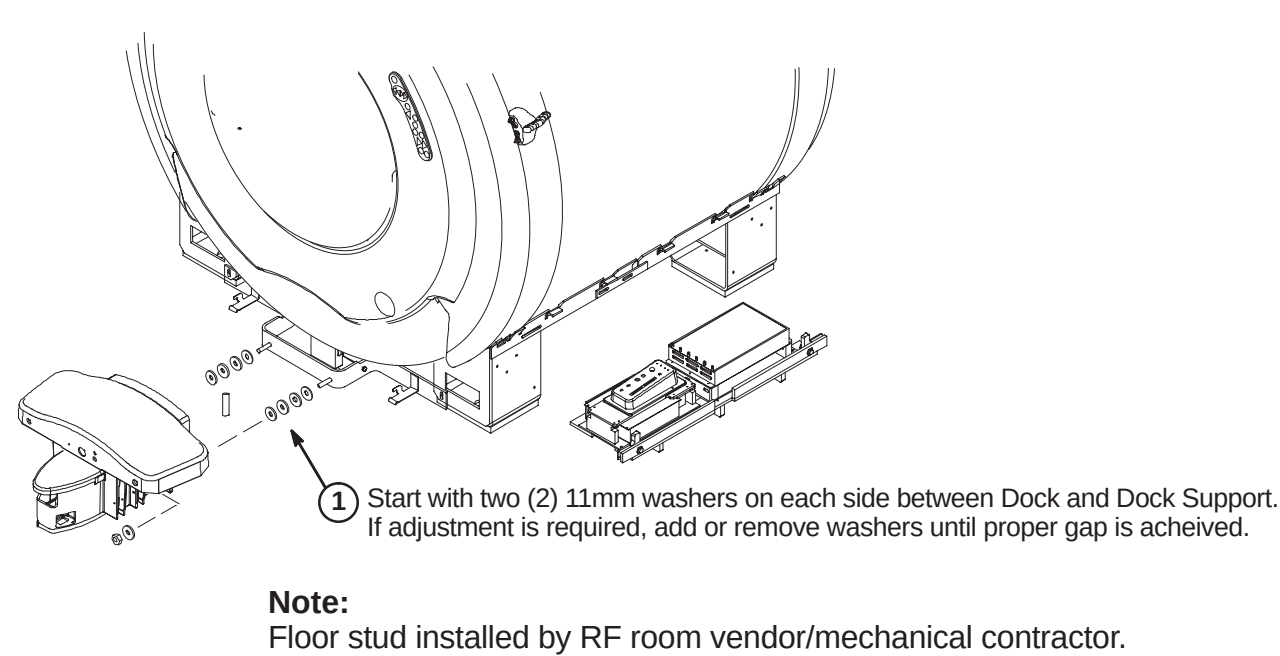
- J1 – Dock Adjustment for Patient Transport Gap
- J2 – Route and Connect Magnet Enclosure Front Area Cables
- J3 – Cable Connection to PAC Module

J1 – DOCK ADJUSTMENT FOR PATIENT TRANSPORT GAP



Note:
Gap between Patient Transport front rubber inlay and front of End Bell must be 0.75 Inches $\pm$ 0.25 (19.0 $\pm$ 6.4mm). See Step 1 below.

WARNING!
DOCK CONTAINS FERROUS MATERIAL.
HANDLE WITH CARE DURING INSTALLATION.



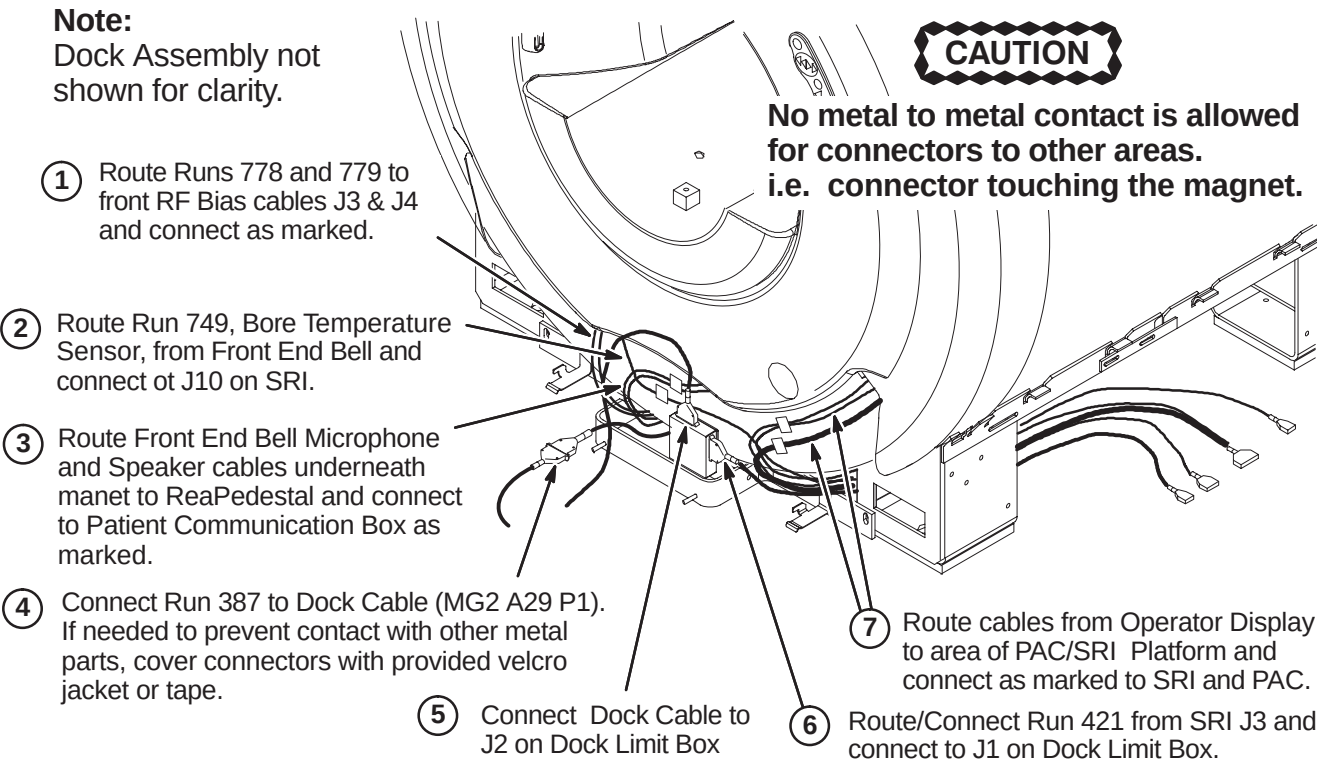
1 Start with two (2) 11mm washers on each side between Dock and Dock Support. If adjustment is required, add or remove washers until proper gap is achieved.

Note:
Floor stud installed by RF room vendor/mechanical contractor.

J2 – ROUTE AND CONNECT MAGNET ENCLOSURE FRONT AREA CABLES

Note:
Dock Assembly not shown for clarity.

CAUTION
No metal to metal contact is allowed for connectors to other areas. i.e. connector touching the magnet.



1 Route Runs 778 and 779 to front RF Bias cables J3 & J4 and connect as marked.

2 Route Run 749, Bore Temperature Sensor, from Front End Bell and connect at J10 on SRI.

3 Route Front End Bell Microphone and Speaker cables underneath magnet to ReaPedestal and connect to Patient Communication Box as marked.

4 Connect Run 387 to Dock Cable (MG2 A29 P1). If needed to prevent contact with other metal parts, cover connectors with provided velcro jacket or tape.

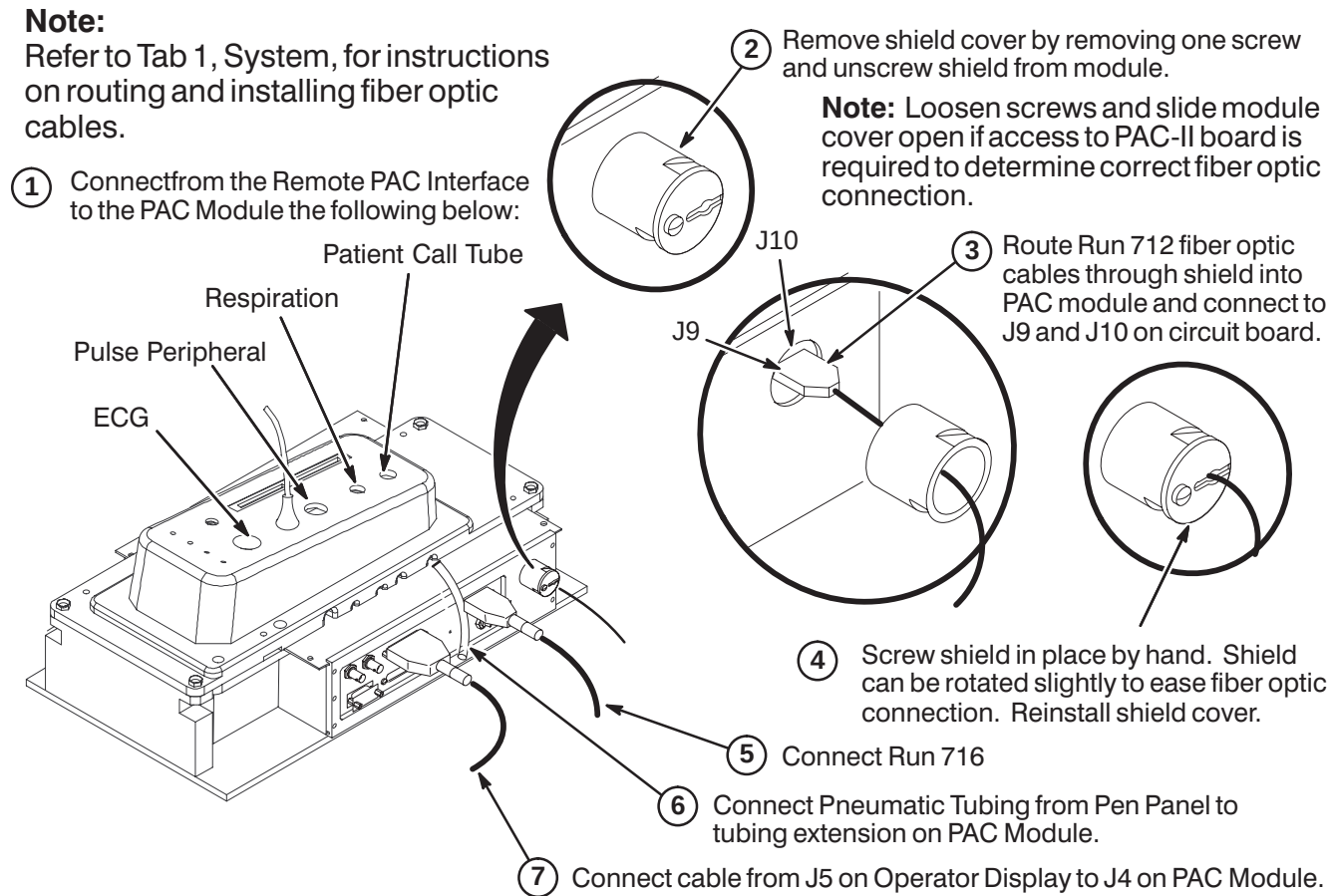
5 Connect Dock Cable to J2 on Dock Limit Box

6 Route/Connect Run 421 from SRI J3 and connect to J1 on Dock Limit Box.

7 Route cables from Operator Display to area of PAC/SRI Platform and connect as marked to SRI and PAC.

J3 – CABLE CONNECTION TO PAC MODULE

Note:
Refer to Tab 1, System, for instructions on routing and installing fiber optic cables.



1 Connect from the Remote PAC Interface to the PAC Module the following below:

2 Remove shield cover by removing one screw and unscrew shield from module.

Note: Loosen screws and slide module cover open if access to PAC-II board is required to determine correct fiber optic connection.

3 Route Run 712 fiber optic cables through shield into PAC module and connect to J9 and J10 on circuit board.

4 Screw shield in place by hand. Shield can be rotated slightly to ease fiber optic connection. Reinstall shield cover.

5 Connect Run 716

6 Connect Pneumatic Tubing from Pen Panel to tubing extension on PAC Module.

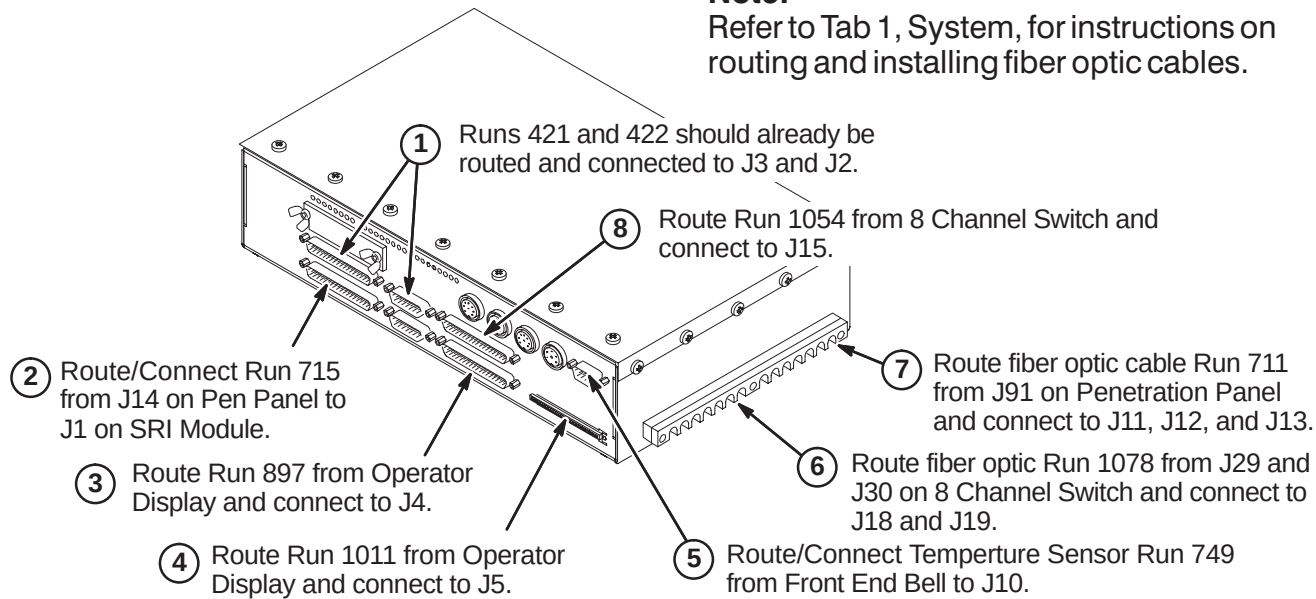
7 Connect cable from J5 on Operator Display to J4 on PAC Module.

INSTALLATION PROCEDURES SUMMARY

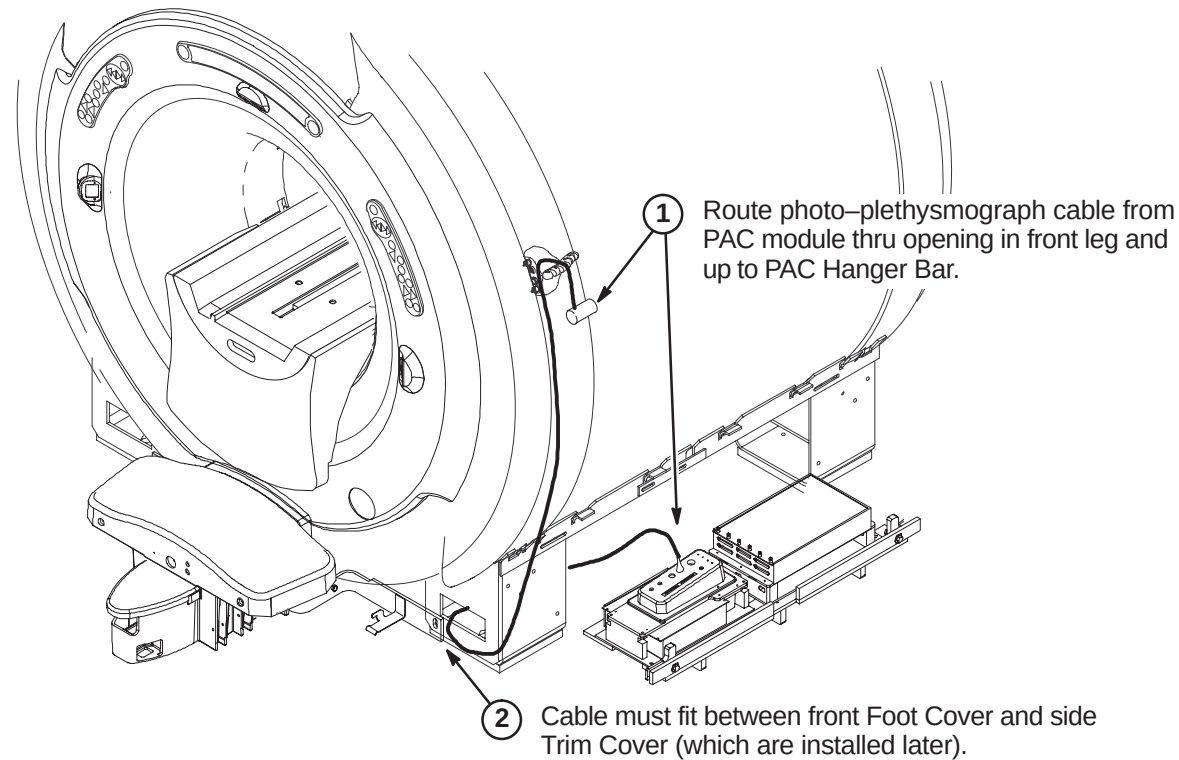
- ❑ K1 – Connect Cables to SRI 2 Module
- ❑ K2 – Routing of Photo-Plethysmograph Cable
- ❑ K3 – Route and Tie-wrap Cables to PAC/SRI Platform
- ❑ K4 – Install PAC/SRI Platform under Magnet

❑ K1 – CONNECT CABLES TO SRI 2 MODULE

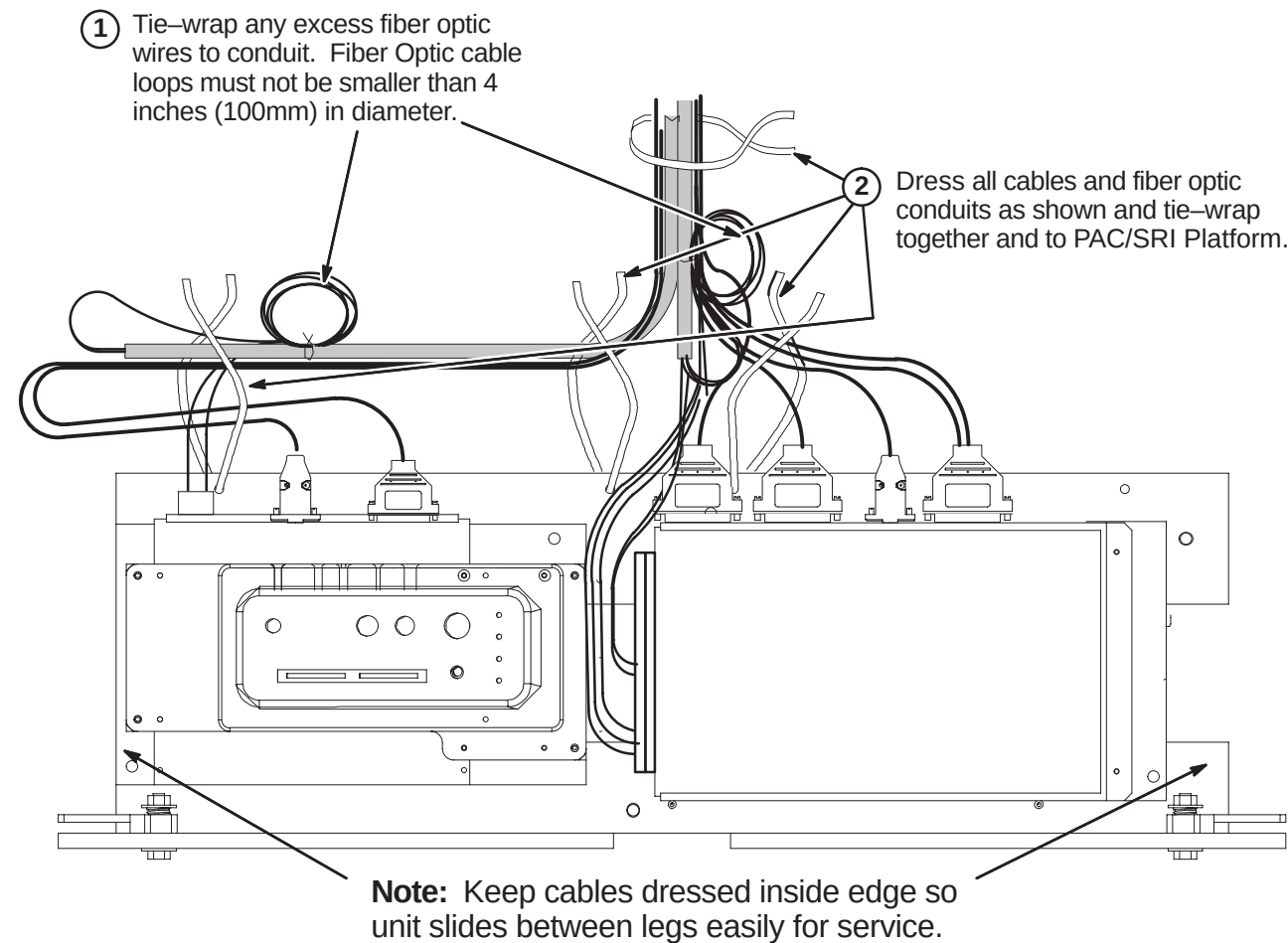
Note:
Refer to Tab 1, System, for instructions on routing and installing fiber optic cables.



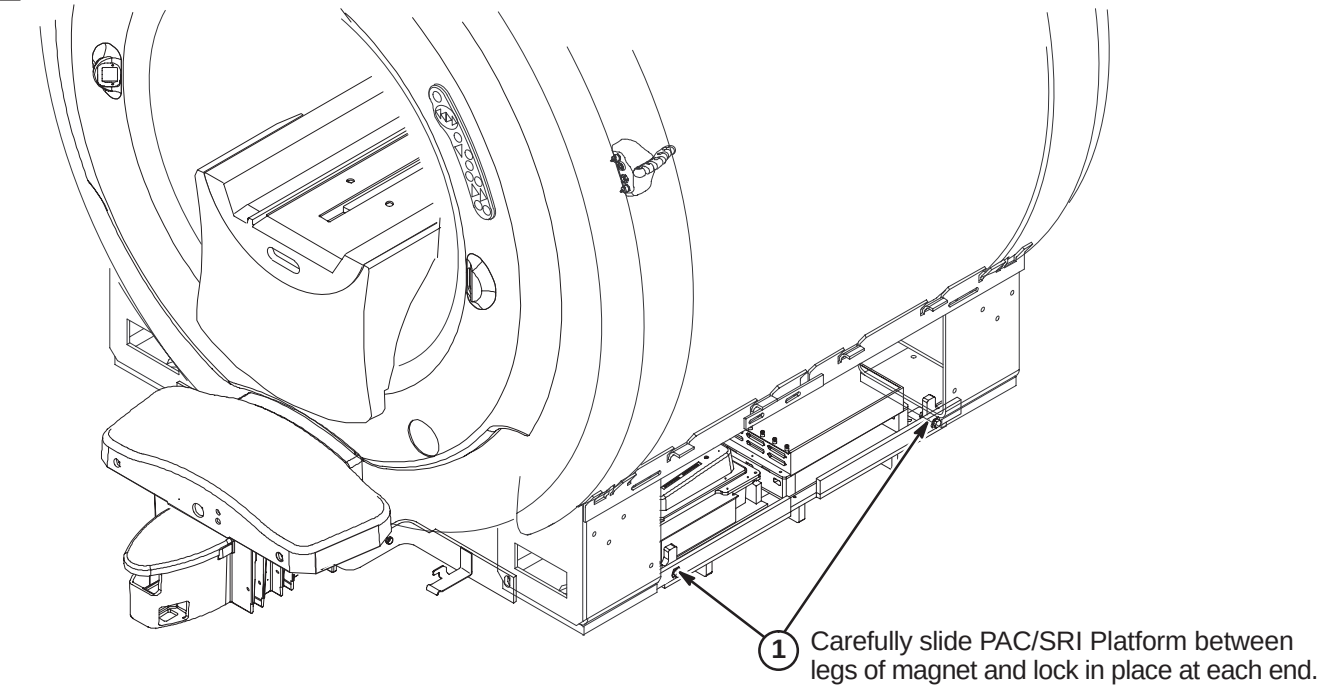
❑ K2 – ROUTING OF PHOTO-PLETHYSMOGRAPH CABLE



❑ K3 – ROUTE AND TIE-WRAP CABLES TO PAC/SRI PLATFORM



❑ K4 – INSTALL PAC/SRI PLATFORM UNDER MAGNET



INSTALLATION PROCEDURES SUMMARY

- ❑ L1 – Install Enclosure Base Trim Covers
- ❑ L2 – Install Enclosure Top Covers
- ❑ L3 – Install Rear Pedestal Covers

❑ L1 – INSTALL ENCLOSURE BASE TRIM COVERS

1 Slide front Foot Covers (2217310 & 2217308) from side, above spring clip, and between End Bell Flange and End Bell Spacers.

2 Install remaining side Base Trim Covers as shown.

Note: The gap from the bottom of the Base Trim to floor should be a nominal 0.50 inch (13mm).

❑ L2 – INSTALL ENCLOSURE TOP COVERS

Note: All enclosure seams and gaps visible to the operator and patient shall be less than 0.25 inches (6.35mm). This applies to the mating seam of the two Side Top Covers to the Front Top Cover

Gap uniformity along any continuous seam shall be ± 0.0625 inch (1.59mm)

1 Install front (2217457) and rear (2217458) Top Covers as shown using M10 x 25mm Low Profile hex screws and blue Loctite.

2 Install side Top Covers as shown.

2 Place a level against vertical surface of Top Covers. If top of covers are leaning towards magnet, cut a notch at this location and reinstall. Repeat as necessary until vertically level.

❑ L3 – INSTALL REAR PEDESTAL COVERS

5 Pedestal Covers require an opening to be cut out depending on routing of cables and hoses. Covers are marked in three places (left side, right side, or rear) for cutting.

6 After cutting out appropriate opening, hang covers on Rear Pedestal.

7 If covers need adjustment, loosen the screws at each end of the support bar and reposition bar as necessary.

8 Fasten covers together with furnished hardware.

1 Remove mating piece of velcro from Rear End Bell Cover Plates (2228970 & 2228970-2) and attach to Rear Pedestal Covers

2 Remove mating velcro and attach to back side of Arc Covers.

3 Attach Rear End Bell Cover Plates (2228970 & 2228970-2) to both sides of Rear Pedestal with furnished hardware – two M4 washers (2109879) and two M4 x 8mm Pan head screws (2109873-13) for each plate.

4 Install left (2287512) and right (2287511) Rear Bridge Covers. The left cover is installed over the vacuum connection on the end bell.

INSTALLATION PROCEDURES SUMMARY

- M1 – Install Laser Alignment Light Warning Labels

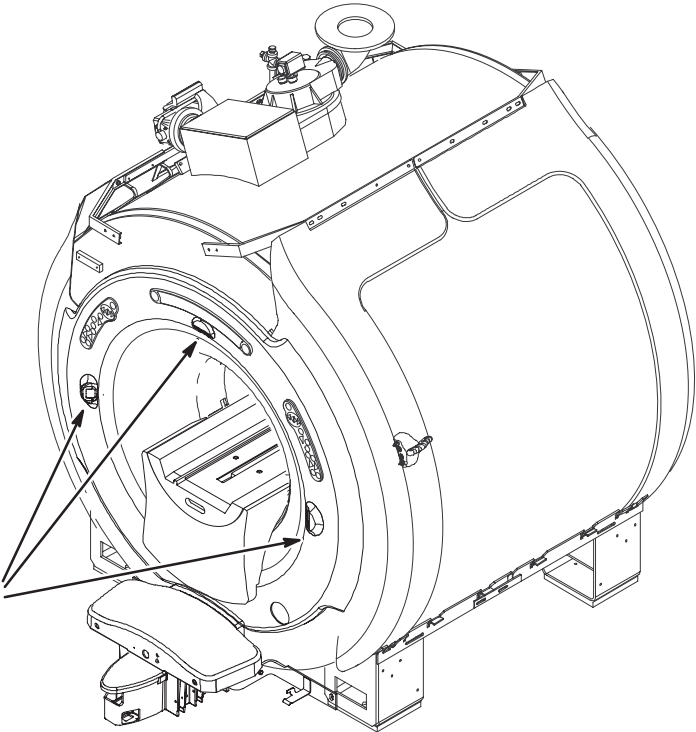
M1 – INSTALL LASER ALIGNMENT LIGHT WARINING LABELS

A sheet of Laser Alignment Light Warning Labels has been supplied with the magnet, attached to the magnet bridge.

The magnet enclosure is shipped with a label in English attached below each of the three laser lights.



English



- 1 For those sites which require a label other than English, peel off the appropriate label and affix over the English labels at the three laser light locations.



French



German



Portuguese



Spanish



Italian



Swedish

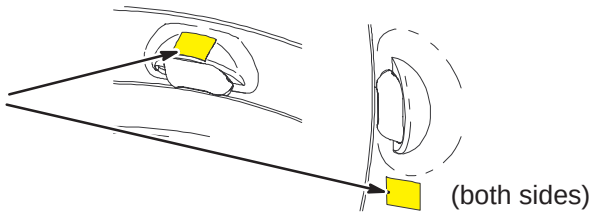


Japanese



Chinese

- 2 If enclosure does not have labels attached, attach new labels as indicated.



COOLING SYSTEM INSTALLATION

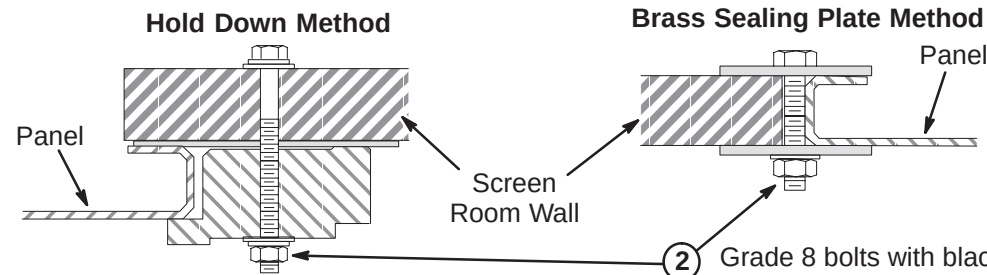
Consult vendor supplied manual, that was delivered with Cooling Cabinet, for Cooling Cabinet installation, start-up, and operating instructions and procedures.

INSTALLATION STEPS SUMMARY

- A – Remove Blank Panel/Prepare Screen Room Opening Surface
- B – Install Penetration Panel
- C – Install Penetration Panel Stand-Off Hardware
- D – Install Penetration Panel Covers

A – REMOVE BLANK PANEL/PREPARE SCREEN ROOM OPENING SURFACE

- 1 If still in place, remove vendor supplied blank panel. Save mounting hardware of either method shown below for installation of Penetration Panel.



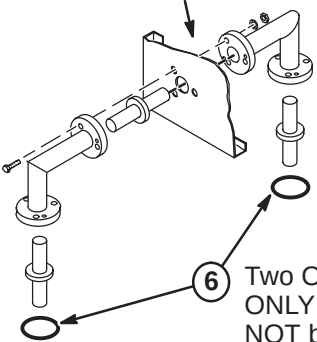
- 2 Grade 8 bolts with black oxide coating should be used as mounting hardware. Tighten to torque of 20 lb-ft (1.5 to 2 turns after lock washer flattens).
- 3 To insure good conductivity, use a fine sandpaper or scouring pad to clean surface of screen room wall which will be in contact with Penetration Panel. **Do not sand surface of Penetration Panel as it is coated.**

B – INSTALL PENETRATION PANEL

- 2 Install Penetration Panel using same method and hardware for mounting of blank panel above.
- 3 Seal all joints with copper foil tape with a conductive adhesive (3M #1181 or 46-258218P4, 5, or 6) for RF-tight integrity.
- 4 Fill inside of J60 and J61 guides with Bronze Wool (46-318068P1) that was supplied with magnet.

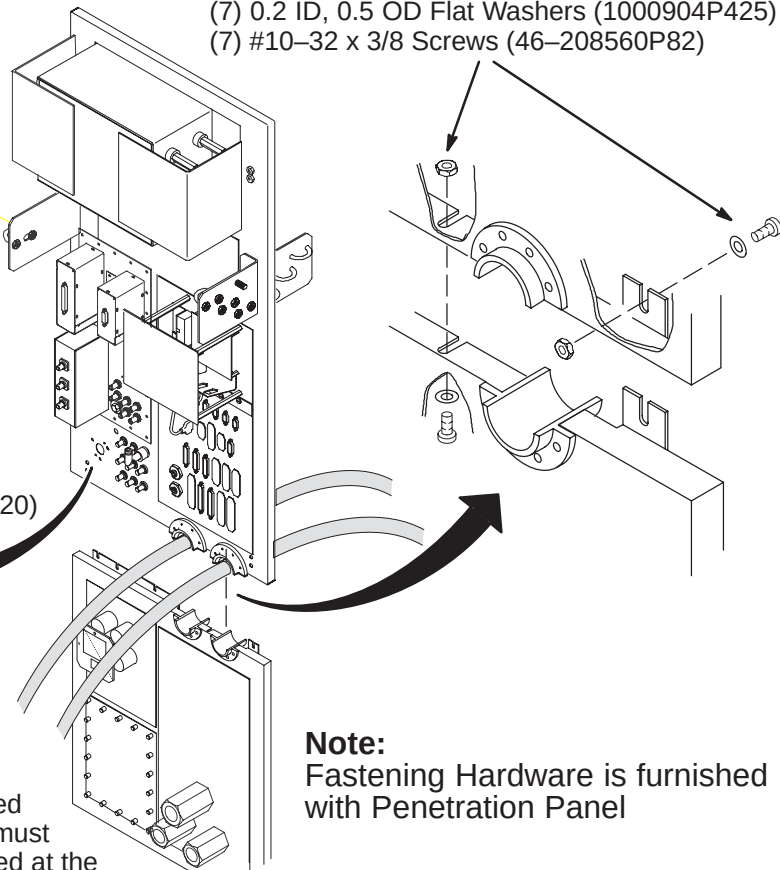
Note: Steps 5 and 6 are not performed if Times Microwave RF cables are to be installed.

- 5 Attach HN Right Angle Flanged Connectors (46-230013P1) to Penetration Panel with:
- (3) 1/4-20 X 7/16 Nuts (46-208935P8)
 - (3) 1/4 Flat Washers (46-221449P6)
 - (3) 1/4-20 X 1-1/4 Screws (46-208939P20)



- 6 Two O-rings to be installed ONLY as shown. There must NOT be an O-ring installed at the Penetration Panel connection.

- 1 Assemble Penetration Panel around helium lines (Runs 621 and 622) using:
- (7) #10-32 Nylock Lock Nuts (46-208937P6)
 - (7) 0.2 ID, 0.5 OD Flat Washers (1000904P425)
 - (7) #10-32 x 3/8 Screws (46-208560P82)



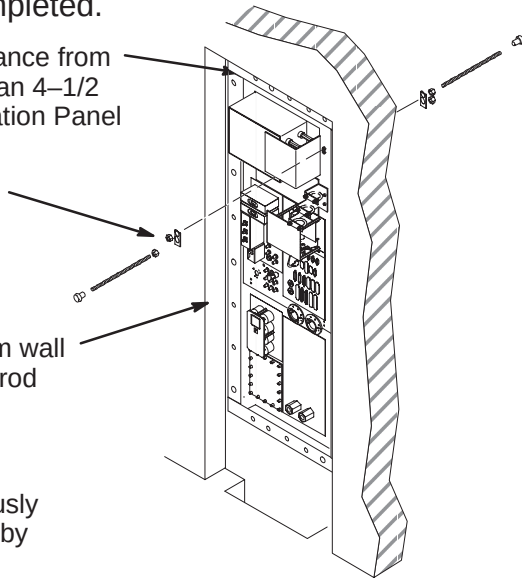
Note: Fastening Hardware is furnished with Penetration Panel

C – INSTALL PENETRATION PANEL STAND-OFF HARDWARE

Note: The Penetration Panel Floor plate that is supplied with the cover kit is to be installed prior to routing and connecting cables to Penetration Panel. Refer to Installation Drawing furnished with the Cover Kit.

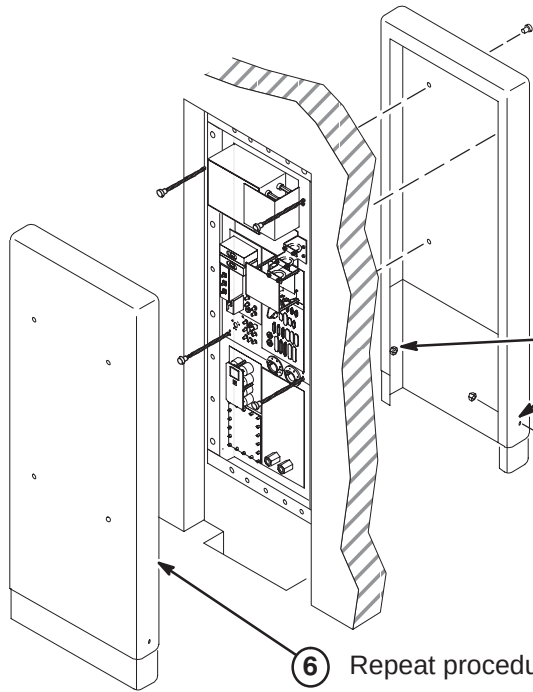
Note: The Penetration Panel Cover can be installed only after all connections and System installation has been completed.

- 1 Verify that cover will enclose finished wall opening. The distance from the top of Penetration Panel to wall opening must be less than 4-1/2 inches (114mm). If not, new mounting stud holes in Penetration Panel may be required.
- 2 Place (8) Reinforcing Plates (46-301211P1) over each hole pair on both sides of Penetration Panel and attach using:
- (16) 5/16-18 x 9/16 Brass Nut (46-208935P9)
 - (8) 5/16-18 x 18 inch Threaded Rod (46-301213P2)
 - (8) 5/16-18 x 1 inch Barrel Nut (46-301212P1)
- 3 Measure distance from face of Pen Panel to equipment room wall surface. Add six inches (152mm) to obtain length threaded rod should extend from panel.
- 4 Repeat step 3 procedure for exam room cover.
- 5 Adjust threaded rods with attached barrel nuts to the previously calculated dimensions from steps 3 and 4. Secure to panel by tightening brass nuts on both sides of Penetration Panel.



D – INSTALL PENETRATION PANEL COVERS

- 1 Remove all barrel nuts.
- 2 In the Magnet room, temporarily install Top (46-306936P1) and Bottom (46-306964P1) Cover assembly on threaded rods and secure with four barrel nuts.
- 3 Adjust position of bottom cover section and mark two lower fastener holes.
- 4 Remove covers and drill hole in lower cover at each mark.
- 5 Attach top and bottom covers with furnished 10-32 nuts and screws.
- 6 Repeat procedure for Equipment room cover.
- 7 After all installation procedures are completed in Magnet and Equipment rooms, install assembled covers on threaded rods and secure with four barrel nuts.



SYSTEM CABINET INSTALLATION INSTRUCTIONS

1-1 **Tab 8, System, contains installation instructions for the following:**

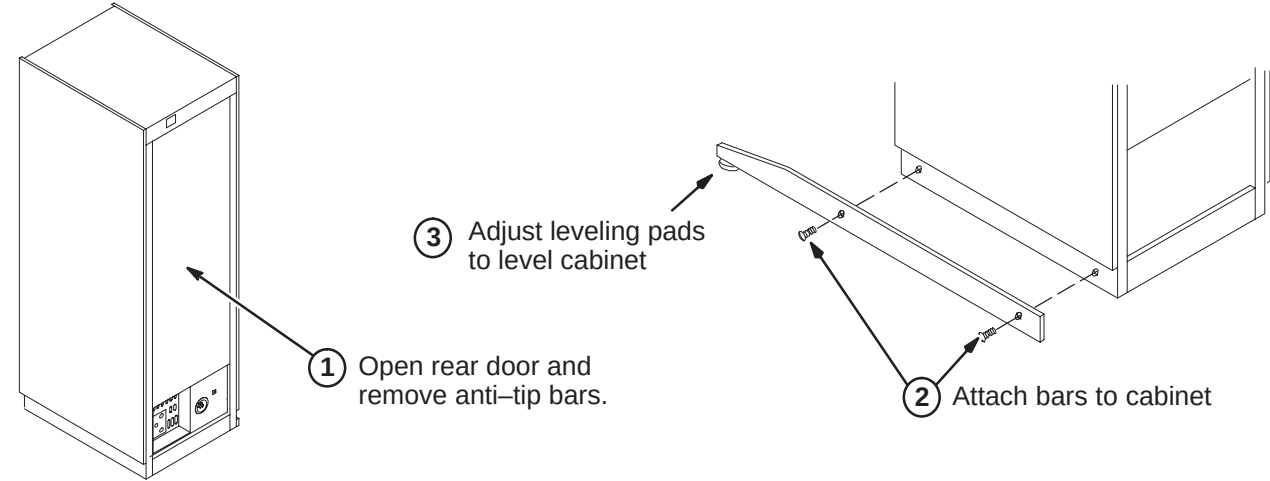
- *DIRECTION 2217004* – 8.x/9.x System Cabinet Installation Instructions
- *DIRECTION 2338502* – Signa EXCITE I System Cabinet Installation Instructions
- *DIRECTION 2398344* – Signa EXCITE II System Cabinet Installation Instructions

INSTALLATION STEPS SUMMARY

- ☐ A – Position/Secure System Cabinet
- ☐ B – Install Magnet Monitoring Cables
- ☐ C – Connect Fiber Optic Runs 708, 710, 711/712, and 836 to I/F Panel
- ☐ D – Route/Connect Fiber Optic Runs 836 to BIT3 Module
- ☐ E – Route/Connect Fiber Optic Runs 708, 710, and 711/712 to Tyme II
- ☐ F – Connect Power, Ground, and Data Cables
- ☐ G – Install InSite “SYSTEMS” Cabinet Magnet

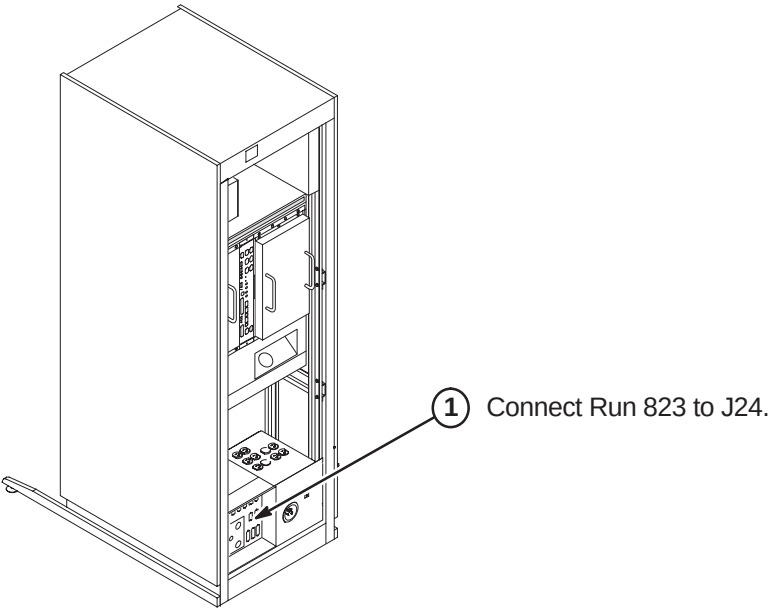
☐ A – POSITION/SECURE SYSTEM CABINET

Comply with UL requirements by either installing anti tip legs (see illustration below) or securing the cabinet to the floor with local supplied brackets and floor anchors.



☐ B – INSTALL MAGNET MONITORING CABLES

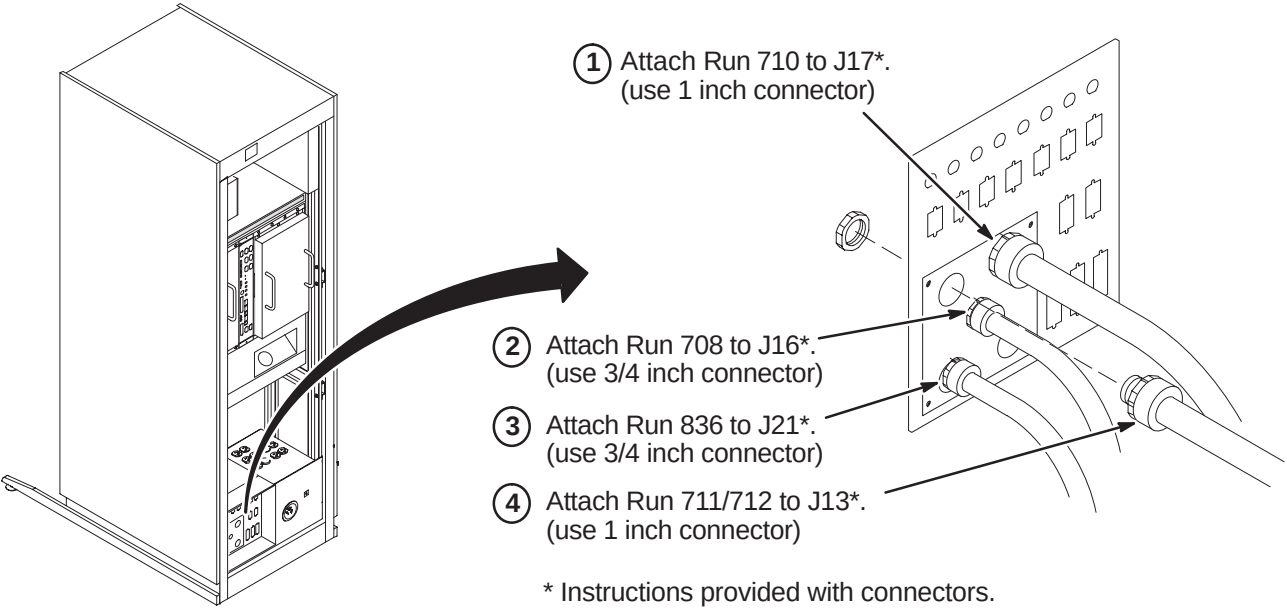
Note: Refer to Direction 2230681, *Magnet Monitor Hardware Installation Manual*, for instructions on connecting cables in the System Cabinet. This manual should be found in the box containing the parts for Magnet Monitoring.



☐ C – CONNECT FIBER OPTIC RUNS 708, 710, 711/712, AND 836 TO I/F PANEL

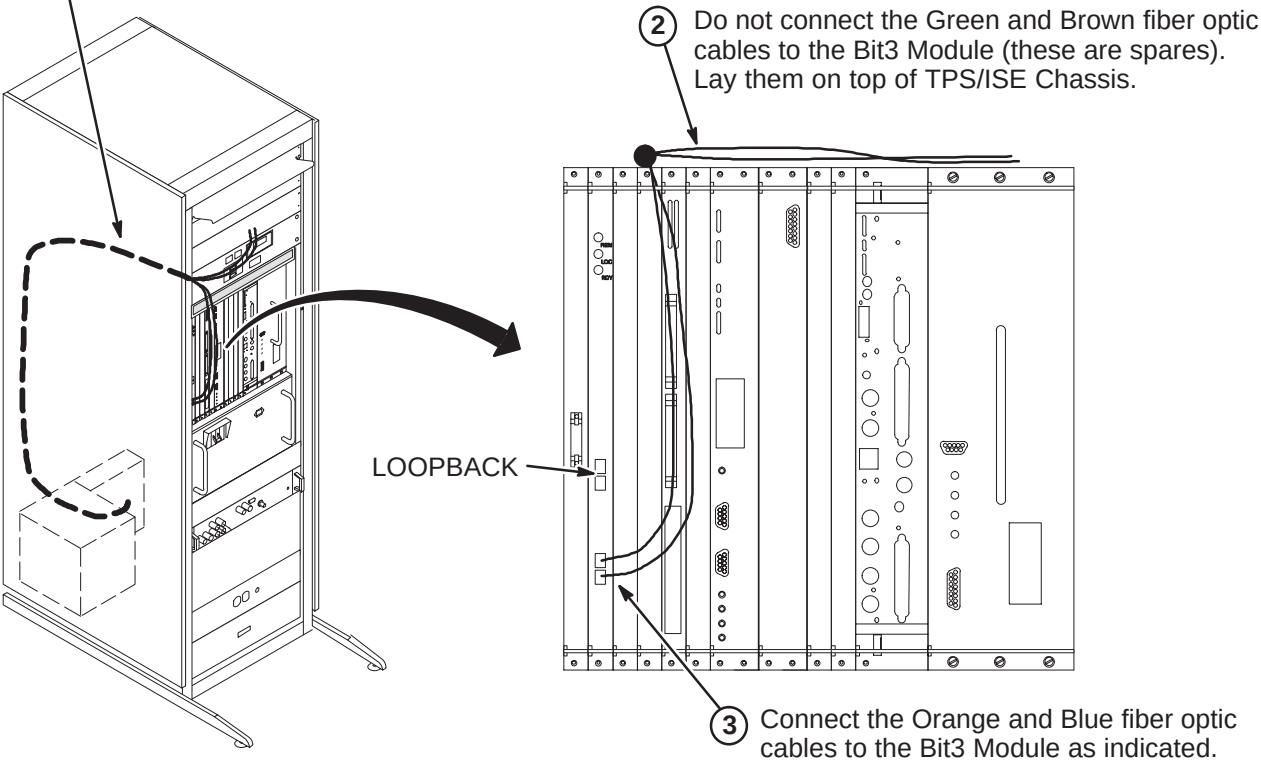
CAUTION

Handle fiber optic cables carefully. Do not bend fiber optic cables to radius smaller than two inches (50mm). Avoid scratching connector ends. Keep connectors protected until ready to connect.



☐ D – ROUTE/CONNECT FIBER OPTIC RUN 836 TO BIT3 MODULE

- 1 Route Run 836 fiber optic cables from cabinet I/F J21 to front of Bit3 module (MR2 A11 A30). Secure cable to side of cabinet. Do not bend fiber optic cables to radius smaller than two inches (50mm).



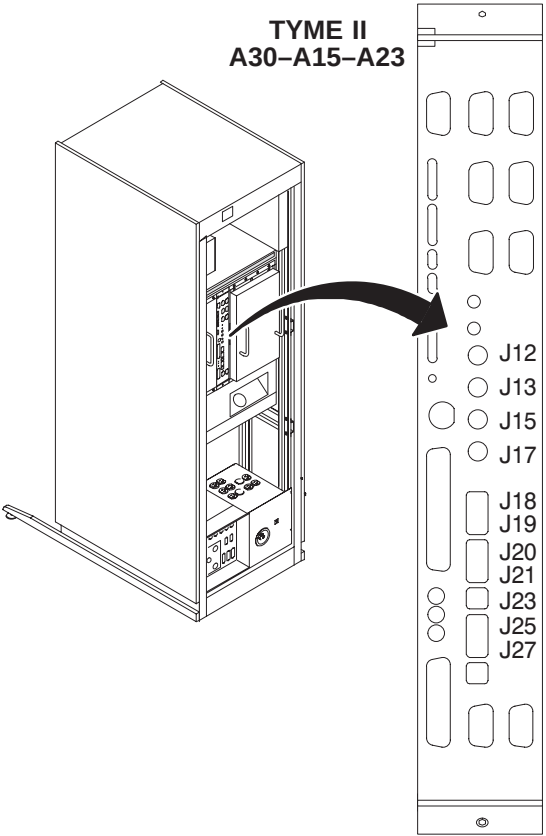
E – ROUTE/CONNECT FIBER OPTIC RUNS 708, 710, AND 711/712 TO TYME II

CAUTION

Handle fiber optic cables carefully. Do not bend fiber optic cables to radius smaller than two inches. Avoid scratching connector ends. Keep connectors protected until ready to connect. Do not coil naked outside of cabinet, or store fiber optic cables in bottom of cabinet.

Note:
Runs 708, 710, 711, and 712 are used for both Horizon 5.x and 8.x. The fiber optic connections to the TYME board are different for 5.x and 8.x. The labels on the cables include both the 5.x and 8.x connections.
The table below indicates the connection labels as they should be marked on this end of the fiber optic cables. **Use only the 8.x connections.**

RUN #	5.x	8.x
708-1	J5	J19
710-1	J13	J12
710-2	J15	J13
710-3	J17	J15
710-4	J14	J17
710-5	J4	J18
711-1	J6,J7	J25,J27
711-2	J8	J23
712-1	J9,J10	J20,J21

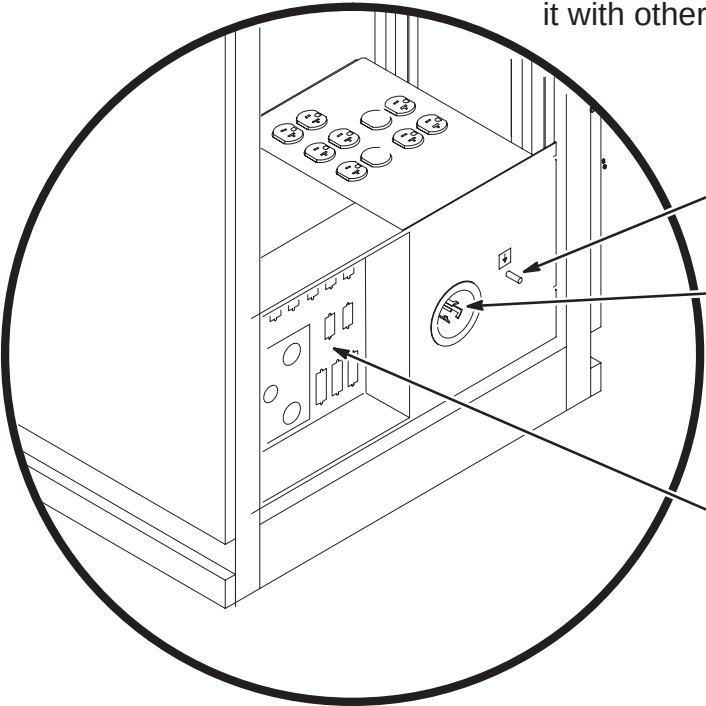


- 1 Route/Connect as marked Run 710 fiber optic cables to J12, J13, J15, J17 and J18 on the TYME II Board.
- 2 Route/Connect as marked Run 708 fiber optic cable to J19 on the TYME II Board.
- 3 Route/Connect as marked Run 711/712 fiber optic cables to J20, J21, J23, J25, and J27 on the TYME II Board.

Note:
Use tie-wraps to secure fiber optic cables to ISE Chassis cables.

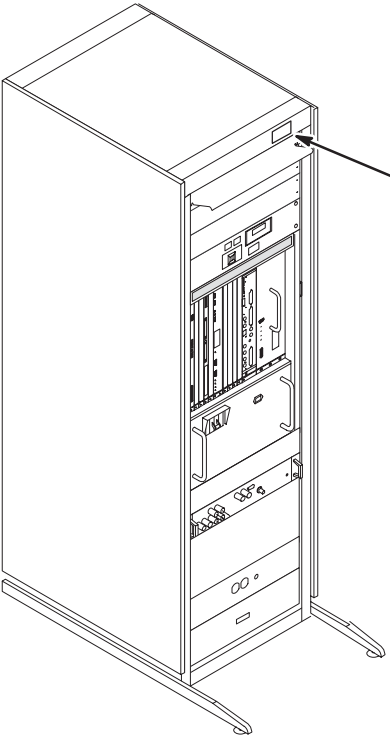
F – CONNECT POWER, GROUND, AND DATA CABLES

Note:
Locate CERD Test Cable, 2168505 (it is packed with the Operator Workspace cable kit, 2154392). Store it with other test cables or in the System Cabinet.



- 1 Connect Run 037 GND cable to ground stud.
- 2 Connect Run 030 or 968 cable plug to "J1". Be sure to push and turn plug for secure connection.
- 3 Perform Ground Resistance Checks before connecting remaining cabinet cables.
- 4 Connect remaining cabinet cables as marked.

G – INSTALL INSITE "SYSTEMS" CABINET MAGNET



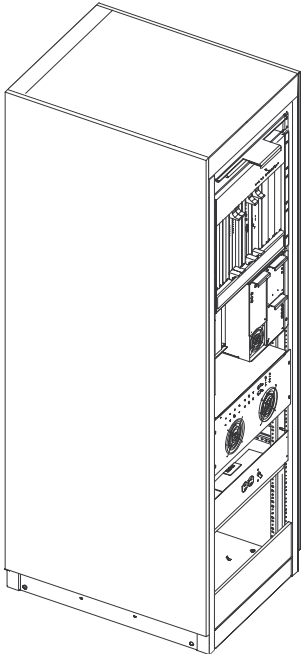
- 1 From the 8.0 LX Insite Kit (46-301708G14), locate the "Systems" cabinet magnet (46-320095P5) and attach to front of cabinet as shown.

INSTALLATION STEPS SUMMARY

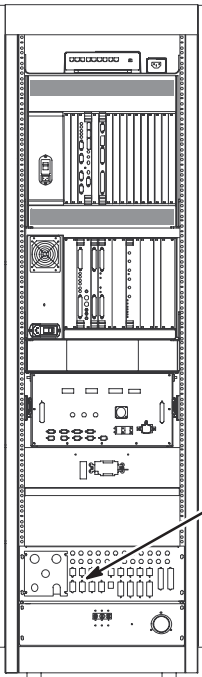
- ☐ A – Position/Secure System Cabinet
- ☐ B – Install Magnet Monitoring Cables
- ☐ C – Connect Fiber Optic Runs 1033, 1034, 1044, and 1045 to I/F Panel
- ☐ D – Route/Connect Fiber Optic Run 1044 to APS Board
- ☐ E – Route/Connect Fiber Optic Runs 1033, 1034, and 1045 to STIF Board
- ☐ F – Connect Power, Ground, and Data Cables
- ☐ G – Install InSite “SYSTEMS” Cabinet Magnet

A – POSITION/SECURE SYSTEM CABINET

Comply with UL requirements or local codes for securing the cabinet to the floor with locally supplied brackets and floor anchors.



B – INSTALL MAGNET MONITORING CABLES



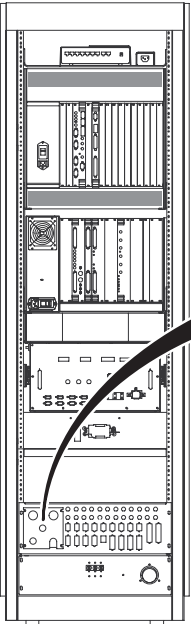
Note: Refer to Direction 2230681, *Magnet Monitor Hardware Installation Manual*, for instructions on connecting cables in the System Cabinet. This manual should be found in the box containing the parts for Magnet Monitoring.

- 1 Connect Run 823 to J24.

C – CONNECT FIBER OPTIC RUNS 1033, 1034, 1044, AND 1045 TO I/F PANEL

CAUTION

Handle fiber optic cables carefully. Do not bend fiber optic cables to radius smaller than two inches (50mm). Avoid scratching connector ends. Keep connectors protected until ready to connect.

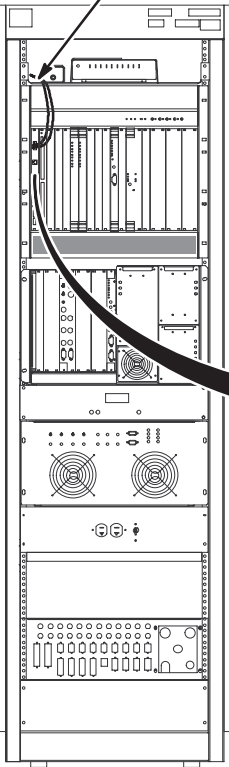


- 1 Attach Run 1034 to J17*.
(use 1 inch connector)
- 2 Attach Run 1045 to J13*.
(use 1 inch connector)
- 3 Attach Run 1033 to J16*.
(use 3/4 inch connector)
- 4 Attach Run 1044 to J22*.
(use 3/4 inch connector)

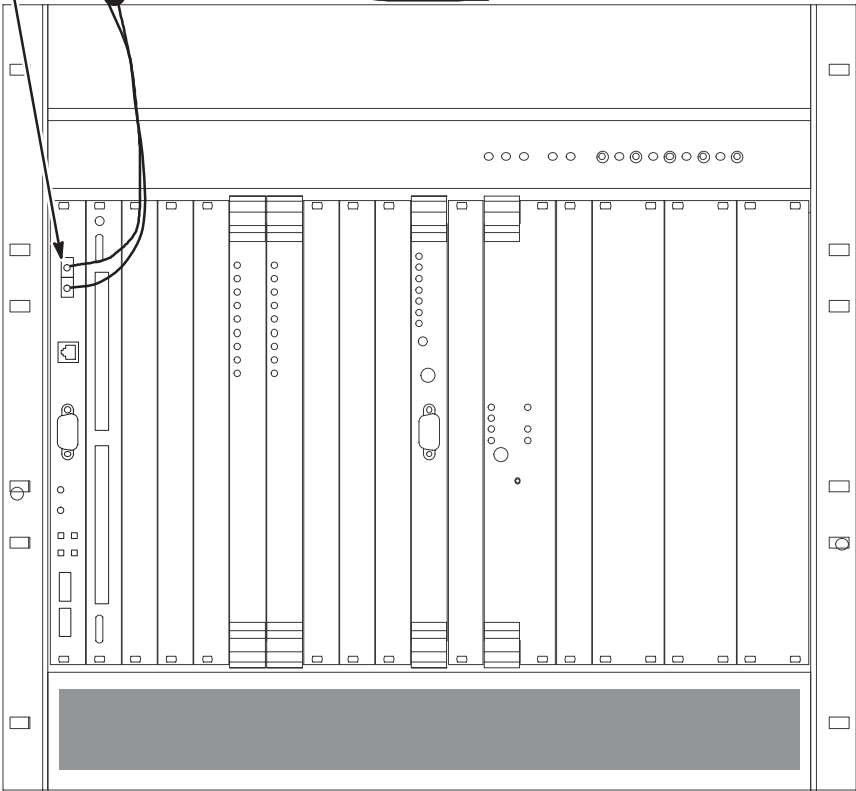
* Instructions provided with connectors.

D – ROUTE/CONNECT FIBER OPTIC RUN 1044 TO APS BOARD

- 1 Route Run 1044 fiber optic cables from cabinet I/F J22 to front of APS module. Secure cable to side of cabinet. Do not bend fiber optic cables to radius smaller than two inches (50mm).



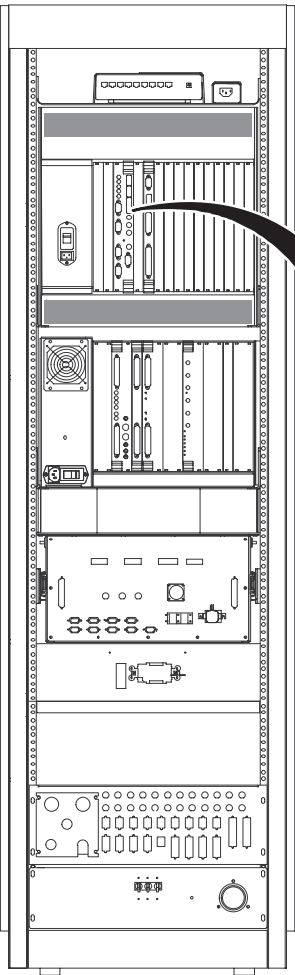
- 2 Connect the Orange and Blue fiber optic cables to the TI-MGD Daughter Board as indicated.
- 3 Do not connect the Green and Brown fiber optic cables to the APS Module (these are spares). Lay them on top of MGD Chassis.



E – ROUTE/CONNECT FIBER OPTIC RUNS 1033, 1034, AND 1045 TO STIF BOARD

CAUTION

Handle fiber optic cables carefully. Do not bend fiber optic cables to radius smaller than two inches. Avoid scratching connector ends. Keep connectors protected until ready to connect. Do not coil naked outside of cabinet, or store fiber optic cables in bottom of cabinet.



NOTE:
Refer to table at right for relabel information if existing cables are to be reused.

New Cables		Existing Cables	
RUN #	10.x	RUN #	8.x/9.x
1033-1	J21	708-1	J19
1034-1	J14	710-1	J12
1034-2	J13	710-2	J13
1034-3	J12	710-3	J15
1034-4	J11	710-4	J17
1034-5	J20	710-5	J18
1045-1	J17	711-2	J23
1045-2	J18,J19	711-3	J20, J21
1045-3	J15,J16	711-1	J25, J27

- ①

Relabel (if old cable re-used) and Route/Connect Run 1033 fiber optic cable to J21 on the STIF Board.
- J21

J20

J19

J18

J17

J16

J15

②

Relabel (if old cable re-used) and Route/Connect Run 1045 fiber optic cables to J15, J16, J17, J18 and J19 on the STIF Board.
- J14

J13

J12

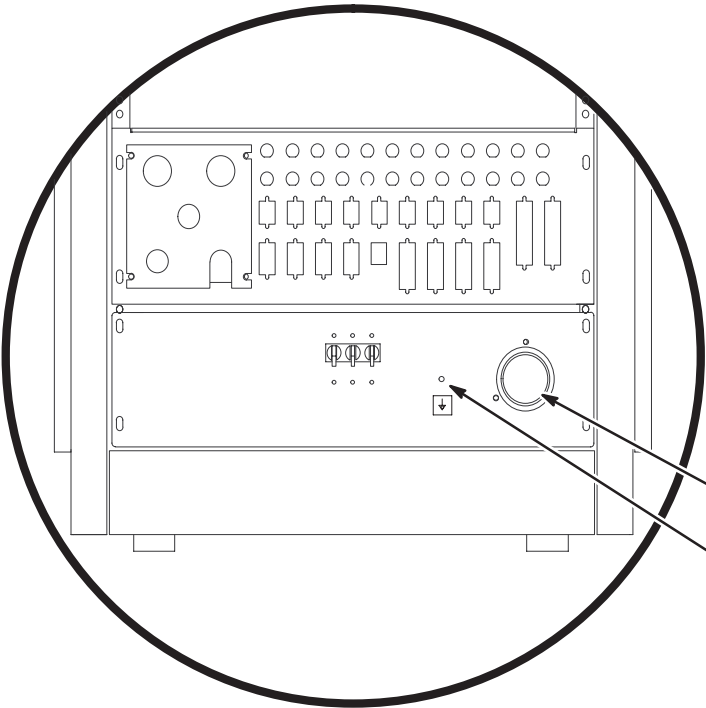
③

Relabel (if old cable re-used) and Route/Connect Run 1034 fiber optic cables to J11, J12, J13, J14, and J20 above on the STIF Board.
- J11

Note:
Use tie-wraps to secure fiber optic cables to MGD Chassis cables.

F – CONNECT POWER, GROUND, AND DATA CABLES

Note:
Locate CERD Test Cable, 2168505 (it is packed with the Operator Workspace cable kit, 2154392). Store it with other test cables or in the System Cabinet.



WARNING!

SYSTEM CABINET POWER AND GROUND CABLES MUST BE CONNECTED TO PDU MODULE LOCATED WITHIN ACGD/PDU CABINET.

- ①

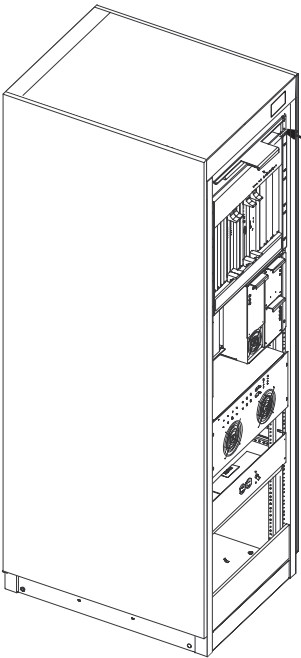
Connect Run 968 cable plug to “J1”. Be sure to push and turn plug for secure connection.
- ②

Connect Run 037 GND cable to ground stud.
- ③

Perform Ground Resistance Checks before connecting remaining cabinet cables.
- ④

Connect remaining cabinet cables as marked.

G – INSTALL INSITE “SYSTEMS” CABINET MAGNET



- ①

From the LX Insite Kit (46-301708G14), locate the “Systems” cabinet magnet (46-320095P5) and attach to front of cabinet as shown.

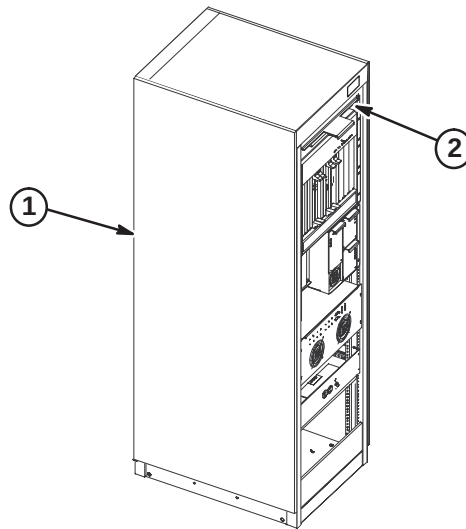
TABLE OF CONTENTS

<u>SECTION</u>	<u>PAGE</u>
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1-2 SYSTEM CABINET POWER AND GROUND CABLE CONNECTIONS	1-2
1-3 CONNECT FIBER OPTIC RUNS TO SYSTEM CABINET I/F PANEL	1-3
1-4 ROUTE/CONNECT FIBER OPTIC RUN 1083 TO APS BOARD	1-4
1-5 ROUTE/CONNECT FIBER OPTIC RUNS 1033, 1034, AND 1045 TO STIF BOARD	1-5
1-6 CONNECT COMMUNICATION CABLES TO SYSTEM CABINET I/F	1-6

1-1 POSITION AND SECURE SYSTEM CABINET

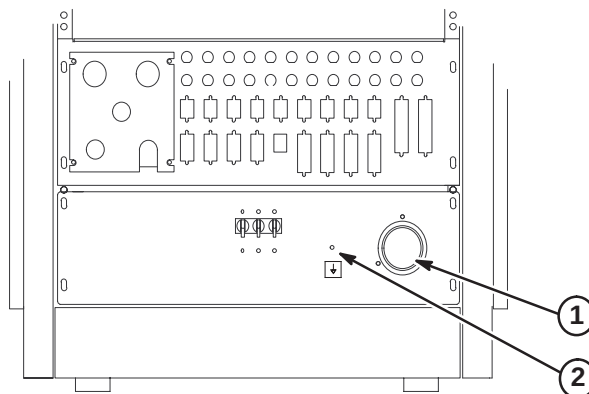
Comply with UL requirements or local codes for securing the cabinet to the floor with locally supplied brackets and floor anchors.

1. Position System cabinet to final position as indicated by site layout drawings.
2. From the LX Insite Kit (46-301708G14), locate the "Systems" cabinet magnet (46-320095P5) and attach to front of cabinet as shown.



1-2 SYSTEM CABINET POWER AND GROUND CABLE CONNECTIONS

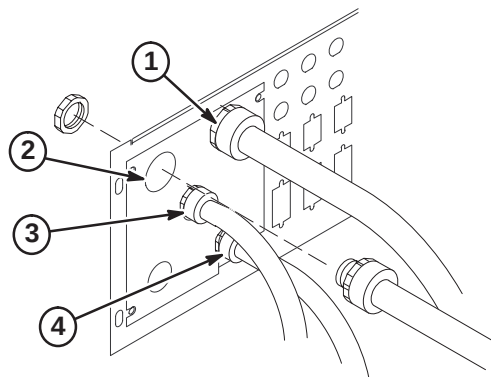
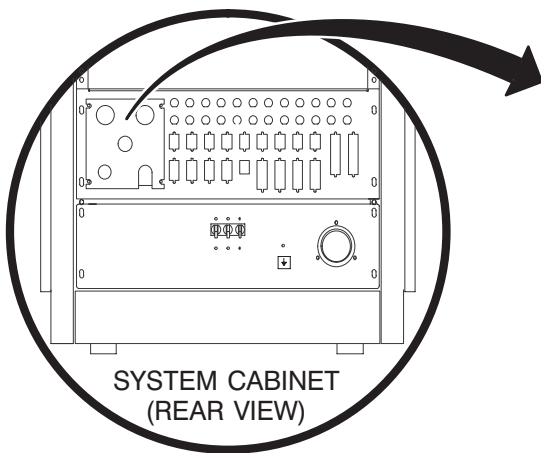
1. Connect Run 968 cable plug to "J1". Be sure to push and turn plug for secure connection.
2. Connect Run 037 GND cable to ground stud.
3. Perform Ground Resistance Checks before connecting remaining cabinet cables.



1-3 CONNECT FIBER OPTIC RUNS 1033, 1034, 1045, AND 1083 TO SYSTEM CABINET I/F PANEL**CAUTION**

Handle fiber optic cables carefully. Do not bend fiber optic cables to radius smaller than two inches (50mm). Avoid scratching connector ends. Keep connectors protected until ready to connect.

1. Attach Run 1034 to J17*. (use 1 inch connector)
2. Attach Run 1045 to J13*. (use 1 inch connector)
3. Attach Run 1033 to J16*. (use 3/4 inch connector)
4. Attach Run 1083 to J22*. (use 3/4 inch connector)

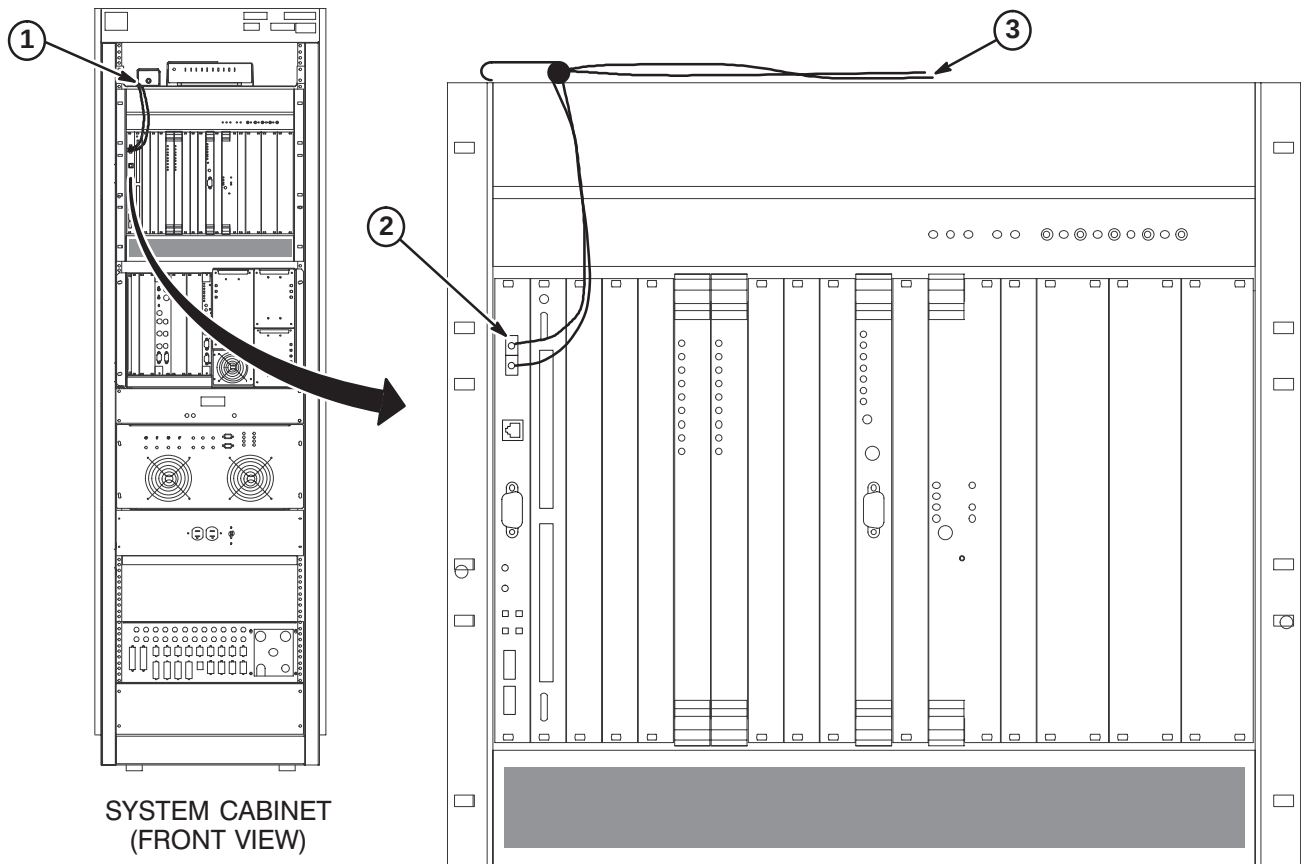


* Instructions provided with connectors.

1-4 ROUTE/CONNECT FIBER OPTIC RUN 1083 TO APS BOARD**CAUTION**

Handle fiber optic cables carefully. Do not bend fiber optic cables to radius smaller than two inches (50mm). Avoid scratching connector ends. Keep connectors protected until ready to connect.

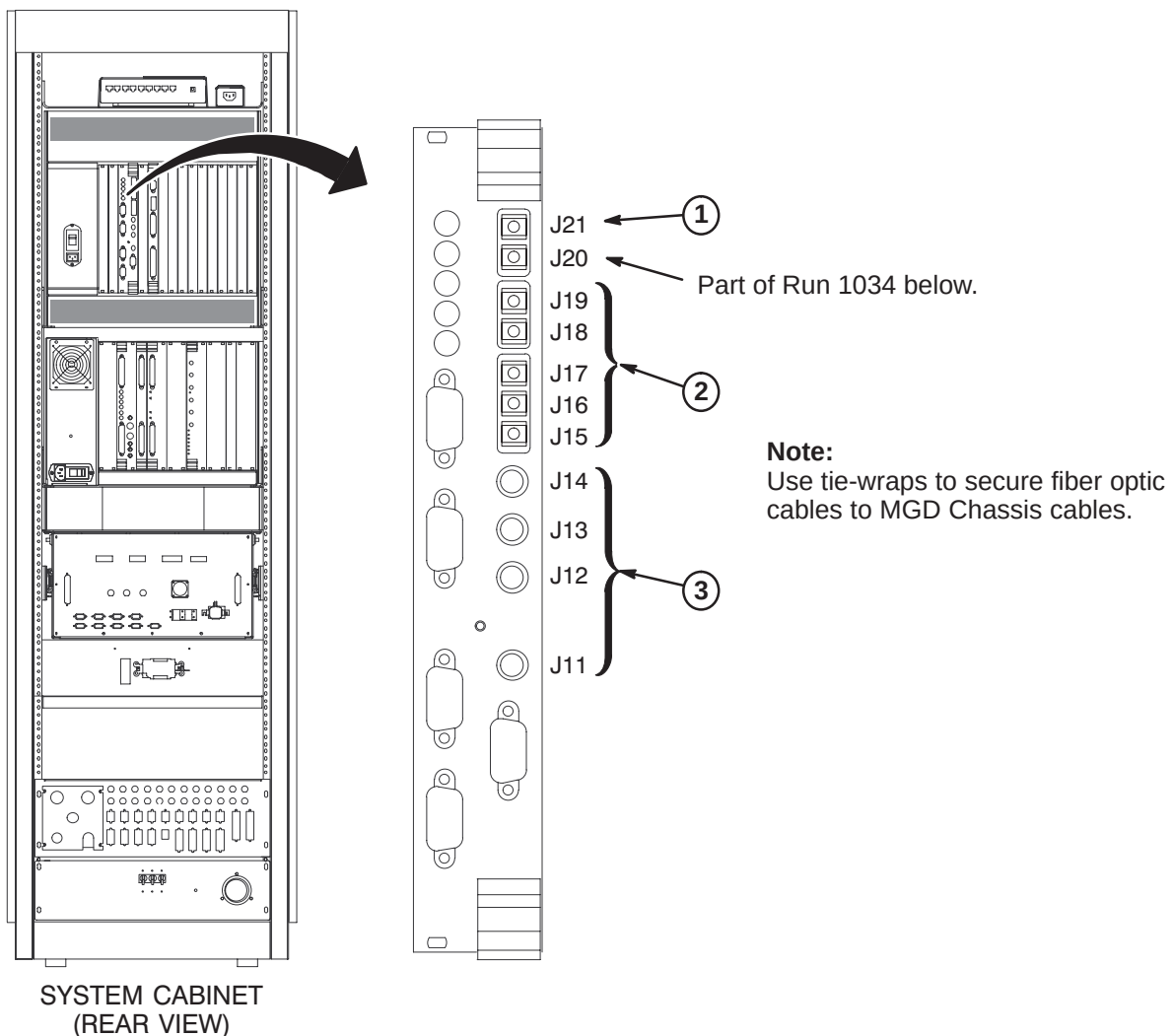
1. Route Run 1083 fiber optic cables from cabinet I/F J22 to front of APS module. Secure cable to side of cabinet. **Do not bend fiber optic cables to radius smaller than two inches (50mm).**
2. Connect the Orange and Blue fiber optic cables to the TI-MGD Daughter Board as indicated.
3. Do not connect the Green and Brown fiber optic cables to the APS Module (these are spares). Lay them on top of MGD Chassis.



1-5 ROUTE/CONNECT FIBER OPTIC RUNS 1033, 1034, AND 1045 TO STIF BOARD**CAUTION**

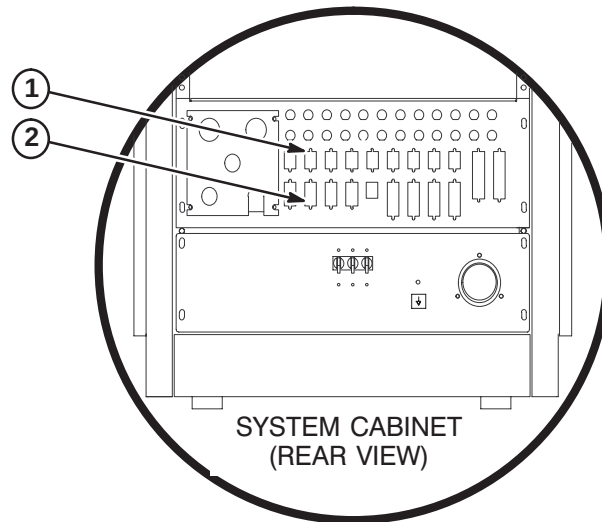
Handle fiber optic cables carefully. Do not bend fiber optic cables to radius smaller than two inches (50mm). Avoid scratching connector ends. Keep connectors protected until ready to connect.

1. Route/Connect Run 1033 fiber optic cable to J21 on the STIF Board.
2. Route/Connect Run 1045 fiber optic cables to J15, J16, J17, J18 and J19 on the STIF Board.
3. Route/Connect Run 1034 fiber optic cables to J11, J12, J13, J14, and J20 above on the STIF Board.



1-6 CONNECT COMMUNICATION CABLES TO SYSTEM CABINET I/F

1. Connect RF Door Switch cable Run 701 to J15
2. Connect Magnet Monitor cable Run 823 to J24
3. Connect remaining cables as marked. Refer to cable map in Cable section of Overview tab.



ACCESSORIES AND OPTIONS INSTALLATION INSTRUCTIONS

1-1 **Tab 11, Accessories & Options, contains installation instructions for the following:**

- *DIRECTION 2123158* – Patient Transport Installation Instructions
- *DIRECTION 2231762* – Signa WideOpen Pneumatic Patient Alert Installation

PATIENT TRANSPORT INSTALLATION INSTRUCTIONS

INSTALLATION STEPS SUMMARY

- ☐ A – Patient Transport Installation
- ☐ B – Head Coil Holder
- ☐ C – Head Coil Adapters

☐ A – PATIENT TRANSPORT INSTALLATION

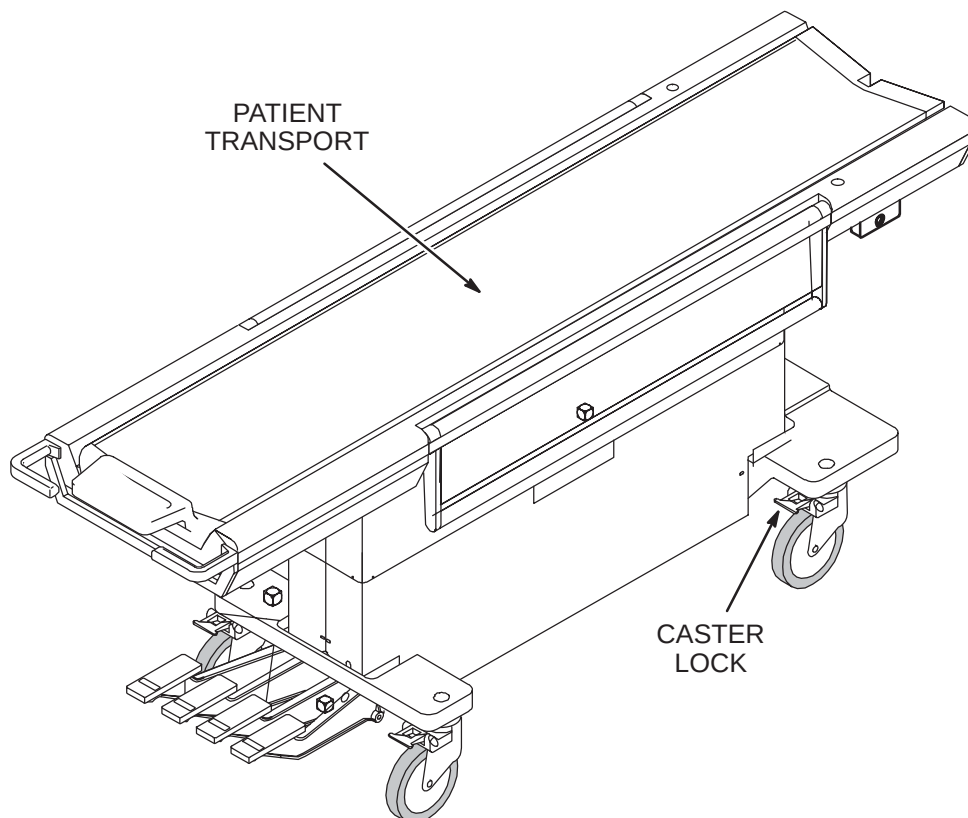
Patient Transport is shipped covered with protective cardboard which should remain in place until ready to be moved into the magnet room. Cover table with plastic previously removed when checks and adjustments are completed.

1. Unlock casters and move Patient Transport to magnet room for adjustments and function checks at the dock.
2. Remove protective plastic sheet from table.
3. Perform mechanical procedures in MR CD-ROM, Set up and Calibration: Patient Transport, LIGHTWEIGHT PATIENT TRANSPORT.

Note

If hardware fastening procedures for installation of the dock and support brackets were followed, the hardware used to fasten the dock and dock supports was left loose. The loose hardware permits a quick and easy dock centering adjustment with table docked.

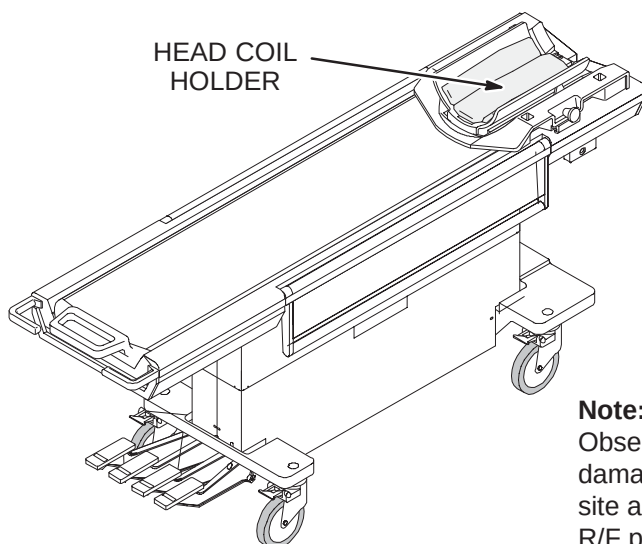
4. If additional Patient Transports are supplied, refer to *Direction 15477, Second Patient Transport Option Installation*, (delivered with M1020BD option).



PATIENT TRANSPORT INSTALLATION INSTRUCTIONS

□ B – HEAD COIL HOLDER

1. Unpack Head Coil Holder and install Head Coil Holder on Patient Transport cradle.
2. With Patient Transport docked, check mechanical operation of Head Coil with holder.

**Note:**

Observe warning Label on head coil. To prevent head coil damage, be sure that all service personnel and users at the site are aware that the head coil must be removed before R/F power is applied to the body coil. This warning also applies to service procedures.

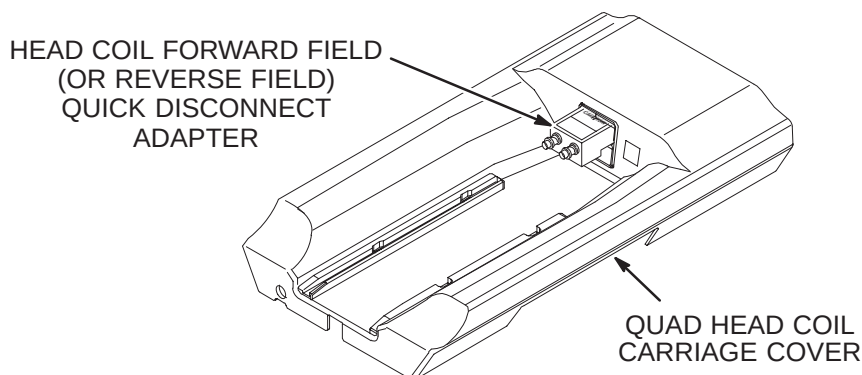
□ C – HEAD COIL ADAPTERS

1. Unpack Head Coil, Coil Adapters (Head Coil Forward Field, Extremity Coil, Surface Coil), and adapter cables.
2. Install Head Coil Adapter (Head Coil Forward or Reverse Field) into upper coil adapter port on carriage cover.

Note

If magnet is configured with reverse polarity, “reverse field” Head and Surface Coil Adapters are required to be ordered. The “forward field” Coil Adapters must be returned. Since an unaware operator could use the wrong adapter, do not leave both adapters on site.

3. Remove head coil cables from bag attached to Head Coil. Place Head Coil on carriage cover and connect cables to verify connections. Connect blue cable to connector indicated with blue.
3. Set Head Coil with cables, holder, and adapters aside for safe storage (outside of magnet room) until needed during set-up calibrations, functional checks, and system tests.



INSTALLATION STEPS SUMMARY

- ☐ A1 – Preinstallation Planning
- ☐ A2 – Squeeze Bulb Hanger Installation
- ☐ A3 – Squeeze Bulb Installation
- ☐ A4 – Magnet Room Routing of Tubing & Wave Guide Installation
- ☐ A5 – Determining Need of Extender Box

☐ A1 – PREINSTALLATION PLANNING

The customer and users should be involved in the following decisions:

Routing of Pneumatic Tubing:

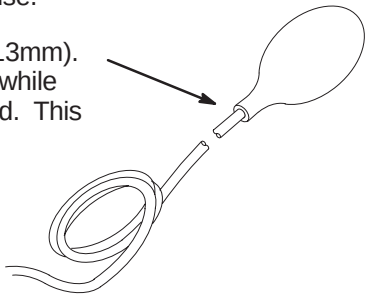
Location of Control Box



Do not substitute other types of tubing for the pneumatic tubing supplied with this Installation Kit. Substitution of other types of tubing may cause a control box malfunction.

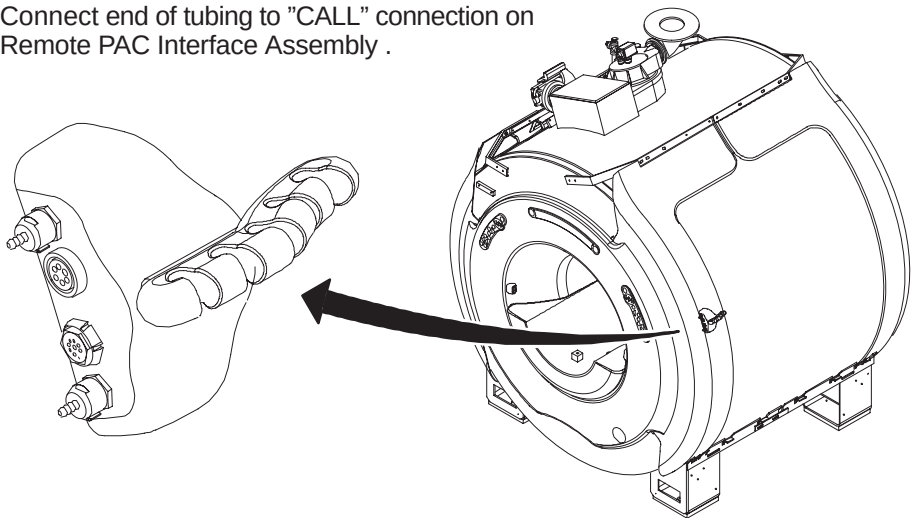
☐ A2 – SQUEEZE BULB INSTALLATION

- ② Before cutting pneumatic tubing from roll, verify that when the squeeze bulb is held by a patient being scanned, the tubing length can be routed to the “CALL” connector on the Remote PAC Interface Assembly without getting in the way of operator or patient during use.
- ① Insert end of Pneumatic Tubing into Squeeze Bulb approximately 1/2 inch (13mm). For ease of pneumatic tubing insertion, pinch pneumatic tubing in one hand while squeezing squeeze bulb and pushing pneumatic tubing in with the other hand. This will create a positive pressure and expand squeeze bulb opening.



☐ A3 – CONNECT TUBE TO REMOTE PAC INTERFACE ASSEMBLY

- ① Connect end of tubing to “CALL” connection on Remote PAC Interface Assembly .



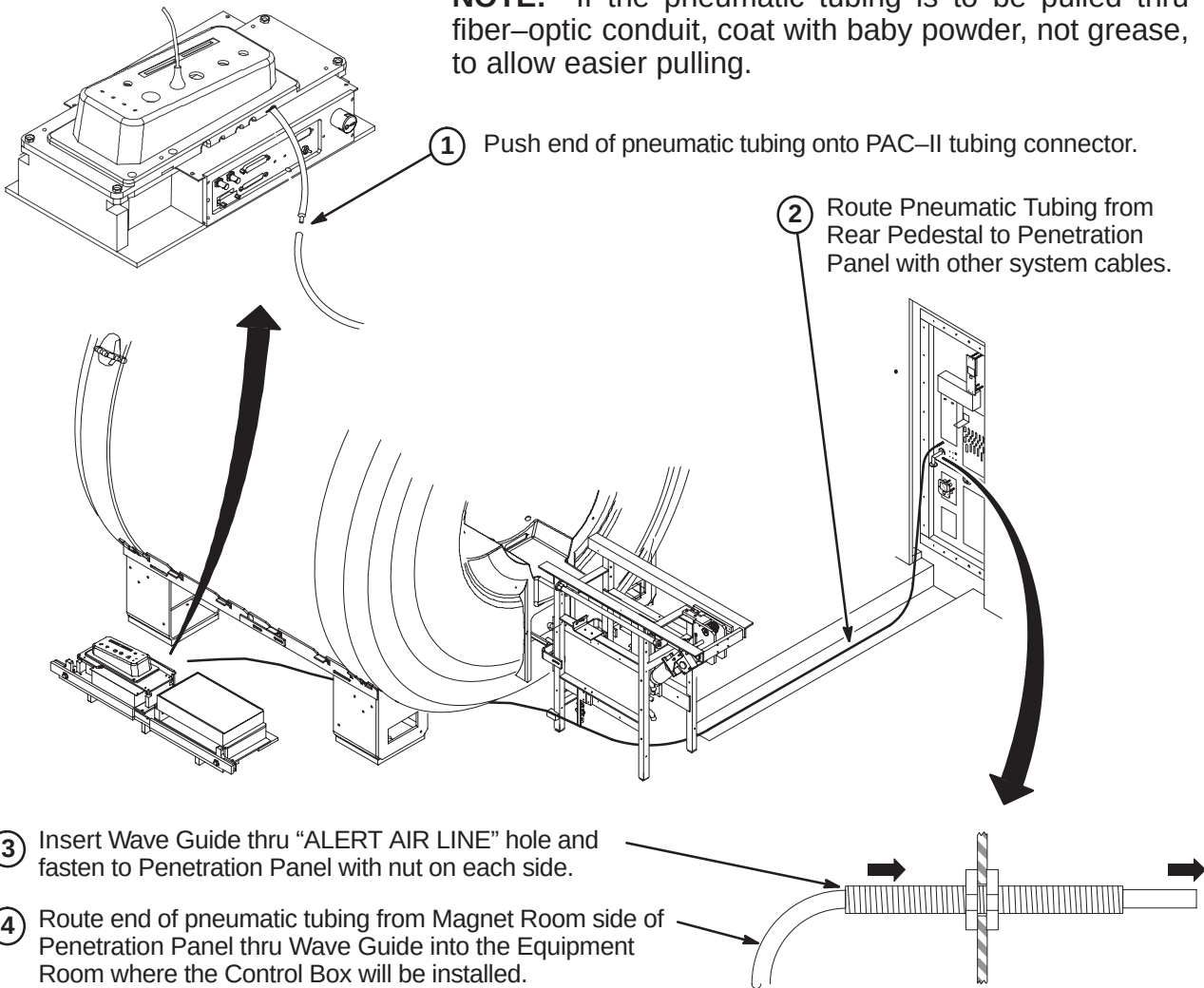
☐ A4 – MAGNET ROOM ROUTING OF TUBING & WAVE GUIDE INSTALLATION



Make sure that pneumatic tubing is not routed where it can be stepped on, pinched, or have an object set on it. The control box alarm will not work if pneumatic tubing is pinched.

NOTE: Also, make sure that the pneumatic tubing can be routed from the PAC connector through the Rear Pedestal, and to the Penetration panel without being pinched along the route by other cables.

NOTE: If the pneumatic tubing is to be pulled thru fiber-optic conduit, coat with baby powder, not grease, to allow easier pulling.



☐ A5 – DETERMINING NEED OF EXTENDER BOX

Note: If installation requires greater than 115 feet of pneumatic tubing between the squeeze bulb (hanging on Magnet Enclosure) and control box (near Operator’s Console), Extender Kit, 46–317758P2, must be ordered.

If the pneumatic tubing is long enough to reach control box:
The extender is not needed. Proceed to Step B3, CONTROL BOX INSTALLATION.

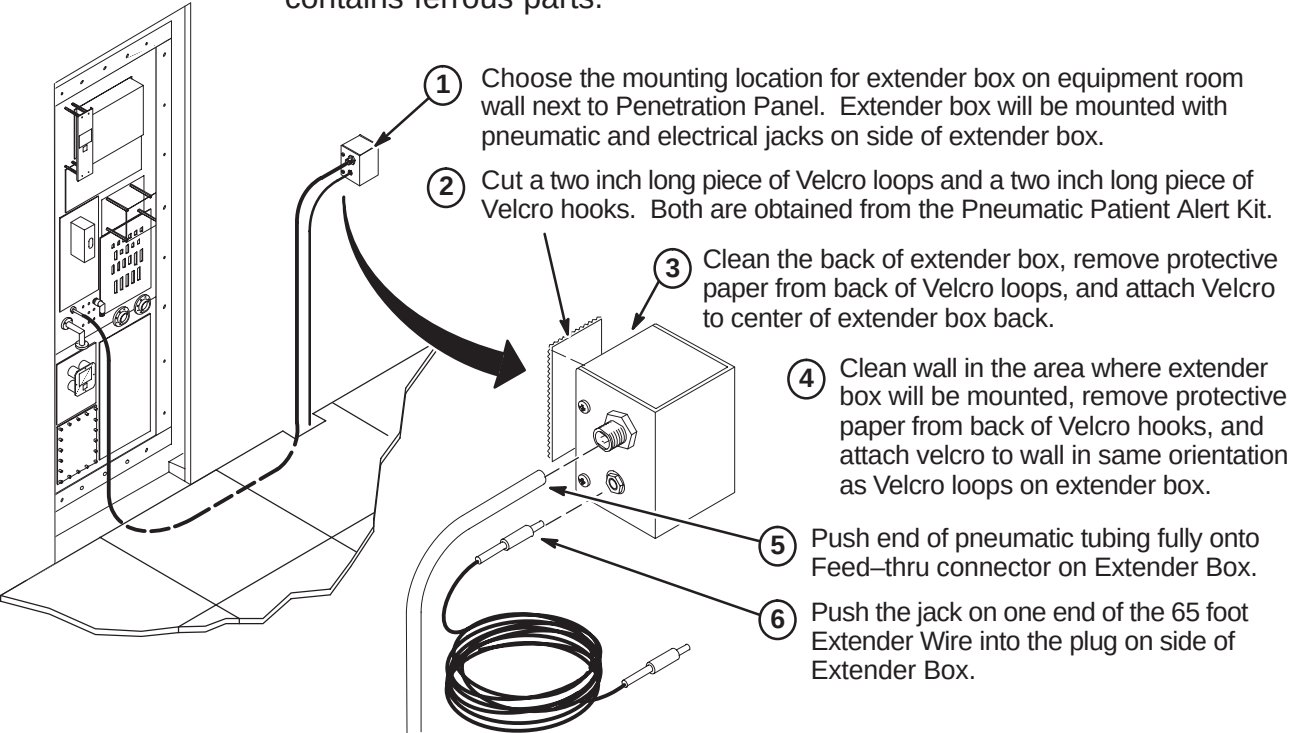
If the pneumatic tubing is not long enough to reach control box:
The extender is needed. Proceed to Step B1, INSTALLATION OF EXTENDER KIT.

INSTALLATION STEPS SUMMARY

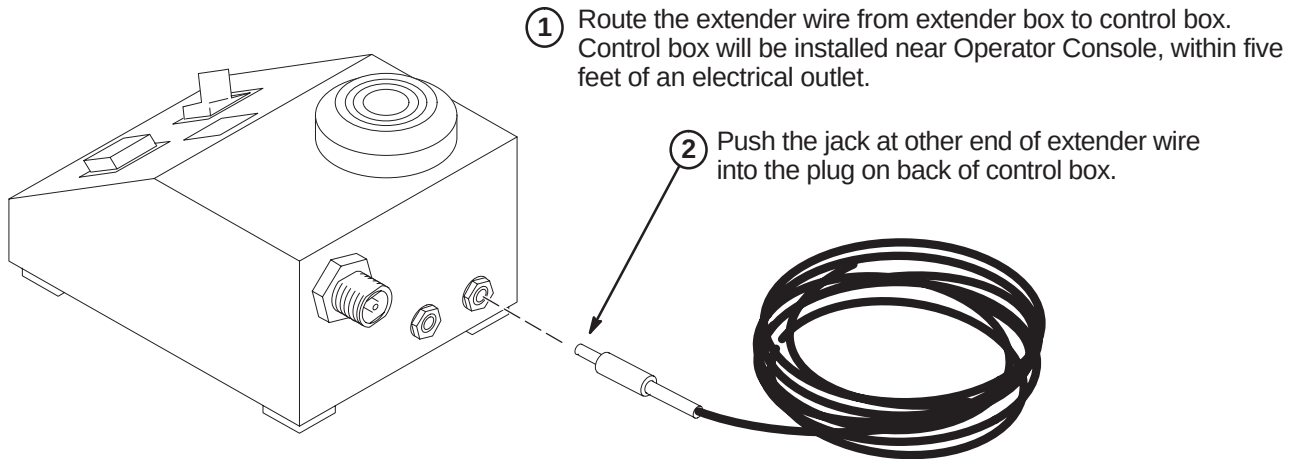
- ☐ B1 – Installation of Extender Kit (46–317758P2)
- ☐ B2 – Connection of Extender Wire to Control Box
- ☐ B3 – Control Box Installation
- ☐ B4 – Control Box Installation on Wall or Under Shelf
- ☐ B5 – Control Box Mounted with Velcro

☐ B1 – INSTALLATION OF EXTENDER KIT (46–317758P2)

Note: Do not mount Extender Box in Magnet Room because the box contains ferrous parts.



☐ B2 – CONNECTION OF EXTENDER WIRE TO CONTROL BOX



☐ B3 – CONTROL BOX INSTALLATION

Consult with the operator in choosing the mounting orientation for the control box. Some key factors to consider are ease of use by operator, remaining within sight of operator, and remaining within five feet of an electrical outlet. Choose one of the following control box locations:

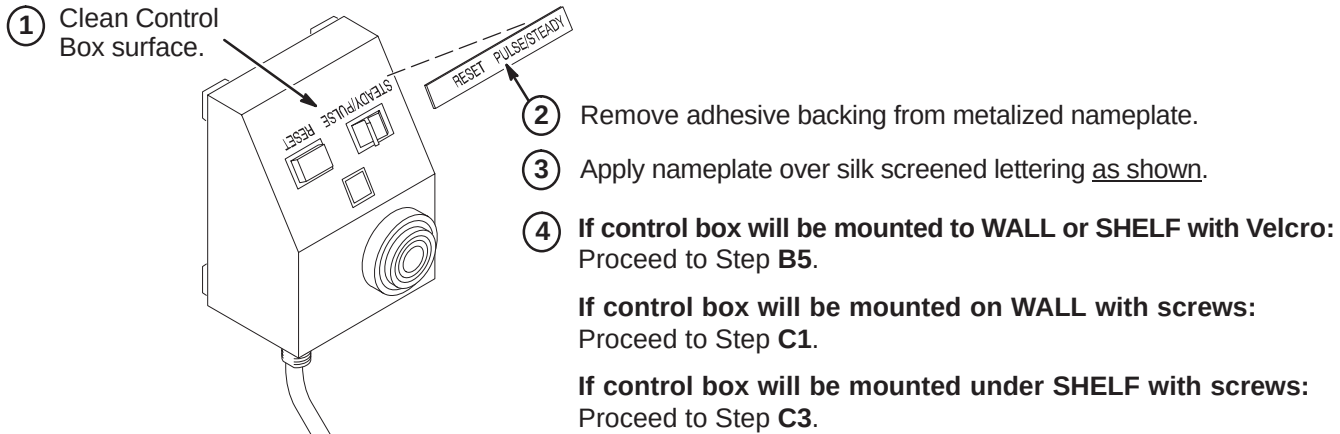
Mount control box on a wall or other vertical surface per Step **B4**, CONTROL BOX INSTALLATION ON WALL OR UNDER SHELF.

Mount control box under shelf per Step **B4**, CONTROL BOX INSTALLATION ON WALL OR UNDER SHELF.

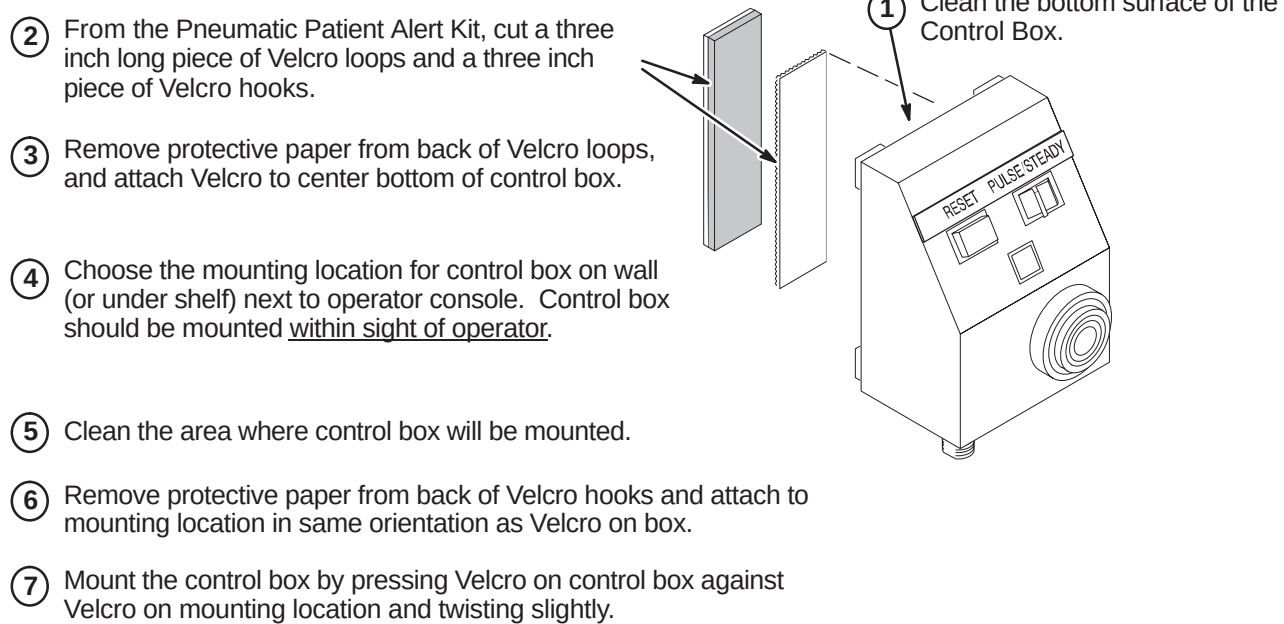
Place Control Box on a counter top, desk top, or other horizontal surface. Set the Control Box near Operator Console and within five feet of electrical outlet. Control Box should be placed within sight of operator. Proceed to Step **C5**, FINAL CONNECTIONS TO CONTROL BOX.

Note: The installer must provide additional mounting hardware if supplied screws or adhesive backed Velcro are not adequate for installation of the control box.

☐ B4 – CONTROL BOX INSTALLATION ON WALL OR UNDER SHELF



☐ B5 – CONTROL BOX MOUNTED WITH VELCRO

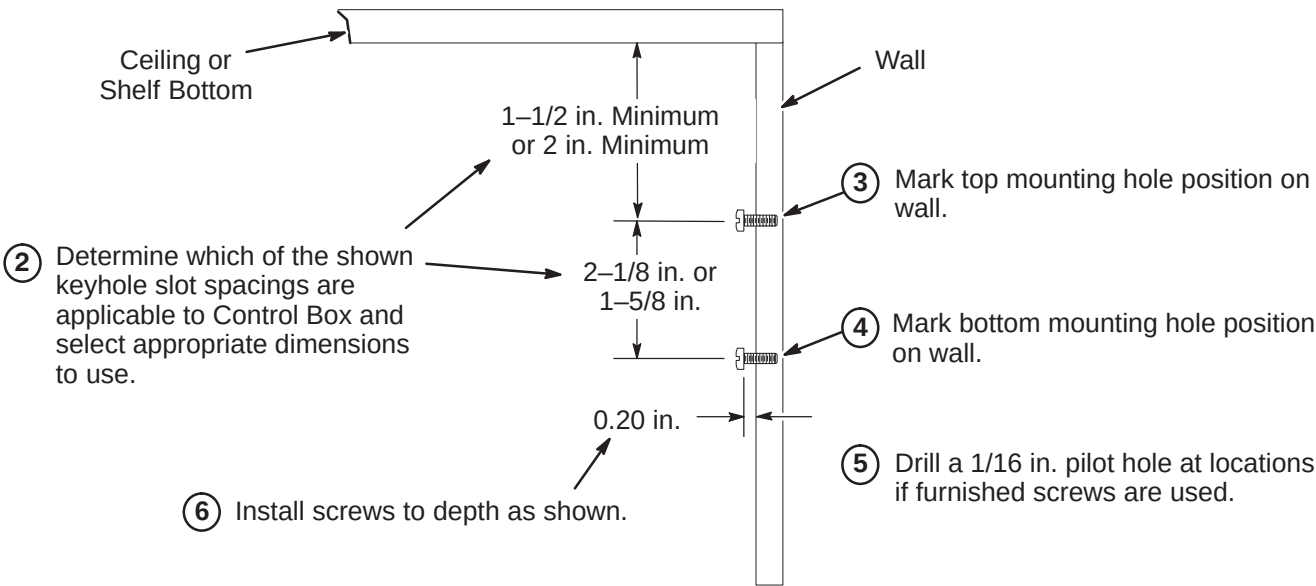


INSTALLATION STEPS SUMMARY

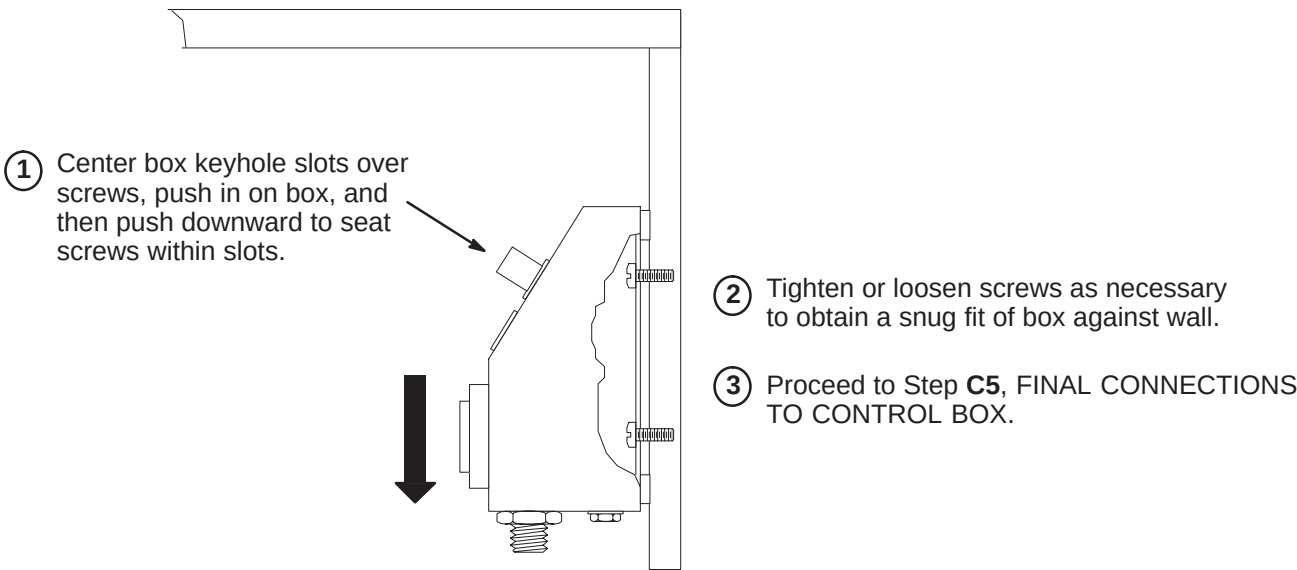
- C1 – Installing Mounting Screws in Vertical Surface
- C2 – Installing Control Box on Vertical Surface
- C3 – Installing Mounting Screws Under Shelf
- C4 – Installing Control Box Under Shelf
- C5 – Final Connections to Control Box

C1 – INSTALLING MOUNTING SCREWS IN VERTICAL SURFACE

- 1 Choose the mounting area near the Operator Console and install mounting screws (item 9). Control box should be mounted within sight of operator.



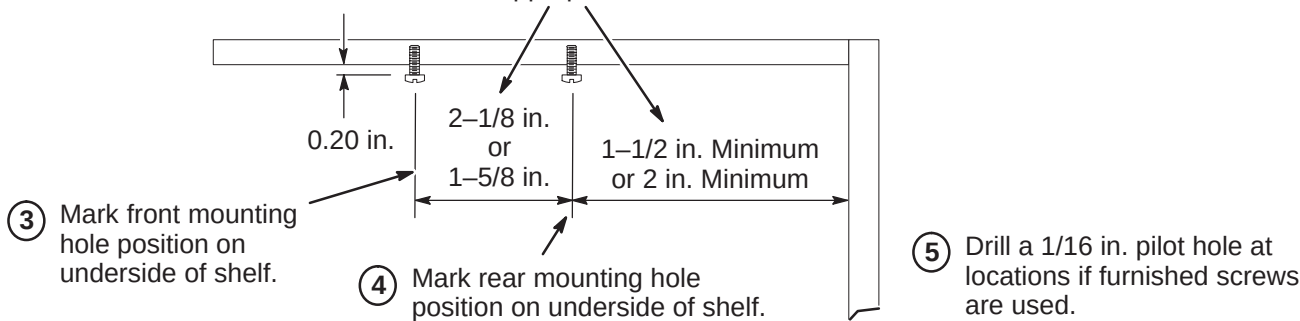
C2 – INSTALLING CONTROL BOX ON VERTICAL SURFACE



C3 – INSTALLING MOUNTING SCREWS UNDER SHELF

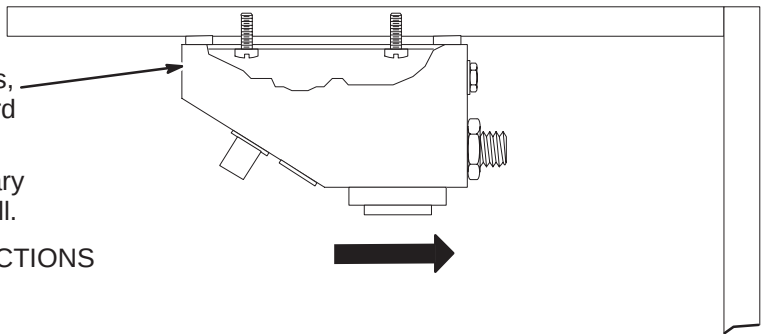
- 1 Choose the mounting area near the Operator Console and install mounting screws (item 9). Control box should be mounted within sight of operator.

- 2 Determine which of the shown keyhole slot spacings are applicable to Control Box and select appropriate dimensions to use.



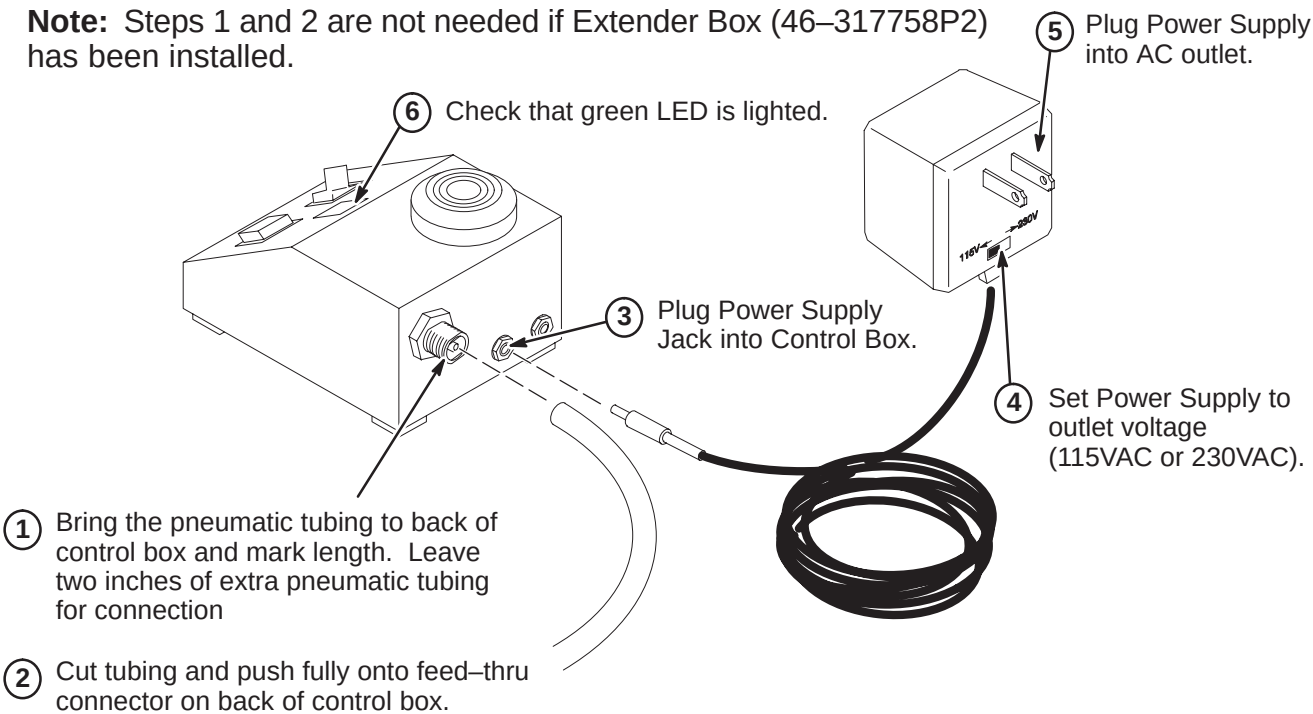
C4 – INSTALLING CONTROL BOX UNDER SHELF

- 1 Center box keyhole slots over screws, push up on box, and then push inward to seat screws within slots.
- 2 Tighten or loosen screws as necessary to obtain a snug fit of box against wall.
- 3 Proceed to Step C5, FINAL CONNECTIONS TO CONTROL BOX.



C5 – FINAL CONNECTIONS TO CONTROL BOX

Note: Steps 1 and 2 are not needed if Extender Box (46-317758P2) has been installed.



OPERATOR WORKSPACE INSTALLATION INSTRUCTIONS

1-1 **Tab 13, Operator Workspace, contains installation instructions for the following:**

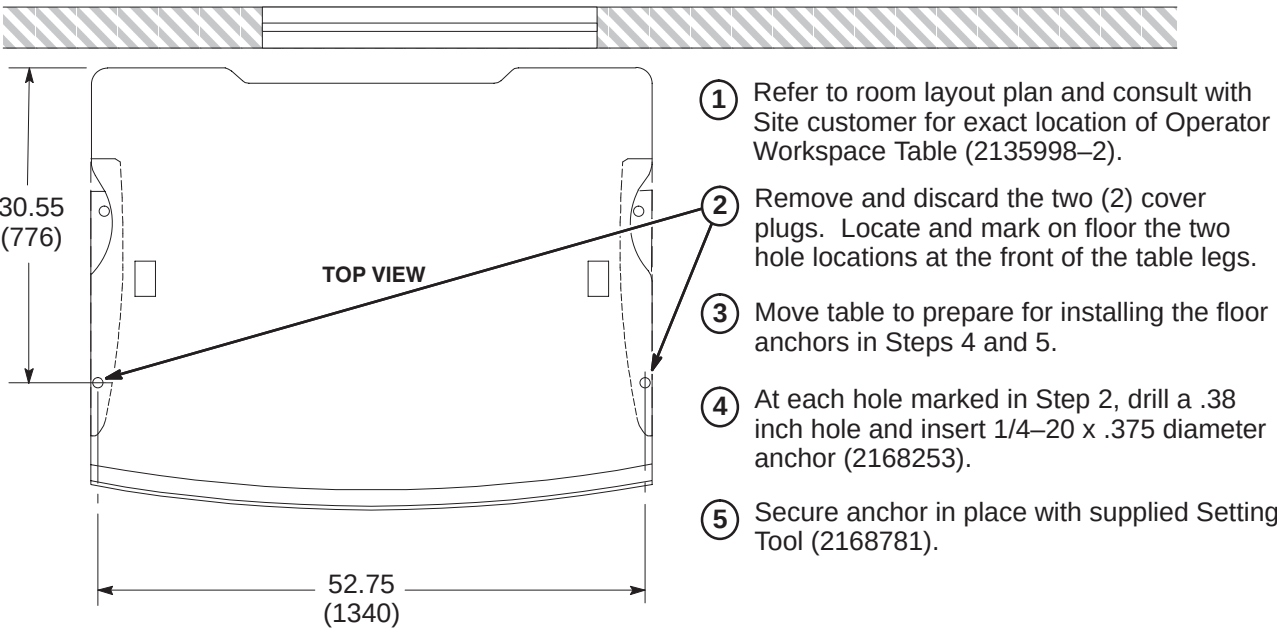
- *DIRECTION 2210982* – Signa Octane Operator Workspace Installation Instructions
- *DIRECTION 2386129* – Signa Linux Operator Workspace Installation Instructions

INSTALLATION STEPS SUMMARY

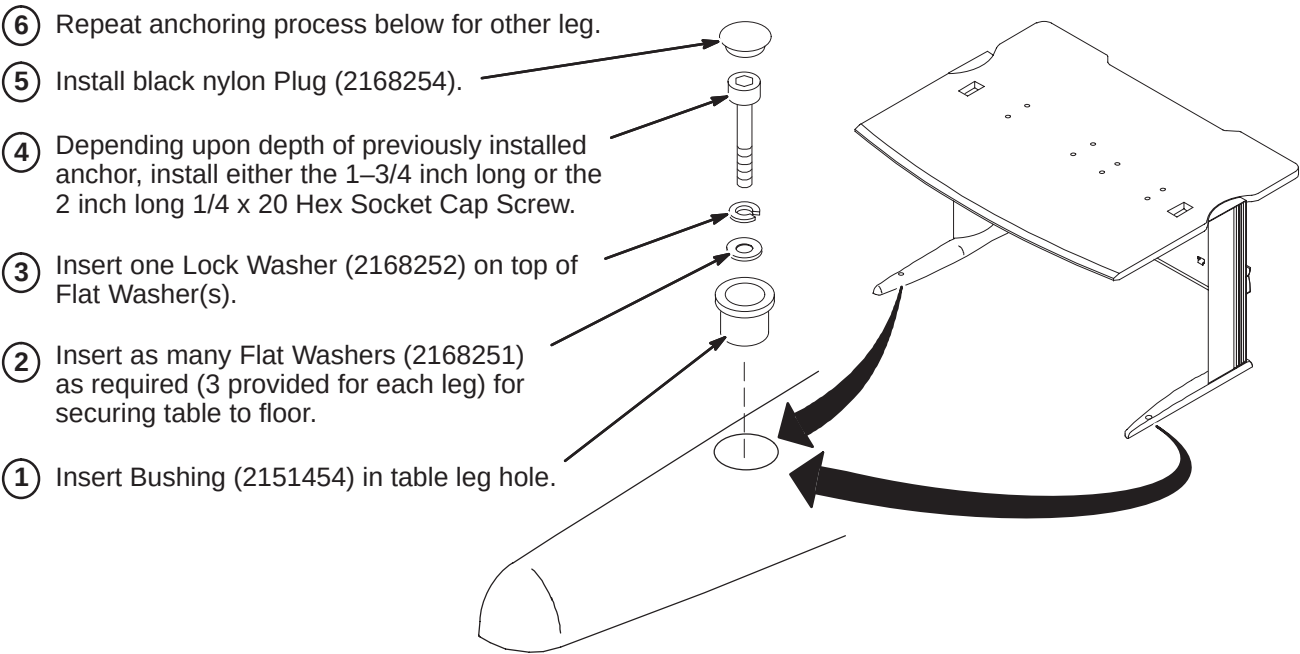
- A1 – Operator Workspace Table Anchor Installation
- A2 – Securing Operator Workspace Table to Floor
- A3 – Attach Workspace Interface Module (2141978 series) to Table
- A4 – Attach Power Distribution Box (46–307830G1) to Table

Note: The CERD Spike Noise Test Cable (2168505) is part of the 8.x Octane Operator Workspace Cable collector (2154392–2). Remove this cable from the collector and place it in the TPS RF Service Interface Kit (46–301927G1).

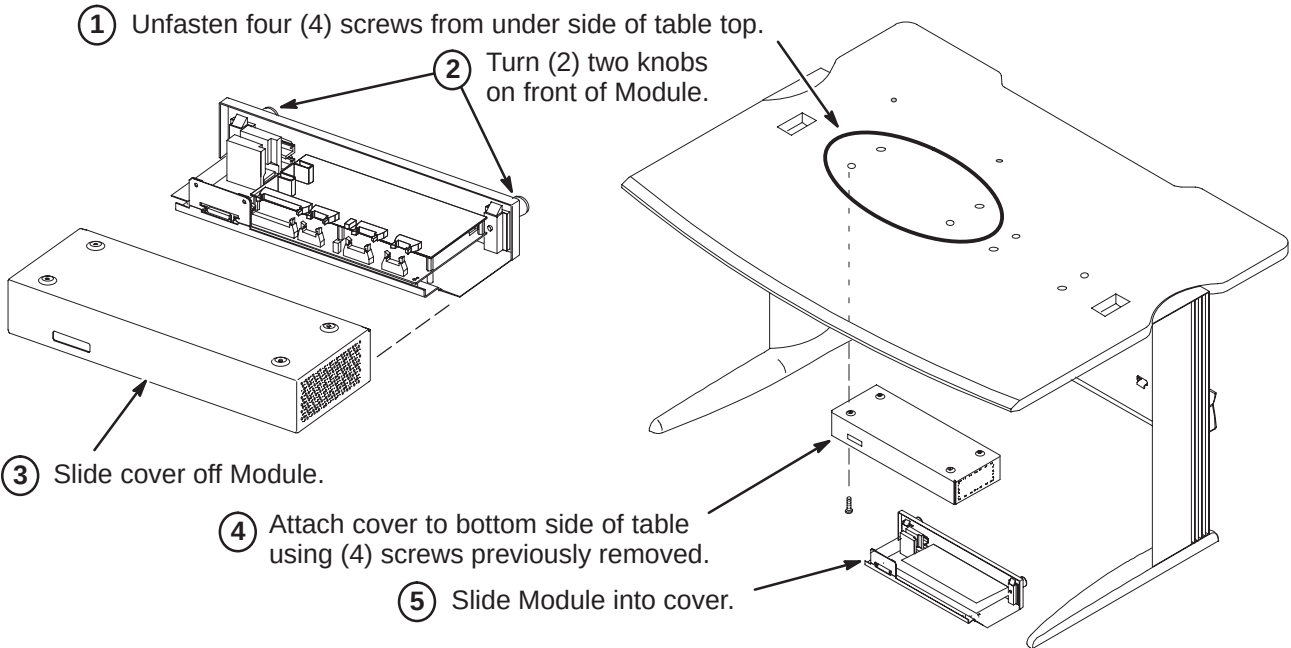
A1 – OPERATOR WORKSPACE TABLE ANCHOR INSTALLATION



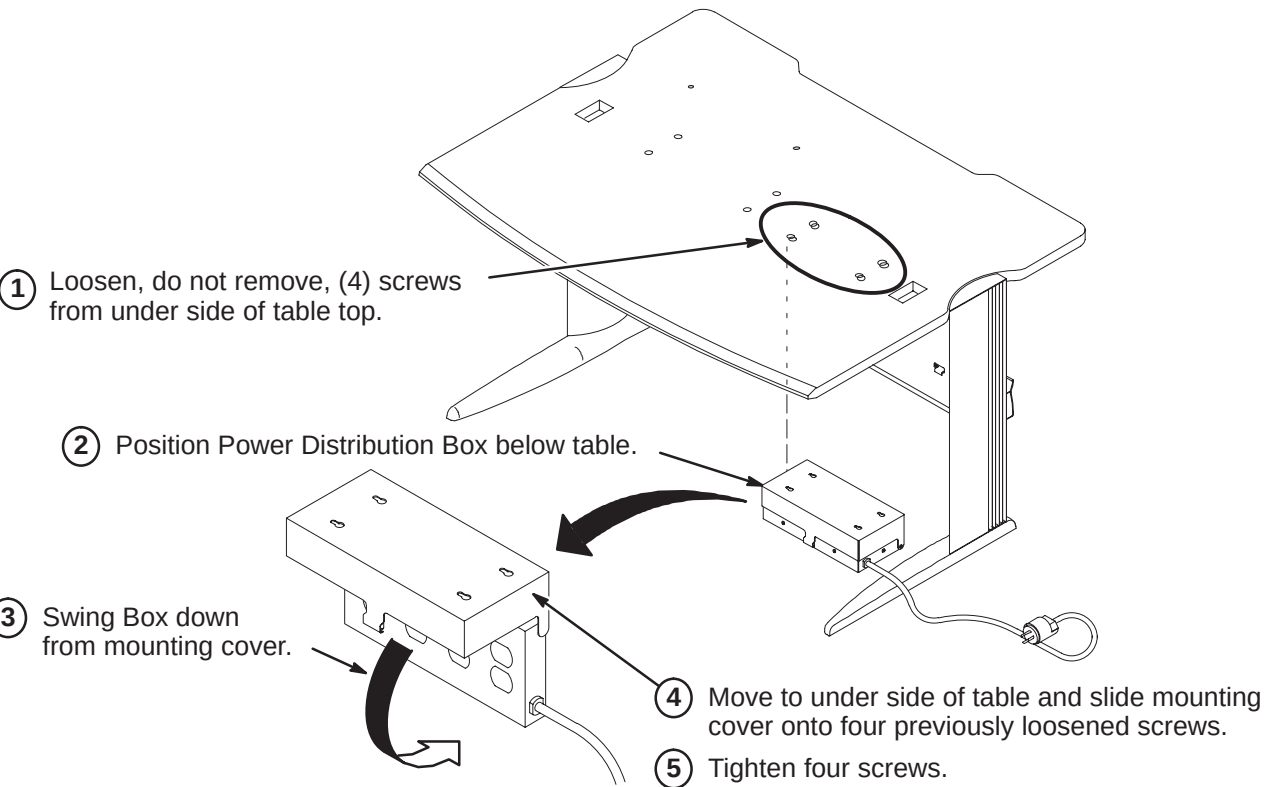
A2 – SECURING OPERATOR WORKSPACE TABLE TO FLOOR



A3 – ATTACH WORKSPACE INTERFACE MODULE (2141978 series) TO TABLE



A4 – ATTACH POWER DISTRIBUTION BOX (46–307830G1) TO TABLE

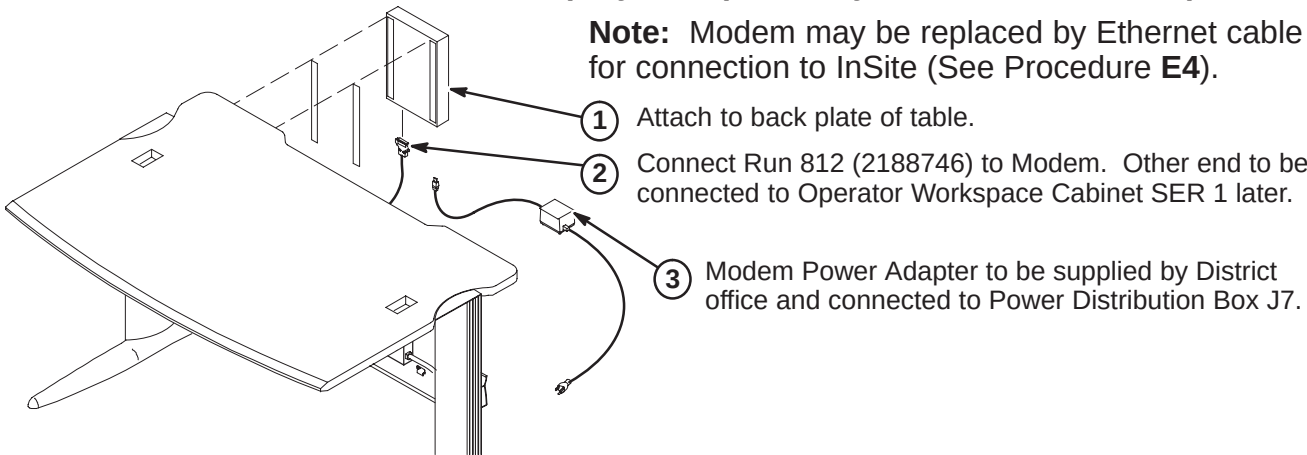


INSTALLATION STEPS SUMMARY

- ☐ B1 – Attach Modem to Table (May be replaced by ethernet connection)
- ☐ B2 – Attach DASM to Table
- ☐ B3 – Attach Serial Expansion Box Asm to Underside of Table Top
- ☐ B4 – Attach Serial Coverter to Underside of Table Top (MGD & OpenSpeed only)

☐ B1 – ATTACH MODEM TO TABLE (May be replaced by ethernet connection)

Note: Modem may be replaced by Ethernet cable for connection to InSite (See Procedure E4).



☐ B2 – ATTACH DASM TO TABLE

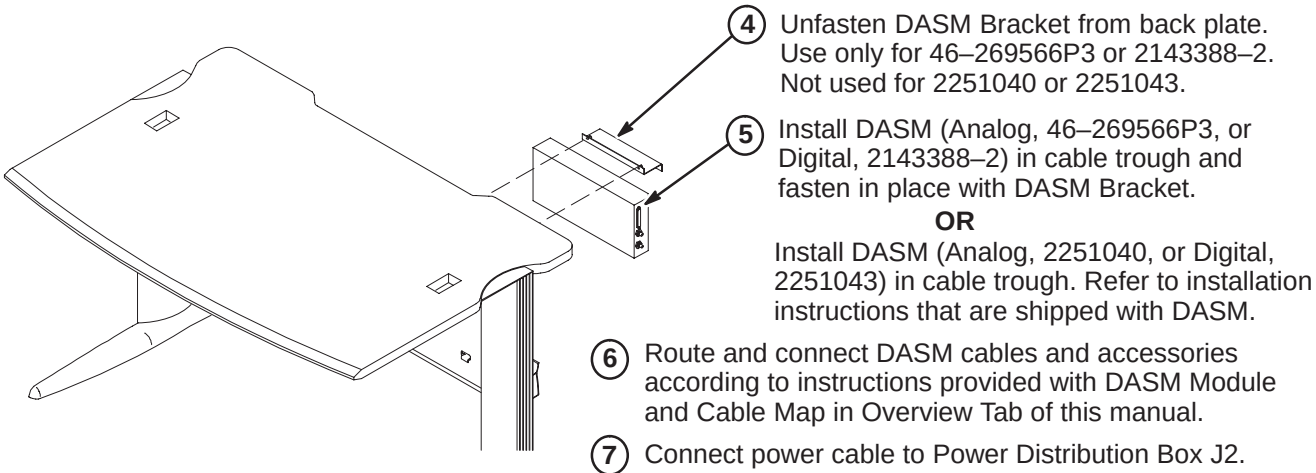
Note: The Analog and Digital DASMs are shipped with the jumper set to SCSI ID=0. This is the correct ID for an AW, but incorrect for Signa Horizon LX (SCSI ID=3).

- 1 Unfasten DASM cover screw and remove cover.
- 2 For SCSI ID = 3, set Configuration Jumper J2 as follows:
5=OFF, 6=OFF, 7=ON
Note: A jumper in the OFF position enables the SCSI ID bit.
- 3 Re-attach DASM cover.

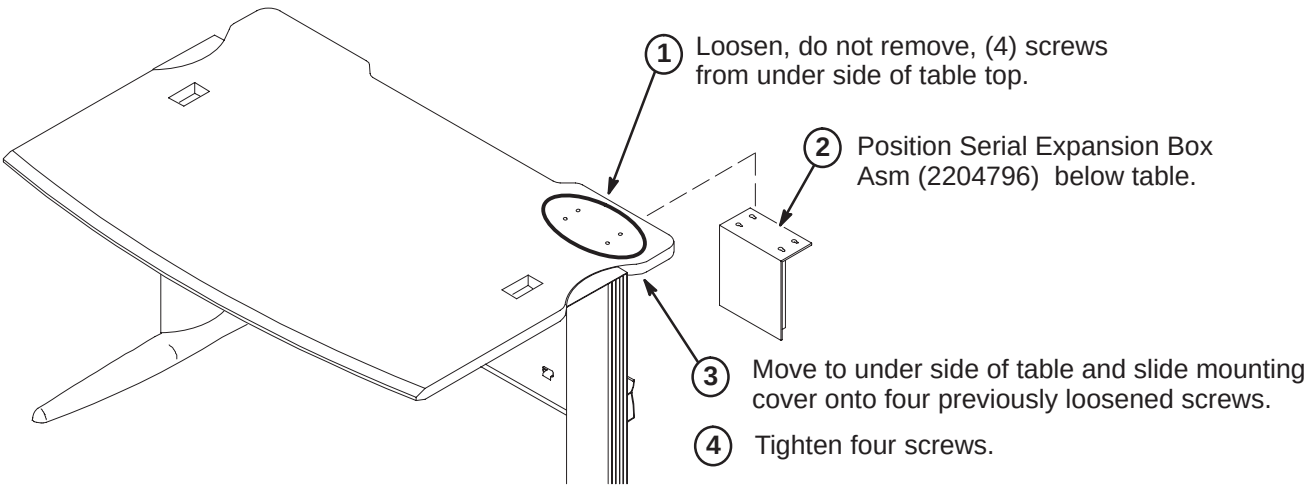


Note: Consult with Site customer for preference of DASM location. This procedure installs DASM in the cable trough. Customer may prefer DASM on top of table.

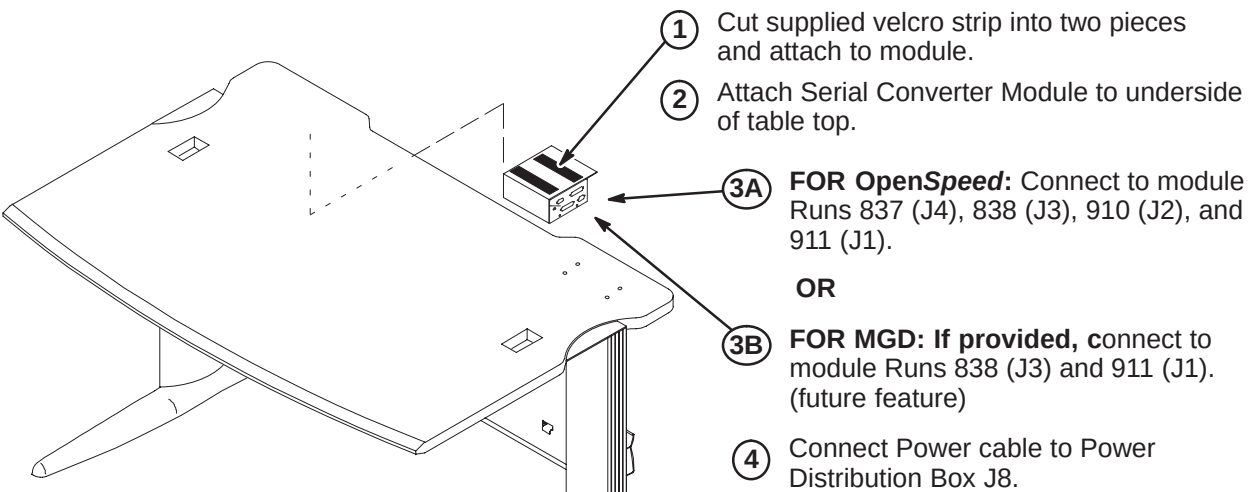
CAUTION: Longer SCSI cables cannot be substituted.



☐ B3 – ATTACH SERIAL EXPANSION BOX ASM TO UNDERSIDE OF TABLE TOP



☐ B4 – ATTACH SERIAL CONVERTER TO BOTTOM OF TABLE (MGD & OpenSpeed)



INSTALLATION STEPS SUMMARY

- ☐ C1 – Install Keyboard on Table
- OR
- ☐ C2 – Install SCIM/Keyboard Data Cable on Table
- ☐ C3 – Install SCIM/Keyboard on Table
- ☐ C4 – Install Flat Screen Color Monitor on Table

NOTE:
Consult with Site customer for exact location of Color Monitor on Operator Workspace Table.

☐ C1 – INSTALL KEYBOARD ON TABLE

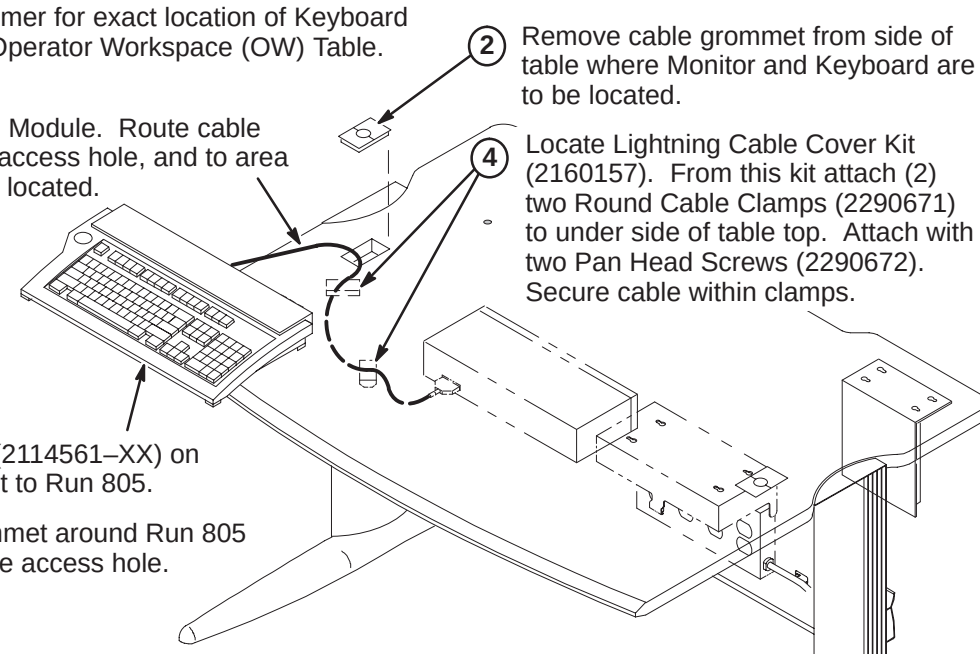
- 1 Consult with Site customer for exact location of Keyboard and Color Monitor on Operator Workspace (OW) Table.

3 Connect Run 805 to I/F Module. Route cable under table, thru cable access hole, and to area where Keyboard will be located.

5 Place Keyboard (2114561-XX) on table and connect to Run 805.

6 Place cable grommet around Run 805 and insert in cable access hole.

2 Remove cable grommet from side of table where Monitor and Keyboard are to be located.

4 Locate Lightning Cable Cover Kit (2160157). From this kit attach (2) two Round Cable Clamps (2290671) to under side of table top. Attach with two Pan Head Screws (2290672). Secure cable within clamps.
- 
- The diagram shows a side view of the operator workspace table. A keyboard is being placed on the table surface. A cable (Run 805) is shown being routed under the table through a cable access hole. The cable is then secured under the table top using two round cable clamps attached with pan head screws. A cable grommet is shown being placed around the cable as it enters the table.

OR

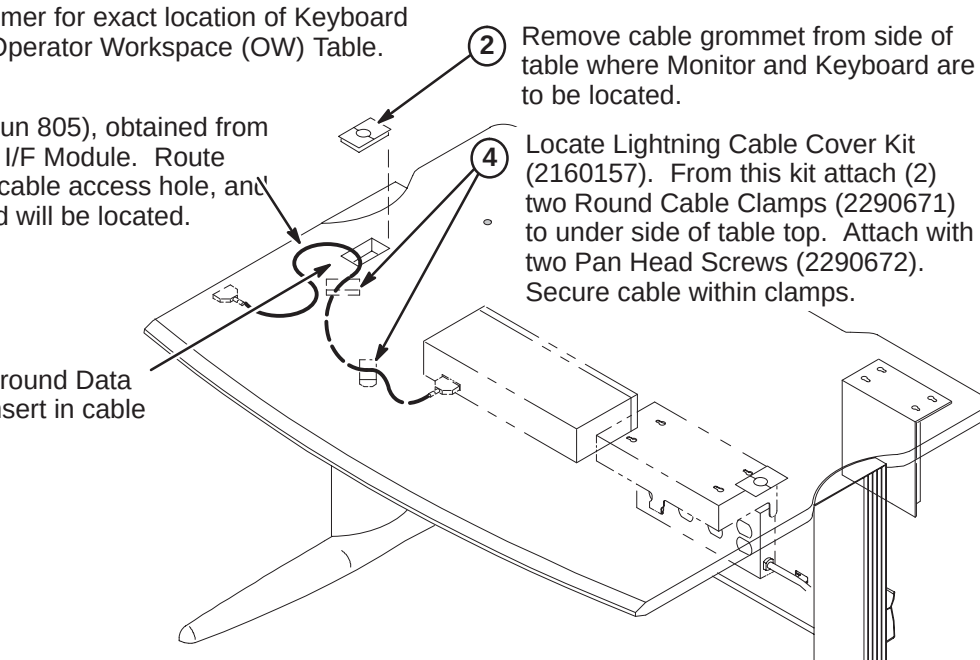
☐ C2 – INSTALL SCIM/KEYBOARD DATA CABLE ON TABLE

- 1 Consult with Site customer for exact location of Keyboard and Color Monitor on Operator Workspace (OW) Table.

3 Connect Data Cable (Run 805), obtained from SCIM Kit (2307175), to I/F Module. Route cable under table, thru cable access hole, and to area where Keyboard will be located.

5 Place cable grommet around Data Cable (Run 805) and insert in cable access hole.

2 Remove cable grommet from side of table where Monitor and Keyboard are to be located.

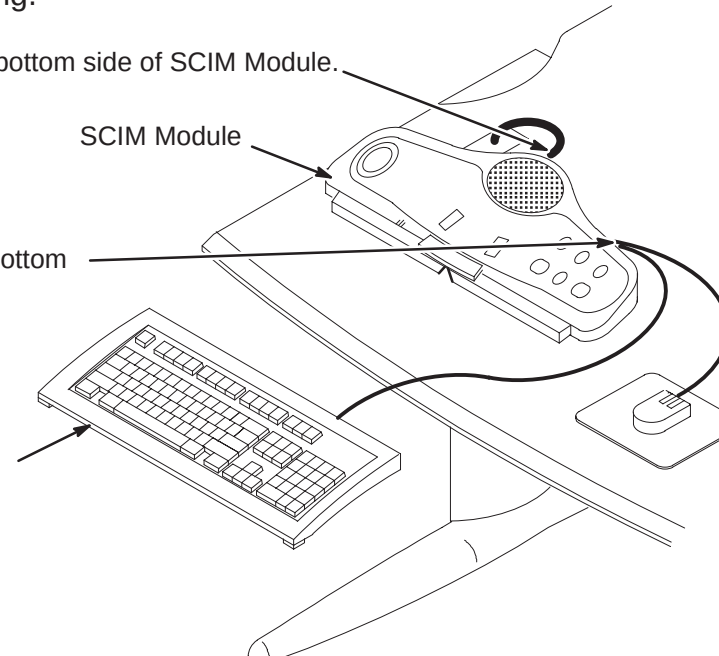
4 Locate Lightning Cable Cover Kit (2160157). From this kit attach (2) two Round Cable Clamps (2290671) to under side of table top. Attach with two Pan Head Screws (2290672). Secure cable within clamps.
- 
- The diagram shows a side view of the operator workspace table. A data cable (Run 805) is being routed under the table through a cable access hole. The cable is then secured under the table top using two round cable clamps attached with pan head screws. A cable grommet is shown being placed around the cable as it enters the table.

☐ C3 – INSTALL SCIM/KEYBOARD ON TABLE

NOTE: Refer to Installation document OW1INA3.DOC on Service Methods CD-ROM or MR Service Engineering web site for installation information on language labels and keyboard mounting.

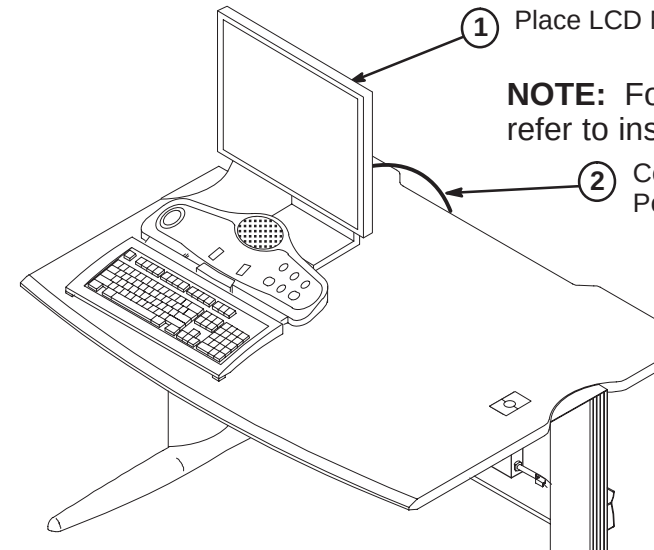
- 1 Connect Data Cable to bottom side of SCIM Module.

2 Connect Keyboard and Mouse cables to bottom side of SCIM Module as marked.

3 Place Keyboard on table in front of SCIM Module.
- 
- The diagram shows a top-down view of the SCIM module on the table. A data cable is connected to the bottom of the module. Keyboard and mouse cables are also connected to the bottom of the module. The keyboard is placed on the table in front of the module.

☐ C4 – INSTALL FLAT SCREEN COLOR MONITOR ON TABLE

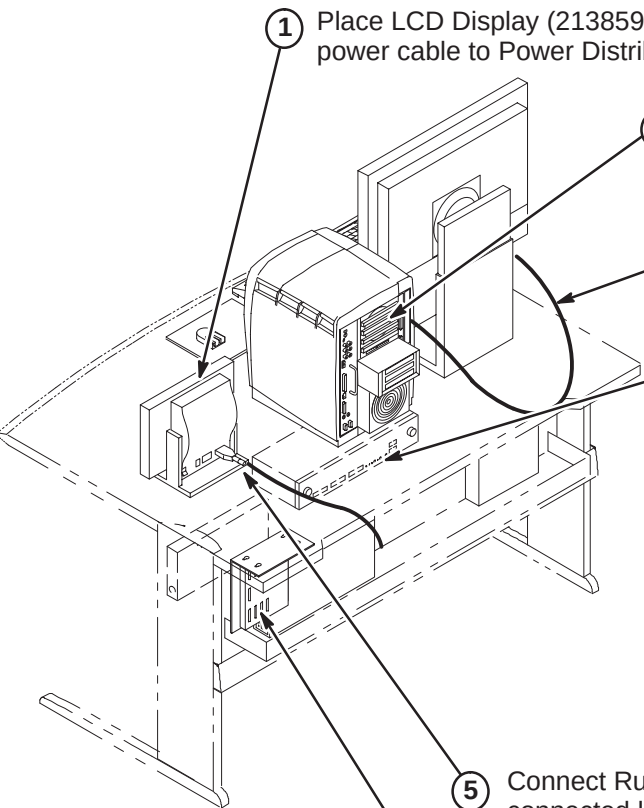
- 1 Place LCD Monitor on table in back of SCIM.

2 Connect power cable to Power Distribution Box J1.
- NOTE:** For fastening monitor to surface, refer to installation drawing 2204493DDW.
- 
- The diagram shows a top-down view of the operator workspace table. A flat screen color monitor is placed on the table in front of the SCIM module. A power cable is connected to the monitor and then to a power distribution box on the table.

INSTALLATION STEPS SUMMARY

- D1 – Install LCD Display on Table and Position Octane Computer
- D2 – Connect Table Component Cables
- D3 – Serial Expansion Box Set Up
- D4 – SGI Octane Computer Cables Connections

□ D2 – CONNECT TABLE COMPONENT CABLES



1 Place LCD Display (2138599) on table. Connect power cable to Power Distribution Box J5.

2 Place Octane Computer on table.
NOTE: Refer to Procedure **D4** for Run connections to SGI Octane Computer.

3 Connect Run 798/821 (2150533–2) to R IN, B IN, G IN on Monitor and route/connect to MONITOR on the Octane Computer module.

4 Connect the following cables from the OW I/F Module (OW A1 A4) for later connection to Operator Workspace Cabinet:

RUN #	CONNECTION	PART #
805	J6	2114561–38
810	J8	2141980–2
814	J9	2196990
817	J17	46–328000G958

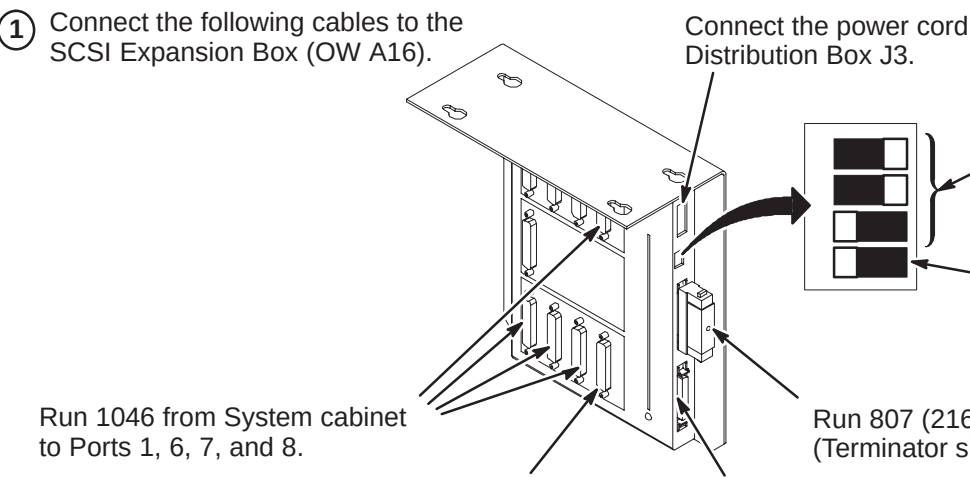
Connect the following cables to OW I/F Module. These cables will be routed to the System Cabinet and Penetration Panel:

RUN #	CONNECTION	PART #
789	J18	46–317068G938
788	J7	46–328000G937

5 Connect Run 797 (2138599–2) to LCD Display. Other end is connected later to PC in the Operator Workspace Cabinet.

NOTE: Refer to Procedure **D3** for Run connections and DIP switch settings.

□ D3 – SERIAL EXPANSION BOX SET UP FOR MGD (10.x)



1 Connect the following cables to the SCSI Expansion Box (OW A16).

Connect the power cord to Power Distribution Box J3.

2 Set SCSI ID switches on side of module as shown. (Addr = 4)

Termination Enabled
NOTE: If DASM not installed, move switch setting to disabled

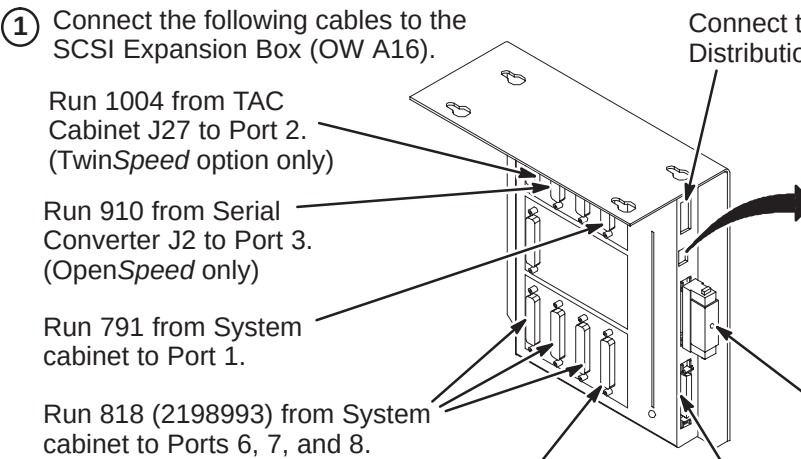
Run 1046 from System cabinet to Ports 1, 6, 7, and 8.

Run 911 from Serial Converter J1 to Port 5. (Future)

Run 807 (2163886) from DASM. (Terminator shown)

Run 815 (2198983) from Octane computer

□ D3 – SERIAL EXPANSION BOX SET UP (9.x)



1 Connect the following cables to the SCSI Expansion Box (OW A16).

Run 1004 from TAC Cabinet J27 to Port 2. (TwinSpeed option only)

Run 910 from Serial Converter J2 to Port 3. (OpenSpeed only)

Run 791 from System cabinet to Port 1.

Run 818 (2198993) from System cabinet to Ports 6, 7, and 8.

Run 911 from Serial Converter J1 to Port 5. (OpenSpeed only)

Connect the power cord to Power Distribution Box J3.

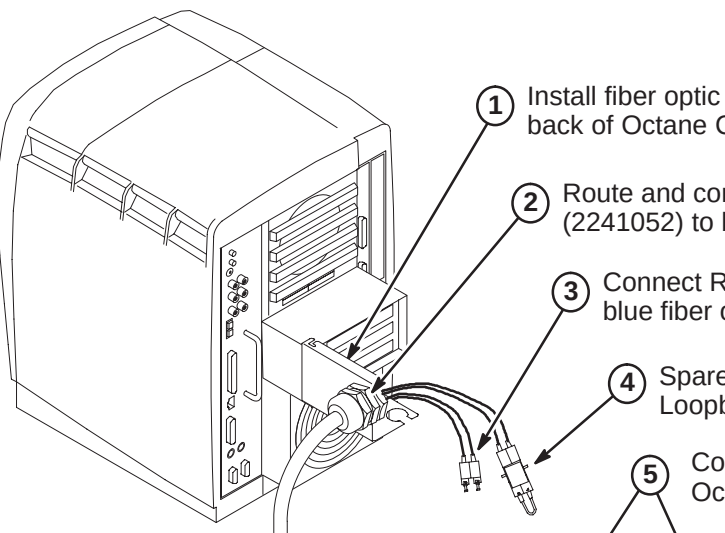
2 Set SCSI ID switches on side of module as shown. (Addr = 4)

Termination Enabled
NOTE: If DASM not installed, move switch setting to disabled

Run 807 (2163886) from DASM. (Terminator shown)

Run 815 (2198983) from Octane computer

□ D4 – SGI OCTANE COMPUTER CABLES CONNECTIONS



1 Install fiber optic Strain Relief Bracket (2245994) to back of Octane Computer using existing screws.

2 Route and connect Bit3 Fiber Optic cable Run 836 (2241052) to bracket.

3 Connect Run 836 (or Run 1044, MGD) orange and blue fiber optic cables as indicated in Step 5 below.

4 Spare Fiber Optic cable and Loopback Assembly (2241052–6).

5 Connect the following cables to the SGI Octane Computer (OW A15) as marked.

Run 798 or 821 (2150533–2) OR Run 799 or 820 (2150533)

Run 1031 (MGD)

Run 819

Run 811 (2153396–3)

Run 810 (2141980–2)

Run 813 (2188747)

Run 812 (2188746)

Run 836 (2241052) OR Run 1044 (MGD)

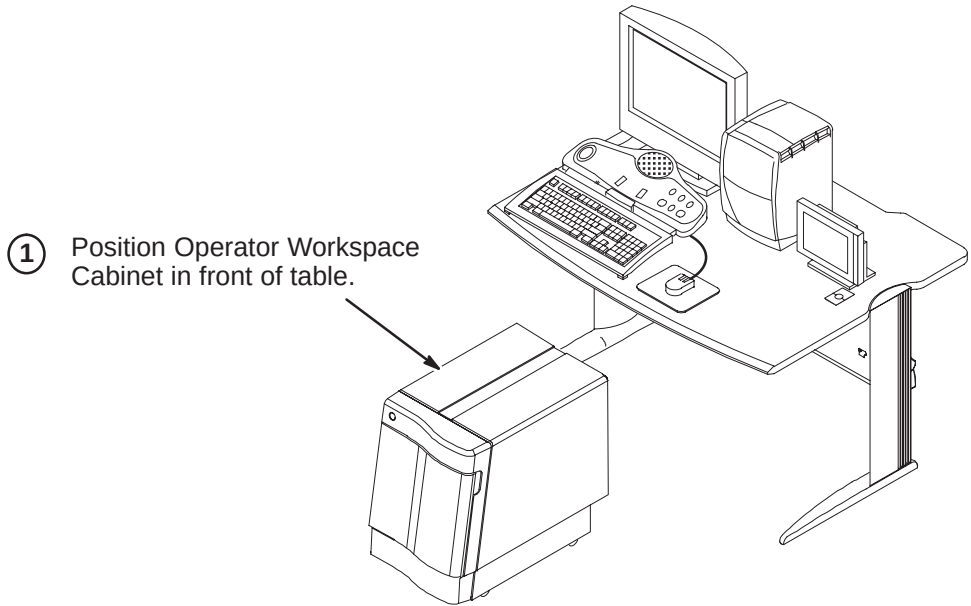
Run 815 (2198983)

Connect power cable to Power distribution Box J4.

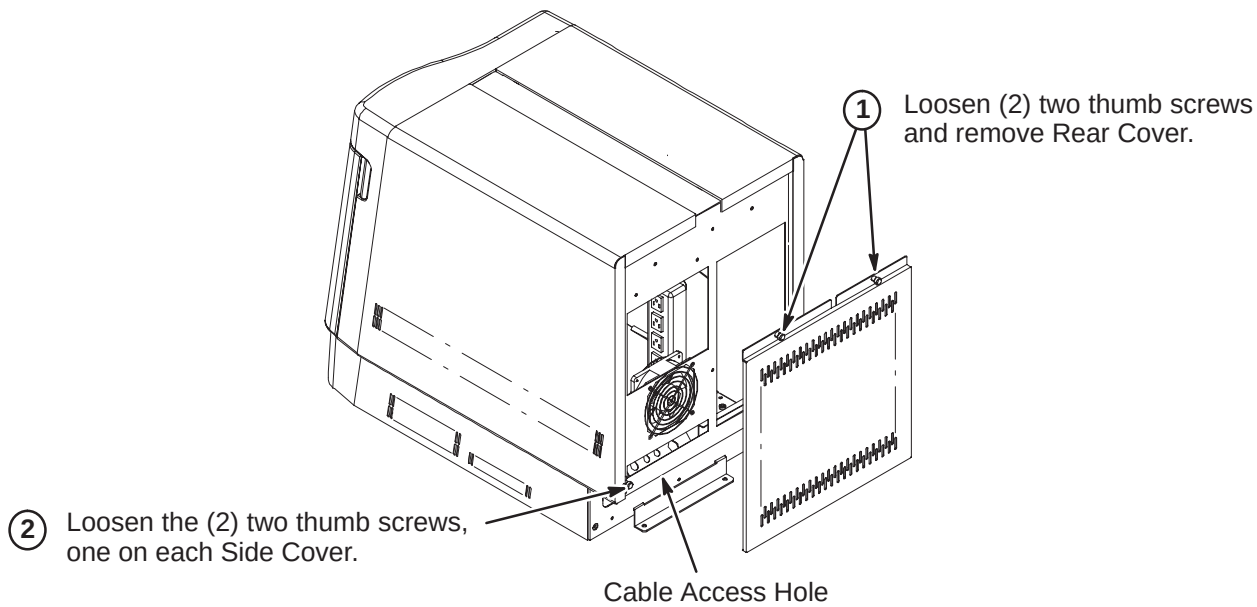
INSTALLATION STEPS SUMMARY

- E1 – Locate Operator Workspace Cabinet in Front of Table
- E2 – Remove Operator Workspace Cabinet Rear Cover
- E3 – Remove Operator Workspace Cabinet Front And Side Covers
- E4 – Connect Ethernet Cables
- E5 – Route/Connect Runs 792, 797, 813, and 814 to PC Module

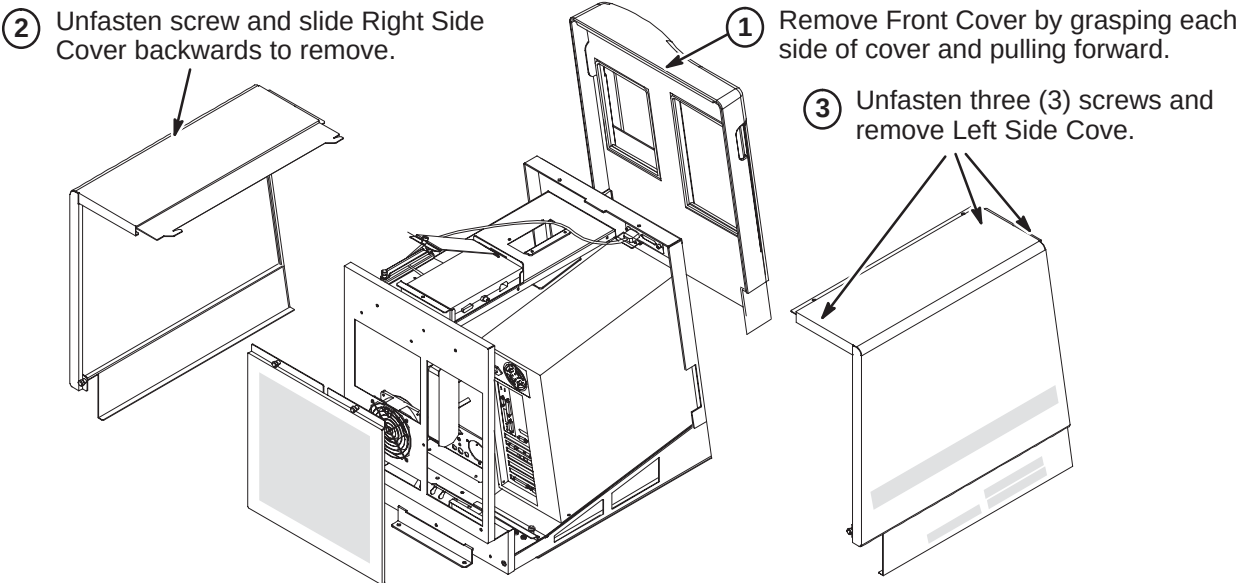
E1 – LOCATE OPERATOR WORKSPACE CABINET IN FRONT OF TABLE



E2 – REMOVE OPERATOR WORKSPACE CABINET REAR COVER

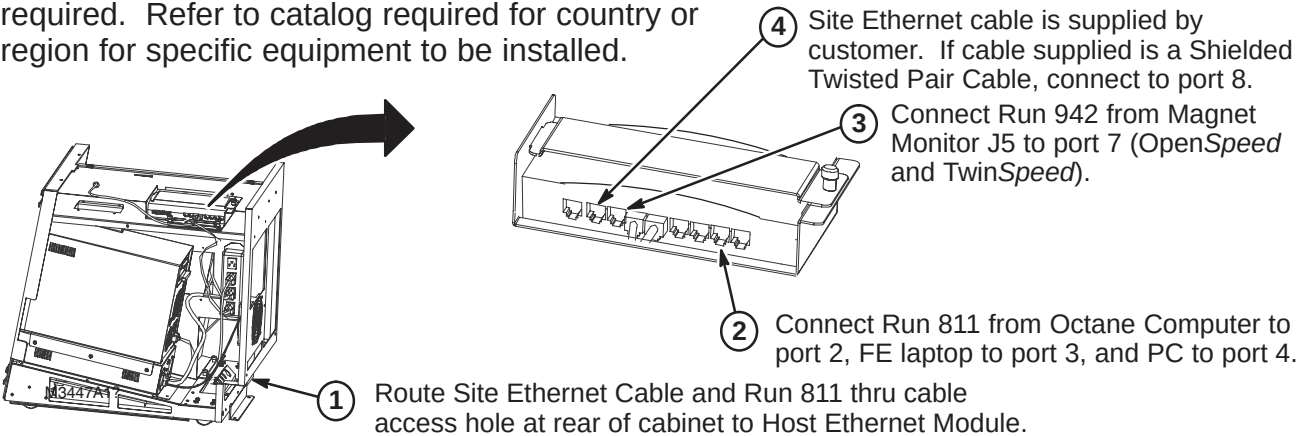


E3 – REMOVE OPERATOR WORKSPACE CABINET FRONT AND SIDE COVERS



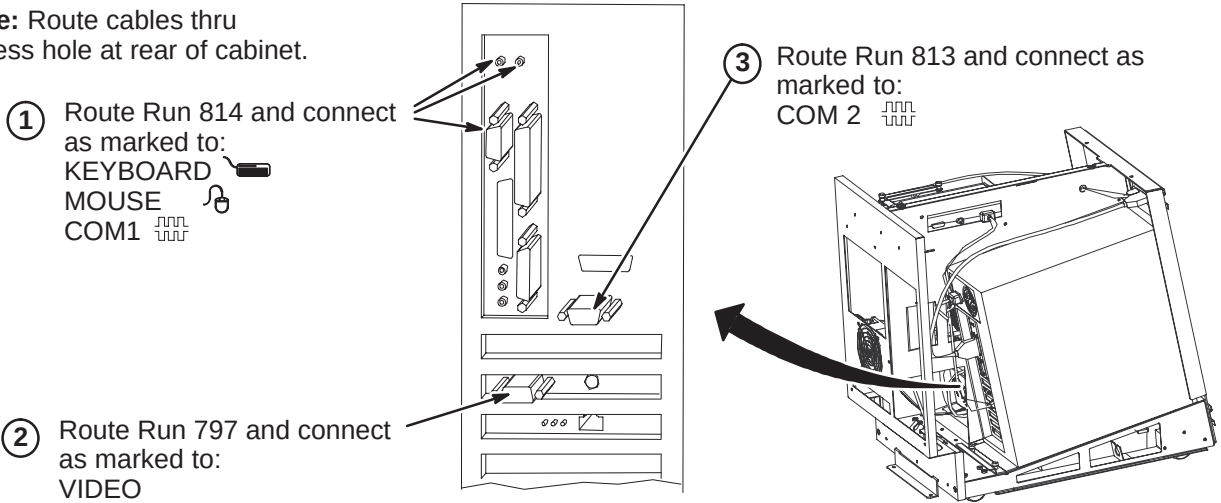
E4 – CONNECT ETHERNET CABLES

Note: Site Ethernet cable for InSite connection is supplied by customer. For Europe, connection to ISDN router is required. Refer to catalog required for country or region for specific equipment to be installed.



E5 – ROUTE/CONNECT RUNS 792, 797, 813, AND 814 TO PC MODULE

Note: Route cables thru access hole at rear of cabinet.



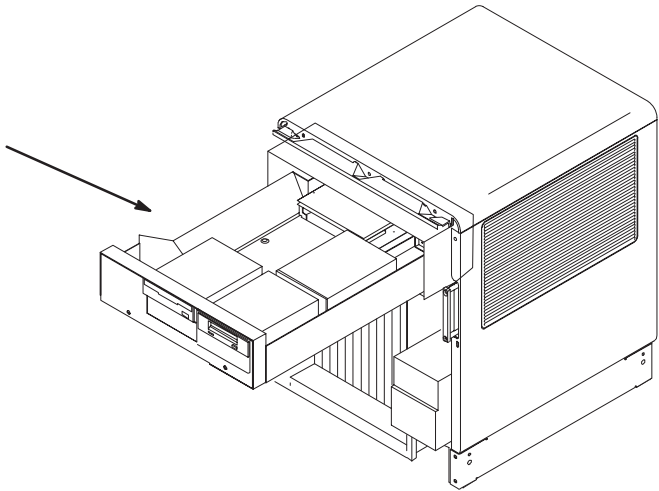
PROCEDURES F1 THRU F5 ARE USED ONLY WHEN UPGRADING FROM 5.x PLATFORMS, **DO NOT PERFORM PROCEDURES F1 THRU F5 FOR NEW OCTANE INSTALLATIONS**

INSTALLATION STEPS SUMMARY

- ☐ F1 – Removal of DAT and MOD Drives from Genesis Computer
- ☐ F2 – Remove SCSI Expansion Box
- ☐ F3 – DAT Drive Settings
- ☐ F4 – MOD Drive Settings
- ☐ F5 – Install DAT and MOD Drives (Legacy Media) in SCSI Expansion Box

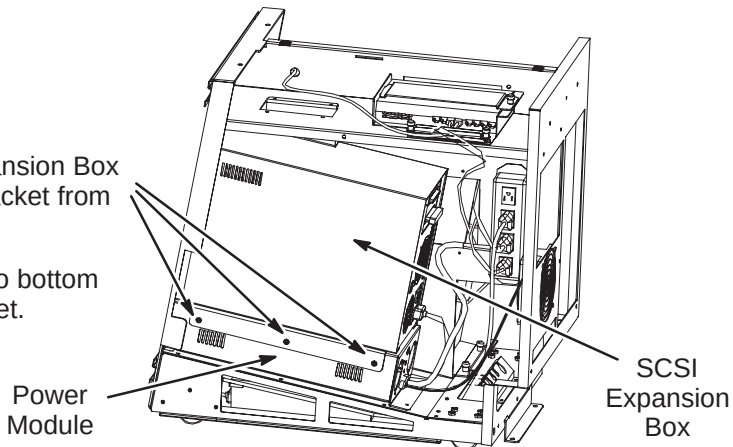
☐ F1 – REMOVAL OF DAT AND MOD DRIVES FROM GENESIS COMPUTER

UPGRADE FROM 5.x OPTIONS:
For transferring Legacy Media DAT and MOD Drives from Genesis Computer – refer to Tab 13, Directions 2174955 and 2174956, steps **A1** thru **A4** only for removal of existing drives.



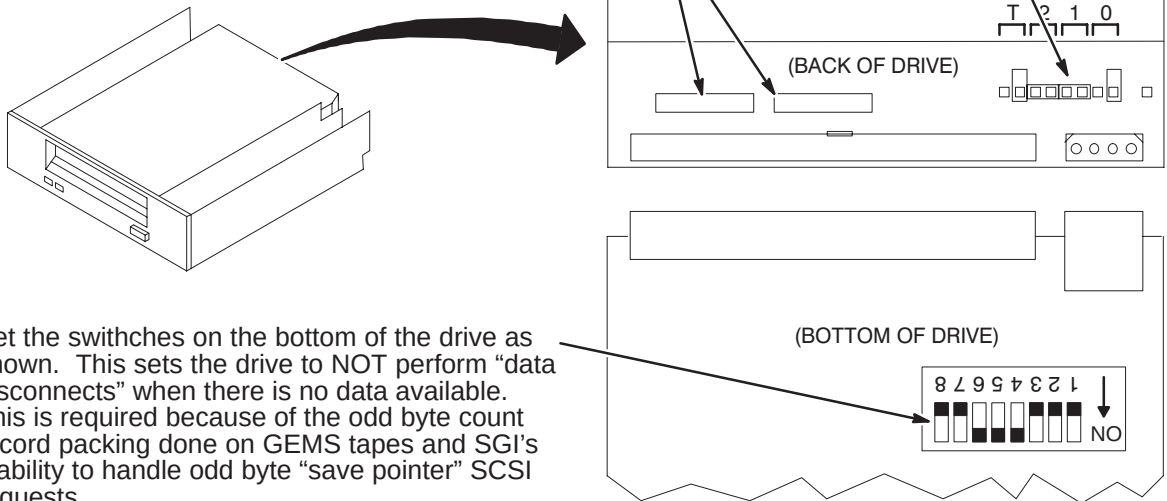
☐ F2 – REMOVE SCSI EXPANSION BOX

- ① Unfasten three (3) screws from SCSI Expansion Box mounting bracket and remove Box and bracket from power module.
- ② Unfasten six (6) screws that hold bracket to bottom of SCSI Expansion Box and remove bracket.
- ③ Remove SCSI Expansion Box cover.



☐ F3 – DAT DRIVE SETTINGS

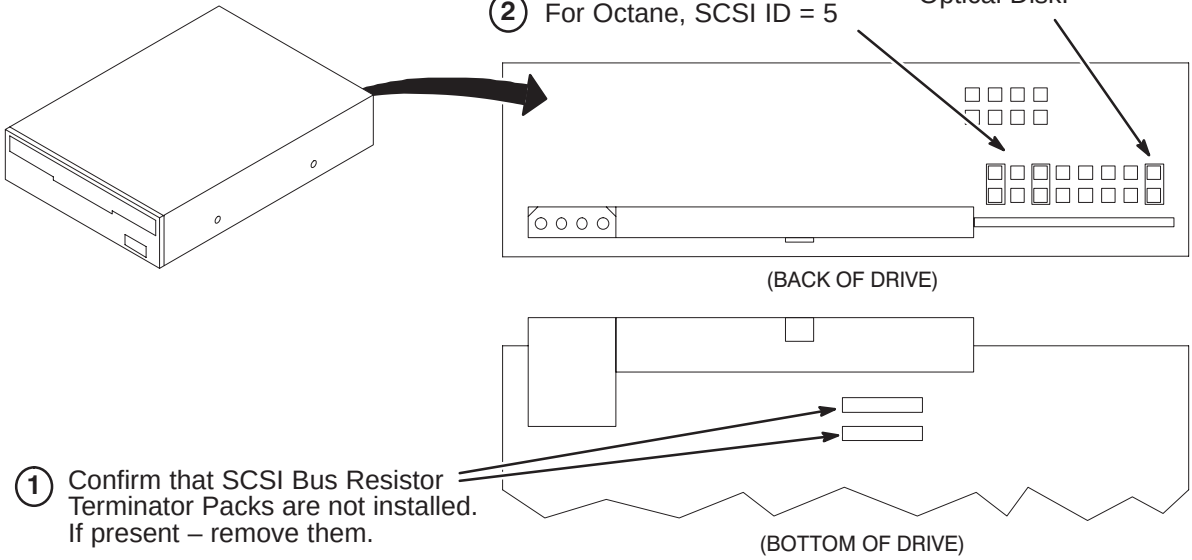
- ① Confirm that SCSI Bus Resistor Terminator Packs are not installed. If present – remove them.
- ② For Octane, set SCSI ID = 6



- ③ Set the switches on the bottom of the drive as shown. This sets the drive to NOT perform “data disconnects” when there is no data available. This is required because of the odd byte count record packing done on GEMS tapes and SGI's inability to handle odd byte “save pointer” SCSI requests.

☐ F4 – MOD DRIVE SETTINGS

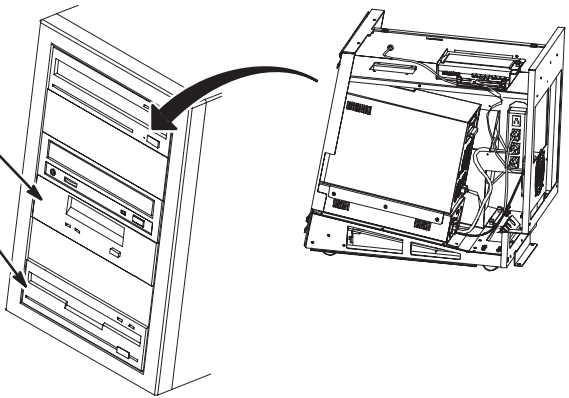
- ② For Octane, SCSI ID = 5
- ③ Make sure jumper is installed on pins at position 8 for SCSI Optical Disk.



- ① Confirm that SCSI Bus Resistor Terminator Packs are not installed. If present – remove them.

☐ F5 – INSTALL DAT AND MOD DRIVES (LEGACY MEDIA) IN SCSI EXPANSION BOX

- ① Install DAT Drive in slot 4 with existing hardware.
- ② Install MOD Drive in slot 6 with existing hardware.
- ③ Connect power cords to DAT and MOD Drives.
- ④ Connect ribbon cable to DAT and MOD Drives.
- ⑤ Install SCSI Expansion Box cover. Reattach bracket to bottom of Box. Reattach bracket and Box assembly to power module.

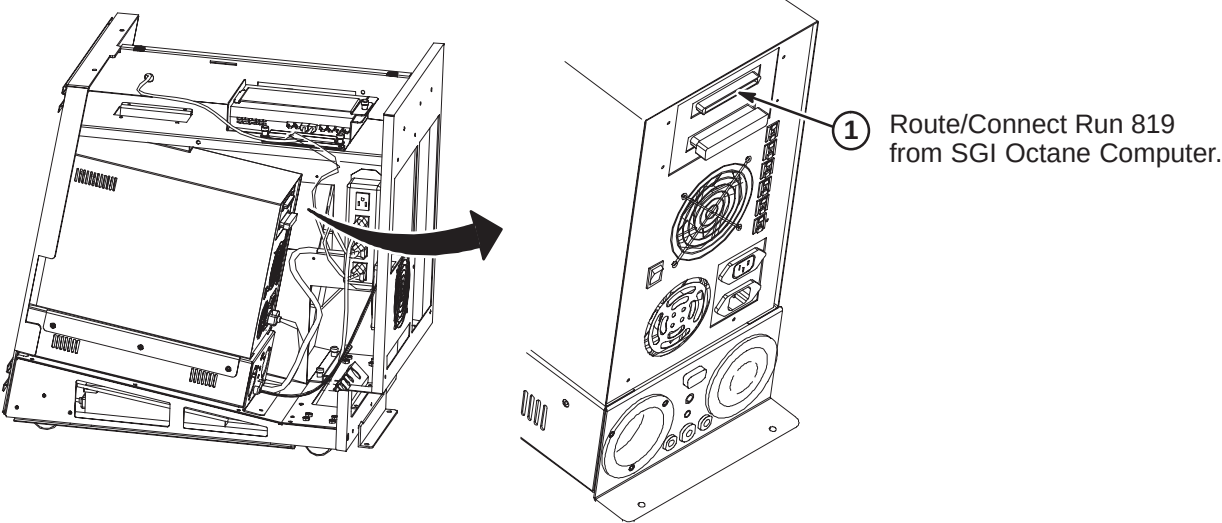


INSTALLATION STEPS SUMMARY

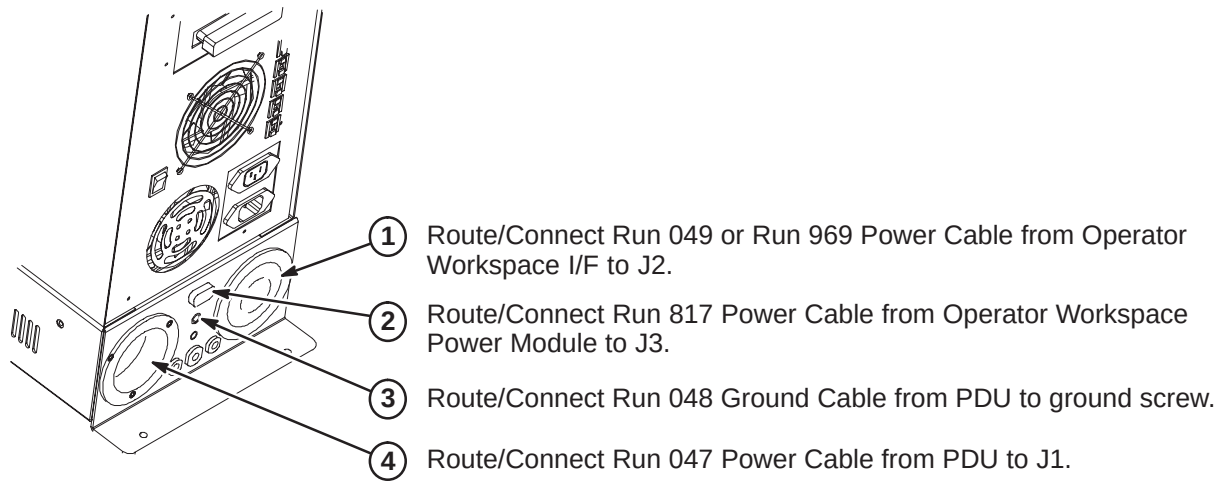
- G1 – Route/Connect Run 819 to SCSI Expansion Box
- G2 – Route/Connect Runs 047, 048, 049/969, and 817 to Power Module
- G3 – Align Operator Workspace Cabinet Cables & Attach InSite “Computer” Magnet
- G4 – Bundle Operator Workspace Cabinet Cables

G1 – ROUTE/CONNECT RUN 819 TO SCSI EXPANSION BOX

Note: Route all cables from SGI Octane Computer thru cable access hole at rear of cabinet.

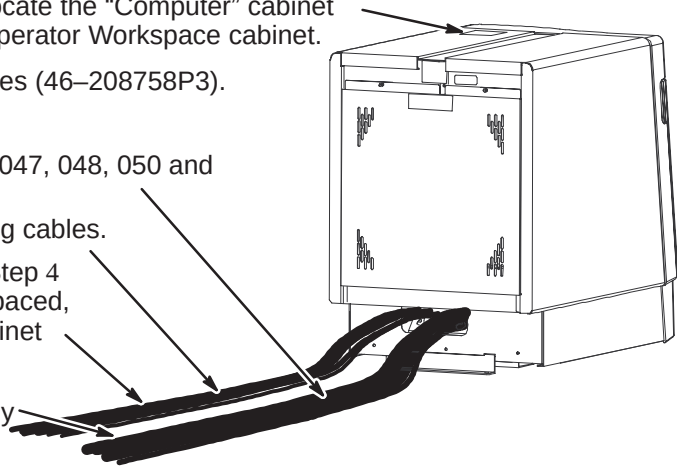


G2 – ROUTE/CONNECT RUNS 047, 048, 049/969, AND 817 TO POWER MODULE



G3 – ALIGN CABINET CABLES & ATTACH InSite “COMPUTER” MAGNET

- 1 Install Operator Workspace Cabinet right and left Side covers, Front cover and Rear cover.
- 2 From the 8.0 LX InSite Kit (46–301708G14), locate the “Computer” cabinet magnet (46–320095P1) and attach to top of Operator Workspace cabinet.
- 3 Locate the twenty (20) seven inch long cable ties (46–208758P3).
- 4 Separate cables into two bundles;
The first bundle will include power cable Runs 047, 048, 050 and fiber optic cable Run 790.
The second bundle will include all the remaining cables.
- 5 Align the cables of the first bundle created in Step 4 and attach ten (10) of the cable ties, equally spaced, around the cable bundle between the OW Cabinet and the OW Table.
- 6 Attach the remaining ten (10) cable ties, equally spaced, around the second cable bundle.



G4 – BUNDLE OPERATOR WORKSPACE CABINET CABLES

- 1 For each tie–wrapped bundle, place Cable Cover (46–307782P1) underneath.
- 2 Attach mating VelcroT to outside surface of Operator Workspace cabinet.
- 3 Wrap Cable Cover around bundle attaching the VelcroT hooks and loops.
- 4 Attach cable bundle Clamp (46–307873P1) to center of Cable Cover.
- 5 Carefully roll OW Cabinet into final position. Make sure cable bundles are unobstructed and slide easily. Attach to mating Velcro on cabinet.
- 6 Repeat Step 5 as often as necessary.

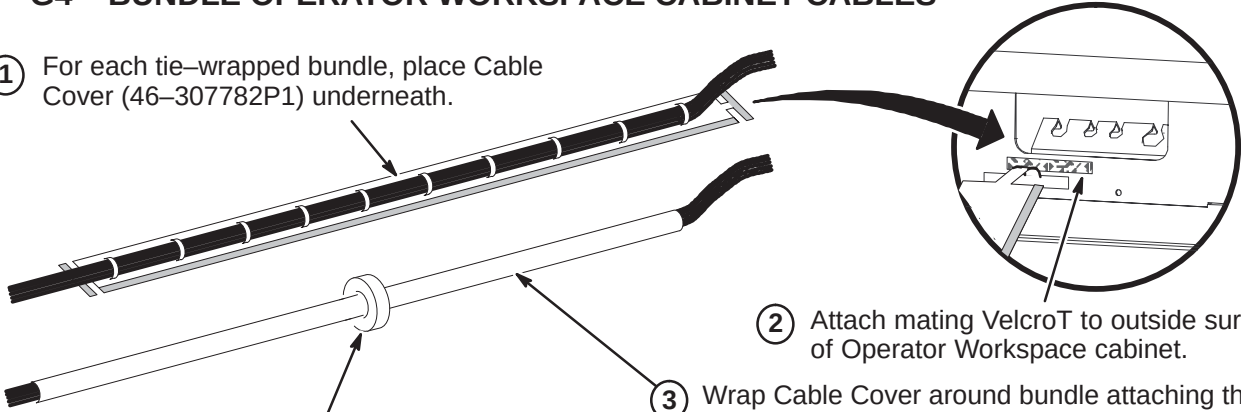


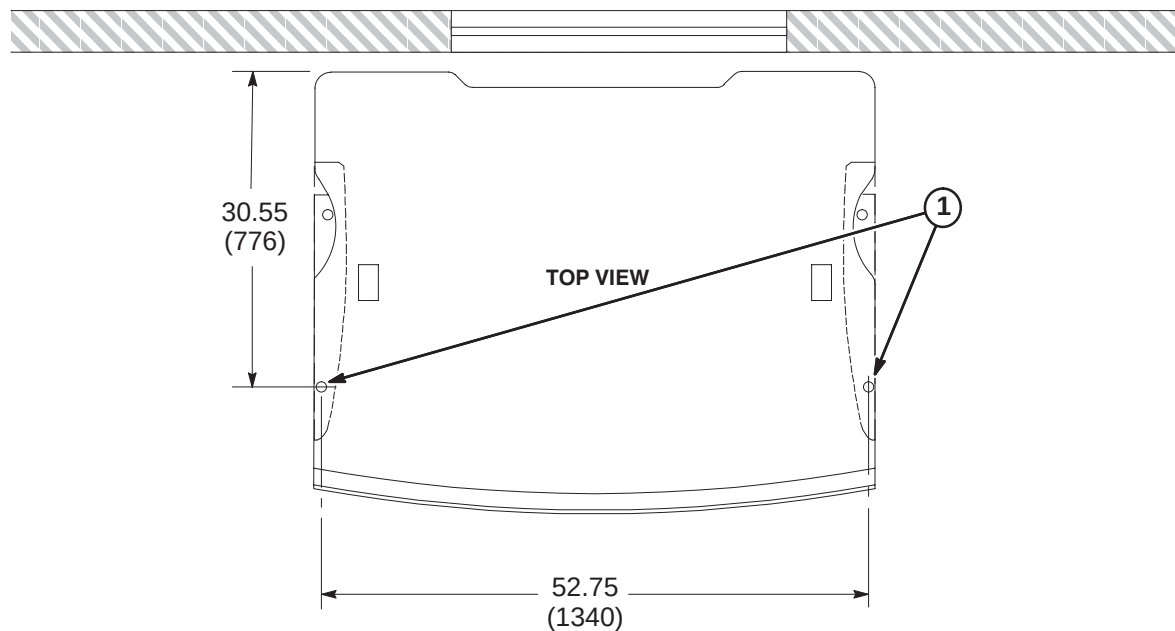
TABLE OF CONTENTS

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1-2 SECURE OPERATOR WORKSPACE TABLE TO FLOOR	1-3
1-3 POSITION COMPUTER CABINET	1-4
1-4 SEISMIC ANCHORING (IF REQUIRED)	1-4
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1-9 FINAL POSITIONING OF GOC CABINET	1-9
1-10 INSTALL SCIM/KEYBOARD AND MONITOR	1-10
1-11 INSTALL SCSI EXPANSION BOX AND LCD DISPLAY	1-11

1-1 OPERATOR WORKSPACE TABLE ANCHOR INSTALLATION

Refer to room layout plan and consult with Site customer for exact location of Operator Workspace Table. Items installed are part of Color Monitor Table Tie Down Kit, 2154928.

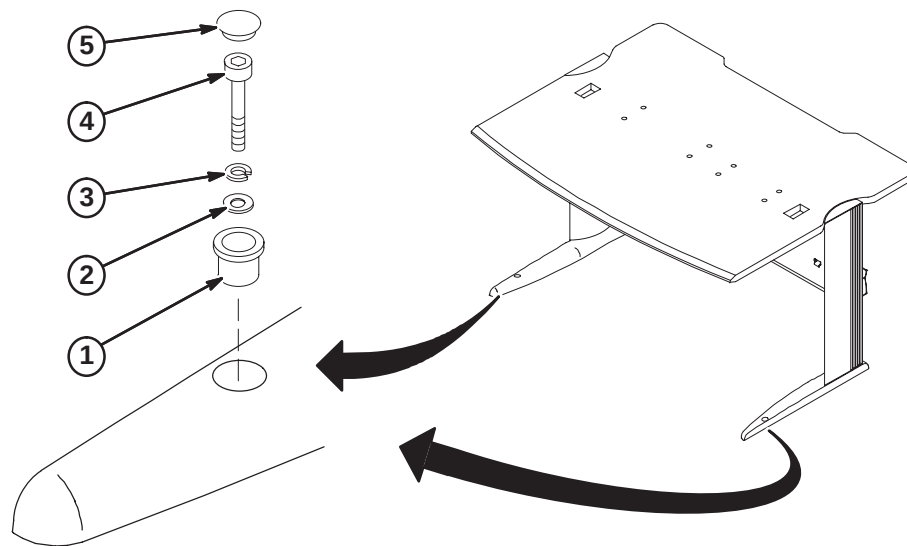
1. Remove and discard the two (2) cover plugs. Locate and mark on floor the two hole locations at the front of the table legs.
2. Move table to prepare for installing the floor anchors in Steps 3 and 4.
3. At each hole marked in Step 2, drill a .38 inch hole and insert 1/4-20 x .375 diameter anchor (2168253).
4. Secure anchor in place with supplied Setting Tool (2168781).



1-2 SECURE OPERATOR WORKSPACE TABLE TO FLOOR

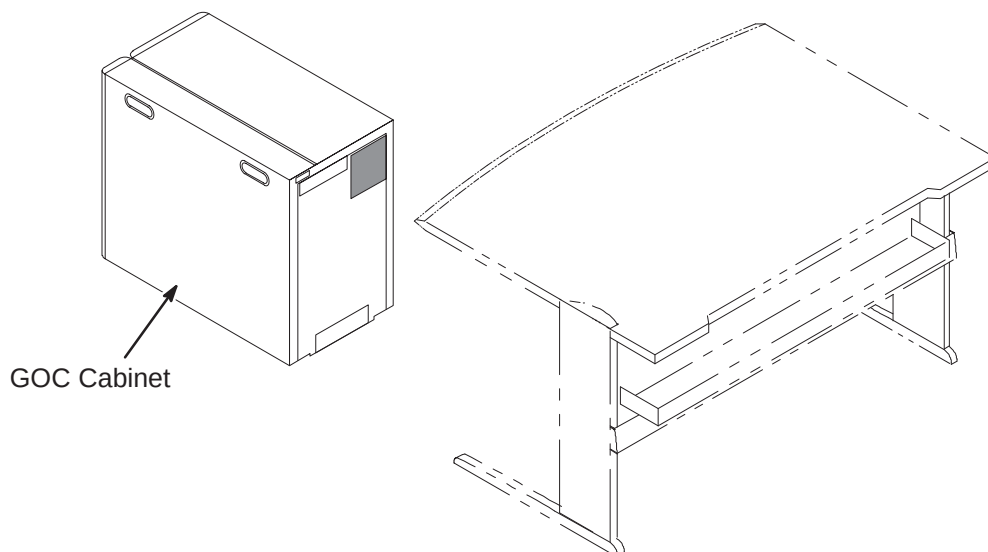
Items installed are part of Color Monitor Table Tie Down Kit, 2154928.

1. Insert Bushing (2151454) in table leg hole.
2. Insert as many Flat Washers (2168251) as required (3 provided for each leg) for securing table to floor.
3. Insert one Lock Washer (2168252) on top of Flat Washer(s).
4. Depending upon depth of previously installed anchor, install either the 1-3/4 inch long or the 2 inch long 1/4 x 20 Hex Socket Cap Screw.
5. Install black nylon Plug (2168254).
6. Repeat anchoring process below for other leg.

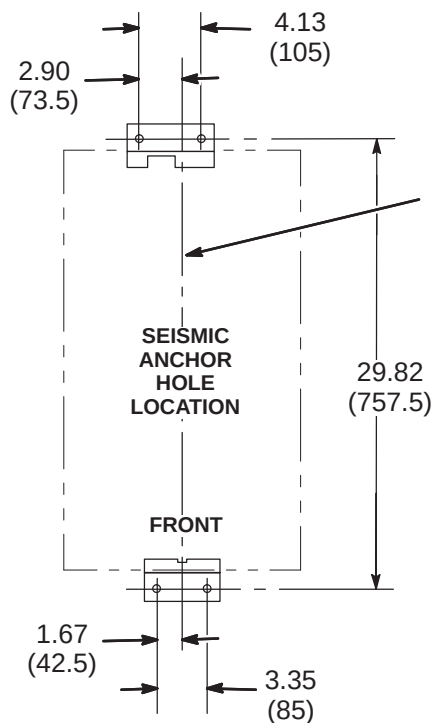


1-3 POSITION COMPUTER CABINET

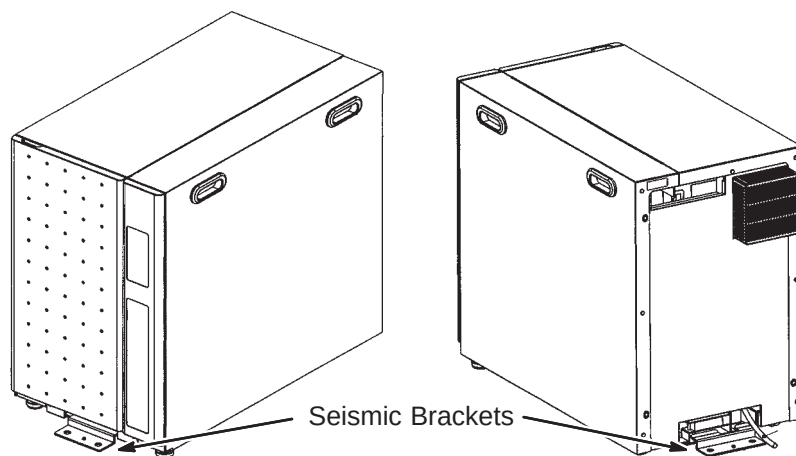
Consult with Site customer for exact location of Global Operator Cabinet (GOC). The location could be underneath the left or right side of the Operator Workspace table, or on the outside left or right side of the table.

**1-4 SEISMIC ANCHORING (IF REQUIRED)**

If seismic anchoring is required, refer to site architectural drawings for fastener type details or applicable Pre-installation manual. Illustration below indicates hole location for installing brackets.

**NOTE:**

- ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.



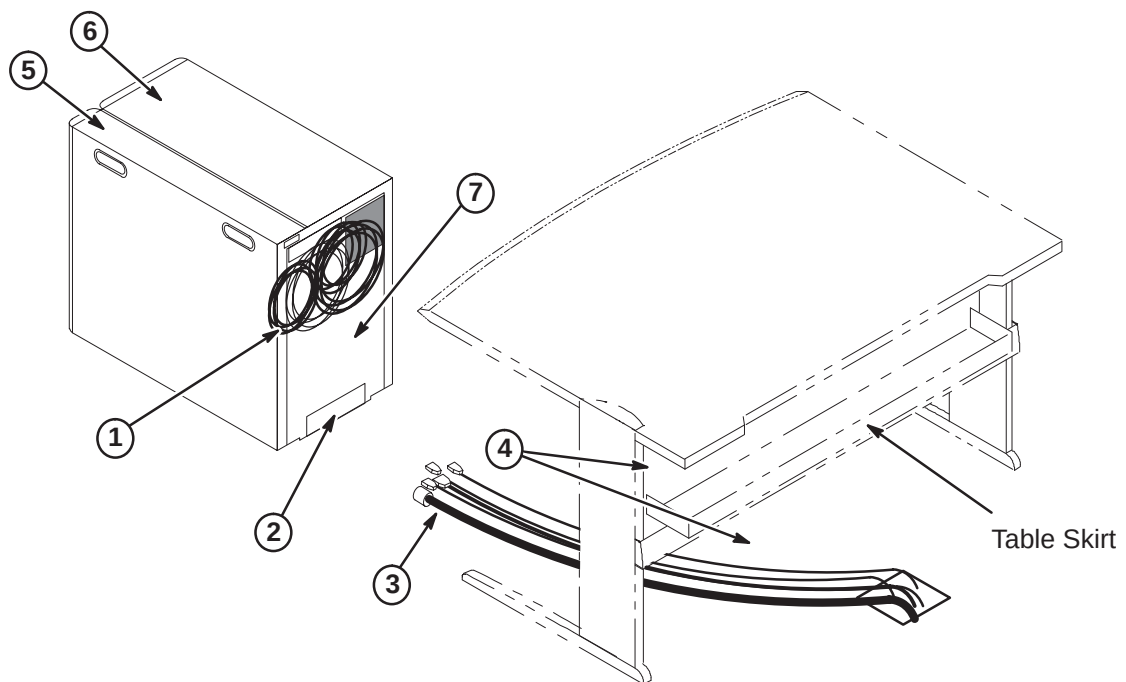
1-5 COMPUTER CABINET CABLE ROUTING AND CONNECTIONS

The GOC (Computer) Cabinet is shipped with seven cables connected to the computer that will be routed and connected to equipment that will be installed on the top of the OW Table. The cables are:

- Three power cords
- Two video output cables
- Keyboard output cable
- SCSI output cable

These cables are already connected to the computer and routed thru the opening located at the top of the rear of the GOC cabinet as shown in the illustration below. The remainder of the cables to be connected to designated modules within the GOC Cabinet will have to be routed thru the lower opening.

1. Runs 1062, 1063, 1064, 1065, 1068, 1069, and 1070 are pre-connected to computer and are routed thru top opening.
2. Runs 1077, 1081, 1082, 1083, 1084, 1085, power cord for the Patient Alert system, and any other option cables will be routed thru the lower opening.
3. Route all cables before connecting any cables. Refer to cable map for connection information.
4. If GOC Cabinet is to be located under OW Table, the cable coming out of the top opening of the cabinet will be routed above the table skirt. The cables coming out of the lower opening will be routed below the table skirt.



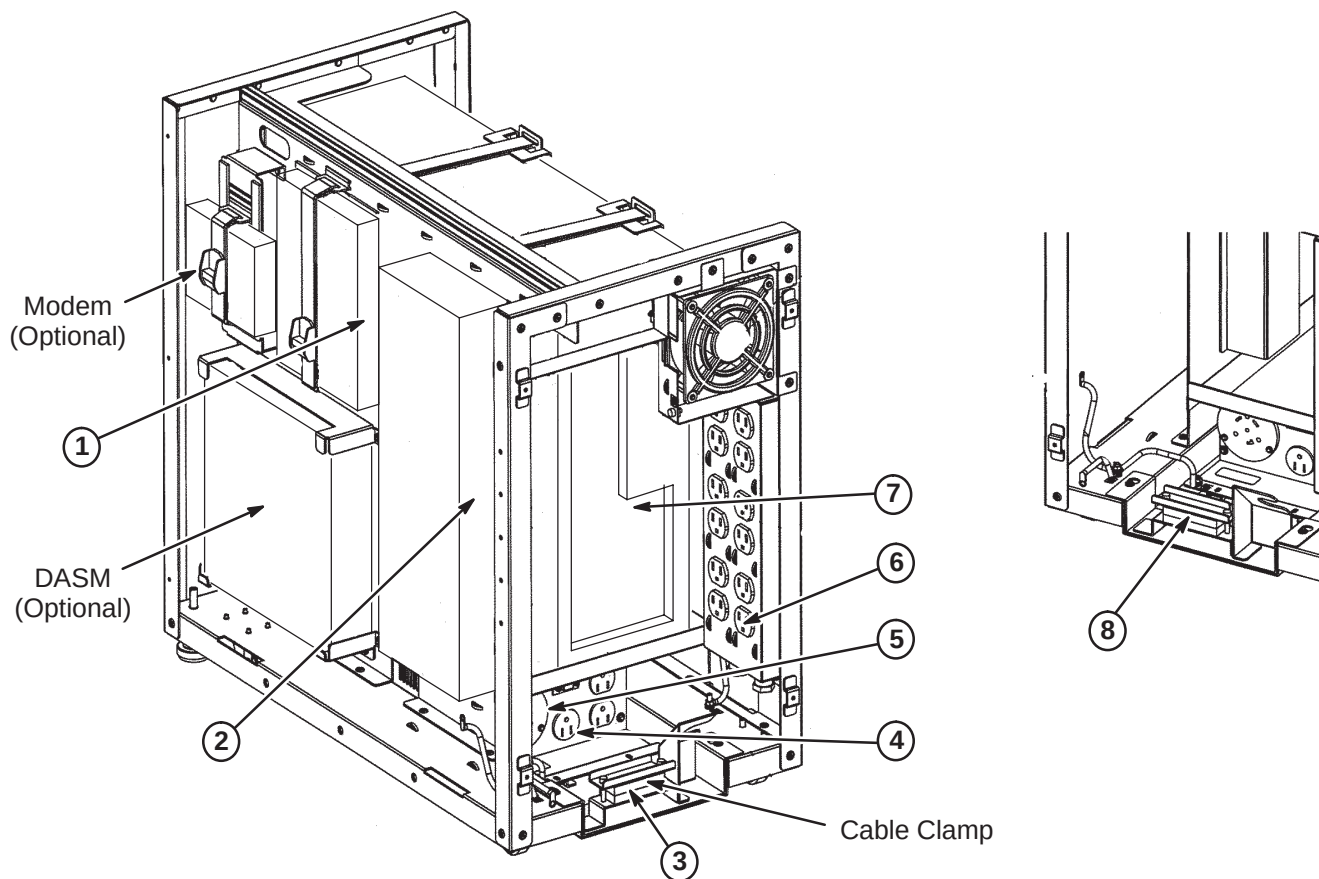
The Computer Cabinet covers need to be removed to access the cable connection points.

5. For right side cover, remove three screws at rear of cover, slide toward rear of cabinet and then lift up.
6. For left side cover, remove four screws at rear of cover, slide toward rear of cabinet and then lift up.
7. Loosen four screws holding the back cover. Lift and remove cover.

1-6 ROUTE CABLES FOR LOWER OPENING

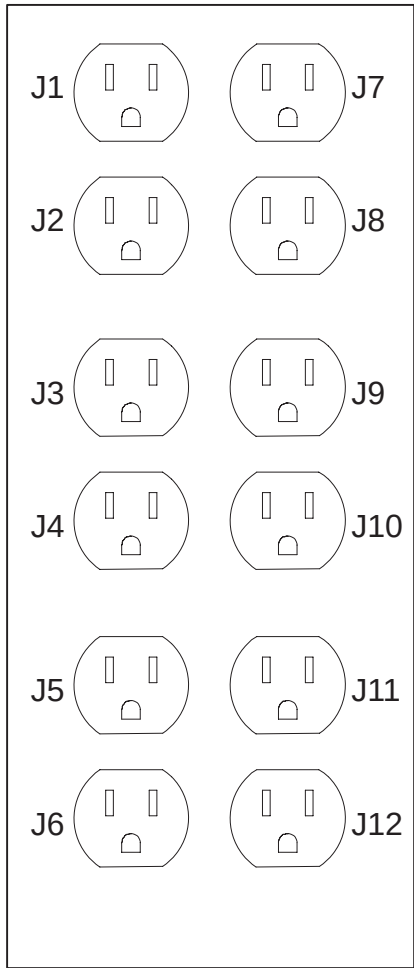
Refer to cable map for information on cables to be connected. Route cables thru clamp at lower rear of cabinet and connect cable runs as shown in illustration below.

1. Run 1077 to Lan Hub.
2. Run 1085 to WIM.
3. Run 1084 routes from GOC Cabinet thru clamp to J26 on System Cabinet.
4. Make sure Modem power cable is connected to J5 on Computer Cabinet PDU.
5. Run 1081 to PDU J1.
6. Patient Alert Power cable to power strip at J12. Refer to Section 1-7 for other power cable connections to power strip.
7. Runs 1082 and 1083 to Linux PC computer. Refer to Section 1-8 for all cables connected to Linux PC computer.
8. Tighten clamp to secure cables in place.



1-7 POWER STRIP CONNECTIONS

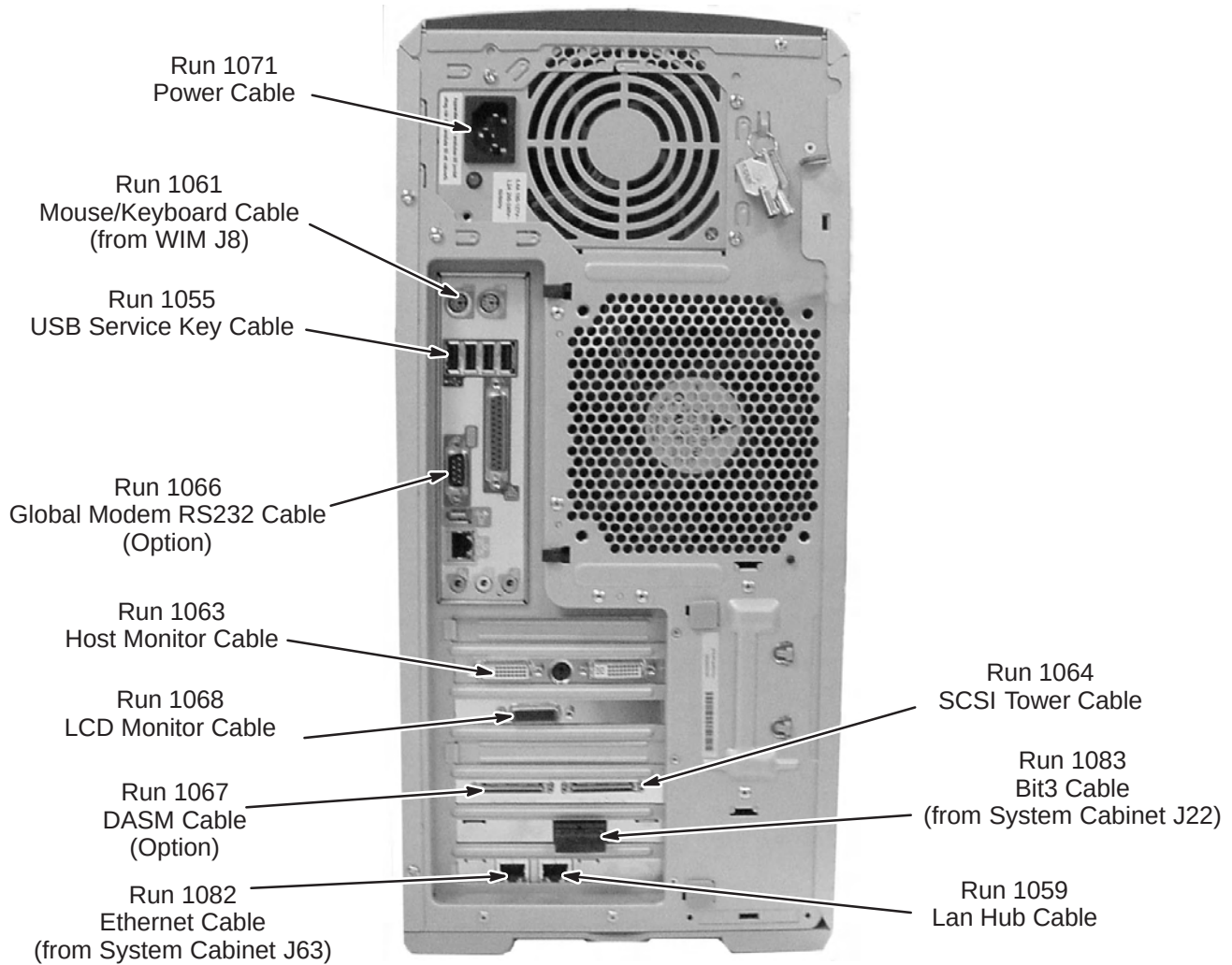
The illustration below show the power cable connection locations to the power strip in the GOC cabinet. Make sure power cables are attached as shown.



POSITION NAME		MAX AMP	
		CIRCUIT 1	CIRCUIT 2
J1	Host	6.4	
J2	Host LCD	0.8	
J3	Fan	0.2	
J4			
J5			
J6			
J7	Hub		0.2
J8	2nd LCD		0.8
J9			
J10	DASM		0.6
J11	SCSI Tower		7.0
J12	Patient Alert		0.2
TOTAL		7.4	8.8

1-8 LINUX PC CABLE CONNECTIONS

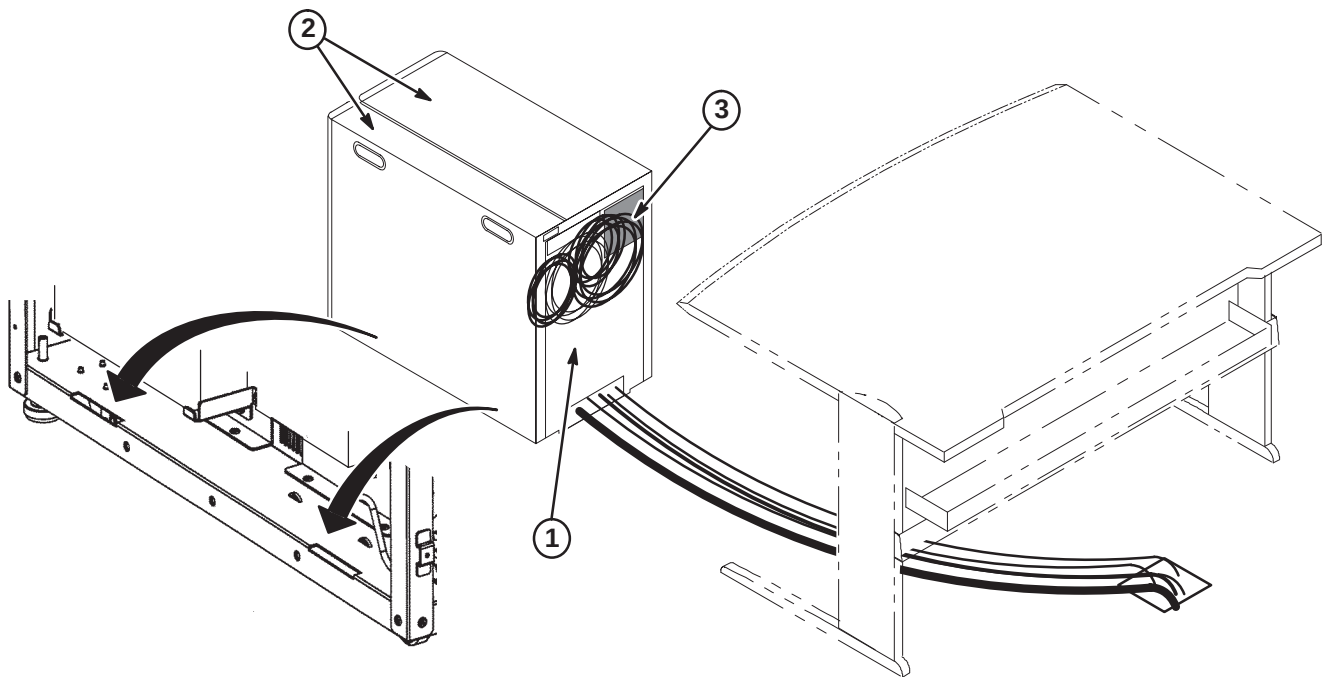
The illustration below shows the connection points of cables routed to the Linux PC. All cables shown are preconnected except for Runs 1082 and 1083, which have to be routed and connected at site.



1-9 FINAL POSITIONING OF GOC CABINET

Complete positioning of GOC Cabinet.

1. Re-attach rear cover and tighten screws.
2. Install both side covers by inserting tabs at bottom of covers into slots (as indicated in illustration below) and slide forward. Install screws at rear of cover that were removed in earlier procedure.
3. Route remaining cables from top opening at rear of GOC Cabinet thru opening of table skirt.
4. Place cabinet in final position as determined by customer.



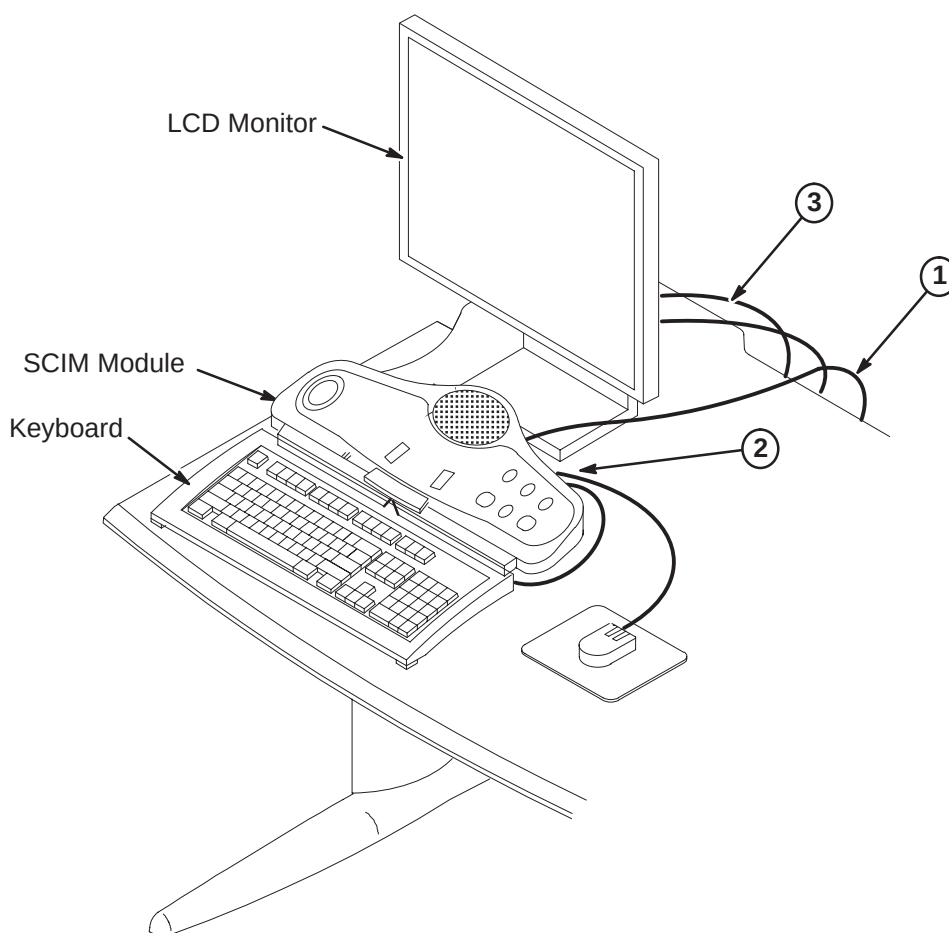
1-10 INSTALL SCIM/KEYBOARD AND MONITOR

Consult with Site customer for exact location of Keyboard and Monitor on Operator Workspace (OW) Table.

Note

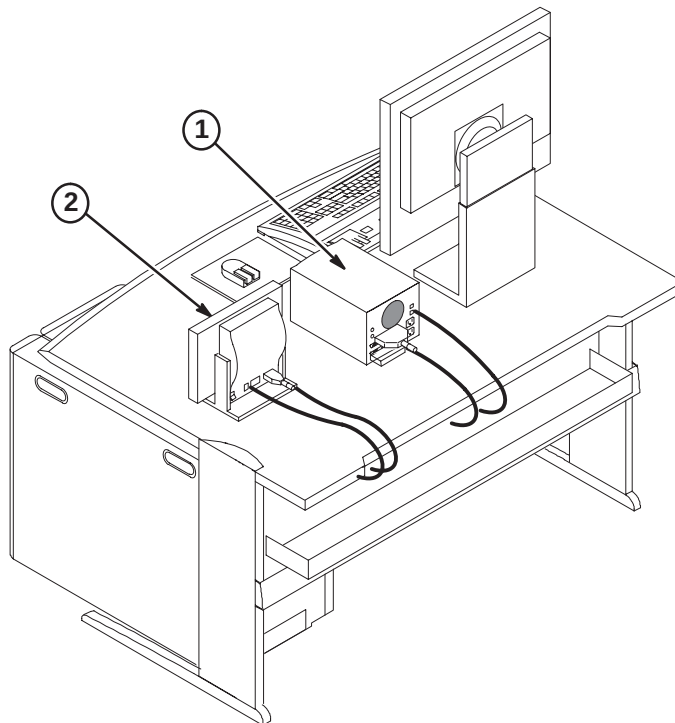
Refer to Installation document OW1INA3.DOC on Service Methods CD-ROM or MR Service Engineering web site for installation information on language labels and keyboard mounting.

1. Route data cable Run 1062 from Computer Cabinet to bottom side of SCIM Module and connect as marked. (Run 805, supplied with SCIM Kit (2307175) is not used for EXCITE) Run 1062 is preconnected to the Computer Cabinet and is located on back side of cabinet.
2. Place Keyboard and Mouse on table and connect cables to bottom side of SCIM Module as marked.
3. Place LCD Host Monitor on table in back of SCIM. Route Runs 1063 and 1065 from Computer Cabinet and connect to monitor.



1-11 INSTALL SCSI EXPANSION BOX AND LCD DISPLAY

1. Place the SCSI Tower on the OW table and connect Runs 1064 and 1070 from GOC Cabinet. Install terminator.
2. Place LCD Display on OW table and connect Runs 1068 and 1069 from GOC Cabinet.



INSTALLATION STEPS SUMMARY

- A1 – Position ACGD/PDU Cabinet
- A2 – Install Fan Assembly and Attach InSite Cabinet Magnet
- A3 – Input Voltage Selection
- A4 – Circuit Breaker DIP Switch Settings

A1 – POSITION ACGD/PDU CABINET

- 1 Move cabinet to final position.

WARNING!

DUE TO WEIGHT OF CABINET, AT LEAST TWO PEOPLE ARE REQUIRED TO MOVE CABINET

- 2 Anti-tip bars for the ACGD/PDU Cabinet are shipped in the SRF Cabinet. Remove from SRF Cabinet for this procedure.
- 3 Attach bars to cabinet.
- 4 Adjust leveling pads to level cabinet.

A2 – INSTALL FAN ASSEMBLY AND ATTACH INSITE MAGNET

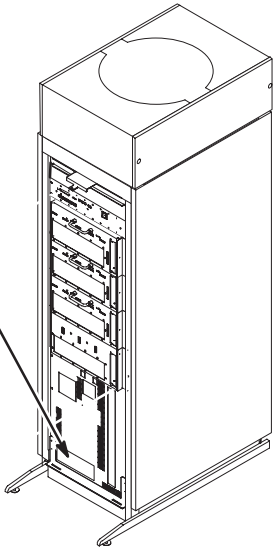
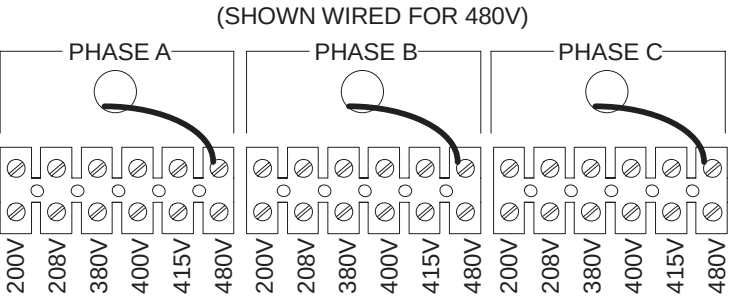
- 1 Remove Fan Assembly from shipping container.
- 2 Remove two side trim panels (two screws from each).
- 3 Requiring two people, place the Fan Assembly on top of cabinet with the cable connector facing the rear of the cabinet.
- 4 Secure Fan Assembly in place with (4) four screws and washers, two on each side of cabinet.
- Note:** The fasteners are provided in a plastic bag located in PDU Module. Rear PDU covers will need to be removed for access to plastic bag with fasteners.
- 5 Install two side trim panels with previously removed screws.
- 6 Connect cable from SGA Power Supply to the fan connector.

DANGER!!

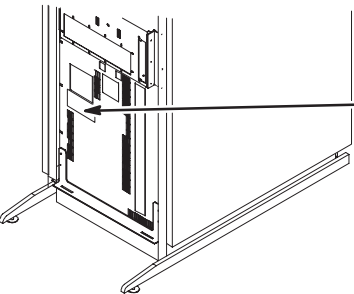
LETHAL VOLTAGES ARE PRESENT WITHIN THE PDU. CHECK THAT POWER AT THE MAIN DISCONNECT IS OFF, LOCKED, AND TAGGED BEFORE PROCEEDING.

A3 – INPUT VOLTAGE SELECTION

- 1 Using a voltmeter or other voltage indicating device, check to be sure that no voltage is being applied to the input of the PDU.
- 2 Remove the plate from the front of the PDU marked INPUT CIRCUIT BREAKER CURRENT SELECT / INPUT VOLTAGE SELECT.
- 3 For each of the three phases, be sure the wire is connected to the proper terminal for the actual input voltage at the site and that the terminal screw is tight to make a good connection.



A4 – CIRCUIT BREAKER DIP SWITCH SETTINGS



- 1 Remove small cover plate under the main circuit breaker to access DIP switches controlling the circuit breaker operation.
- 2 Table below shows the position of the DIP switches for each input voltage selection. The switches immediately under the letter “L” and “I” are the only ones that should be changed.

INPUT VOLTAGE	DIP SWITCH “L” SETTING	L	I	DIP SWITCH “I” SETTING	INPUT VOLTAGE
200V					200V
208V					208V
380V					380V
400V					400V
415V					415V
480V					480V

INSTALLATION STEPS SUMMARY

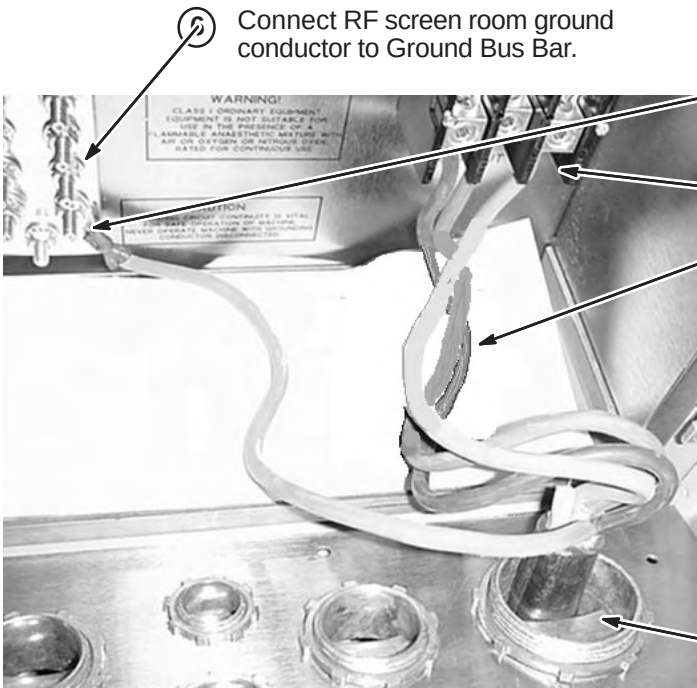
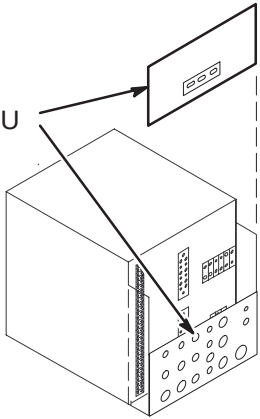
- ☐ B1 – Facility Power and Ground Connention
- ☐ B2 – Route Power Cables Thru Interface Panel
- ☐ B3 – Run 967 Power Cable Alteration
- ☐ B4 – Connect Power Cables to PDU Module

B1 – FACILITY POWER AND GROUND CONNECTION

WARNING!

IF 3 PHASE WYE WITH NEUTRAL AND GROUND (5 WIRE SYSTEM) INPUT IS USED THE NEUTRAL MUST BE TERMINATED INSIDE THE MAIN DISCONNECT CONTROL AND NOT BROUGHT TO THE ACGD/PDU CABINET.

- 1 Remove rear PDU module covers.



- 2 Connect RF screen room ground conductor to Ground Bus Bar.

- 5 Connect facility input dedicated ground conductor to Ground (GRN/YEL) Bus Bar.

- 4 Connect L1–L3 conductors to terminal block.

- 3 NOTE: Air flow thru cabinet must exit thru top of cabinet. If not, fan at top of cabinet is rotating in wrong direction. To reverse rotation of fan, switch any two of the three incoming L1–L3 power wires.

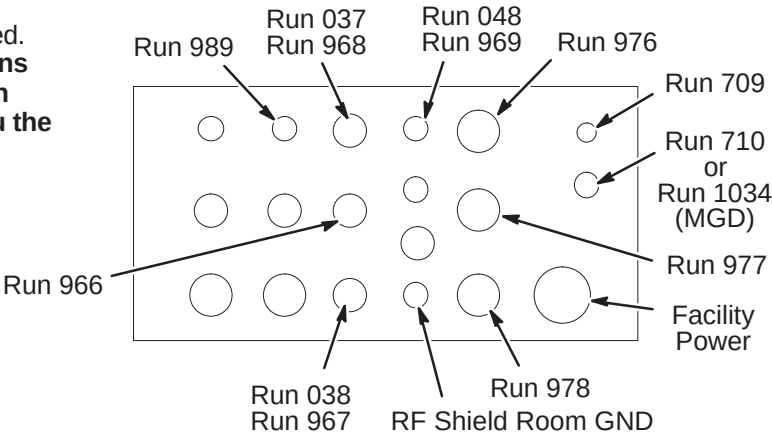
DANGER!!!

MAKE SURE ALL POWER TO CABINET IS OFF, LOCKED, & TAGGED BEFORE SWITCHING ANY ABOVE WIRES.

- 2 Route facility input power cable and ground wire thru interface panel.

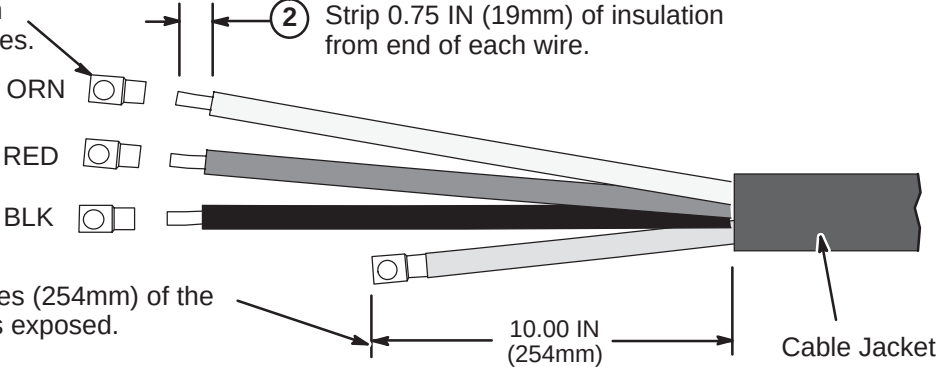
B2 – ROUTE POWER CABLES THRU INTERFACE PANEL

- 1 Route Runs thru interface panel as marked. Note: The GRN/YEL ground wires (Runs 037, 038, and 048) MUST be routed with their respective power cable Runs thru the same port.
- 2 Route any option Runs as marked.



B3 – RUN 967 POWER CABLE ALTERATION

- 1 Cut off ring terminal from ORN, RED, and BLK wires.
- 2 Strip 0.75 IN (19mm) of insulation from end of each wire.

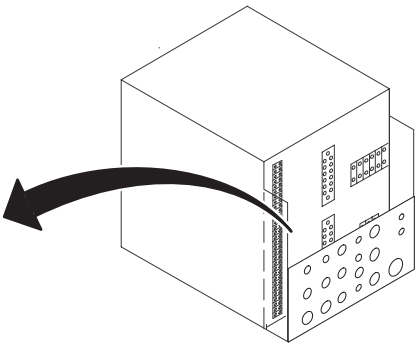


- 3 Strip jacket until 10 Inches (254mm) of the GRN/YEL ground wire is exposed.

B4 – CONNECT POWER CABLES TO PDU MODULE

WARNING!

ALL POWER CABLES LISTED BELOW MUST BE CONNECTED TO PDU MODULE LOCATED WITHIN ACGD/PDU CABINET.



- 1 Route/connect power cables to pressure connectors and studs on back of PDU module as shown in Table below.
- 2 Trim and connect ground wire cables (Runs 037, 038, and 048) to ground bus connectors.
- 3 Tighten clamps of interface panel to secure cables in place.

1	ACGD POWER
2	
3	
4	SYSTEM CABINET
5	
6	
7	SYSTEM SUPPORT MODULE
8	
9	
10	
11	RF/AMP
12	
13	
14	MNS/ SPARE #2
15	
16	TWIN ACCESSORY CABINET
17	
18	PC (BLK)
19	HOST (BRN)
20	SPARE #1
21	SPARE #2
22	UPS IN
23	UPS OUT
24	REGULATOR IN
25	
26	REGULATOR OUT
27	
28	
29	
30	TGWC INDOOR CAB

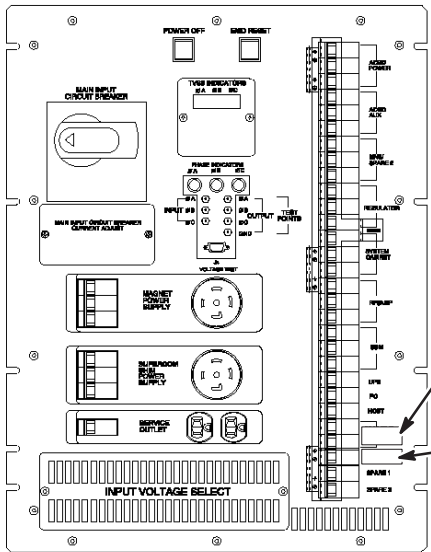
CONNECT TO	RUN	PHASE	DESTINATION
4,5,6	968	φ A, B, C	System Cabient 3φ, Neutral, GND
7,8,9	966	φ A, B, C	SSM 3φ Neutral, GND
10,11,12	967	φ A, B, C	RFI/AMP 3φ, GND
16,17	989	φ A,C	TAC Cabinet Single φ Neutral, GND
18,19	969	φ C,A	Operator Workspace Single φ, Neutral, GND
30		φ A	TGWC Indoor Water Cooled Cabinet Single φ, Neutral, GND
GRN/YEL	037		System Cabinet Ground Wire
GRN/YEL	038		SRF Cabinet Ground Wire
GRN/YEL	048		Operator Workspace Ground Wire
GRN/YEL			RF Shield Room Ground (Customer Supplied)

COLOR CODE: for remainder of cables:
3φ= Black, Red, Orange Neutral = Light blue Gnd = Grn/Yel
Single φ = Brown, Black

INSTALLATION STEPS SUMMARY

- ❑ C1 – Apply New Circuit Breaker Labels to Front of PDU
- ❑ C2 – Route/Connect Ribbon Cable 2315915 Inside ACGD/PDU Cabinet
- ❑ C3 – Route/Connect Run 974 to GP Module (If Required)
- ❑ C4 – Install Ferrite Filters on Gradient Cables (Required for TwinSpeed)

❑ C1 – APPLY NEW CIRCUIT BREAKER LABELS TO FRONT OF PDU



ACGD CABINET
(FRONT VIEW)

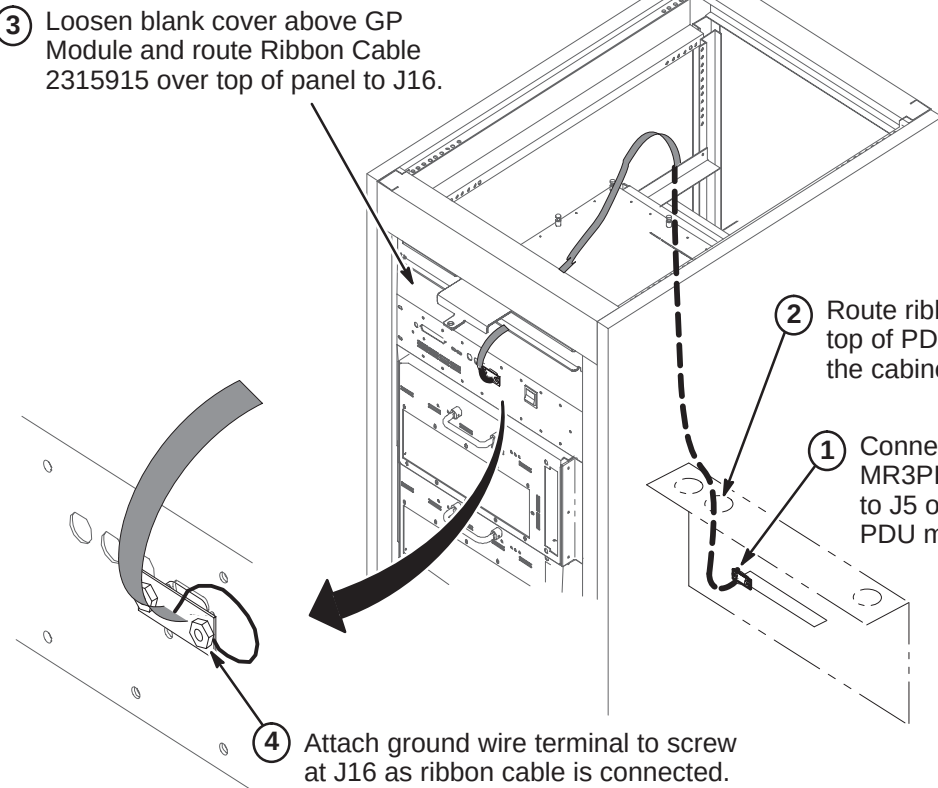
1 Apply TWIN ACCESSORY CABINET label over GRADIENT WATER CHILLER

2 If system being installed has a TGWC Indoor Water Cooled Cabinet, apply TGWC COOLING CAB label over GRADIENT FAN.
If system being installed does not have a TGWC indoor water cooled cabinet, apply "NOT USED" label over Gradient Fan.

❑ C2 – ROUTE/CONNECT RIBBON CABLE 2315915 INSIDE ACGD/PDU CABINET

NOTE: Grounded Ribbon Cable may not be present on early versions. If not available to install, go to Procedure C3 and complete routing of Run 974 to J16.

3 Loosen blank cover above GP Module and route Ribbon Cable 2315915 over top of panel to J16.



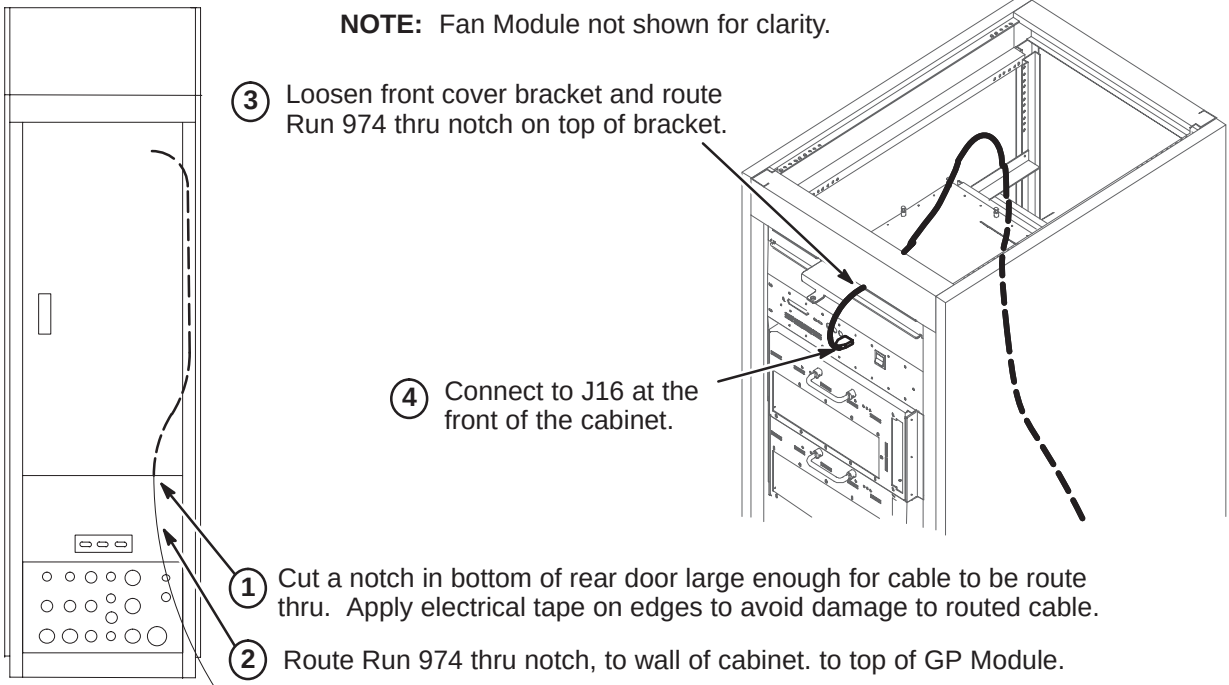
2 Route ribbon cable thru opening in top of PDU cover and up the side of the cabinet to top of modules.

1 Connect end of ribbon cable labeled MR3PDU-J5 (J4 in early versions) to J5 on board inside back cover of PDU module.

4 Attach ground wire terminal to screw at J16 as ribbon cable is connected.

❑ C3 – ROUTE/CONNECT RUN 974 TO GP MODULE (IF REQUIRED)

NOTE: COMPLETE THIS PROCEDURE ONLY IF RUN 974 CAN NOT BE CONNECTED TO J5 ON THE REAR INTERFACE PANEL. THIS WOULD INDICATE THAT THE INTERNAL CABINET CABLE THAT IS ROUTED IN PROCEDURE C2 FROM J5 TO J16 ON THE GP MODULE IS NOT PRESENT. THIS PROCEDURE INSTRUCTS YOU TO ROUTE THE CABLE DIRECTLY TO J16 ON THE GP MODULE.



NOTE: Fan Module not shown for clarity.

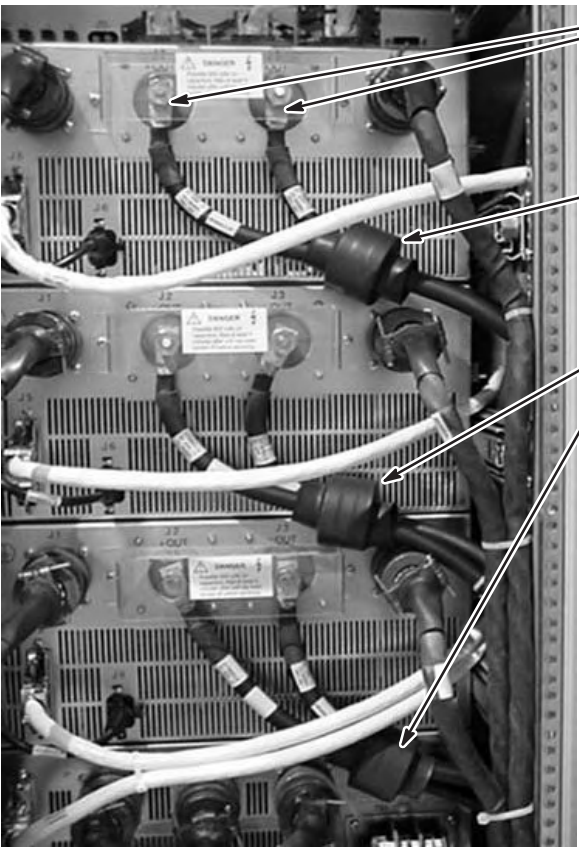
3 Loosen front cover bracket and route Run 974 thru notch on top of bracket.

4 Connect to J16 at the front of the cabinet.

1 Cut a notch in bottom of rear door large enough for cable to be route thru. Apply electrical tape on edges to avoid damage to routed cable.

2 Route Run 974 thru notch, to wall of cabinet. to top of GP Module.

❑ C4 – INSTALL FERRITE FILTERS ON GRADIENT CABLES (TwinSpeed REQUIRED)



1 Unfasten lug nuts holding gradient cables to studs on back of amplifier.

2 Slide Ferrite Filter (2308879) over gradient cables as shown and refasten cables to studs using previously removed lug nuts.

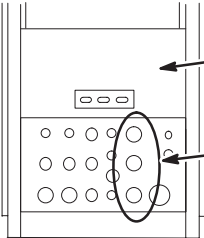
3 Repeat steps 1 and 2 for the other two sets of gradient cables.

INSTALLATION STEPS SUMMARY

- D1 – Gradient Output Cable Interface Panel Access
- D2 – Gradient Output Cable Routing
- D3 – Route/Connect Fiber Optics Cables to GP Module
- D4 – Connect Runs 971 and 972 to PDU Module
- D5 – Attach InSite Cabinet Magnets

D1 – GRADIENT OUTPUT CABLE INTERFACE PANEL ACCESS

ACGD CABINET
(REAR VIEW)

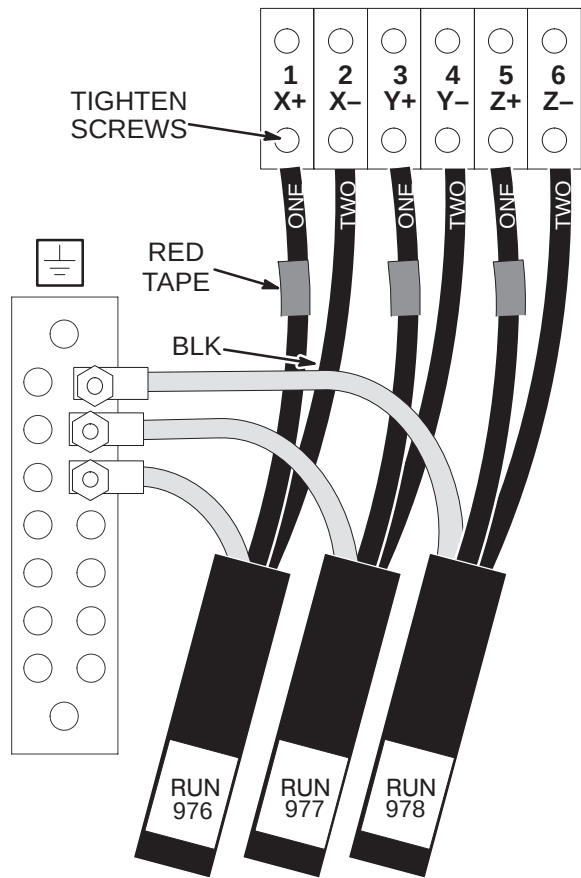


NOTE: Output cables (Runs 976, 977, and 978) were previously routed, cut to length, and terminated.

- 1 Remove Rear Cover Plate by removing seven screws that hold it in place.
- 2 Loosen screws on clamps for gradient output cables. Insert ends of gradient output cables into Cabinet Interface as marked. (Runs 976 into J3, Run 977 into J4, and Run 978 into J5)

D2 – GRADIENT OUTPUT CABLE ROUTING

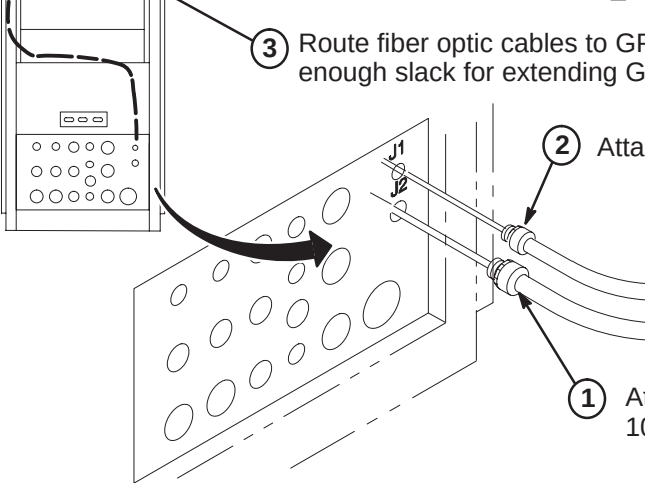
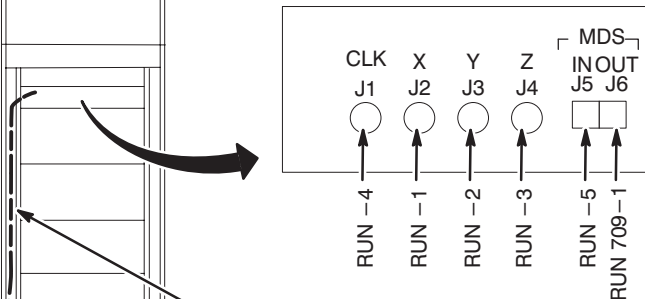
GRADIENT CABLE CONNECTIONS



- 1 Connect gradient output cables with red-taped (+) #1 wires to + output terminals and black (-) #2 wires to - output terminals.
NOTE: OVER TIGHTENING OF SCREWS IN STEP 2 CAN DAMAGE TERMINAL BLOCK.
- 2 Tighten screws to 52–65 inch-lbs (5.88–7.34 N-M). This is about what can be applied using only your thumb and little finger on wrench. Loosen 1/2 turn and re-tighten to the same torque. Repeat approximately 6 times or until wrench returns to the same spot after torquing.
- 3 Connect the GND leads of each run to the Ground Bus Bar studs.
- 4 Tighten clamps on Interface Panel to secure gradient output cables.

D3 – ROUTE/CONNECT FIBER OPTICS TO GP MODULE

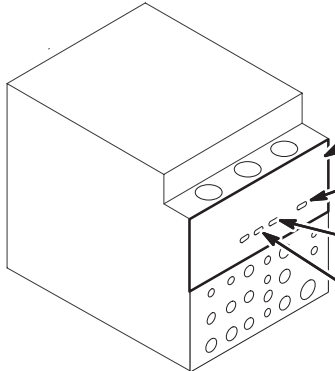
GP INTERFACE PANEL



SYSTEM		GP J#
10.x	LX	
709-1	709-1	J6
1034-5	710-5	J5
1034-3	710-3	J4
1034-2	710-2	J3
1034-1	710-1	J2
1034-4	710-4	J1

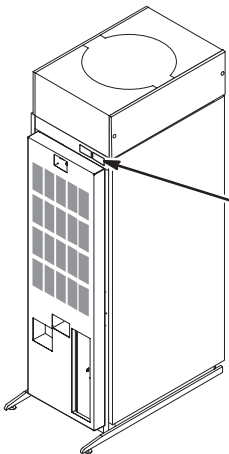
- 3 Route fiber optic cables to GP Module. Leave enough slack for extending GP module forward.
- 2 Attach Run 709 to I/F J1
- 1 Attach Run 710 or Run 1034 to I/F J2
- 4 Connect fiber optics to GIP module according to table.

D4 – CONNECT RUNS 971 AND 972 TO PDU MODULE



- 1 Attach rear PDU module covers removed in Procedure B1.
- 2 Connect Run 974 to J5 if ribbon cable 2315915 is installed in cabinet.
- 3 Connect Run 972 to J3.
- 4 Connect Run 971 to J2.

D5 – ATTACH INSITE CABINET MAGNETS



- 1 Attach front and rear covers.
- 2 From the 8.0 LX InSite Kit (46-301708G14), locate the “Gradient” cabinet magnet (46-320095P7) and the “PDU” cabinet magnet (46-320095P4). Attach to front of cabinet as shown.

SRF CABINET INSTALLATION INSTRUCTIONS

1-1 Tab 16, SRF Cabinet, contains installation instructions for the following:

- *DIRECTION* 2286812 – Signa SRF Cabinet Installation Instructions
- *DIRECTION* 2346942 – Signa SRFD2 Cabinet Installation Instructions

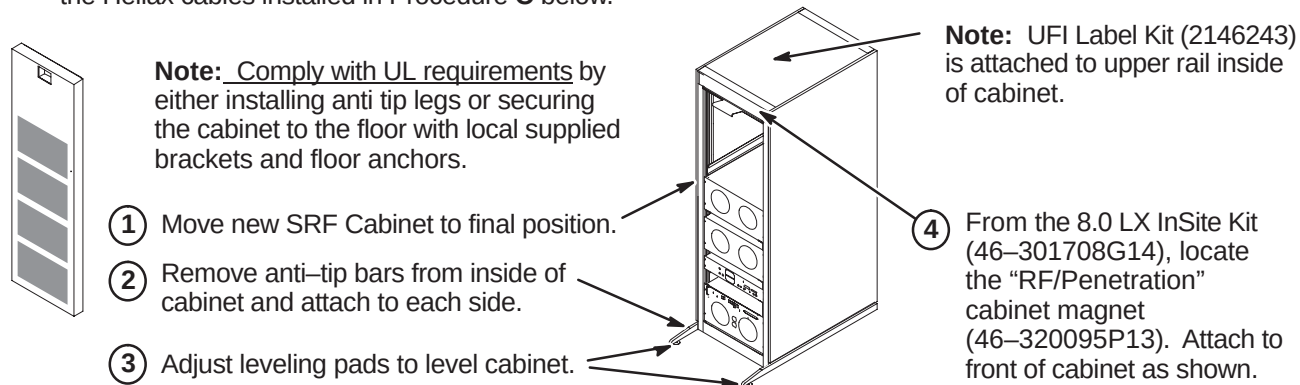
Use only the applicable SRF Cabinet Installation Instruction Direction for the site being upgraded

INSTALLATION STEPS SUMMARY

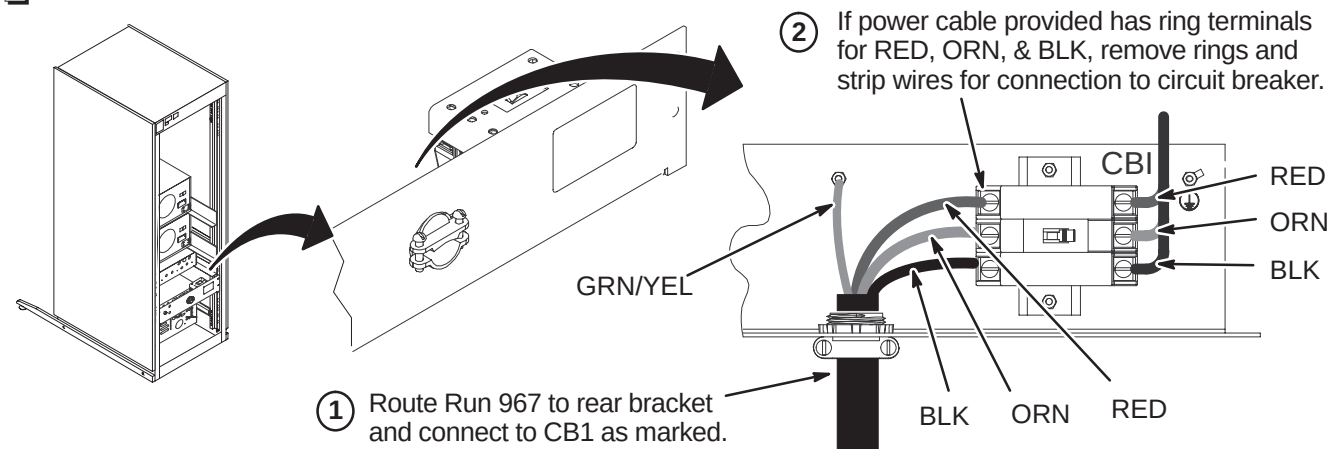
- ☐ A – Install New SRF Cabinet
 - ☐ B1 – Route/Connect Run 967 to Circuit Breaker
 - OR
 - ☐ B2 – Route/Connect Run 967 to Terminal Strip
 - ☐ C – Connect Runs 235/745 and 236/748 to RF Interface (RFI) Module
 - ☐ D – Connect Fiber Optics and Run 229
 - ☐ E – System Support Module Cable Connections
- } Use the applicable procedure.

A – INSTALL NEW SRF CABINET

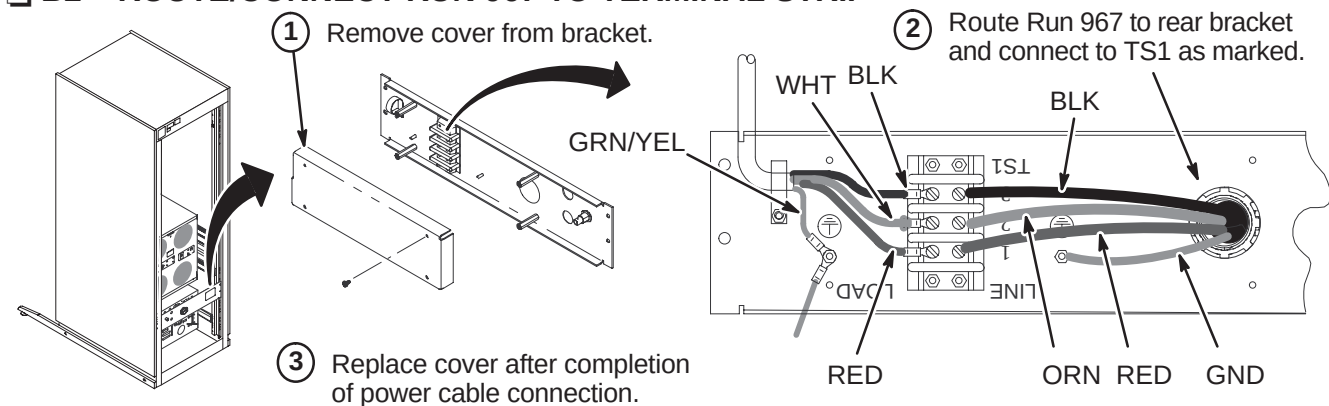
Note: The 1.5T SRF (Scaleable RF) cabinet is shown in these procedures. The 1.0T is the same except for the Helix cables installed in Procedure **C** below.



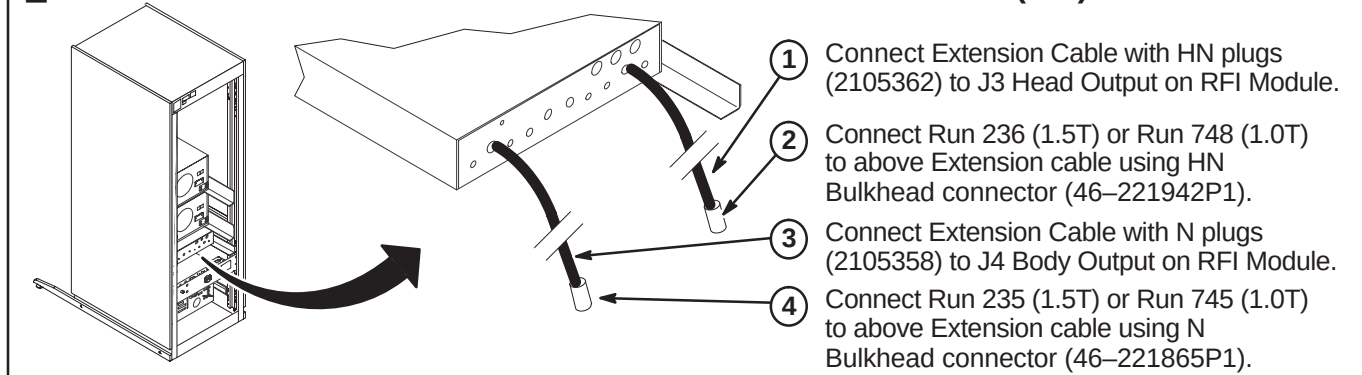
B1 – ROUTE/CONNECT RUN 967 TO CIRCUIT BREAKER



☐ B2 – ROUTE/CONNECT RUN 967 TO TERMINAL STRIP



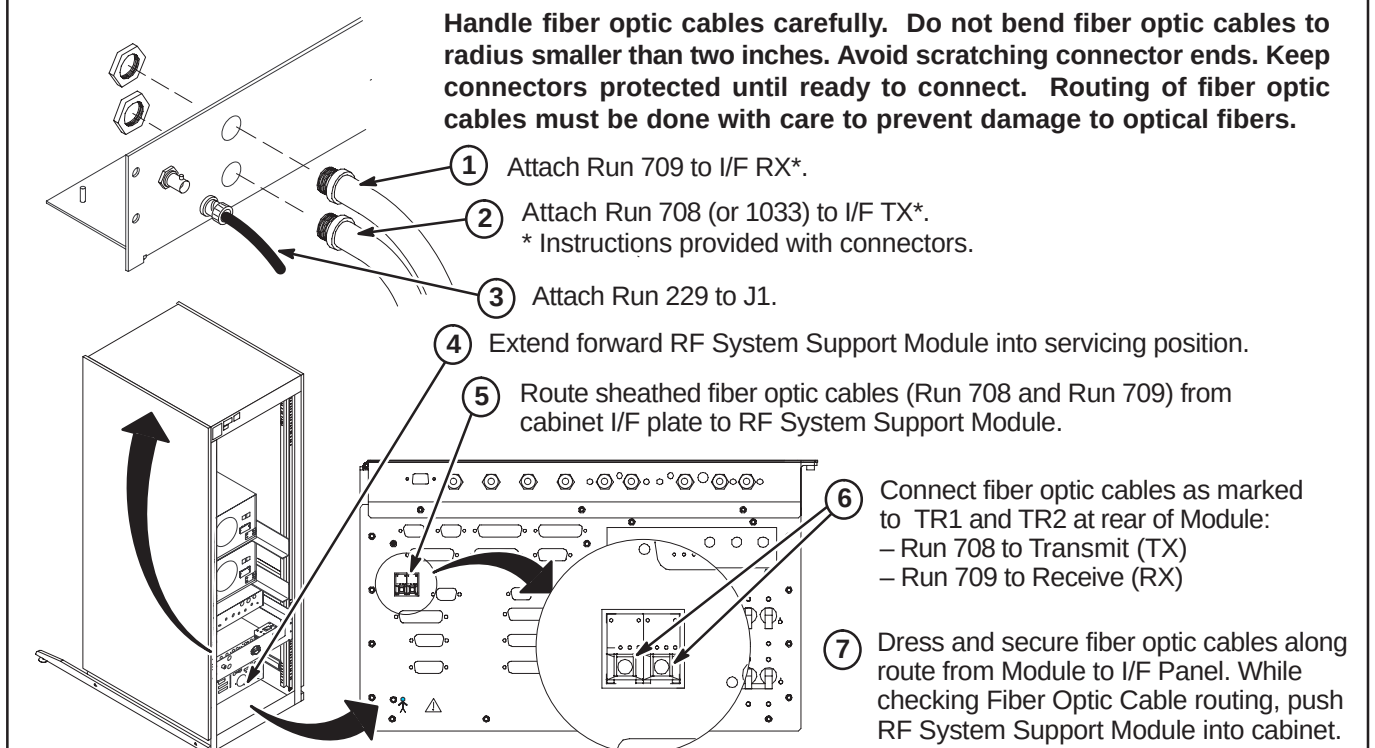
❑ C – CONNECT RUNS 235/745 AND 236/748 TO RF INTERFACE (RFI) MODULE



 D – ROUTE/CONNECT FIBER OPTICS AND RUN 229

CAUTION

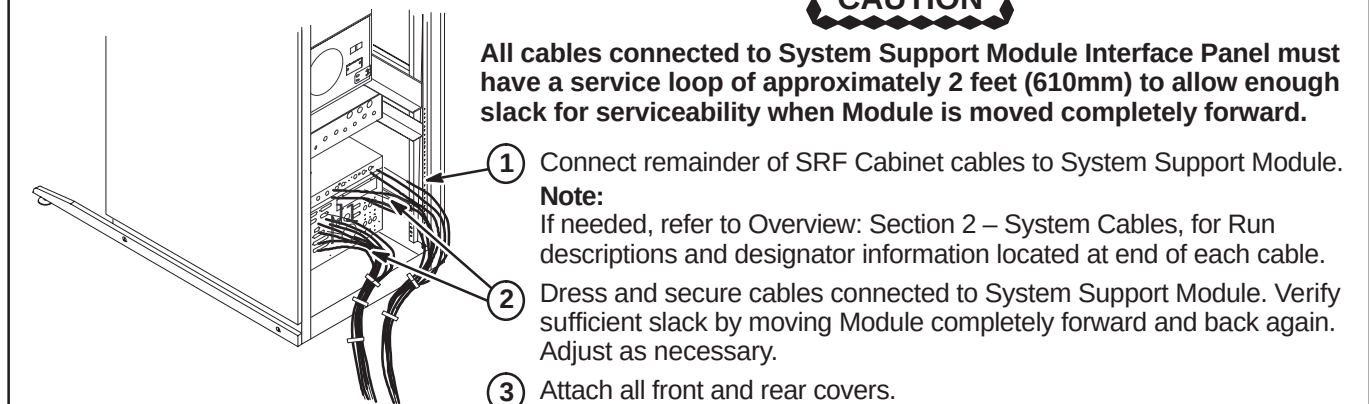
Handle fiber optic cables carefully. Do not bend fiber optic cables to radius smaller than two inches. Avoid scratching connector ends. Keep connectors protected until ready to connect. Routing of fiber optic cables must be done with care to prevent damage to optical fibers.



E – SYSTEM SUPPORT MODULE CABLE CONNECTIONS

CAUTION

All cables connected to System Support Module Interface Panel must have a service loop of approximately 2 feet (610mm) to allow enough slack for serviceability when Module is moved completely forward.



INSTALLATION STEPS SUMMARY

- A – Install New SRFD2 Cabinet
- B – Route/Connect Run 967
- C – Connect Runs 235 and 236 to RF Amp & Interface Module
- D – Connect Fiber Optics and Run 229
- E – System Support Module Cable Connections
- F – Attach InSite Cabinet Magnets

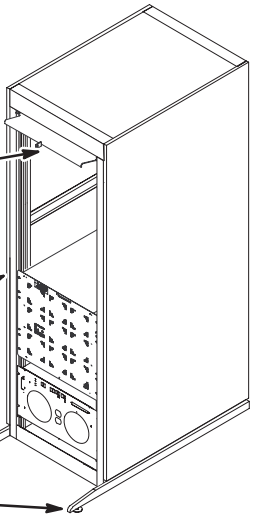
A – INSTALL NEW SRFD2 CABINET

Note: Comply with UL requirements by either installing anti tip legs or securing the cabinet to the floor with local supplied brackets and floor anchors.

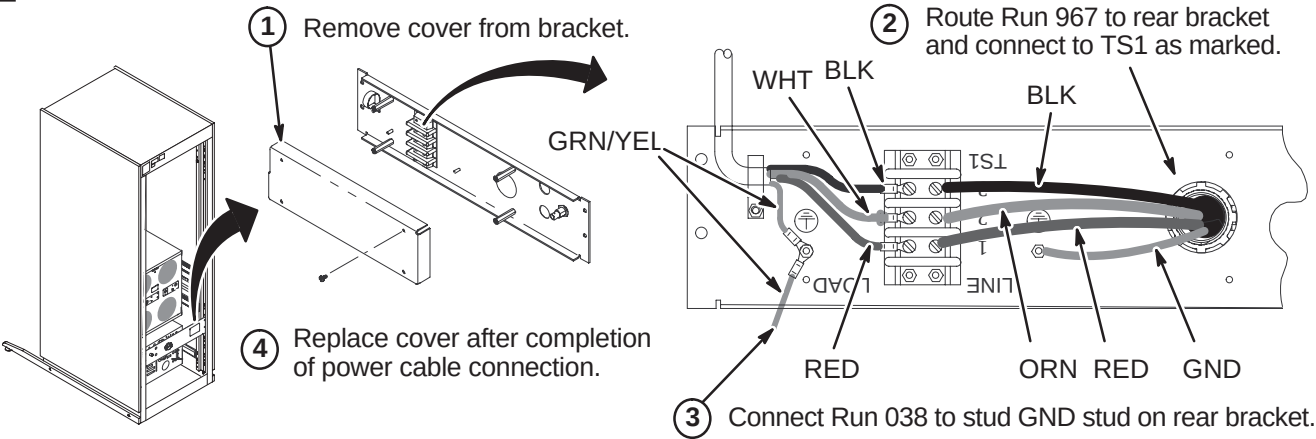


Note: UFI Label Kit (2146243) is attached to upper rail inside of cabinet.

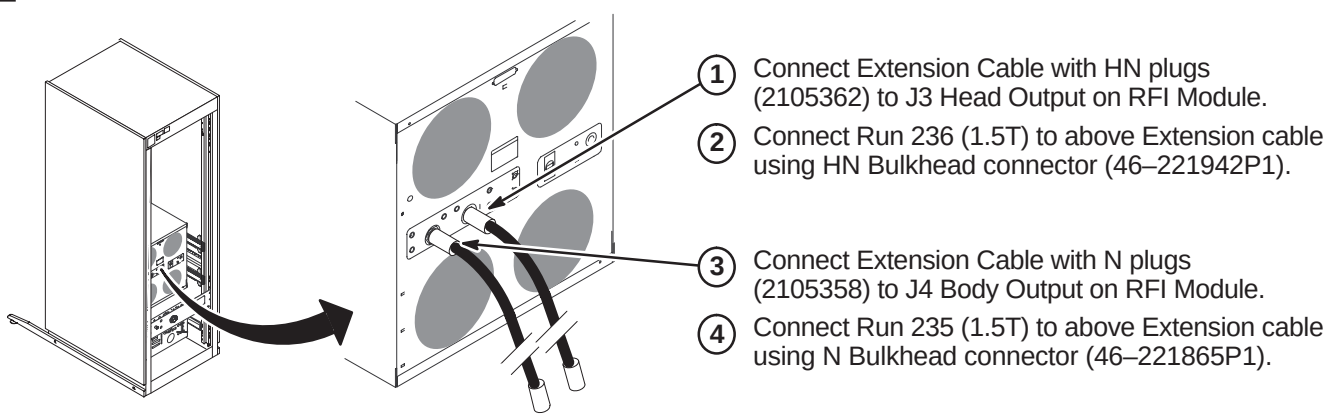
- 1 Move new SRFD2 Cabinet to final position.
- 2 Remove anti-tip bars from inside of cabinet and attach to each side.
- 3 Adjust leveling pads to level cabinet.



B – ROUTE/CONNECT RUN 967



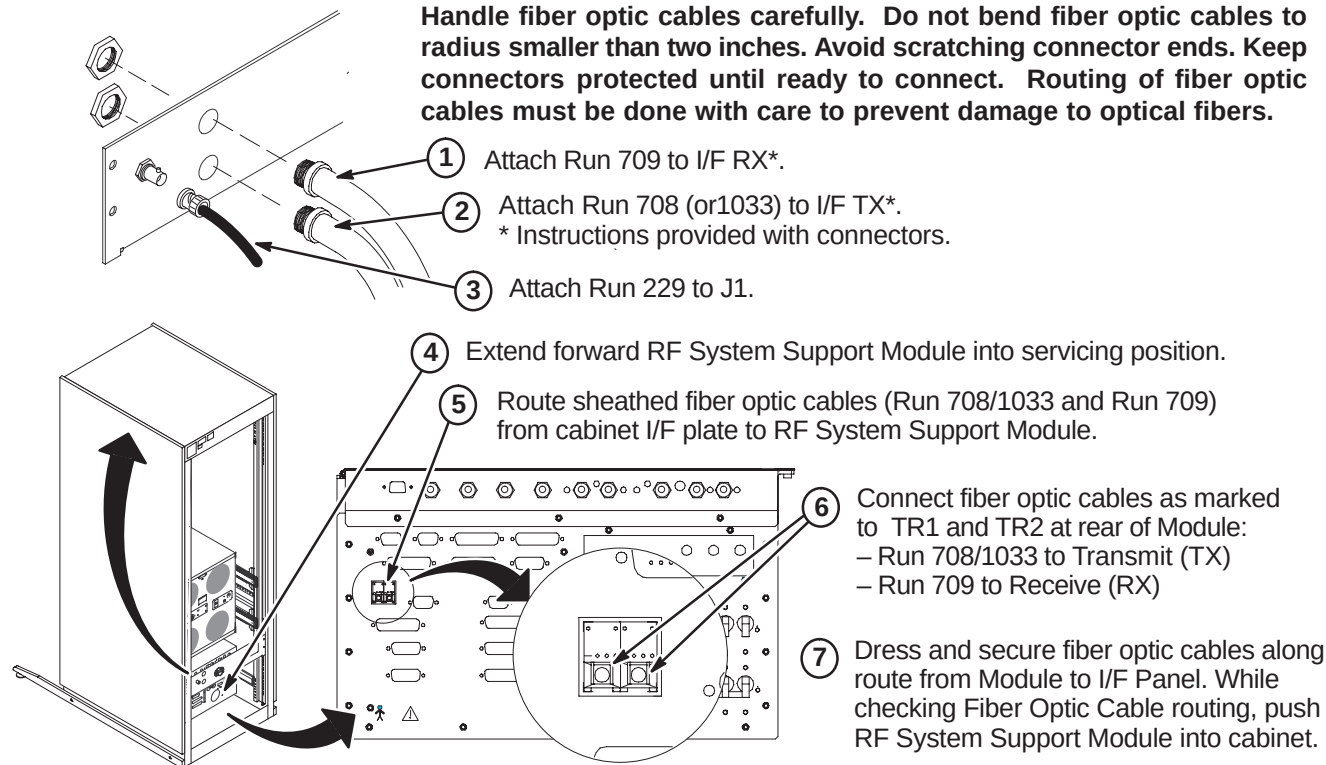
C – CONNECT RUNS 235 AND 236 TO RF AMP & INTERFACE MODULE



D – ROUTE/CONNECT FIBER OPTICS AND RUN 229

CAUTION

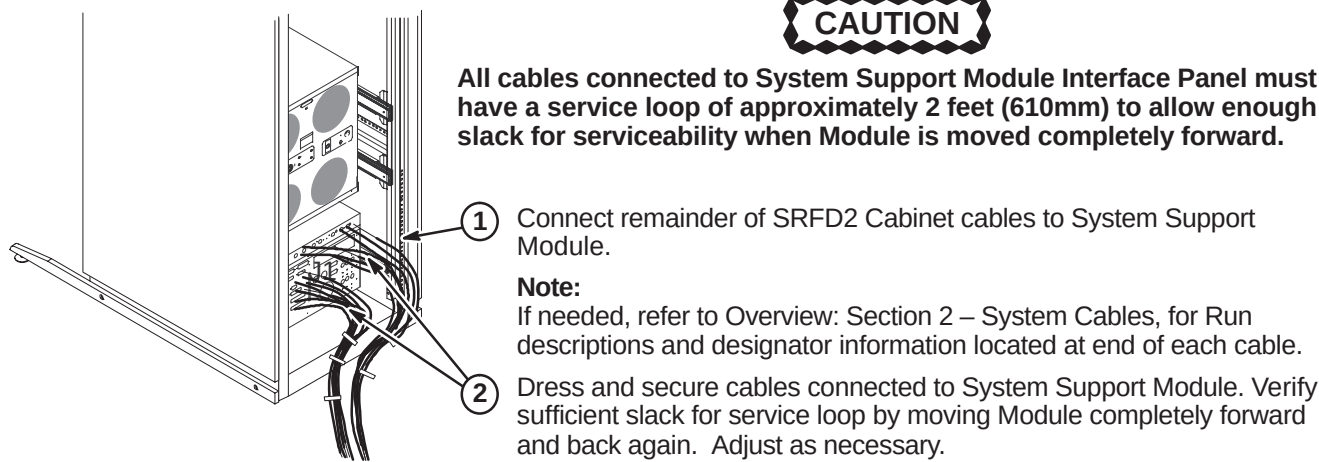
Handle fiber optic cables carefully. Do not bend fiber optic cables to radius smaller than two inches. Avoid scratching connector ends. Keep connectors protected until ready to connect. Routing of fiber optic cables must be done with care to prevent damage to optical fibers.



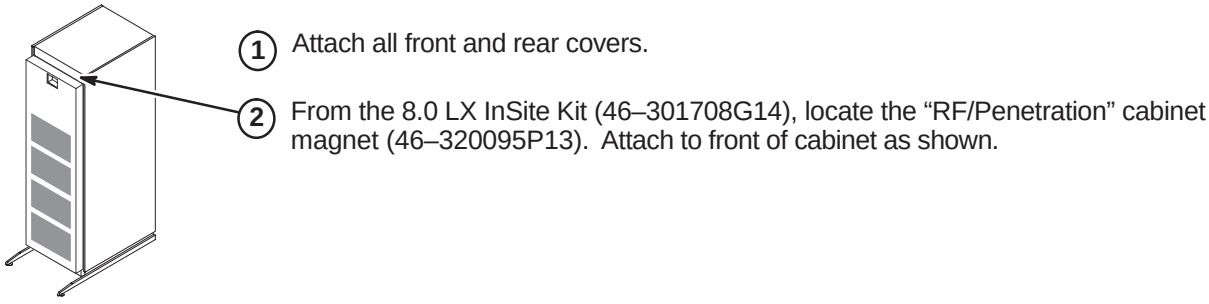
E – SYSTEM SUPPORT MODULE CABLE CONNECTIONS

CAUTION

All cables connected to System Support Module Interface Panel must have a service loop of approximately 2 feet (610mm) to allow enough slack for serviceability when Module is moved completely forward.



F – ATTACH INSITE CABINET MAGNET



INSTALLATION STEPS SUMMARY

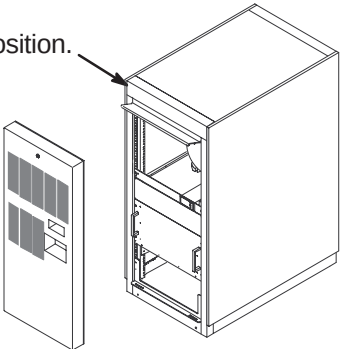
- A – Install TAC Cabinet
- B – Route/Connect Runs 976, 977, and 978 from ACGD/PDU Cabinet
- C – Route/Connect Runs 979, 980, and 981 from Penetration Panel
- D – Connect Runs 568, 611, 974, 989, 991, 992, and Vacuum to TAC Interface Panel

A – INSTALL TAC CABINET

- 1 Move new TAC Cabinet to final position.

Note: Comply with UL requirements by securing the cabinet to the floor with local supplied brackets and floor anchors.

Note: RUN 975 is a service tool cable. It will not be part of installation.

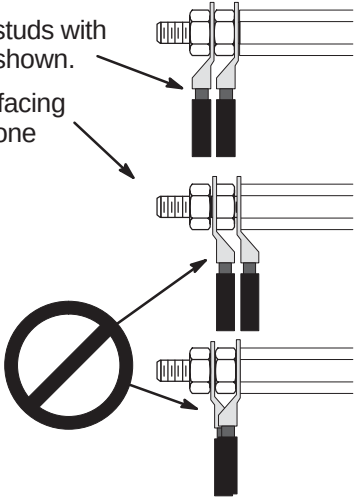


B – ROUTE/CONNECT RUNS 976, 977, AND 978 FROM ACGD/PDU CABINET

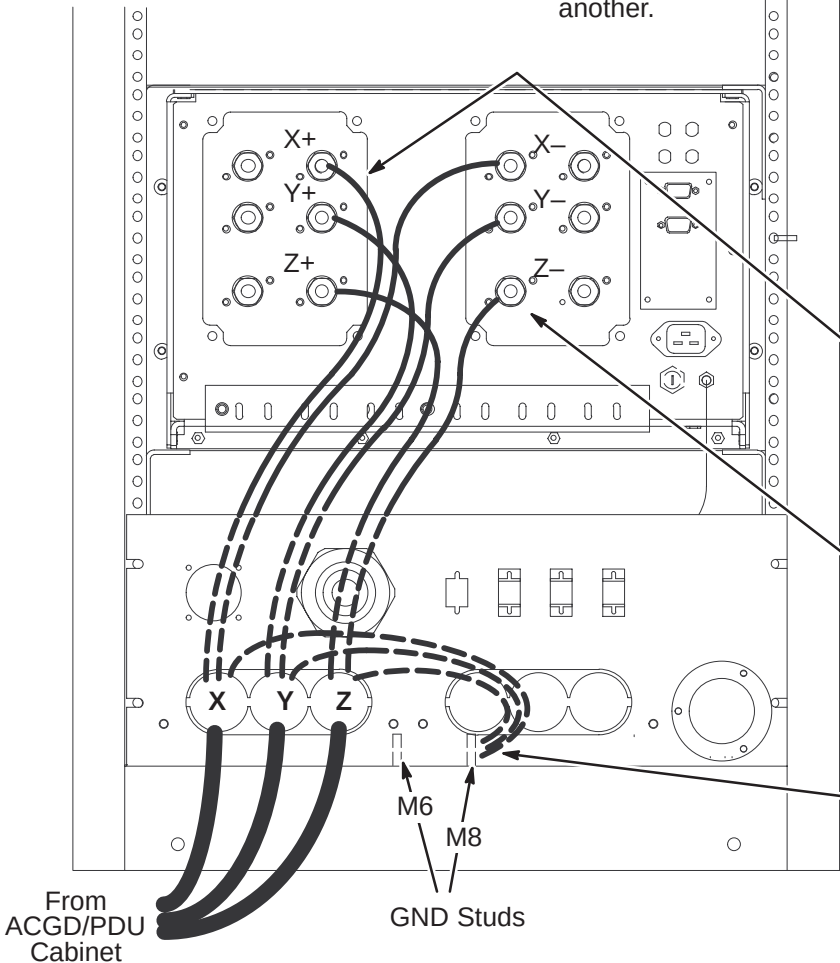
Note: To make installation easier, route and connect Run 978 (Z terminals) first, then Run 977 (Y), and last Run 976 (X).

Note: Connect cables to all studs with terminals facing outward as shown.

Do not install with terminals facing toward the module or facing one another.



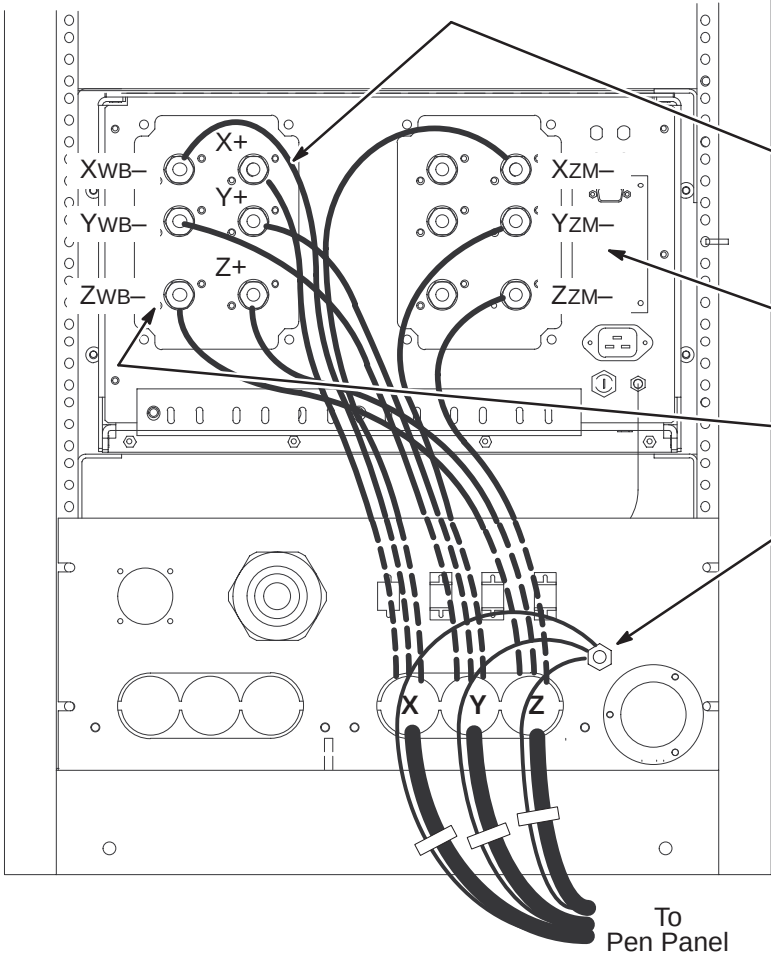
- 1 Connect X+, Y+, and Z+ cables as marked. Cables should have red tape applied towards end of cable.
- 2 Make sure X+, Y+, and Z+ labels are applied to respective "ONE" conductor cables.
- 3 Connect X-, Y-, and Z- cables as marked.
- 4 Make sure X-, Y-, and Z- labels are applied to respective "TWO" conductor cables.
- 5 Connect Ground leads from each cable to M8 GND stud located at bottom of cabinet.
- 6 Apply GND label to each "THREE" conductor cable.



C – ROUTE/CONNECT RUNS 979, 980, AND 981 FROM PENETRATION PANEL

Note: To make installation easier, route and connect Run 981 (Z terminals) first, then Run 980 (Y), and last Run 979 (X).

- 1 Connect X+, Y+, and Z+ cables as marked. Cables should have red tape applied towards end of cable.
- 2 Connect XZM-, YZM-, and ZZM- cables as marked.
- 3 Connect XWB-, YWB-, and ZWB- cables as marked.
- 4 Route and connect the three Run 988 Ground wires to GND stud on interface panel, (each ground wire has been ty-wrapped to a gradient cable, refer to Tab 1, Direction 2308287).



D – CONNECT RUNS 568, 611, 974, 989, 991, 992, AND VACUUM TO TAC I/F PANEL

- 1 Connect cables as marked.
- 2 Connect vacuum tube to J11 using supplied Chain Clamp (2293618-2) and Centering Ring (2293608). Apply vacuum grease to centering ring and outside of flanges. These parts come from the Twin Vacuum Hose Asm Kit (2295226).

Note: The end of the vacuum tube should already have been cut to length and the 1" Hose Connector (2293616) installed in the end. It's attached with a worm-drive clamp (2294010-3).

